eRAN

Inter-RAT Mobility Management in Connected Mode Feature Parameter Description

Issue 03

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Contents

1 About This Document 1
1.1 Scope 1
1.2 Intended Audience 2
1.3 Change History 2
1.4 Differences Between eNodeB Types 4
2 Overview 6
2.1 Definition 6
2.2 Benefits 6
2.3 Architecture 6
3 Inter-RAT Mobility Management Between E-UTRAN and UTRAN 7
3.1 Inter-RAT Mobility Between E-UTRAN and UTRAN 8
3.1.1 Triggering and Stopping of Inter-RAT Measurement 8
3.1.1.1 Event A2 9
3.1.1.2 Event A1 11
3.1.2 Inter-RAT Measurement 13
3.1.3 Inter-RAT Handover Triggering 14
3.1.3.1 Event B1 14
3.1.3.2 Event B2 17
3.1.4 Blind Handover 18
3.2 Service-based Inter-RAT Handover to UTRAN 19
3.2.1 Triggering and Stopping Inter-RAT Measurement 20
3.2.2 Inter-RAT Measurement 21
3.2.3 Inter-RAT Handover Triggering 21
3.3 Distance-based Inter-RAT Handover to UTRAN 22
3.3.1 Triggering and Stopping Inter-RAT Measurement 22
3.3.2 Inter-RAT Measurement 23
3.3.3 Inter-RAT Handover Triggering 23
3.4 UL-Quality-based Inter-RAT Handover to UTRAN 23
3.4.1 Triggering and Stopping Inter-RAT Measurement 24
3.4.2 Inter-RAT Measurement 24
3.4.3 Inter-RAT Handover Triggering 24
3.4.4 Blind Handover 24

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3.5 E-UTRAN to UTRAN CS/PS Steering....25
3.5.1 Inter-RAT Measurement....25
3.5.2 Blind Handover....25
3.6 Handover Decision....25
3.6.1 Basic Handover Decision....26
3.6.2 Decision Based on System Information....27
3.7 Handover Execution....27
3.7.1 Handover Policy Selection....27
3.7.2 Separate Mobility Policies to UTRAN for Multiple PLMNs....29
3.7.3 Special Signaling Retransmission Optimization....31
3.7.4 Retry and Penalty....31
3.8 Signaling Flow....31
4 RIM-Based LTE Target Cell Selection....32
4.1 RIM Procedure....33
4.2 E-UTRAN Cell Load Status....34
5 Inter-RAT Mobility Management Between E-UTRAN and GERAN....35
5.1 PS Inter-RAT Mobility Between E-UTRAN and GERAN....36
5.1.1 Triggering and Stopping Inter-RAT Measurement....36
5.1.2 Inter-RAT Measurement....36
5.1.3 Inter-RAT Handover Triggering....37
5.1.4 Blind Handover....39
5.2 Service-based Inter-RAT Handover to GERAN....40
5.2.1 Triggering and Stopping Inter-RAT Measurement....40
5.2.2 Inter-RAT Measurement....40
5.2.3 Inter-RAT Handover Triggering....40
5.3 Distance-based Inter-RAT Handover to GERAN....41
5.3.1 Triggering and Stopping Inter-RAT Measurement....41
5.3.2 Inter-RAT Measurement....41
5.3.3 Inter-RAT Handover Triggering....41
5.4 UL-Quality-based Inter-RAT Handover to GERAN....42
5.4.1 Triggering and Stopping Inter-RAT Measurement....42
5.4.2 Inter-RAT Measurement....42
5.4.3 Inter-RAT Handover Triggering....42
5.4.4 Blind Handover....42
5.5 Handover Decision....43
5.6 Handover Execution....43
5.6.1 Handover Policy Selection....43
5.6.2 Retry and Penalty....45
5.7 Signaling Procedure....45
6 Related Features....47
6.1 Features Related to PS Inter-RAT Mobility Between E-UTRAN and UTRAN....47

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6.2 Features Related to Service-based Inter-RAT Handover to UTRAN....47
6.3 Features Related to Distance-based Inter-RAT handover to UTRAN....48
6.4 Features Related to E-UTRAN to UTRAN CS/PS Steering....48
6.5 RIM-Based LTE Target Cell Selection....48
6.6 Separate Mobility Policies to UTRAN for Multiple PLMNs....49
6.7 PS Inter-RAT Mobility Between E-UTRAN and GERAN....49
6.8 Service based inter-RAT handover to GERAN....50
6.9 Distance based Inter-RAT handover to GERAN....50

7 Network Impact....51

7.1 PS Inter-RAT Mobility between E-UTRAN and UTRAN....51
7.2 Service-based Inter-RAT Handover to UTRAN....51
7.3 Distance-based Inter-RAT Handover to UTRAN....51
7.4 E-UTRAN to UTRAN CS/PS Steering....52
7.5 RIM Based LTE Target Cell Selection....52
7.6 Separate Mobility Policies to UTRAN for Multiple PLMNs....52
7.7 PS Inter-RAT Mobility between E-UTRAN and GERAN....52
7.8 Service based inter-RAT handover to GERAN....53
7.9 Distance based Inter-RAT handover to GERAN....53

8 Engineering Guidelines....54

8.1 Inter-RAT Mobility Between E-UTRAN and UTRAN....55
8.1.1 When to Use Inter-RAT Mobility Between E-UTRAN and UTRAN....55
8.1.2 Required Information....55
8.1.3 Deployment Requirements....56
8.1.4 Data Preparation....56
8.1.5 Activation....69
8.1.5.1 Using the CME to Perform Batch Configuration for Newly Deployed eNodeBs....70
8.1.5.2 Using the CME to Perform Batch Configuration for Existing eNodeBs....71
8.1.5.3 Using the CME to Perform Single Configuration....72
8.1.5.4 Using MML Commands....73
8.1.6 Activation Observation....73
8.1.7 Reconfiguration....74
8.1.8 Deactivation....74
8.1.9 Performance Monitoring....75
8.1.10 Parameter Optimization....76
8.1.11 Troubleshooting....83
8.2 Service-based Inter-RAT Handover to UTRAN....84
8.2.1 When to Use Service-based Inter-RAT Handover to UTRAN....84
8.2.2 Required Information....85
8.2.3 Deployment Requirements....85
8.2.4 Data Preparation....85
8.2.5 Activation....88
8.2.5.1 Using the CME to Perform Batch Configuration for Newly Deployed eNodeBs....88

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8.2.5.2 Using the CME to Perform Batch Configuration for Existing eNodeBs 90
8.2.5.3 Using the CME to Perform Single Configuration 91
8.2.5.4 Using MML Commands 92
8.2.6 Activation Observation 92
8.2.7 Reconfiguration 92
8.2.8 Deactivation 93
8.2.9 Performance Monitoring 93
8.2.10 Parameter Optimization 94
8.2.11 Troubleshooting 95
8.3 Distance-based Inter-RAT Handover to UTRAN 95
8.3.1 When to Use Distance-based Inter-RAT Handover to UTRAN 95
8.3.2 Required Information 96
8.3.3 Deployment Requirements 96
8.3.4 Data Preparation 96
8.3.5 Activation 98
8.3.6 Activation Observation 100
8.3.7 Reconfiguration 101
8.3.8 Deactivation 101
8.3.9 Performance Monitoring 101
8.3.10 Parameter Optimization 102
8.3.11 Troubleshooting 104
8.4 UL-Quality-based Inter-RAT Handover to UTRAN 104
8.4.1 When to Use UL-Quality-based Inter-RAT Handover to UTRAN 104
8.4.2 Required Information 104
8.4.3 Deployment Requirements 104
8.4.4 Data Preparation 105
8.4.5 Activation 105
8.4.6 Activation Observation 108
8.4.7 Reconfiguration 109
8.4.8 Deactivation 109
8.4.9 Performance Monitoring 109
8.4.10 Parameter Optimization 110
8.4.11 Troubleshooting 111
8.5 E-UTRAN to UTRAN CS/PS Steering 111
8.5.1 When to Use E-UTRAN to UTRAN CS/PS Steering 111
8.5.2 Required Information 112
8.5.3 Deployment Requirements 112
8.5.4 Data Preparation 112
8.5.5 Activation 114
8.5.6 Activation Observation 117
8.5.7 Reconfiguration 117
8.5.8 Deactivation 118

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8.5.9 Performance Monitoring....118
8.5.10 Parameter Optimization....119
8.5.11 Troubleshooting....119
8.6 Separate Mobility Policies to UTRAN for Multi PLMN....119
8.6.1 When to Use Separate Mobility Policies to UTRAN for Multi PLMN....119
8.6.2 Required Information....119
8.6.3 Deployment Requirements....119
8.6.4 Data Preparation....120
8.6.5 Activation....122
8.6.6 Activation Observation....125
8.6.7 Reconfiguration....125
8.6.8 Deactivation....125
8.6.9 Performance Monitoring....126
8.6.10 Parameter Optimization....126
8.6.11 Troubleshooting....126
8.7 RIM Based LTE Target Cell Selection....126
8.7.1 When to Use RIM Based LTE Target Cell Selection....126
8.7.2 Required Information....126
8.7.3 Requirements....126
8.7.4 Data Preparation....127
8.7.5 Activation....128
8.7.6 Activation Observation....131
8.7.7 Reconfiguration....131
8.7.8 Deactivation....131
8.7.9 Performance Monitoring....132
8.7.10 Parameter Optimization....133
8.7.11 Troubleshooting....133
8.8 PS Inter-RAT Mobility Between E-UTRAN and GERAN....135
8.8.1 When to Use PS Inter-RAT Mobility Between E-UTRAN and GERAN....135
8.8.2 Required Information....135
8.8.3 Requirements....136
8.8.4 Data Preparation....136
8.8.5 Activation....150
8.8.5.1 Using the CME to Perform Batch Configuration for Newly Deployed eNodeBs....151
8.8.5.2 Using the CME to Perform Batch Configuration for Existing eNodeBs....152
8.8.5.3 Using the CME to Perform Single Configuration....153
8.8.5.4 Using MML Commands....154
8.8.6 Activation Observation....154
8.8.7 Reconfiguration....155
8.8.8 Deactivation....155
8.8.9 Performance Monitoring....156
8.8.10 Parameter Optimization....156

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8.8.11 Troubleshooting....160
8.9 Service-based Inter-RAT Handover to GERAN....160
8.9.1 When to Use Service-based Inter-RAT Handover to GERAN....160
8.9.2 Required Information....161
8.9.3 Deployment Requirements....161
8.9.4 Data Preparation....161
8.9.5 Activation....162
8.9.5.1 Using the CME to Perform Batch Configuration for Newly Deployed eNodeBs....162
8.9.5.2 Using the CME to Perform Batch Configuration for Existing eNodeBs....164
8.9.5.3 Using the CME to Perform Single Configuration....165
8.9.5.4 Using MML Commands....166
8.9.6 Activation Observation....167
8.9.7 Reconfiguration....167
8.9.8 Deactivation....167
8.9.9 Performance Monitoring....168
8.9.10 Parameter Optimization....168
8.9.11 Troubleshooting....169
8.10 Distance-based Inter-RAT Handover to GERAN....169
8.10.1 When to Use Distance based Inter-RAT handover to GERAN....169
8.10.2 Required Information....170
8.10.3 Deployment Requirements....170
8.10.4 Data Preparation....170
8.10.5 Activation....172
8.10.6 Activation Observation....175
8.10.7 Reconfiguration....175
8.10.8 Deactivation....175
8.10.9 Performance Monitoring....176
8.10.10 Parameter Optimization....176
8.10.11 Troubleshooting....178
8.11 UL-Quality-based Inter-RAT Handover to GERAN....178
8.11.1 When to Use UL-Quality-based Inter-RAT Handover to GERAN....178
8.11.2 Required Information....179
8.11.3 Deployment Requirements....179
8.11.4 Data Preparation....179
8.11.5 Activation....180
8.11.6 Activation Observation....183
8.11.7 Reconfiguration....183
8.11.8 Deactivation....183
8.11.9 Performance Monitoring....184
8.11.10 Parameter Optimization....184
8.11.11 Troubleshooting....185
9 Parameters 186

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10 Counters 312

11 Glossary 375

12 Reference Documents 376

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1 About This Document

1.1 Scope

This document describes inter-RAT mobility management in connected mode, including its technical principles, related features, network impact, and engineering guidelines.

This document covers the following features:

- LOFD-001019 PS Inter-RAT Mobility between E-UTRAN and UTRAN
- LOFD-001043 Service based inter-RAT handover to UTRAN
- LOFD-001072 Distance based Inter-RAT handover to UTRAN
- LOFD-001078 E-UTRAN to UTRAN CS/PS Steering
- LOFD-070216 Separate Mobility Policies to UTRAN for Multi PLMN
- LOFD-070203 RIM Based LTE Target Cell Selection
- LOFD-001020 PS Inter-RAT Mobility between E-UTRAN and GERAN
- LOFD-001046 Service based inter-RAT handover to GERAN
- LOFD-001073 Distance based Inter-RAT handover to GERAN

Any managed objects (MOs), parameters, alarms, or counters described herein correspond to the software release delivered with this document. Any future updates will be described in the product documentation delivered with future software releases.

This document applies only to LTE FDD. Any "LTE" in this document refers to LTE FDD, and "eNodeB" refers to LTE FDD eNodeB.

This document applies to the following types of eNodeBs.

eNodeB Type	Model
Macro	3900 series eNodeB
Micro	BTS3202E and BTS3203E
LampSite	DBS3900

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1.2 Intended Audience

This document is intended for personnel who:

- Need to understand the features described herein
- Work with Huawei products

1.3 Change History

This section provides information about the changes in different document versions. There are two types of changes:

- Feature change

  Changes in features and parameters of a specified version as well as the affected entities.
- Editorial change

  Changes in wording or addition of information and any related parameters affected by editorial changes. Editorial change does not specify the affected entities.

eRAN7.0 03 (2014-09-30)

This issue includes the following changes.

Change Type	Change Description	Parameter Change
Feature change	None	None
Editorial change	Changed blind handling to blind handover. For details, see descriptions about blind handover in this document.	None

eRAN7.0 02 (2014-07-25)

This issue includes the following changes.

Change Type	Change Description	Parameter Change	Affected Entity
Feature change	Added the description of special signaling retransmission optimization. For details, see **3.7.3 Special Signaling Retransmission Optimization**.	Added the **ENodeBAlgoSwitch**. ***HighLoadNetOptSwitch*** parameter.	Macro/Micro/Lampsite

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Change Type	Change Description	Parameter Change	Affected Entity
Editorial change	Revised descriptions about inter-RAT measurement object selection in inter-RAT handovers. For details, see **3.1.2 Inter-RAT Measurement**.	None	N/A

eRAN7.0 01 (2014-04-26)

This issue does not include any changes.

eRAN7.0 Draft C (2014-03-07)

This issue includes the following changes.

Change Type	Change Description	Parameter Change	Affected Entity
Feature change	None	None	N/A
Editorial change	Added **1.4 Differences Between eNodeB Types**, which describes differences in feature support between eNodeB types.	None	N/A

eRAN7.0 Draft B (2014-02-28)

This issue includes the following changes.

Change Type	Change Description	Parameter Change
Feature change	The following features can be batch activated: ● LOFD-001019 PS Inter-RAT Mobility between E-UTRAN and UTRAN ● LOFD-001043 Service based inter-RAT handover to UTRAN ● LOFD-001020 PS Inter-RAT Mobility between E-UTRAN and GERAN ● LOFD-001046 Service based inter-RAT handover to GERAN	None
Editorial change	None	None

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eRAN7.0 Draft A (2014-01-20)

This is a draft for eRAN7.0.

1.4 Differences Between eNodeB Types

Features Support by Macro, Micro, and LampSite eNodeBs

Feature ID	Feature Name	Supported by Macro eNodeBs	Supported by Micro eNodeBs	Supported by LampSite eNodeBs
LOFD-001019	PS Inter-RAT Mobility between E-UTRAN and UTRAN	Yes	Yes	Yes
LOFD-001043	Service based inter-RAT handover to UTRAN	Yes	Yes	Yes
LOFD-001072	Distance based inter-RAT handover to UTRAN	Yes	No	No
LOFD-001078	E-UTRAN to UTRAN CS/PS Steering	Yes	Yes	Yes
LOFD-070216	Separate Mobility Policies to UTRAN for Multi PLMN	Yes	Yes	Yes
LOFD-070203	RIM Based LTE Target Cell Selection	Yes	Yes	Yes
LOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN	Yes	Yes	Yes
LOFD-001046	Service based inter-RAT handover to GERAN	Yes	Yes	Yes
LOFD-001073	Distance based inter-RAT handover to GERAN	Yes	No	No

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Function Implementation in Macro, Micro, and LampSite eNodeBs

Each feature is implemented in the same way on micro, macro, and LampSite sites.

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2 Overview

2.1 Definition

Inter-RAT mobility management involves mobility management between E-UTRAN and UTRAN and mobility management between E-UTRAN and GERAN.

Mobility management between E-UTRAN and UTRAN refers to management of UEs in connected mode during handovers from an E-UTRAN cell to a UTRAN cell. Handover policies from E-UTRAN to UTRAN involve PS handover, single radio voice call continuity (SRVCC), and redirection.

Mobility management between E-UTRAN and GERAN refers to management of UEs in connected mode during handovers from an E-UTRAN cell to a GERAN cell. Handover policies from E-UTRAN to UTRAN involve PS handover, SRVCC, redirection, and cell change order with or without network assisted cell change (CCO/NACC).

For details about handover policy selection, see *Mobility Management in Connected Mode*.

2.2 Benefits

Inter-RAT mobility management in connected mode provides the following benefits:

- Ensures the continuity of radio network coverage and provides uninterrupted communication services for UEs.
- Provides a means to transfer UEs and supports flexible networking to meet service steering and load balancing requirements of operators.

2.3 Architecture

For the network architecture for inter-RAT mobility management in connected mode, see *Overview of Mobility Management in Connected Mode*.

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3 Inter-RAT Mobility Management Between E-UTRAN and UTRAN

Inter-RAT mobility management between E-UTRAN and UTRAN can be implemented in multiple ways. This chapter describes the following features and functions:

- LOFD-001019 PS Inter-RAT Mobility between E-UTRAN and UTRAN
- LOFD-001043 Service based inter-RAT handover to UTRAN
- LOFD-001072 Distance based Inter-RAT handover to UTRAN
- UL-quality-based inter-RAT handover to UTRAN
- LOFD-001078 E-UTRAN to UTRAN CS/PS Steering
- LOFD-070216 Separate Mobility Policies to UTRAN for Multi PLMN

The triggering conditions and processes for the preceding features and functions are different, but the following general processes apply:

1. Target cell/frequency selection

   For a measurement, the eNodeB generates a candidate cell list based on inter-RAT measurement results.

   For a blind handover, the eNodeB selects a blind handover target based on the blind handover priority or frequency priority of neighboring cells.
2. Handover decision

   In the handover decision phase, the eNodeB checks the candidate cell list. Based on the check result, the eNodeB determines whether a handover needs to be initiated and, if so, to which cell the UE is to be handed over.
3. Handover execution

   The eNodeB controls the UE to be handed over from the serving cell to the target cell.

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3.1 Inter-RAT Mobility Between E-UTRAN and UTRAN

This section describes the optional feature LOFD-001019 PS Inter-RAT Mobility between E-UTRAN and UTRAN, which is implemented by coverage-based handovers from E-UTRAN to UTRAN. For engineering guidelines, see **8.1 Inter-RAT Mobility Between E-UTRAN and UTRAN**. The coverage-based handover from E-UTRAN to UTRAN is enabled or disabled by the **UtranRedirectSwitch** or **UtranPsHoSwitch** switch under the **ENodeBAlgoSwitch**.***HoModeSwitch*** parameter.

During a coverage-based handover from E-UTRAN to UTRAN, measurements or blind handover can be triggered and stopped by different events, as shown in **Table 3-1**.

Table 3-1 Events for triggering/stopping measurements or blind handover during a coverage-based handover from E-UTRAN to UTRAN

Procedure	Subprocedure	Triggering Events	Stopping Events
Measurement	Inter-RAT measurement	A2	A1
	Inter-RAT handover	B1 or B2	-
Blind handover	-	A2	A1

The eNodeB delivers measurement configuration for event A2 for inter-RAT measurement or event A2 for blind handover as follows:

- If the UE is in RRC-connected mode, the eNodeB delivers measurement configuration for both event A2 for inter-RAT measurement and event A2 for blind handover.
  - If the signal quality in the serving cell is lower than the specified threshold, the UE reports event A2 for inter-RAT measurement. After receiving the report, the eNodeB delivers an inter-RAT measurement configuration.
  - If the signal quality in the serving cell further deteriorates and the eNodeB does not perform a handover for the UE, the UE reports event A2 for blind handover. After receiving the report, the eNodeB considers the serving cell no longer capable of poviding services for the UE and performs blind handover.
- In the following scenarios, the eNodeB delivers measurement configuration for event A2 only for blind handover:
  - The UE does not support inter-RAT measurements.
  - The configured event A2 threshold for inter-RAT measurements is less than or equal to that for blind handover.

3.1.1 Triggering and Stopping of Inter-RAT Measurement

Inter-RAT measurements are triggered by event A2 and stopped by event A1.

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3.1.1.1 Event A2

Event A2 indicates that the signal quality of the serving cell is lower than a threshold.

Event A2 for Inter-RAT Measurement

The entering and leaving conditions of event A2 are as follows, with the same principle as that for event A2 for inter-frequency measurement described in *Intra-RAT Mobility Management in Connected Mode*. For details about event A2, see section 5.5.4.3 "Event A2 (Serving becomes worse than threshold)" in 3GPP TS 36.331 V10.1.0.

- Entering condition: *Ms* + *Hys* < *Thresh*

  If the entering condition is met, the UE reports an event A2. After receiving the event A2 report, the eNodeB delivers a measurement configuration to start inter-RAT measurement.

- Leaving condition: *Ms* - *Hys* > *Thresh*

  If the leaving condition of event A2 is met, the UE no longer reports event A2.

The variables in the preceding formulas are described in **Table 3-2** and **Table 3-3**. Thresholds related to event A2 vary according to events that are used to trigger inter-RAT handovers.

Table 3-2 Parameters related to event A2

Variables in the Formula	Parameter Name	Parameter ID	Parameter Description
Ms	-	-	*Ms* is the measurement result of the serving cell. The measurement result type is specified by the **InterRatHoComm.*InterRatHoA1A2TrigQuan*** parameter.
Hys	InterRAT A1A2 hysteresis	**InterRatHoCommGroup.*InterRatHoA1A2Hyst***	This parameter is set for each QCI.

Table 3-3 Thresholds related to event A2

Parameter Name of the Threshold	Parameter ID of the Threshold	Parameter Description
InterRAT A2 RSRP trigger threshold	**InterRatHoCommGroup.*InterRatHoA2ThdRsrp***	This parameter is set for each QCI. The **InterRatHoComm.*InterRatHoA1A2TrigQuan*** parameter determines which threshold is to be used. ● If **InterRatHoComm.*InterRatHoA1***

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Parameter Name of the Threshold	Parameter ID of the Threshold	Parameter Description
InterRAT A2 RSRQ trigger threshold	**InterRatHoCommGroup**.***InterRatHoA2ThdRsrq***	***A2TrigQuan*** is set to **BOTH**, two A2 events are reported, one RSRP-based and the other RSRQ-based. When the entering condition for either of the A2 events is met, the UE reports the A2 event. ● If **InterRatHoComm**.***InterRatHoA1A2TrigQuan*** is set to **RSRP** or **RSRQ**, the UE reports an A2 event only when the entering condition for the related A2 event is met.

Table 3-4 lists other parameters related to the event A2 report.

Table 3-4 Other parameters related to the event A2 report

Parameter Name	Parameter ID	Parameter Description
InterRAT A1A2 time to trigger	**InterRatHoCommGroup**.***InterRatHoA1A2TimeToTrig***	This parameter is set for each QCI. The UE reports the cell measurement result associated with the event to the eNodeB only when the entering condition is continuously met during a specified period defined by the time-to-trigger parameter, therefore preventing unnecessary handovers.
InterRat A1A2 measurement trigger quantity	**InterRatHoComm**.***InterRatHoA1A2TrigQuan***	This is a common parameter for inter-RAT handover and is set for each eNodeB. This parameter specifies the Ms type and triggering quantity for events A1 and A2.

Event A2 for RAT-based Measurement

Event A2 specific to a target RAT can be configured on the eNodeB. This ensures that the eNodeB can preferentially perform a coverage-based handover to a UTRAN or GERAN cell.

When **EmcInterFreqBlindHoSwitch** under the **ENodeBAlgoSwitch**.***HoAlgoSwitch*** parameter is turned off, blind handover is performed for coverage-based inter-RAT handover.

The variables Ms and Hys for event A2 for RAT-based measurement are the same as those for event A2 for inter-frequency measurement. Table 3-5 lists the thresholds related to event A2 for RAT-based measurement.

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Table 3-5 Thresholds related to event A2 for RAT-based measurement

RAT	Parameter ID of the Threshold Offset	Parameter ID of the Threshold	Parameter Description
UTRAN	**CnOperatorHoCfg**.***UtranA2ThdRsrpOffset***	**InterRatHoCommGroup**.***InterRatHoA2ThdRsrp***	The threshold for event A2 is the sum of the RSRP threshold (set by **InterRatHoCommGroup**.***InterRatHoA2ThdRsrp***) and RSRP offset (set by **CnOperatorHoCfg**.***UtranA2ThdRsrpOffset*** or **CnOperatorHoCfg**.***GeranA2ThdRsrpOffset***). Only the RSRP threshold for event A2 is accompanied by an offset, while the RSRQ threshold for event A2 is not.
GERAN	**CnOperatorHoCfg**.***GeranA2ThdRsrpOffset***		

The eNodeB delivers measurement configurations for event A2 for RAT-based measurement as follows:

- If **CnOperatorHoCfg**.***UtranA2ThdRsrpOffset*** and **CnOperatorHoCfg**.***GeranA2ThdRsrpOffset*** are set to different values, the eNodeB delivers two measurement configurations for events A2, with one corresponding to UTRAN and the other corresponding to GERAN. After a UE reports an event A2 related to an RAT, the eNodeB delivers the measurement configuration of the related RAT.
- If **CnOperatorHoCfg**.***UtranA2ThdRsrpOffset*** and **CnOperatorHoCfg**.***GeranA2ThdRsrpOffset*** are set to the same value, the eNodeB delivers one measurement configuration for event A2, without distinguishing between UTRAN and GERAN. After a UE reports an event A2, the eNodeB delivers two measurement configurations, one for UTRAN and the other for GERAN.

Pay attention to the following points related to event A2 for RAT-based measurement:

- If one threshold offset is set to -100, the eNodeB does not trigger inter-RAT measurement.
- If both threshold offsets are set to -100, the effect is the same as for those set to 0. In this case, the eNodeB sends an inter-RAT event A2, with the threshold value set to the event A2 threshold.
- If the calculated RSRP threshold is lower than the protocol-specified minimum value, the protocol-specified minimum value is used as the RSRP threshold.
- If the calculated RSRP threshold is higher than the protocol-specified maximum value, the protocol-specified maximum value is used as the RSRP threshold.

3.1.1.2 Event A1

Event A1 indicates that the signal quality of the serving cell is higher than a threshold.

The entering and leaving conditions of event A1 are as follows, with the same principle as that for event A1 for inter-frequency measurement described in *Intra-RAT Mobility*

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Management in Connected Mode. For details about event A1, see section 5.5.4.2 "Event A1 (Serving becomes better than threshold)" in 3GPP TS 36.331 V10.1.0.

- Entering condition: *Ms* - *Hys* > *Thresh*

  When event A1 is triggered, the eNodeB receives an event A1 from the UE.
- Leaving condition: *Ms* + *Hys* < *Thresh*

  When the leaving condition of event A1 is met, the eNodeB no longer reports event A1.

The variables in the formulas for event A1 are the same as those in the formulas for event A2. **Table 3-6** describes thresholds related to event A1.

Table 3-6 Thresholds related to event A1

Parameter Name of the Threshold	Parameter ID of the Threshold	Parameter Description
InterRAT A1 RSRP trigger threshold	**InterRatHoCommGroup.*InterRatHoA1ThdRSRP***	This parameter is set for each QCI. The **InterRatHoComm.*InterRatHoA1A2TrigQuan*** parameter determines which threshold is to be used. ● If **InterRatHoComm.*InterRatHoA1A2TrigQuan*** is set to **BOTH**, two A1 events are reported, with one RSRP-based and the other RSRQ-based. The eNodeB stops coverage-based inter-RAT measurement only when both RSRP-based and RSRQ-based measurement results meet the Entering condition. ● If **InterRatHoComm.*InterRatHoA1A2TrigQuan*** is set to **RSRP** or **RSRQ**, the UE reports an A1 event only when the entering condition for the related A1 event is met.
InterRAT A1 RSRQ trigger threshold	**InterRatHoCommGroup.*InterRatHoA1ThdRSRQ***	

Other parameters related to the event A1 report are the same as those related to event A2 report, as listed in **Table 3-4**.

To ensure that coverage-based inter-RAT measurements can be stopped based on event A1, the event A1 threshold must be greater than the event A2 threshold. If the **UtranA2ThdRsrpOffset** or **GeranA2ThdRsrpOffset** parameter is specified, the event A1 threshold must be greater than or equal to the event A2 threshold plus the parameter value.

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3.1.2 Inter-RAT Measurement

Table 3-7 lists the parameters in the Measurement Configuration message exchanged during coverage-based E-UTRAN to UTRAN handovers. 3.1.3 Inter-RAT Handover Triggering lists information contained in the reporting configuration.

Table 3-7 Parameters in the E-UTRAN-to-UTRAN handover measurement configuration

Measurement Configuration	Parameter Name	Parameter ID	Parameter Description
Measurement objects	Downlink UARFCN	**UtranNFreq**.***UtranDlArfcn***	This is a cell-level parameter. If an inter-RAT neighboring cell is a multi-band cell, the eNodeB calculates and delivers a UARFCN based on the band supporting capability of the UE.
	Frequency offset	**UtranNFreq**.***OffsetFreq***	This is a frequency-level parameter.
UTRAN FDD filter coefficients	UTRAN RSCP filter coefficient	**HoMeasComm**.***UtranFilterCoeffRSCP***	This is a common parameter for inter-RAT handover and is set for each eNodeB. This is an eNodeB-level common parameter for UTRAN measurement.
	UTRAN ECN0 filter coefficient	**HoMeasComm**.***UtranFilterCoeffECN0***	
UTRAN TDD filter coefficient	UTRAN RSCP filter coefficient	**HoMeasComm**.***UtranFilterCoeffRSCP***	
Measurement gap configuration	GAP measurement pattern	**HoMeasComm**.***GAPPatternType***	

The parameters related to measurement object selection are described as follows:

- The frequency measurement priority is specified by the **UtranNFreq**.***ConnFreqPriority*** parameter. A larger value indicates a higher priority. The eNodeB selects the frequencies for measurement in descending order of frequency priority till the number of selected frequencies reaches the maximum. If all frequencies with the same priority are selected and the number of frequencies for measurement exceeds the maximum, the eNodeB randomly selects frequencies with this priority till the number of selected priorities reaches the maximum.
- The eNodeB delivers neighboring cells based on neighboring UTRAN cell configurations. The neighboring cell selection and frequency selection are similar to each other.

  The cell measurement priority is specified by the **UtranNCell**.***CellMeasPriority*** parameter.

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- For details about the maximum number of frequencies and the maximum number of neighboring cells in a measurement configuration message, see *Overview of Mobility Management in Connected Mode*.

If the optional feature LOFD-001078 E-UTRAN to UTRAN CS/PS Steering is enabled, the UTRAN frequency measurement priorities will change. For details, see **3.5 E-UTRAN to UTRAN CS/PS Steering**.

3.1.3 Inter-RAT Handover Triggering

Coverage-based inter-RAT handovers can be triggered by event B1 or B2:

- If the UE supports both events B1 and B2, the **InterRatHoComm**.***InterRatHoEventType*** parameter specifies which event triggers coverage-based inter-RAT handovers.
- If the UE supports only event B2, the eNodeB sends measurement configuration for event B2.

When the eNodeB receives the event A2 that triggers inter-RAT measurement, the eNodeB sends related inter-RAT handover events according to parameter configurations.

3.1.3.1 Event B1

Event B2 indicates that the signal quality of an inter-RAT neighboring cell is higher than a threshold.

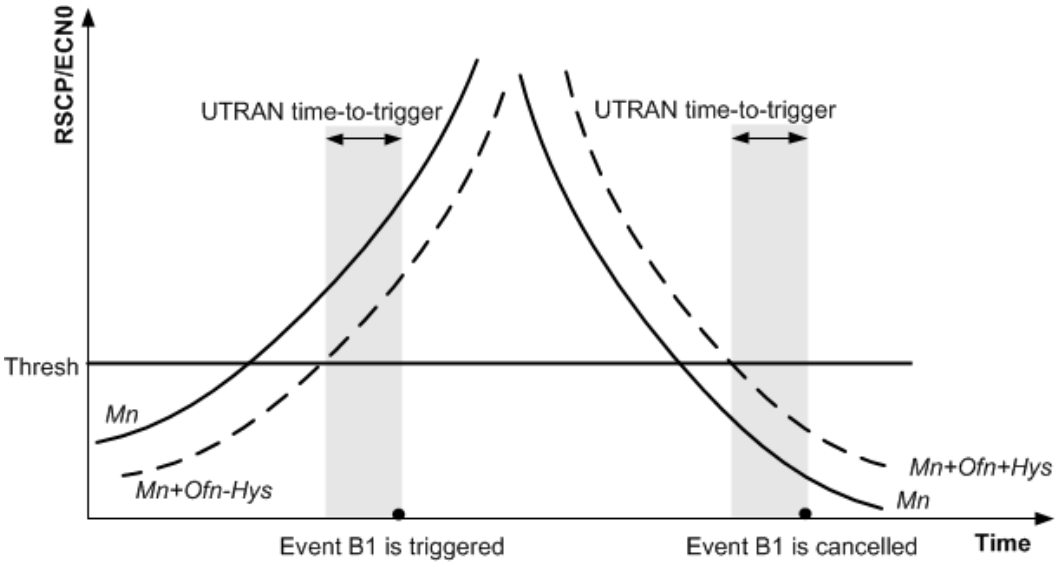
The entering and leaving conditions of event B1 are shown in **Figure 3-1**. For details about event B1, see section 5.5.4.7 "Event B1 (Inter RAT neighbour becomes better than threshold)" in 3GPP TS 36.331 V10.1.0.

- Entering condition: $Mn + Ofn - Hys > Thresh$

  If the entering condition is met, the UE reports an event B1. After receiving the event B1 report, the eNodeB delivers a measurement configuration to start inter-RAT handover decision.
- Leaving condition: $Mn + Ofn + Hys < Thresh$

  If the leaving condition of event B1 is met, the UE no longer reports event B1.

Figure 3-1 Entering and leaving conditions of event B1



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The variables in the preceding formulas are described in **Table 3-8**.

Table 3-8 Parameters related to event B1 for UTRAN

Variables in the Formula	Parameter Name	Parameter ID	Parameter Description
Mn	-	-	*Mn* is the measurement result of the neighboring cell. The measurement result type is specified by the **InterRatHoComm.*InterRATHoUtranB1MeasQuan*** parameter. The measurement result type for UTRAN TDD is RSCP
Ofn	Frequency offset	**UtranNFreq.*OffsetFreq***	This parameter is set for each frequency. This parameter is contained in the associated measurement configuration message.
Hys	UTRAN handover hysteresis	**InterRatHoUtranGroup.*InterRATHoUtranB1Hyst***	This parameter is set for each QCI.
Thresh	CoverageBased UTRAN RSCP trigger threshold	**InterRatHoUtranGroup.*InterRATHoUtranB1ThdRSCP***	This parameter is set for each QCI. The **InterRatHoComm.*InterRATHoUtranB1MeasQuan*** parameter determines which threshold is to be used. ● If **InterRatHoComm.*InterRATHoUtranB1MeasQuan*** is set to **BOTH**, two B1 events are reported by R10 UEs, with one RSCP-based and the other ECN0-based. When the entering condition for either of the B1 events is met, the UE reports the event B1. UEs with a version earlier than

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Variables in the Formula	Parameter Name	Parameter ID	Parameter Description
	CoverageBased UTRAN ECN0 trigger threshold	**InterRatHoUtranGroup**.***InterRATHoUtranB1ThdEcN0***	R10 report only RSCP-based event B1. ● If **InterRatHoComm**.***InterRATHoUtranB1MeasQuan*** is set to **RSCP** or **ECN0**, the UE reports an event B1 only when the entering condition for the related event B1 is met.

Table 3-9 lists other parameters related to the event B1 report.

Table 3-9 Other parameters related to the event B1 report

Parameter Name	Parameter ID	Parameter Description	
UTRAN time to trigger	**InterRatHoUtranGroup**.***InterRATHoUtranB1TimeToTrig***	This parameter is set for each QCI. The time-to-trigger parameters for UTRAN and GERAN are different, and the parameters can be set to different values. For example, if the time-to-trigger parameter for UTRAN is set to a value smaller than that for GERAN, a UE preferentially reports event B1 for a neighboring UTRAN cell, given the same network conditions, and therefore the eNodeB preferentially sends a handover request to a neighboring UTRAN cell.	
Max report cell number	**InterRatHoComm**.***InterRATHoMaxRprtCell***	Reported neighboring cells are sequenced according to the value of Mn that triggers event B1.	This is a common parameter for inter-RAT handover and is set for each eNodeB.
Measurement report amount	**InterRatHoComm**.***InterRATHoRprtAmount***	This parameter specifies the number of events to be reported. Setting this parameter to a proper value reduces the number of handover retries.	

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Parameter Name	Parameter ID	Parameter Description	
Utran measurement trigger quantity	**InterRatHoComm**. ***InterRATHoUtranB1MeasQuan***	This parameter specifies the measurement quantity type of the Mn value.	
UTRAN FDD measurement report interval	**InterRatHoComm**. ***InterRATHoUtranRprtInterval***	This parameter specifies the interval between two event B1 measurement reports from the UE. Setting this parameter to a proper value reduces signaling traffic on the air interface.	

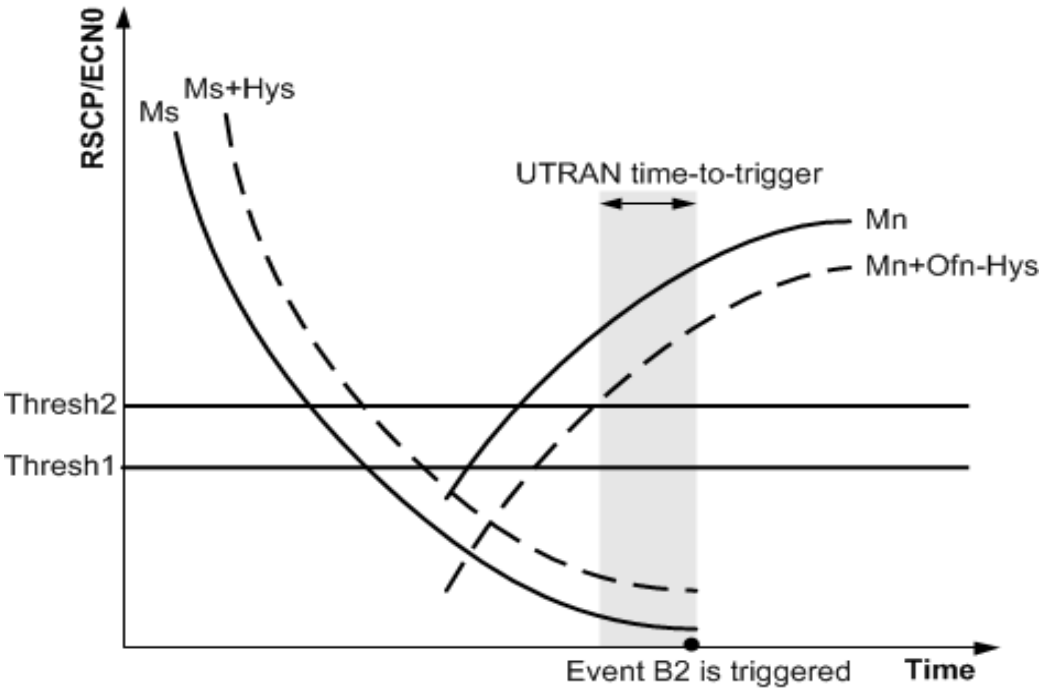
3.1.3.2 Event B2

Event B2 indicates that the signal quality of the serving cell is lower than a threshold and the signal quality of a neighboring cell is higher than another threshold.

The entering and leaving conditions of event B2 are shown in **Figure 3-2**. For details about event B2, see section 5.5.4.8 "Event B2 (PCell becomes worse than threshold1 and inter RAT neighbour becomes better than threshold2)" in 3GPP TS 36.331 V10.1.0.

- Entering conditions: $Ms + Hys < Thresh1$ and $Mn + Ofn - Hys > Thresh2$
- Leaving condition: $Ms - Hys > Thresh1$ or $Mn + Ofn + Hys < Thresh2$

Figure 3-2 Entering condition of event B2



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Except threshold parameters, the variables in the formulas for event B2 are the same as those in the formulas for event B1 shown in **Table 3-8** and **Table 3-9**.

When event B2 is used to trigger inter-RAT handovers, the value of *Thresh1* is the same as the threshold value used for event A2 in inter-RAT handovers, and the value of *Thresh2* is the same as the threshold value used for event B1 in coverage-based inter-RAT handovers, as shown in **Table 3-10**.

Table 3-10 Thresholds related to event B2 in UTRAN

Variables in the Formula	Parameter Name	Parameter ID	Parameter Description	
Thresh1	InterRAT A2 RSRP trigger threshold	**InterRatHoCommGroup.*InterRatHoA2ThdRSRP***	The type of the *Ms* measurement result and the threshold to be used are determined by the **InterRatHoComm.*InterRatHoA1A2TrigQuan*** parameter.	This parameter is set for each QCI.
	InterRAT A2 RSRQ trigger threshold	**InterRatHoCommGroup.*InterRatHoA2ThdRSRQ***		
Thresh2	CoverageBased UTRAN RSCP trigger threshold	**InterRatHoUtranGroup.*InterRATHoUtranB1ThdRSCP***	The type of the *Mn* measurement result and the threshold to be used are determined by the **InterRatHoComm.*InterRATHoUtranB1MeasQuan*** parameter.	
	CoverageBased UTRAN ECN0 trigger threshold	**InterRatHoUtranGroup.*InterRATHoUtranB1ThdEcN0***		

3.1.4 Blind Handover

Triggering and Stopping of Blind Handover

Events A2 and A1 for blind handover trigger and stop coverage-based inter-RAT blind handover in the same procedures as they trigger and stop coverage-based inter-frequency blind handover. Coverage-based inter-frequency blind handover and inter-RAT blind handover share the same thresholds related to events A2 and A1 and have the same working principles. For details, see *Intra-RAT Mobility Management in Connected Mode*.

Target Selection

The eNodeB selects the target RAT based on the RAT priorities specified by the following parameters:

- **CSFallBackBlindHoCfg.*InterRatHighestPri***: specifies the RAT with the highest priority.

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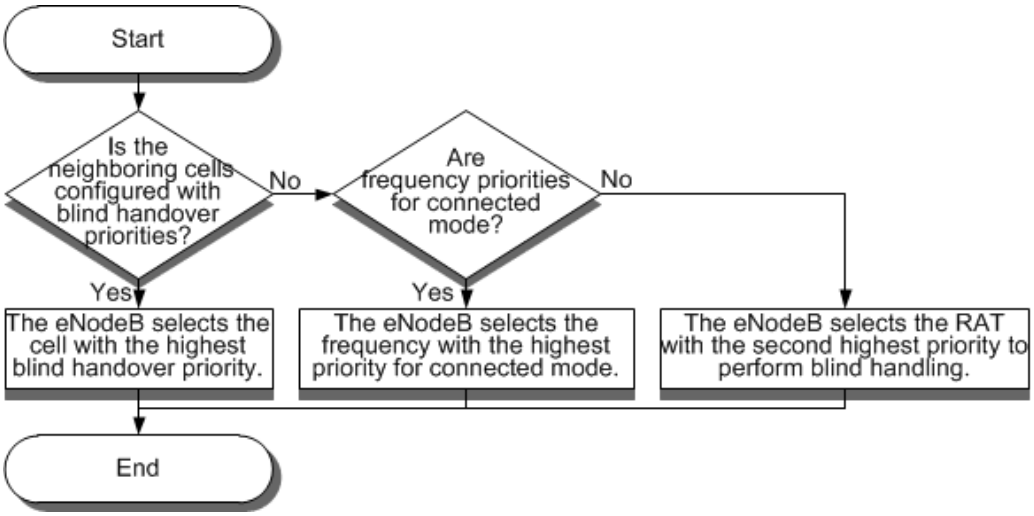
- **CSFallBackBlindHoCfg**.***InterRatSecondPri***: specifies the RAT with the second-highest priority.
- **CSFallBackBlindHoCfg**.***InterRatLowestPri***: specifies the RAT with the lowest priority.

Figure 3-3 shows the target selection procedure when the target RAT with the highest priority is UTRAN. Parameters in the figure are as follows:

- **UtranNCell**.***BlindHoPriority*** specifies the blind handover priority, with the neighboring cell whose blind handover priority set to 1 to 32 is selected. A greater value indicates a higher priority.
- The frequency that has the greatest **UtranNFreq**.***ConnFreqPriority*** is selected. A greater value indicates a higher priority. If this parameter is set to 0 for a frequency, this frequency is not selected as the target frequency for blind redirection.
- If the priorities of neighboring cells or the priorities of frequencies are the same, the eNodeB randomly selects the target frequencies or neighboring cells. Due to uncertainty of random selection, you are not advised to set the priorities to the same to ensure the blind handover success.

Figure 3-3 Target selection procedure in blind handover



If the optional feature LOFD-001078 E-UTRAN to UTRAN CS/PS Steering is enabled, the target UTRAN frequency priorities for blind handover change. For details, see **3.5 E-UTRAN to UTRAN CS/PS Steering**.

3.2 Service-based Inter-RAT Handover to UTRAN

This section describes the optional feature LOFD-001043 Service based inter-RAT handover to UTRAN. For engineering guidelines, see **8.2 Service-based Inter-RAT Handover to UTRAN**. The switch **UtranServiceHoSwitch** under the **ENodeBAlgoSwitch**.***HoAlgoSwitch*** parameter specifies whether to enable service-based handover from E-UTRAN to UTRAN.

Service-based inter-RAT handovers to UTRAN are applicable when the E-UTRAN and UTRAN cover the same area. Based on the QCIs of the services that are running on a UE, the eNodeB can divert the UE to an appropriate UTRAN frequency to implement service steering.

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Table 3-11 describes conditions for triggering/stopping measurement and handover during a service-based inter-RAT handover to UTRAN. Blind handover is not applied to service-based inter-RAT handover.

Table 3-11 Conditions for triggering/stopping measurement and handover during a service-based inter-RAT handover to UTRAN

Procedure	Triggering Condition	Stopping Condition
Inter-RAT measurement	UE service with the highest-priority QCI is allowed to be handed over to a frequency on a different RAT.	The UE service is terminated.
Inter-RAT handover	An event B1 is reported.	-

3.2.1 Triggering and Stopping Inter-RAT Measurement

To implement service-based inter-RAT handovers, operators must configure service-based inter-RAT handover policies. Since the services of each operator are classified by QCIs, each QCI can be associated with a service-based inter-RAT handover policy so that the RAT specified by the policy preferentially carries the services with this QCI. The eNodeB triggers a service-based inter-RAT handover for a UE based on the highest-priority QCI of the services running on the UE.

Service-based inter-RAT handovers enable the eNodeB to hand over a UE to a different RAT.

1. Set the service-based inter-RAT handover policy (specified by the **ServiceIrHoCfgGroup**.***InterRatHoState*** parameter) for an operator.
2. Associate a QCI (specified by **CnOperatorStandardQci**.***Qci*** or **CnOperatorExtendedQci**.***ExtendedQci***) with the service-based inter-RAT handover policy.
3. If the UE performs a service with this QCI, the eNodeB handles the service based on the handover policy.

For services with QCIs 1 to 9, three handover policies are available, specified by the **ServiceIrHoCfgGroup**.***InterRatHoState*** parameter:

- If the parameter is set to **MUST_HO** for a QCI, the eNodeB delivers the event B1 related measurement configuration for service-based inter-RAT handover to a UE when the UE initiates a service with that QCI. When the service with that QCI is interrupted, the service-based inter-RAT measurement is stopped.
- If this parameter is set to **NO_HO** for a QCI, the eNodeB does not deliver the service-based inter-RAT handover measurement configuration to a UE running a service with that QCI, even if the UE initiates a service for which this parameter is set to **MUST_HO**.
- If this parameter is set to **PERMIT_HO** for a QCI, the eNodeB does not deliver the service-based inter-RAT handover measurement configuration to a UE when the UE initiates a service with that QCI. If a service with the same QCI is running on the UE and the UE initiates a service for which this parameter is set to **MUST_HO**, the eNodeB delivers the event B1 related measurement configuration for service-based inter-RAT

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handover to the UE. When the service with that QCI is interrupted, the service-based inter-RAT measurement is stopped.

For example, an operator plans to use bearers with a QCI of 1 to carry voice over IP (VoIP) services, bearers with a QCI of 5 to carry IMS signaling, and bearers with a QCI of 9 as default bearers. The operator also expects that UEs performing only VoIP services are served by an inter-RAT system to ensure service continuity while sparing E-UTRAN resources. The recommended value of the **ServiceIrHoCfgGroup**.***InterRatHoState*** parameter varies according to QCIs:

- The value **MUST_HO** is recommended for QCI 1.
- The value **NO_HO** is recommended for QCI 2 to QCI 8, except QCI 5.
- The value **PERMIT_HO** is recommended for QCI 5 and QCI 9.

3.2.2 Inter-RAT Measurement

Before delivering the measurement configuration related to event B1 and the measurement gap configuration, the eNodeB checks whether the measurement gap has been activated for use in other types of handovers. If the measurement gap has been activated, the eNodeB delivers only the measurement configuration related to event B1. If not, the eNodeB delivers both.

If a UE has performed gap-assisted measurements for a service-based inter-RAT handover for a period of time (internally set to 3s) but fails to trigger such a handover, the eNodeB will stop the service-based inter-RAT measurements.

Parameters in the measurement configuration for service-based inter-RAT measurement are the same as those for coverage-based inter-RAT measurement. For details, see **3.1.2 Inter-RAT Measurement**.

3.2.3 Inter-RAT Handover Triggering

Service-based inter-RAT handovers can be triggered by event B1. Working principles of event B1 for triggering service-based inter-RAT handovers are the same as those of event B1 for triggering coverage-based inter-RAT handovers. For details, see **3.1.3.1 Event B1**.

All parameters (other than thresholds) for event A4 that triggers service-based inter-RAT handovers are the same as those for event A4 that triggers coverage-based inter-RAT handovers. **Table 3-12** lists the thresholds related to event B1.

Table 3-12 Thresholds related to event B1 that triggers service-based inter-RAT handovers

Parameter Name	Parameter ID	Parameter Description
Load Service Based UTRAN EventB1 RSCP trigger threshold	**InterRatHoUtranGroup**.***LdSvBasedHoUtranB1ThdRSCP***	This parameter is set for each QCI. The **InterRatHoComm**.***InterRATHoUtranB1MeasQuan*** parameter determines which threshold is to be used.
Load Service Based UTRANB1 ECN0 threshold	**InterRatHoUtranGroup**.***LdSvBasedHoUtranB1ThdECN0***	

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Parameters related to the event B1 report for service-based inter-RAT measurement are the same as parameters related to the event B1 report for coverage-based inter-RAT measurement. For details, see **3.1.3.1 Event B1**.

3.3 Distance-based Inter-RAT Handover to UTRAN

This section describes the optional feature LOFD-001072 Distance based inter-RAT handover to UTRAN. For engineering guidelines, see **8.3 Distance-based Inter-RAT Handover to UTRAN**. The **CellAlgoSwitch**.***DistBasedHoSwitch*** parameter specifies whether to enable distance-based handover from E-UTRAN to UTRAN.

Besides the preceding parameter setting, distance-based inter-RAT handover to UTRAN can be triggered only when the **DistBasedHO**.***DistBasedMeasObjType*** parameter is set to **UTRAN**.

Distance-based inter-RAT handover applies to scenarios where coverage of the E-UTRAN overlaps with that of another RAT and the RF signals from an E-UTRAN cell travel distances that are significantly longer than the inter-site spacing specified in the network plan, causing severe cross-cell coverage to cells of another RAT.

The principle of distance-based inter-RAT handovers is the same as that of distance-based inter-frequency handovers. For details, see *Intra-RAT Mobility Management in Connected Mode*.

Table 3-13 describes conditions for triggering/stopping measurement and handover during a distance-based inter-RAT handover to UTRAN. Blind handover does not apply to distance-based inter-RAT handover.

Table 3-13 Conditions for triggering/stopping measurement and handover during a distance-based inter-RAT handover to UTRAN

Procedure	Triggering Condition	Stopping Condition
Inter-RAT measurement	Distance between the UE and eNodeB exceeds the specified threshold.	Distance between the UE and eNodeB falls below the specified threshold.
Inter-RAT handover	An event B1 is reported.	-

3.3.1 Triggering and Stopping Inter-RAT Measurement

Distance-based inter-RAT handovers are triggered by the eNodeB based on the distance between the eNodeB and the UE.

The eNodeB continues monitoring the distance to all UEs. Upon detecting that the measured distance to a UE exceeds the threshold specified by the **DistBasedHO**.***DistBasedHOThd*** parameter for 10 seconds (which is a fixed value), the eNodeB delivers the measurement configuration for event B1 to the UE.

During the measurements for a distance-based inter-RAT handover, if the TA reported by the UE falls below the threshold for 10 seconds, the measurements are immediately stopped.

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3.3.2 Inter-RAT Measurement

Before delivering the measurement configuration related to event B1 and the measurement gap configuration, the eNodeB checks whether the measurement gap has been activated for use in other types of handovers. If the measurement gap has been activated, the eNodeB delivers only the measurement configuration related to event B1. If not, the eNodeB delivers both.

If a UE has performed gap-assisted measurements for a distance-based inter-RAT handover for a period of time (internally set to 3s) but fails to trigger such a handover, the eNodeB will stop the distance-based inter-RAT measurements.

Parameters in the measurement configuration for distance-based inter-RAT measurement are the same as those for coverage-based inter-RAT measurement. For details, see **3.1.2 Inter-RAT Measurement**.

3.3.3 Inter-RAT Handover Triggering

Distance-based inter-RAT handovers can be triggered by event B1. Working principles of event B1 for triggering distance-based inter-RAT handovers are the same as those of event B1 for triggering coverage-based inter-RAT handovers. For details, see **3.1.3.1 Event B1**.

3.4 UL-Quality-based Inter-RAT Handover to UTRAN

This section describes UL-quality-based inter-RAT handover to UTRAN. For engineering guidelines, see **8.1 Inter-RAT Mobility Between E-UTRAN and UTRAN**. The **UlQualityInterRATHoSwitch** switch under the **ENodeBAlgoSwitch**.***HoAlgoSwitch*** parameter specifies whether to enable UL-quality-based inter-RAT handover to UTRAN.

A UL-quality-based inter-RAT handover prevents service drops caused by deteriorated UL quality.

Table 3-14 describes conditions for triggering and stopping inter-RAT measurement and handover during an UL-quality-based inter-RAT handover to UTRAN.

Table 3-14 Conditions for triggering and stopping measurement and handover during an UL-quality-based inter-RAT handover to UTRAN

Procedure	Subprocedure	Triggering Condition	Stopping Condition
Measurement handling	Inter-RAT measurement	UL quality worsens.	UL quality improves.
	Inter-RAT handover	An event B1 is reported.	-
Blind handover	-	UL quality deteriorates further, but the eNodeB does not receive any event B1 reports.	-

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3.4.1 Triggering and Stopping Inter-RAT Measurement

UL-quality-based inter-RAT handovers are triggered based on UL signal quality. When UL signal quality is unsatisfactory, call drops may occur if handovers are not performed promptly. The eNodeB checks the uplink signal quality. If the eNodeB finds that the uplink signal quality is poor, it delivers the measurement configuration for event B1.

An eNodeB instructs a UE to stop an uplink quality-based inter-RAT measurement when detecting that the UE's uplink quality improves.

3.4.2 Inter-RAT Measurement

Before delivering the measurement configuration related to event B1 and the measurement gap configuration, the eNodeB checks whether the measurement gap has been activated for use in other types of handovers. If the measurement gap has been activated, the eNodeB delivers only the measurement configuration related to event B1. If not, the eNodeB delivers both.

If a UE has performed gap-assisted measurements for a UL-quality-based inter-RAT handover for a period of time (internally set to 3s) but fails to trigger such a handover, the eNodeB releases the measurement gaps for UL-quality-based inter-RAT handovers to stop the measurements.

Parameters in the measurement configuration for UL-quality-based inter-RAT measurement are the same as those for coverage-based inter-RAT measurement. For details, see **3.1.2 Inter-RAT Measurement**.

3.4.3 Inter-RAT Handover Triggering

UL-quality-based inter-RAT handovers can be triggered by event B1. Working principles of event B1 for triggering UL-quality-based inter-RAT handovers are the same as those of event B1 for triggering coverage-based inter-RAT handovers. For details, see **3.1.3.1 Event B1**.

3.4.4 Blind Handover

Triggering Blind Handover

If the UL quality for the UE further deteriorates but the eNodeB does not receive an event B1 measurement report, the eNodeB proceeds as if service drops occur due to poor UL quality. In this situation, the eNodeB performs blind handover for the UE.

Target Selection

The target selection procedure in UL-quality-based blind handover from E-UTRAN to UTRAN is similar to that in inter-frequency blind handover. For details, see *Intra-RAT Mobility Management in Connected Mode*. The target neighboring cell or frequency for blind handover is selected as follows:

- The neighboring cell with **UtranNCell**.***BlindHoPriority*** set to 1 to 32 is selected.
- The frequency that has the greatest **UtranNFreq**.***ConnFreqPriority*** value is selected.

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3.5 E-UTRAN to UTRAN CS/PS Steering

This section describes PS steering function in the optional feature LOFD-001078 E-UTRAN to UTRAN CS/PS Steering. For engineering guidelines, see **8.5 E-UTRAN to UTRAN CS/PS Steering**.

The LOFD-001078 feature is an enhanced feature for coverage-based inter-RAT handover to UTRAN, with the following improvement:

- During inter-RAT measurement on the UTRAN, frequencies with a high CS/PS service steering priority are preferentially measured.

  The **UtranFreqLayerMeasSwitch** switch under the **ENodeBAlgoSwitch**.***FreqLayerSwtich*** parameter specifies whether to enable this function.

- During blind handover, cells working on frequencies with a high CS/PS service steering priority are preferentially selected for handovers.

  The **UtranFreqLayerBlindSwitch** switch under the **ENodeBAlgoSwitch**.***FreqLayerSwtich*** parameter specifies whether to enable this function.

For details about the CS steering function in the optional feature LOFD-001078 E-UTRAN to UTRAN CS/PS Steering, see *CS Fallback*.

3.5.1 Inter-RAT Measurement

If the **UtranFreqLayerMeasSwitch** switch under the **ENodeBAlgoSwitch**.***FreqLayerSwtich*** parameter is turned on, the inter-RAT measurement target selection procedure, for details, see **3.1.2 Inter-RAT Measurement**, with the frequency priority specified by the **UtranNFreq**.***PsPriority*** parameter. A greater parameter value indicates a higher priority. If the parameter is set to **Priority_0** for a frequency, that frequency is not measured. Other procedures are the same as those in coverage-based inter-RAT measurement.

When SRVCC is used, the frequency priority for measurement is determined in a different way from what is described in this chapter. For details, see *SRVCC*.

3.5.2 Blind Handover

If the **UtranFreqLayerBlindSwitch** switch under the **ENodeBAlgoSwitch**.***FreqLayerSwtich*** parameter is turned on, the frequency with the greatest value of the **UtranNFreq**.***PsPriority*** parameter is preferentially selected as the target frequency for E-UTRAN-to-UTRAN PS steering.A larger value of this parameter indicates a higher priority. If this parameter is set to **Priority_0(Priority 0)** for a frequency, the eNodeB does not select this frequency.

The target cell selection procedure is the same as that in coverage-based E-UTRAN-to-UTRAN blind handover. For details, see **3.1.4 Blind Handover**.

3.6 Handover Decision

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3.6.1 Basic Handover Decision

When the handover policy is PS HO, SRVCC, or redirection (excluding flash redirection), the eNodeB does not need to obtain system information of the peer and performs the basic handover decision.

Target Decision

In the handover decision phase, the eNodeB checks the cells in the candidate cell list and determines whether a handover needs to be initiated and, if so, to which cell the UE is to be handed over. Based on the check result, the eNodeB determines whether a handover needs to be initiated and, if so, to which cell the UE is to be handed over. If the eNodeB receives measurement reports about different RATs, it processes the reports on an FIFO basis.

The eNodeB excludes the following cells from the neighboring cell list:

- Blacklisted neighboring cells
- Neighboring cells with a handover prohibition flag
- Neighboring cells that have a different PLMN from the serving cell in the neighboring cell list

  If the inter-PLMN handover switch is turned on, such cells are not excluded.
- Neighboring cells in the areas indicated by the IE Handover Restriction List in the INITIAL CONTEXT SETUP REQUEST message sent from the MME

The eNodeB then sends a handover request to the target cell at the top of the filtered candidate cell list. If the handover request fails, the eNodeB sends the handover request to the next target cell, as described in **Table 3-15**.

Table 3-15 Sequence of handover requests to be sent by the eNodeB

Candidate Cell List Generated by	Sequence of Handover Requests
Measurement	A handover request is sent to the cell with the best signal quality.
Blind handover	A handover request is sent to the cell or frequency with the highest priority. If multiple cells have equally high priority, the eNodeB randomly selects a cell for blind handover.

If the handover request fails in all candidate cells:

- For a measurement procedure, the eNodeB waits until the UE sends the next measurement report.
- For a blind handover procedure, the eNodeB finishes the handover attempt.

Admission Decision

During SPID-based inter-RAT handovers to HPLMN in combined service scenarios, admission decision is related to **MultiQciHoExecOptSwitch** under the **ENodeBAlgoSwitch**.***HoSignalingOptSwitch*** parameter. For details about SPID-based inter-RAT handovers to HPLMN, see *Flexible User Steering*.

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- When this switch is turned on, handover is performed only after the services of all QCIs are successfully admitted.
- When this switch is turned off, handover is performed after the services of any QCI are successfully admitted.

3.6.2 Decision Based on System Information

When the handover policy requires that the eNodeB obtain system information about the peer (for example, flash redirection), handover decision based on system information is performed. If the handover decision is based on system information, the eNodeB includes system information for the target cell of the corresponding RAT. Reading cell system information is not required so that the UE can quickly access the target network.

Decision based on system information adheres to the following principles:

- In blind handover scenarios:
  a. The target cell list for blind handover is selected, including other cells under the target frequency for redirection.
  b. Basic handover decision is applied. For details, see **3.6.1 Basic Handover Decision**.
  c. Cells whose system information is not obtained are filtered out.
  d. Cells whose PLMN is not on the list of PLMNs that the UE is allowed to access are filtered out.
- In measurement scenarios:
  a. Cells in the candidate cell list generated by measurement are selected, plus cells that are not in measurement reports but work on the target frequency for redirection.
  b. Basic handover decision is applied. For details, see **3.6.1 Basic Handover Decision**.
  c. Cells whose system information is not obtained are filtered out.
  d. Cells whose PLMN is not on the list of PLMNs that the UE is allowed to access are filtered out.

You can specify the number of UTRAN cells contained in the redirection message by setting the **InterRatHoComm**.***CellInfoMaxUtranCellNum*** parameter. Assume that this parameter is set to N.

- If the number of target cells after flash redirection decision is greater than N, the eNodeB selects the first N cells.
- If the number of target cells after flash redirection decision is smaller than N, the eNodeB selects target cells after flash redirection decision.

The eNodeB obtains system information for target cells in the RAN information management (RIM) procedure. If a target cell does not support the RIM procedure, the eNodeB cannot obtain system information for that cell.

3.7 Handover Execution

3.7.1 Handover Policy Selection

Four policies are available for E-UTRAN-to-UTRAN handovers: PS handover, SRVCC, redirection, and flash redirection. **Figure 3-4** shows the handover policy selection procedure.

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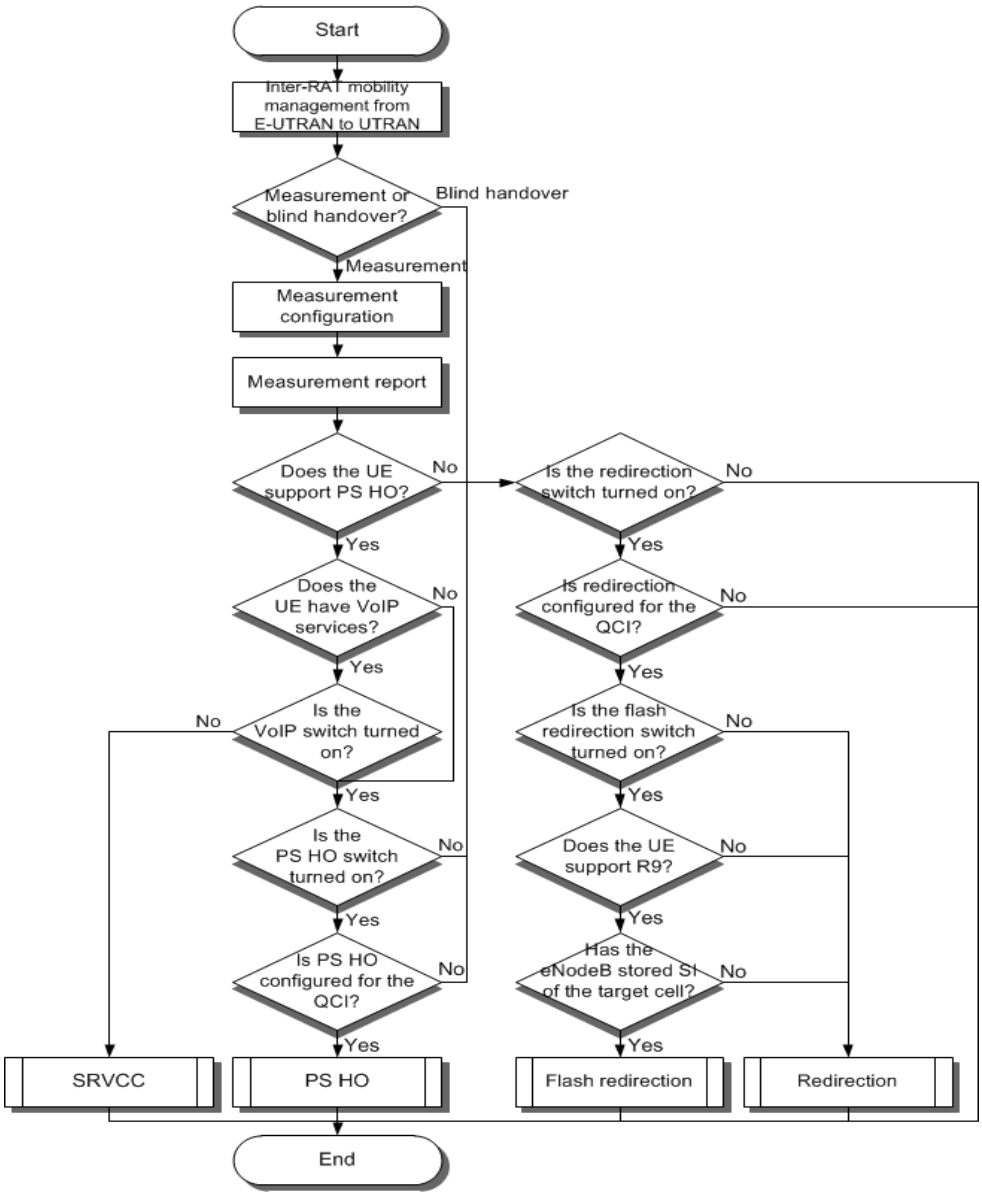
For details, see *SRVCC*. This handover policy selection procedure is based on the assumption that neighboring frequency or neighboring cell configurations are proper.

Figure 3-4 E-UTRAN-to-UTRAN handover policy selection procedure



The switches in **Figure 3-4** are specified by the following parameters:

- The UTRAN capability is specified by the following switches under the **ENodeBAlgoSwitch**.***HoModeSwitch*** parameter.
  - The VoIP supporting capability is specified by **UtranVoipCapSwitch**.
  - The PS handover supporting capability is specified by **UtranPsHoSwitch**.
  - The redirection supporting capability is specified by **UtranRedirectSwitch**.
- The flash redirection supporting capability is specified by the **UranFlashRedirectSwitch** switch under the **ENodeBAlgoSwitch**.***RedirectSwitch*** parameter.

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When the handover policy is flash redirection, the eNodeB should obtain UTRAN cell information through the Multi Report RIM procedure to reduce delay. For details about the RIM procedure, see *CS Fallback*.

- The handover policy for each QCI is specified by the **InterRatPolicyCfgGroup**.***UtranHoCfg*** parameter.
  a. Handover policies are configured in the **InterRatPolicyCfgGroup** MO.
  b. The index of a handover policy (specified by the **InterRatPolicyCfgGroup**.***InterRatPolicyCfgGroupId*** parameter) is mapped to a QCI (specified by the **StandardQci**.***InterRatPolicyCfgGroupId*** or **ExtendedQci**.***InterRatPolicyCfgGroupId*** parameter).

  NOTE

  Different QCIs correspond to different services. The eNodeB may not select a correct handover policy if the default value is changed.

  If QCI 1 is used for VoIP services and **REDIRECTION** is selected under the **InterRatPolicyCfgGroup**.***UtranHoCfg*** parameter, call drop occurs during redirection. Do not enable redirection for VoIP services with QCI 1.

For coverage-based inter-RAT handovers, when the signal quality of the serving cell meets the inter-RAT handover event A2, the eNodeB delivers event B1 measurement. If the signal quality of the serving cell continues to degrade and meets the inter-RAT blind handover event A2 (the inter-RAT blind handover event A2 threshold is lower than the inter-RAT handover event A2 threshold), the eNodeB performs a blind redirection.

- The eNodeB preferentially selects a RAT that has not been measured. For example, if the UTRAN has been measured, the eNodeB preferentially selects GERAN for redirection. This is because due to problems such as poor coverage of the measured RAT, the eNodeB does not receive measurement report of the measured RAT. Therefore, the eNodeB preferentially selects a RAT that has not been measured.
- If a cell to which the eNodeB has never attempted to hand over the UE is reported, the eNodeB preferentially selects the operating frequency of the cell for redirection.
- If the eNodeB cannot select the target frequency for redirection, the eNodeB stops the procedure.
- In this situation, the eNodeB selects the target cell for redirection as it does during blind handover. For details about how the eNodeB performs target selection during blind handover, see **3.1.4 Blind Handover**.

The eNodeB processes uplink-quality-based inter-RAT blind handovers the same way as it processes coverage-based inter-RAT blind handovers.

3.7.2 Separate Mobility Policies to UTRAN for Multiple PLMNs

This section describes the optional feature LOFD-070216 Separate Mobility Policies to UTRAN for Multi PLMN. For engineering guidelines, see **8.6 Separate Mobility Policies to UTRAN for Multi PLMN**.

This feature is an enhanced handover policy selection function, which enables the eNodeB to obtain the PS handover capability of UTRAN cells through PLMN and RNC configuration. Based on the obtained PS handover capability information, the eNodeB determines whether to use PS handovers.

The **UtranSepOpMobilitySwitch** switch under the **ENodeBAlgoSwitch**.***MultiOpCtrlSwitch*** parameter specifies whether to enable this feature.

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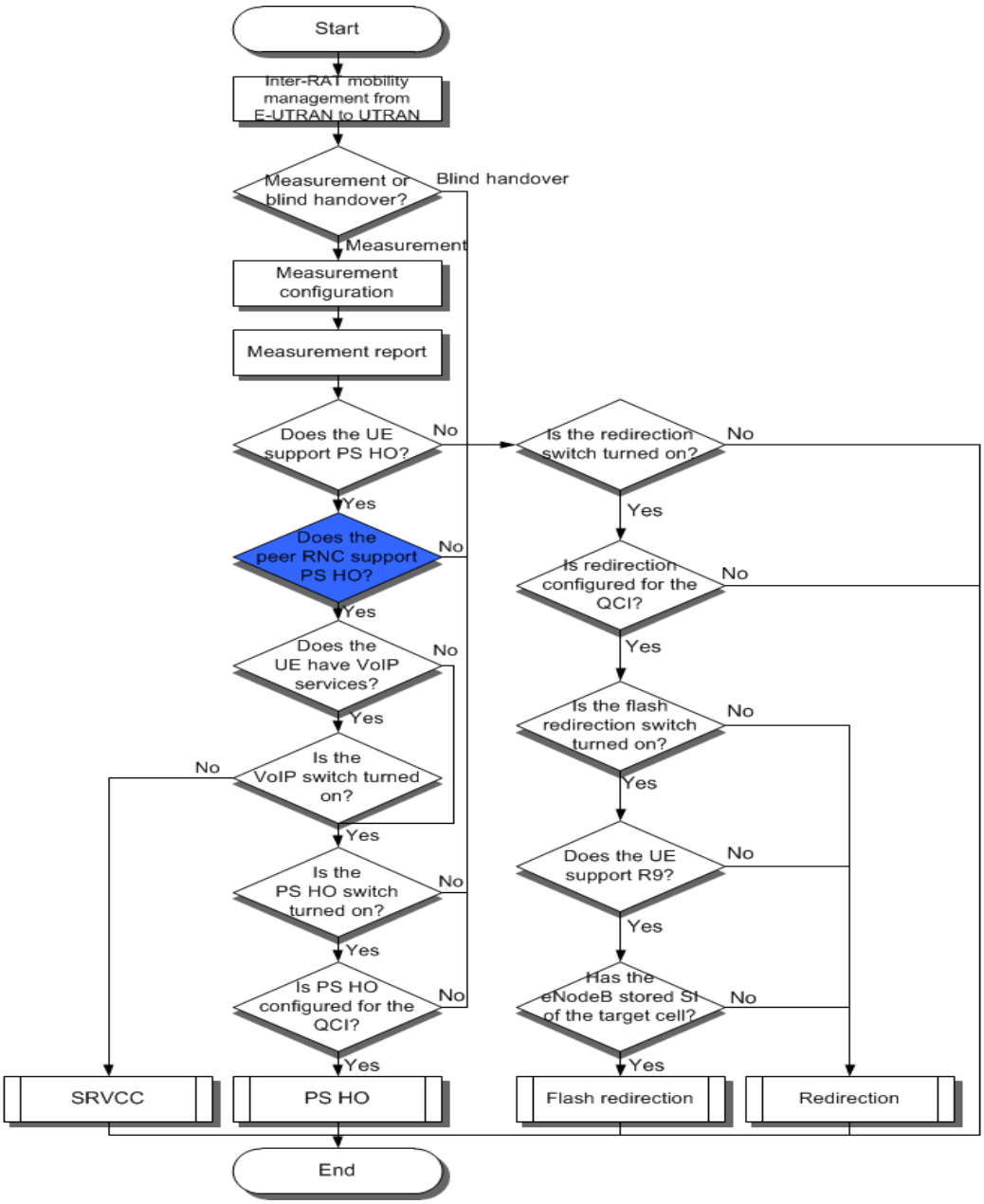
If this feature is enabled, based on the basic handover policy selection procedure, the eNodeB checks whether the RNC supports PS handover, as shown in **Figure 3-5**.

Figure 3-5 Enhanced E-UTRAN-to-UTRAN handover policy selection procedure



The **PsHoCapCfg** value of the **UtranNetworkCapCfg**.***NetworkCapCfg*** parameter specifies whether the peer RNC supports PS handover.

- If **PsHoCapCfg** is selected, the peer RNC supports PS handover.
- If **PsHoCapCfg** is not selected, the peer RNC does not support PS handover.

If **UtranNetworkCapCfg**.***RncId*** is set to **4294967295**, all RNCs under the target PLMN use the same handover policy.

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3.7.3 Special Signaling Retransmission Optimization

Certain UEs do not reply radio link control (RLC) status reports after outgoing handovers from E-UTRAN. This causes the RLC layer to repeatedly issue unnecessary handover commands. In large traffic volume scenarios such as important sports events, turning on the special signaling retransmission optimization switch reduces the number of unnecessary handover commands sent by the RLC layer. This also reduces radio resource overheads.

SPECSIGRETRANSOPTSWITCH under the parameter **ENodeBAlgoSwitch**.***HighLoadNetOptSwitch*** controls the special signaling retransmission optimization switch.

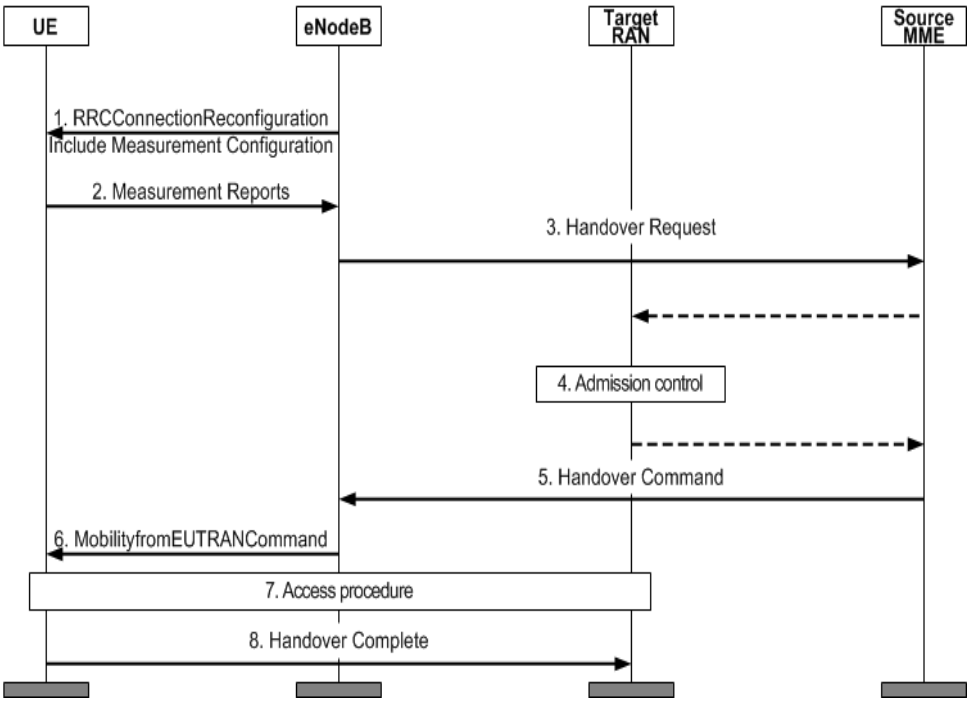
3.7.4 Retry and Penalty

To avoid UEs initiating many handover attempts with a target cell that has insufficient resources, the source eNodeB applies a penalty on UEs that failed in admission. The penalty timer length for mobility management from E-UTRAN to UTRAN is specified by the **HoMeasComm**.***OptHoPreFailPunishTimer*** parameter.

3.8 Signaling Flow

Figure 3-6 shows the signaling procedure for a PS handover from E-UTRAN to UTRAN. The dashed lines in the figure indicate the procedures in the core network. For details, see section 5.5.2 "Inter RAT handover" in 3GPP TS 23.401 V10.3.0.

Figure 3-6 Signaling procedure for a PS handover from E-UTRAN to UTRAN



The RRC connection re-establishment procedure performed after an inter-RAT handover failure is the same as that performed after an intra-frequency handover failure. For details, see *Intra-RAT Mobility Management in Connected Mode*.

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4 RIM-Based LTE Target Cell Selection

This chapter describes the LOFD-070203 RIM Based LTE Target Cell Selection feature. For engineering guidelines, see **8.7 RIM Based LTE Target Cell Selection**.

This feature implements RIM-based load message transfer between the E-UTRAN and UTRAN. When the cell load status of the E-UTRAN changes, the eNodeB informs the RNC of load status changes using the RIM procedure. This feature helps the UTRAN to select a proper target E-UTRAN cell during handovers.

The **RimUtranSwitch** switch in the **GlobalProcSwitch**.***EutranLoadTransSwitch*** parameter specifies whether to enable the feature LOFD-070203 RIM Based LTE Target Cell Selection.

During the RIM procedure, RNC requires that the E-UTRAN cell load information from the eNodeB, and the eNodeB transfers the E-UTRAN cell load information to the RNC. When the cell load status of the E-UTRAN changes, the eNodeB informs the RNC of load status changes using the RIM procedure.

When **RimUtranSwitch** under the **GlobalProcSwitch**.***EutranLoadTransSwitch*** parameter is turned off, the RNC sends RIM messages to the eNodeB to request the LTE cell load information, but the eNodeB does not reply.

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4.1 RIM Procedure

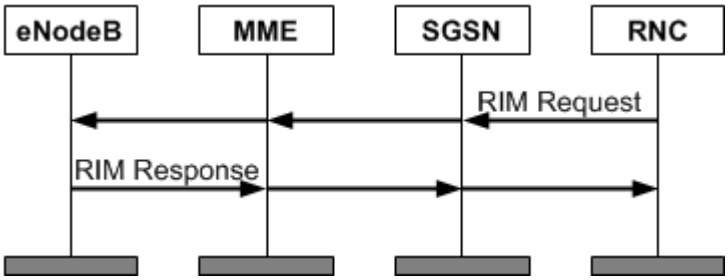
The RIM procedure exchanges information between the UTRAN and E-UTRAN through the core networks or the eCoordinator. The **ENodeBAlgoSwitch**.***RimOnEcoSwitch*** parameter specifies the information exchange mode:

- When this parameter is set to **OFF(Off)**, the RIM procedure is performed through core networks.
- When this parameter is set to **ON(On)**, the RIM procedure is performed through the eCoordinator.

RIM Procedure Based on Core Networks

When information in the RIM procedure is exchanged through core networks as shown in **Figure 4-1**, the involved NEs include eNodeB, MME, SGSN, and RNC. Among these NEs, the MME and the SGSN transfer but do not resolve information. For details about the RIM procedure, see section 11.3.63.1.4 "RAN-INFORMATION-REQUEST Application Container for the SON Transfer Application" in 3GPP TS 48.018 V10.2.0.

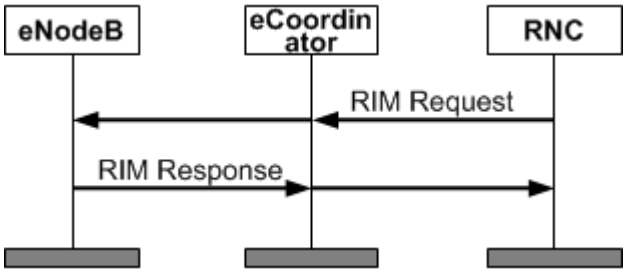
Figure 4-1 RIM procedure based on the eCoordinator



RIM Procedure Based on the eCoordinator

When information in the RIM procedure is exchanged through the eCoordinator as shown in **Figure 4-2**, the involved NEs include eNodeB, eCoordinator, and RNC. Among these NEs, the eCoordinator transfers but does not resolve information.

Figure 4-2 RIM procedure based on the eCoordinator



In the eCoordinator-based RIM procedure, turn on all the switches of involved NEs related to the RIM procedure based on the eCoordinator. Otherwise, information cannot be exchanged through the eCoordinator.

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In the eCoordinator-based RIM procedure, NEs neither send RIM messages to core networks nor respond to RIM messages from core networks.

4.2 E-UTRAN Cell Load Status

The **CellMLB**.***InterRatMlbUeNumThd*** parameter indicates the E-UTRAN cell load status:

- When the average number of UEs served by an E-UTRAN cell is greater than or equals this parameter value added the **CellMLB**.***InterRatMlbUeNumOffset*** parameter value, the E-UTRAN cell status is considered congested.
- When the average number of UEs served by an E-UTRAN cell is less than this parameter value, the E-UTRAN cell status is considered normal.

After obtaining E-UTRAN cell status, the RNC performs as follows if service- or load-based UTRAN to E-UTRAN handovers (including PS handover and redirection) is triggered:

- If there are any target E-UTRAN cells whose status is normal, these cells are used for PS handover or redirection.
- If there is no target E-UTRAN cell whose status is normal, PS handover or redirection is not performed.

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5 Inter-RAT Mobility Management Between E-UTRAN and GERAN

Inter-RAT mobility management between E-UTRAN and GERAN can be implemented in multiple ways. This chapter describes the following features and functions:

- LOFD-001020 PS Inter-RAT Mobility between E-UTRAN and GERAN
- LOFD-001046 Service based inter-RAT handover to GERAN
- LOFD-001073 Distance based Inter-RAT handover to GERAN
- UL-Quality-based Inter-RAT Handover to GERAN

The triggering conditions and processes for the preceding features and functions are different, but the following general processes apply:

1. Target cell/frequency selection

   For a measurement, the eNodeB generates a candidate cell list based on inter-RAT measurement results.

   For a blind handover, the eNodeB selects a blind handover target based on the blind handover priority or frequency priority of neighboring cells.
2. Handover decision

   In the handover decision phase, the eNodeB checks the candidate cell list. Based on the check result, the eNodeB determines whether a handover needs to be initiated and, if so, to which cell the UE is to be handed over.
3. Handover execution

   The eNodeB controls the UE to be handed over from the serving cell to the target cell.

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5.1 PS Inter-RAT Mobility Between E-UTRAN and GERAN

This section describes the optional feature LOFD-001020 PS Inter-RAT Mobility between E-UTRAN and GERAN, which is implemented by coverage-based handovers from E-UTRAN to GERAN. For engineering guidelines, see **8.8 PS Inter-RAT Mobility Between E-UTRAN and GERAN**. The coverage-based handover from E-UTRAN to GERAN is enabled or disabled by the **GeranRedirectSwitch** or **GeranPsHoSwitch** switch under the **ENodeBAlgoSwitch**.***HoModeSwitch*** parameter.

The principles of this feature are the same as those for LOFD-001019 PS Inter-RAT Mobility between E-UTRAN and UTRAN. For details, see **3.1 Inter-RAT Mobility Between E-UTRAN and UTRAN**.

5.1.1 Triggering and Stopping Inter-RAT Measurement

The principles for triggering and stopping measurements or blind handover for this feature are the same as those for LOFD-001019 PS Inter-RAT Mobility between E-UTRAN and UTRAN. For details, see **3.1.1 Triggering and Stopping of Inter-RAT Measurement**.

5.1.2 Inter-RAT Measurement

Table 5-1 lists the parameters in the Measurement Configuration message exchanged during coverage-based E-UTRAN to GERAN handovers. **5.1.3 Inter-RAT Handover Triggering** lists information contained in the reporting configuration.

Table 5-1 Parameters in the E-UTRAN to GERAN handover measurement configuration

Measurement Configuration	Parameter Name	Parameter ID	Parameter Description
Measurement objects	Starting ARFCN	**GeranNfreqGroup**.***StartingArfcn***	This is a cell-level parameter. The three parameters constitute a neighboring GERAN ARFCN group.
	Band indicator	**GeranNfreqGroup**.***BandIndicator***	
	BCCH ARFCN	**GeranNfreqGroupArfcn**.***GeranArfcn***	
	Frequency offset	**GeranNfreqGroup**.***OffsetFreq***	This parameter is set for each frequency.
Filtering coefficient	GERAN measurement filter coefficient	**HoMeasComm**.***GeranFilterCoeff***	This is a common parameter for inter-RAT handover and is set for each eNodeB.

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Measurement Configuration	Parameter Name	Parameter ID	Parameter Description
Measurement gap configuration	GAP measurement mode	**HoMeasComm**.***GAPPatternType***	

The measurement object selection procedure in GERAN is similar to that in UTRAN. The differences are described as follows: The frequency priority is specified by the **GeranNfreqGroup**.***ConnFreqPriority*** parameter. A larger value indicates a higher priority.

If the sum of the number of GERAN frequencies, which can be delivered, in the highest-priority GERAN frequency group and the number of delivered GERAN frequencies exceeds the maximum number of GERAN frequencies allowed for measurement, all the GERAN frequencies in this GERAN frequency group are not delivered in the Measurement Configuration message. The eNodeB determines whether the GERAN frequencies in the frequency group with the second highest priority can be delivered till the number of delivered frequencies equals the maximum number of GERAN frequencies allowed for measurement or all frequency groups are determined.

5.1.3 Inter-RAT Handover Triggering

The principles for coverage-based E-UTRAN to GERAN handovers are the same as those for E-UTRAN to UTRAN. For details, see **3.1.3 Inter-RAT Handover Triggering**. However, the parameters for each event are different.

Event B1

Table 5-2 lists the variables in the formula of event B1 for triggering coverage-based E-UTRAN to GERAN handovers.

Table 5-2 Parameters for event B1 in the GERAN

Variables in the Formula	Parameter Name	Parameter ID	Parameter Description
Mn	Measurement result of the neighboring cell	-	For GERAN, the measurement result type is RSSI.
Ofn	Frequency offset	**GeranNfreqGroup**.***OffsetFreq***	This parameter is set for each frequency. This parameter is contained in the associated measurement configuration message.

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Variables in the Formula	Parameter Name	Parameter ID	Parameter Description
Hys	GERAN handover hysteresis	**InterRatHoGeranGroup.*InterRatHoGeranB1Hyst***	This parameter is set for each QCI.
Thresh	CoverageBased GERAN trigger threshold	**InterRatHoGeranGroup.*InterRatHoGeranB1Thd***	This parameter is set for each QCI.

Table 5-3 lists other parameters related to the event B1 report.

Table 5-3 Other parameters related to the event B1 report

Parameter Name	Parameter ID	Parameter Description	
GERAN time to trigger	**InterRatHoGeranGroup.*InterRatHoGeranB1TimeToTrig***	This parameter is set for each QCI. The time-to-trigger parameters for UTRAN and GERAN are different, and the parameters can be set to different values. For example, if the time-to-trigger parameter for UTRAN is set to a value greater than that for GERAN, a UE preferentially reports event B1 for a neighboring UTRAN cell, given the same network conditions, and therefore the eNodeB preferentially sends a handover request to a neighboring UTRAN cell.	
Max report cell number	**InterRatHoComm.*InterRATHoMaxRprtCell***	Reported neighboring cells are sequenced according to the value of Mn that triggers event B1.	This is a common parameter for inter-RAT handover and is set for each eNodeB.
Measurement report amount	**InterRatHoComm.*InterRATHoRprtAmount***	This parameter specifies the number of periodic reports to be sent after event B1 is triggered to restrict handover retries.	

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Parameter Name	Parameter ID	Parameter Description	
Geran measurement report interval	**InterRatHoComm**.***InterRatHoGeranRprtInterval***	This parameter specifies the interval between two event B1 measurement reports from the UE. Setting this parameter to a proper value reduces signaling traffic on the air interface.	

Event B2

Table 5-4 lists Thresh1 and Thresh2 in the formula of event B2 for triggering coverage-based E-UTRAN to GERAN handovers.

Table 5-4 Thresh1 and Thresh2 for event B2 in the GERAN

Variables in the Formula	Parameter Name	Parameter ID	Parameter Description	
Thresh1	InterRAT A2 RSRP trigger threshold	**InterRatHoCommGroup**.***InterRatHoA2ThdRSRP***	The type of *Ms* measurement result and the threshold to be used are determined by the **InterRatHoComm**.***InterRatHoA1A2TrigQuan*** parameter.	This parameter is set for each QCI.
	InterRAT A2 RSRQ trigger threshold	**InterRatHoCommGroup**.***InterRatHoA2ThdRSRQ***		
Thresh2	CoverageBased GERAN trigger threshold	**InterRatHoGeranGroup**.***InterRATHoGeranB1Thd***	For GERAN, the triggering type is RSSI.	

5.1.4 Blind Handover

Triggering and Stopping Blind Handover

Events A2 and A1 for blind handover trigger and stop coverage-based inter-RAT blind handover in the same way that they trigger and stop coverage-based inter-frequency blind handover. Coverage-based inter-frequency blind handover and inter-RAT blind handover share the same thresholds related to events A2 and A1 and have similar working principles. For details, see *Intra-RAT Mobility Management in Connected Mode*.

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Target Selection

The target selection procedure in coverage-based blind handover from E-UTRAN to GERAN is similar to that in inter-frequency blind handover. For details, see *Intra-RAT Mobility Management in Connected Mode*. The target neighboring cell or frequency for blind handover is selected as follows:

- The neighboring cell with **GeranNcell**.***BlindHoPriority*** set to 1 to 32 is selected. A larger value of this parameter indicates a higher priority.
- The frequency that has the greatest **GeranNfreqGroup**.***ConnFreqPriority*** value is selected. A larger value of this parameter indicates a higher priority. If this parameter is set to 0 for a frequency, this frequency is not selected as the target frequency for blind redirection.

5.2 Service-based Inter-RAT Handover to GERAN

This section describes the optional feature LOFD-001046 Service based inter-RAT handover to GERAN. For engineering guidelines, see **8.9 Service-based Inter-RAT Handover to GERAN**. The switch **GeranServiceHoSwitch** under the **ENodeBAlgoSwitch**.***HoAlgoSwitch*** parameter specifies whether to enable service-based handover from E-UTRAN to GERAN.

The principles of this feature are the same as those for LOFD-001043 Service based inter-RAT handover to UTRAN. For details, see **3.2 Service-based Inter-RAT Handover to UTRAN**.

5.2.1 Triggering and Stopping Inter-RAT Measurement

Ttriggering and stopping inter-RAT measurement for this feature is accomplished in the same way as triggering and stopping LOFD-001043 Service based inter-RAT handover to UTRAN. For details, see **3.2.1 Triggering and Stopping Inter-RAT Measurement**.

5.2.2 Inter-RAT Measurement

Before delivering the measurement configuration related to event B1 and the measurement gap configuration, the eNodeB checks whether the measurement gap has been activated for use in other types of handovers. If the measurement gap has been activated, the eNodeB delivers only the measurement configuration related to event B1. If not, the eNodeB delivers both.

If a UE has performed gap-assisted measurements for a service-based inter-RAT handover for a period of time (internally set to 3s) but fails to trigger such a handover, the eNodeB will stop the service-based inter-RAT measurements.

Parameters in the measurement configuration for service-based inter-RAT measurement are the same as those for coverage-based inter-RAT measurement. For details, see **3.1.2 Inter-RAT Measurement**.

5.2.3 Inter-RAT Handover Triggering

The principles for service-based E-UTRAN to GERAN handovers are the same as those for E-UTRAN to UTRAN. For details, see **3.1.3 Inter-RAT Handover Triggering**. **Table 5-5** describes the threshold parameter differences.

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Table 5-5 Event B1 threshold for service-based E-UTRAN to GERAN handovers

Parameter Name	Parameter ID	Parameter Description
Load Service Based Geran EventB1 trigger threshold	**InterRatHoGeranGroup.*LdSvBasedHoGeranB1Thd***	This parameter is set for each QCI.

5.3 Distance-based Inter-RAT Handover to GERAN

This section describes the optional feature LOFD-001073 Distance based Inter-RAT handover to GERAN. For engineering guidelines, see **8.10 Distance-based Inter-RAT Handover to GERAN**. The **CellAlgoSwitch.*DistBasedHoSwitch*** parameter specifies whether to enable distance-based handover from E-UTRAN to GERAN.

When distance-based handover switch is turned on, also set **DistBasedHO.*DistBasedMeasObjType*** to **GERAN** to enable distance-based handover from E-UTRAN to GERAN.

The principles of this feature are the same as those for LOFD-001072 Distance based inter-RAT handover to UTRAN. For details, see **3.3 Distance-based Inter-RAT Handover to UTRAN**.

5.3.1 Triggering and Stopping Inter-RAT Measurement

Triggering and stopping inter-RAT measurement for this feature is accomplished in the same way as triggering and stopping LOFD-001072 Distance based inter-RAT handover to UTRAN. For details, see **3.3.1 Triggering and Stopping Inter-RAT Measurement**.

5.3.2 Inter-RAT Measurement

Before delivering the measurement configuration related to event B1 and the measurement gap configuration, the eNodeB checks whether the measurement gap has been activated for use in other types of handovers. If the measurement gap has been activated, the eNodeB delivers only the measurement configuration related to event B1. If not, the eNodeB delivers both.

If a UE has performed gap-assisted measurements for a distance-based inter-RAT handover for a period of time (internally set to 3s) but fails to trigger such a handover, the eNodeB will stop the distance-based inter-RAT measurements.

Parameters in the measurement configuration for distance-based inter-RAT measurement are the same as those for coverage-based inter-RAT measurement. For details, see **3.1.2 Inter-RAT Measurement**.

5.3.3 Inter-RAT Handover Triggering

Distance-based inter-RAT handovers can be triggered by event B1. Working principles of event B1 for triggering distance-based inter-RAT handovers are the same as those of event B1 for triggering coverage-based inter-RAT handovers. For details, see **3.1.3.1 Event B1**.

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5.4 UL-Quality-based Inter-RAT Handover to GERAN

This section describes UL-quality-based inter-RAT handover to GERAN. For engineering guidelines, see **8.5 E-UTRAN to UTRAN CS/PS Steering**. The **UlQualityInterRATHoSwitch** switch under the **ENodeBAlgoSwitch**.***HoAlgoSwitch*** parameter specifies whether to enable UL-quality-based inter-RAT handover to GERAN.

5.4.1 Triggering and Stopping Inter-RAT Measurement

UL-quality-based inter-RAT handovers are triggered based on UL signal quality. When the UL signal quality is unsatisfactory, call drops may occur if handovers are not performed promptly. The eNodeB checks the uplink signal quality. If the eNodeB finds that the uplink signal quality is poor, it delivers the measurement configuration for event B1.

An eNodeB instructs a UE to stop an uplink quality-based inter-RAT measurement when detecting that the UE's uplink quality improves.

5.4.2 Inter-RAT Measurement

Before delivering the measurement configuration related to event B1 and the measurement gap configuration, the eNodeB checks whether the measurement gap has been activated for use in other types of handovers. If the measurement gap has been activated, the eNodeB delivers only the measurement configuration related to event B1. If not, the eNodeB delivers both.

If a UE has performed gap-assisted measurements for a UL-quality-based inter-RAT handover for a period of time (internally set to 3s) but fails to trigger such a handover, the eNodeB releases the measurement gaps for UL-quality-based inter-RAT handovers to stop the measurements.

Parameters in the measurement configuration for UL-quality-based inter-RAT measurement are the same as those for coverage-based inter-RAT measurement. For details, see **5.1.2 Inter-RAT Measurement**.

5.4.3 Inter-RAT Handover Triggering

UL-quality-based inter-RAT handovers can be triggered by event B1. Working principles of event B1 for triggering UL-quality-based inter-RAT handovers are the same as those of event B1 for triggering coverage-based inter-RAT handovers. For details, see **5.1.3 Inter-RAT Handover Triggering**.

5.4.4 Blind Handover

Triggering Blind Handover

If UL quality for the UE further deteriorates but the eNodeB does not receive an event B1 measurement report, the eNodeB proceeds as if service drops occur due to poor UL quality. In this situation, the eNodeB performs blind handover for the UE.

Target Selection

The target selection procedure in UL-quality-based blind handover from E-UTRAN to GERAN is similar to that in inter-frequency blind handover. For details, see *Intra-RAT*

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Mobility Management in Connected Mode. The target neighboring cell or frequency for blind handover is selected as follows:

- The neighboring cell with **GeranNcell**.***BlindHoPriority*** set to 1 to 32 is selected. A larger value of this parameter indicates a higher priority.
- The frequency that has the greatest **GeranNfreqGroup**.***ConnFreqPriority*** value is selected. A larger value of this parameter indicates a higher priority. If this parameter is set to 0 for a frequency, this frequency is not selected as the target frequency for blind redirection.

5.5 Handover Decision

The handover decision for E-UTRAN to GERAN mobility management is the same as that for E-UTRAN to UTRAN mobility management. For details, see **3.6 Handover Decision**.

The **InterRatHoComm**.***CellInfoMaxGeranCellNum*** parameter specifies the number of GERAN cells included in a redirection message.

5.6 Handover Execution

5.6.1 Handover Policy Selection

Five policies are available for E-UTRAN-to-GERAN handovers: PS handover, CCO/NACC, SRVCC, redirection, and flash redirection. **Figure 5-1** shows the handover policy selection procedure. For details, see *SRVCC*. This handover policy selection procedure is based on the assumption that neighboring frequency or neighboring cell configurations are proper.

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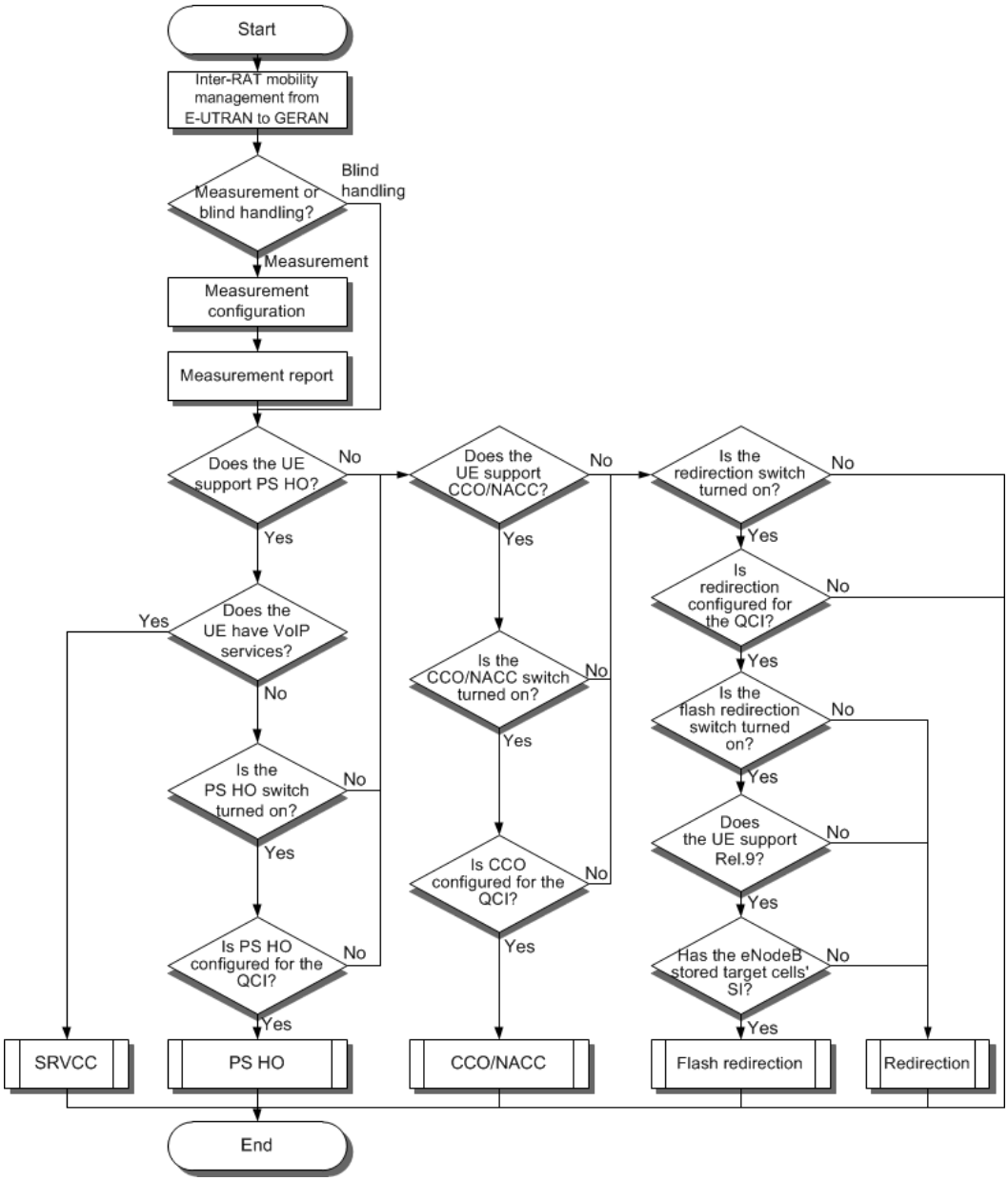
Figure 5-1 E-UTRAN-to-GERAN handover policy selection procedure



The switches in **Figure 5-1** are specified by the following parameters:

- The GERAN capability is specified by the following switches under the **ENodeBAlgoSwitch**.***HoModeSwitch*** parameter.
  - The PS handover supporting capability is specified by **GeranPsHoSwitch**.
  - The CCO supporting capability is specified by **GeranCcoSwitch**.
  - The NACC supporting capability is specified by **GeranNaccSwitch**.
  - The redirection supporting capability is specified by **GeranRedirectSwitch**.
- The flash redirection supporting capability is specified by the **GeranFlashRedirectSwitch** switch under the **ENodeBAlgoSwitch**.***RedirectSwitch*** parameter.

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When the handover policy is flash redirection, the eNodeB should obtain GERAN cell information through the Multi Report RIM procedure to reduce delay. For details about the RIM procedure, see *CS Fallback*.

- The handover policy for each QCI is specified by the **InterRatPolicyCfgGroup**.***GeranGsmHoCfg*** or **InterRatPolicyCfgGroup**.***GeranGprsEdgeHoCfg*** parameter.
  a. Handover policies are configured in the **InterRatPolicyCfgGroup** MO.
  b. The index of a handover policy (specified by the **InterRatPolicyCfgGroup**.***InterRatPolicyCfgGroupId*** parameter) is mapped to a QCI (specified by the **StandardQci**.***InterRatPolicyCfgGroupId*** or **ExtendedQci**.***InterRatPolicyCfgGroupId*** parameter).

  NOTE

  Different QCIs correspond to different services. The eNodeB may not select a correct handover policy if the default value is changed.

  If QCI 1 is used for VoIP services and **REDIRECTION** is selected under the **InterRatPolicyCfgGroup**.***GeranGprsEdgeHoCfg*** parameter, call drop occurs during redirection. Do not enable redirection for VoIP services with QCI 1.

5.6.2 Retry and Penalty

The retry and penalty for E-UTRAN to GERAN mobility management are the same as those for E-UTRAN to UTRAN. For details, see **3.7.4 Retry and Penalty**.

5.7 Signaling Procedure

The signaling procedure for E-UTRAN to GERAN PS handovers is the same as that for E-UTRAN to UTRAN PS handovers. For details, see **3.8 Signaling Flow**.

Figure 5-2 shows the signaling procedure for E-UTRAN to GERAN CCO/NACC when information in the RIM procedure is exchanged through core networks. For details, see section 5.6 "Network Assisted Cell Change" in 3GPP TS 23.401 V10.3.0.

Figure 5-2 Signaling procedure for E-UTRAN to GERAN CCO/NACC (core network)

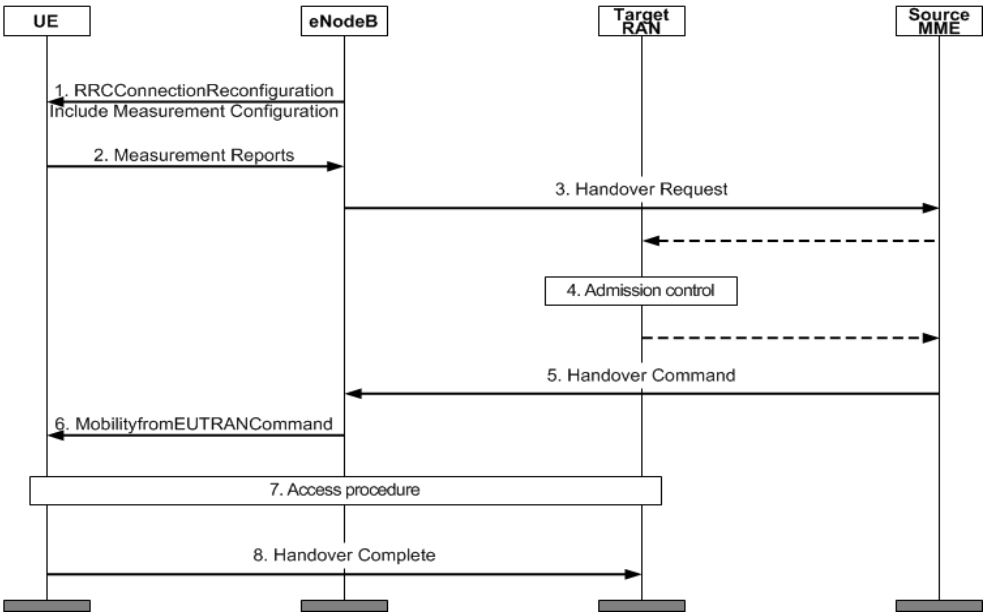

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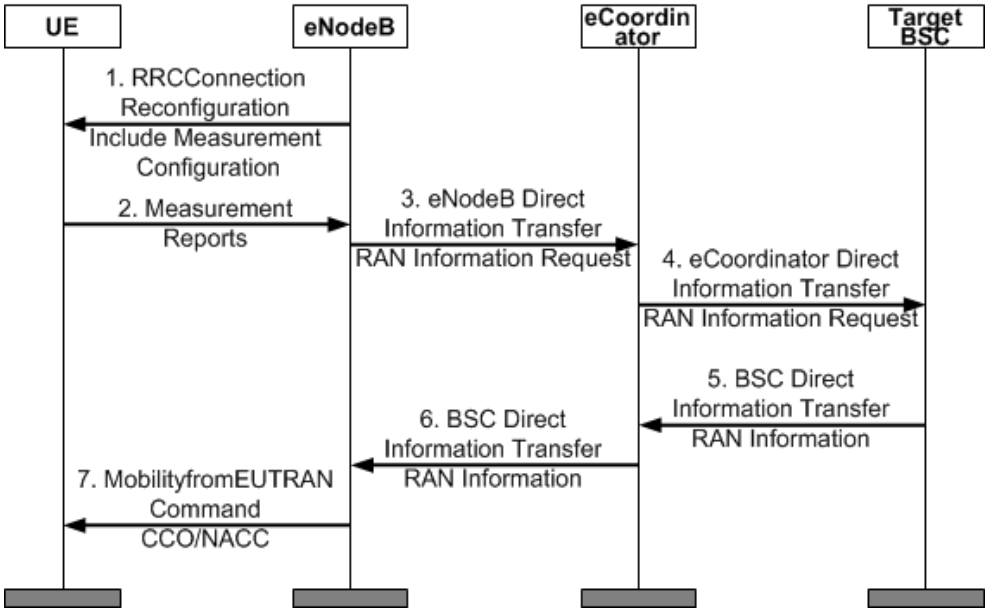
Figure 5-3 shows the signaling procedure for E-UTRAN to GERAN CCO/NACC when information in the RIM procedure is exchanged through the eCoordinator.

Figure 5-3 Signaling procedure for E-UTRAN to GERAN CCO/NACC (eCoordinator)



The RRC connection re-establishment procedure performed after an inter-RAT handover failure is the same as that performed after an intra-frequency handover failure. For details, see *Intra-RAT Mobility Management in Connected Mode*.

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6 Related Features

6.1 Features Related to PS Inter-RAT Mobility Between E-UTRAN and UTRAN

Prerequisite Features

None

Mutually Exclusive Features

Coverage-based inter-RAT handovers to UTRAN support inter-RAT blind redirection to UTRAN in the absence of the configuration of a neighbor relationship with UTRAN. To use inter-RAT blind redirection to UTRAN in the absence of the configuration of a neighbor relationship with UTRAN, ensure that RAN sharing information about the UTRAN frequencies is configured. The parameters for RAN sharing and LOFD-001112 MOCN Flexible Priority Based Camping are mutually exclusive. Therefore, if inter-RAT blind redirection to UTRAN in the absence of the configuration of a neighbor relationship with UTRAN is used, LOFD-001112 MOCN Flexible Priority Based Camping cannot be used.

Impacted Features

If LOFD-001019 PS Inter-RAT Mobility between E-UTRAN and UTRAN is not used, all other features related to interactions between E-UTRAN and UTRAN will be affected.

6.2 Features Related to Service-based Inter-RAT Handover to UTRAN

Prerequisite Features

LOFD-001043 Service based inter-RAT handover to UTRAN requires LOFD-001019 PS Inter-RAT Mobility between E-UTRAN and UTRAN or LOFD-001022 SRVCC to UTRAN.

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Mutually Exclusive Features

None

Impacted Features

None

6.3 Features Related to Distance-based Inter-RAT handover to UTRAN

Prerequisite Features

LOFD-001072 Distance based inter-RAT handover to UTRAN requires LOFD-001019 PS Inter-RAT Mobility between E-UTRAN and UTRAN.

Mutually Exclusive Features

None

Impacted Features

None

6.4 Features Related to E-UTRAN to UTRAN CS/PS Steering

Prerequisite Features

LOFD-001078 E-UTRAN to UTRAN CS/PS Steering requires LOFD-001019 PS Inter-RAT Mobility between E-UTRAN and UTRAN or LOFD-001033 CS Fallback to UTRAN.

Mutually Exclusive Features

None

Impacted Features

None

6.5 RIM-Based LTE Target Cell Selection

Required Features

This feature can be enabled only when one of the following features is enabled:

- WRFD-020129 PS Service Redirection from UMTS to LTE

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- WRFD-140218 Service-Based PS Handover from UMTS to LTE
- WRFD-150216 Load Based PS Redirection from UMTS to LTE
- WRFD-150217 Load Based PS Handover from UMTS to LTE

Mutually Exclusive Features

None

Affected Features

None

6.6 Separate Mobility Policies to UTRAN for Multiple PLMNs

Prerequisite Features

LOFD-070216 Separate Mobility Policies to UTRAN for Multi PLMN requires one of the following features:

- LOFD-001019 PS Inter-RAT Mobility between E-UTRAN and UTRAN
- LOFD-001033 CS Fallback to UTRAN
- LOFD-001052 Flash CS Fallback to UTRAN
- LOFD-001088 CS Fallback Steering to UTRAN
- LOFD-001043 Service based inter-RAT handover to UTRAN
- LOFD-001072 Distance based inter-RAT handover to UTRAN
- LOFD-001078 E-UTRAN to UTRAN CS/PS Steering
- LOFD-001044 Inter-RAT Load Sharing to UTRAN
- LOFD-00105401Camp & Handover Based on SPID

Mutually Exclusive Features

None

Impacted Features

None

6.7 PS Inter-RAT Mobility Between E-UTRAN and GERAN

Prerequisite Features

None

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Mutually Exclusive Features

Coverage-based inter-RAT handovers to GERAN support inter-RAT blind redirection to GERAN in the absence of the configuration of a neighbor relationship with GERAN. To use inter-RAT blind redirection to GERAN in the absence of the configuration of a neighbor relationship with GERAN, ensure that RAN sharing information about the GERAN frequencies is configured. The parameters for RAN sharing and LOFD-001112 MOCN Flexible Priority Based Camping are mutually exclusive. Therefore, if inter-RAT blind redirection to GERAN in the absence of the configuration of a neighbor relationship with GERAN is used, LOFD-001112 MOCN Flexible Priority Based Camping cannot be used.

Impacted Features

If LOFD-001020 PS Inter-RAT Mobility between E-UTRAN and GERAN is not used, all other features related to interactions between E-UTRAN and GERAN will be affected.

6.8 Service based inter-RAT handover to GERAN

Prerequisite Features

LOFD-001046 Service based inter-RAT handover to GERAN requires LOFD-001023 SRVCC to GERAN.

Mutually Exclusive Features

None

Impacted Features

None

6.9 Distance based Inter-RAT handover to GERAN

Prerequisite Features

LOFD-001073 Distance based inter-RAT handover to GERAN requires LOFD-001020 PS Inter-RAT Mobility between E-UTRAN and GERAN.

Mutually Exclusive Features

None

Impacted Features

None

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7 Network Impact

7.1 PS Inter-RAT Mobility between E-UTRAN and UTRAN

System Capacity

No impact.

Network Performance

With coverage-based inter-RAT handovers, areas with weak LTE coverage or coverage holes can be supplemented with inter-RAT neighboring cells, decreasing the call drop rate.

7.2 Service-based Inter-RAT Handover to UTRAN

System Capacity

No impact.

Network Performance

No impact.

7.3 Distance-based Inter-RAT Handover to UTRAN

System Capacity

No impact.

Network Performance

No impact.

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7.4 E-UTRAN to UTRAN CS/PS Steering

System Capacity

No impact.

Network Performance

No impact.

7.5 RIM Based LTE Target Cell Selection

System Capacity

No impact.

Impact on Network Performance

After the RNC obtains the E-UTRAN neighboring cell load information, UEs will not be handed over to congested target E-UTRAN cells during service-based or load-based UTRAN to E-UTRAN handover. In addition, the UE will not be redirected to an E-UTRAN frequency whose neighboring cell status is congested. This lowers the probability of ping-pong handovers.

7.6 Separate Mobility Policies to UTRAN for Multiple PLMNs

System Capacity

No impact.

Network Performance

This feature enables the eNodeB to determine the handover and RIM supporting capability of the UMTS network. This reduces unnecessary signaling exchanges, and therefore reduces the number of handover attempts and failures.

7.7 PS Inter-RAT Mobility between E-UTRAN and GERAN

System Capacity

No impact.

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Network Performance

With coverage-based inter-RAT handovers, areas with weak LTE coverage or coverage holes can be supplemented by inter-RAT neighboring cells, decreasing the call drop rate.

7.8 Service based inter-RAT handover to GERAN

System Capacity

No impact.

Network Performance

No impact.

7.9 Distance based Inter-RAT handover to GERAN

System Capacity

No impact.

Network Performance

No impact.

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8 Engineering Guidelines

This chapter provides engineering guidelines for inter-RAT mobility management in connected mode.

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8.1 Inter-RAT Mobility Between E-UTRAN and UTRAN

8.1.1 When to Use Inter-RAT Mobility Between E-UTRAN and UTRAN

Mobility management is fundamental to service continuity. If the E-UTRAN cannot cover the entire network and therefore requires other RATs, such as UTRAN or GERAN to supplement coverage, you are advised to enable coverage-based inter-RAT handover.

Coverage-based inter-RAT handover is the basis for other types of inter-RAT handover. Before enabling another type of inter-RAT handover, you need to enable coverage-based inter-RAT handover.

Event B1 or event B2 triggers coverage-based inter-RAT handovers. If there are no special requirements for the serving cell signal quality, event B2 may not be used.

8.1.2 Required Information

Inter-RAT handovers have requirements on networks and UEs. Before enabling an inter-RAT handover policy, you need to check whether the networks support inter-RAT operations and whether the UE supports inter-RAT measurements and handovers.

- If the devices in the networks do not support inter-RAT operations, disable the inter-RAT PS handover policies.
- If the devices in the networks do not support SRVCC to inter-RAT systems, disable the inter-RAT SRVCC policies.
- If blind handovers are used to implement inter-RAT handovers, you need to collect the information about the neighboring cells to which blind handovers will be allowed as well as their blind-handover priorities ranging from 1 to 32. These priorities have a positive correlation with the values. Note that the value 0 indicates that blind handovers to the neighboring cell are prohibited.
- If an operator requires flash redirection or NACC, ensure that the equipment on the network side supports RIM. If the equipment does not support RIM, flash redirection cannot be used.

The information to be collected before deploying coverage-based inter-RAT handover is the same as that before deploying coverage-based inter-frequency handover.

Verify that the inter-RAT networks provide continuous coverage in inter-RAT networking scenarios where eNodeBs often share sites with base stations of other RATs. Since the LTE network often provides higher data rates and has a lower threshold for inter-RAT measurement event A2, it is recommended that LTE networks preferentially carry services.

You need to determine whether event B1 or event B2 is used for inter-RAT measurements. If event B2 is configured, event B2 is used for inter-RAT measurements. If event B1 is configured and the UE support event B1, event B1 is used for inter-RAT measurements. If event B1 is configured but the UE only supports event B2, event B2 is used for inter-RAT measurements.

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8.1.3 Deployment Requirements

Operating Environment

The feature LOFD-001019 PS Inter-RAT Mobility between E-UTRAN and UTRAN requires that the core-network equipment should support this feature. If the equipment is provided by Huawei, the version must be SAE1.2. If the core-network equipment is provided by another vendor, check with the vendor whether the equipment supports this feature.

If the MME is provided by another vendor, the **RsvdSwPara1_bit4** option of the **eNBRsvdPara**.***RsvdSwPara1*** parameter must be selected. This option is used to control how the eNodeB fills in the NAS Security Parameters to E-UTRAN IE.

- If the option is selected, the eNodeB fills in the IE from the first byte.
- If the option is cleared, the eNodeB fills in the IE from the second byte.

Transmission Networking

None

License

To use coverage-based inter-RAT handovers, an operator must purchase and activate the license for the feature listed in the following table.

Feature ID	Feature Name	License Control Item	NE	Sales Unit
LOFD-001019	PS Inter-RAT Mobility between E-UTRAN and UTRAN	PS Inter-RAT Mobility between E-UTRAN and UTRAN	eNodeB	per user in RRC connected mode

8.1.4 Data Preparation

This section describes the data that you need to collect for setting parameters. Required data is data that you must collect for all scenarios. Collect scenario-specific data when necessary for a specific feature deployment scenario.

There are three types of data sources:

- Network plan (negotiation not required): parameter values planned and set by the operator
- Network plan (negotiation required): parameter values planned by the operator and negotiated with the EPC or peer transmission equipment
- User-defined: parameter values set by users

Required Data

This section describes the data required for configuring neighboring UTRAN cells.

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NOTE

During coverage-based inter-RAT handovers, the eNodeB can perform a blind handover when there is no neighboring cells. To enable this function, you can configure only UTRAN neighboring frequencies.

To enable other inter-RAT handover features, you must configure required data described in this section.

The following table describes the parameters that must be set in the **UtranNFreq** MO to configure a neighboring UTRAN frequency.

Parameter Name	Parameter ID	Data Source	Setting Notes
Local cell ID	**UtranNFreq**.***LocalCellId***	Network plan (negotiation not required)	Ensure that this parameter has been set in a **Cell** MO based on the network plan.
Downlink UARFCN	**UtranNFreq**.***UtranDlArfcn***	Network plan (negotiation required)	This parameter specifies the DL UARFCN of the cells on the neighboring UTRAN frequency. Set this parameter based on the neighboring relationship configuration. If the peer NodeB is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **UCELL** MO of the peer RNC.

The following table describes the parameters that must be set in the **UtranExternalCell** MO to configure an external UTRAN cell.

Parameter Name	Parameter ID	Data Source	Setting Notes
Mobile country code	**UtranExternalCell**.***Mcc***	Network plan (negotiation required)	This parameter specifies the MCC of the home NodeB of the external UTRAN cell. If this neighboring cell works in RAN sharing mode, set this parameter to the MCC of the primary operator. If the peer NodeB is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **UCNOPERATOR** MO of the peer RNC.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Mobile network code	**UtranExternalCell**.***Mnc***	Network plan (negotiation required)	This parameter specifies the MNC of the home NodeB of the external UTRAN cell. Set this parameter to the MNC of the primary operator if this external cell works in eRAN sharing with a common carrier mode. If the peer NodeB is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **UCNOPERATOR** MO of the peer RNC.
UTRAN cell ID	**UtranExternalCell**.***CellId***	Network plan (negotiation required)	This parameter specifies the ID of the external UTRAN cell. It uniquely identifies a UTRAN cell within a NodeB. If the peer NodeB is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **UCELL** MO of the peer RNC.
Downlink UARFCN	**UtranExternalCell**.***UtranDlArfcn***	Network plan (negotiation required)	This parameter specifies the DL ARFCN of the external UTRAN cell. If the peer NodeB is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **UCELL** MO of the peer RNC.
RNC ID	**UtranExternalCell**.***RncId***	Network plan (negotiation required)	This parameter specifies the home RNC ID of the external UTRAN cell. If the peer NodeB is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **URNCBASIC** MO of the Huawei NodeB.
Routing area code configure indicator	**UtranExternalCell**.***RacCfgInd***	Network plan (negotiation required)	This parameter specifies whether to set the routing area code (RAC) for the external UTRAN cell. This parameter must be set to **CFG(Configure)** if the external UTRAN cell supports PS services.
Routing area code	**UtranExternalCell**.***Rac***	Network plan (negotiation required)	This parameter specifies the RAC of the external UTRAN cell. If **RacCfgInd** is set to **CFG(Configure)**, this parameter must be set. If the peer NodeB is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **UCELL** MO of the peer RNC.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Primary scrambling code	**UtranExternalCell**.***PScrambCode***	Network plan (negotiation required)	This parameter specifies the DL primary scrambling code of the external UTRAN cell. If the peer NodeB is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **UCELL** MO of the peer RNC.
Location area code	**UtranExternalCell**.***Lac***	Network plan (negotiation required)	This parameter specifies the location area code (LAC) of the external UTRAN cell. If the peer NodeB is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **UCELL** MO of the peer RNC.

(Optional, required only if the home NodeB of the external UTRAN cell works in RAN sharing with common carrier mode and multiple operators share the same external UTRAN cell.) The following table describes the parameters that must be set in the **UtranExternalCellPlmn** MO to configure an additional PLMN ID (representing a secondary operator) for an external UTRAN cell.

Parameter Name	Parameter ID	Data Source	Setting Notes
UTRAN cell ID	**UtranExternalCellPlmn**.***CellId***	Network plan (negotiation required)	Set this MO when the neighboring UTRAN cell works in common-carrier mode and the cell is shared by more than one operator. This parameter specifies the ID of the external UTRAN cell. It uniquely identifies a UTRAN cell within a NodeB. If the peer NodeB is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **UCELL** MO of the peer RNC.
Mobile country code	**UtranExternalCellPlmn**.***Mcc***	Network plan (negotiation required)	This parameter specifies the MCC of the home RNC of the external UTRAN cell. If this neighboring cell works in RAN sharing mode, set this parameter to the MCC of the primary operator. If the peer NodeB is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **UCNOPERATOR** MO of the peer RNC.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Mobile network code	**UtranExternalCellPlmn.*Mnc***	Network plan (negotiation required)	This parameter specifies the MNC of the home RNC of the external UTRAN cell. If this external cell works in RAN sharing mode, set this parameter to the MNC of the primary operator. If the peer NodeB is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **URNCBASIC** MO of the peer RNC.
Share mobile country code	**UtranExternalCellPlmn.*ShareMcc***	Network plan (negotiation required)	This parameter specifies the MCC of the secondary operator. If the peer NodeB is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **UCNOPERATOR** MO of the peer RNC.
Share mobile network code	**UtranExternalCellPlmn.*ShareMnc***	Network plan (negotiation required)	This parameter specifies the MNC of the secondary operator. If the peer NodeB is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **UCNOPERATOR** MO of the peer RNC.

(Optional) Set the parameters in the **UtranExternalCellBand** MO to configure the secondary band of the external UTRAN cell if a multi-band cell is configured as the neighboring cell of a UTRAN cell.

Parameter Name	Parameter ID	Data Source	Setting Notes
Mobile country code	**UtranExternalCellBand.*Mcc***	Network plan (negotiation required)	This parameter specifies the MCC of the home RNC of the external UTRAN cell. If this neighboring cell works in RAN sharing mode, set this parameter to the MCC of the primary operator. If the peer NodeB is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **UCNOPERATOR** MO of the peer RNC.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Mobile network code	**UtranExternalCellBand**.***Mnc***	Network plan (negotiation required)	This parameter specifies the MNC of the home RNC of the external UTRAN cell. Set this parameter to the MNC of the primary operator if this external cell works in eRAN sharing with a common carrier mode. If the peer NodeB is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **UCNOPERATOR** MO of the peer RNC.
RNC ID	**UtranExternalCellBand**.***RncId***	Network plan (negotiation required)	This parameter specifies the home RNC ID of the external UTRAN cell. If the peer NodeB is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **URNCBASIC** MO of the Huawei NodeB.
UTRAN cell ID	**UtranExternalCellBand**.***CellId***	Network plan (negotiation required)	This parameter specifies the ID of the external UTRAN cell. It uniquely identifies a UTRAN cell within a NodeB. If the peer NodeB is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **UCELL** MO of the peer RNC.
Frequency band	**UtranExternalCellBand**.***FreqBand***	Network plan (negotiation required)	This parameter specifies the frequency band in which the cell operates. Set the primary and secondary frequency band flags for multi-band inter-RAT neighboring cells.

The following table describes the parameters that must be set in the **UtranNCell** MO to configure the neighbor relationship with a UTRAN cell.

Parameter Name	Parameter ID	Data Source	Setting Notes
Local cell ID	**UtranNCell**.***LocalCellId***	Network plan (negotiation not required)	Ensure that this parameter has been set in a **Cell** MO based on the network plan.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Mobile country code	**UtranNCell**.***Mcc***	Network plan (negotiation required)	This parameter specifies the MCC of the home RNC of the neighboring UTRAN cell. If this neighboring cell works in RAN sharing mode, set this parameter to the MCC of the primary operator. If the peer NodeB is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **UCNOPERATOR** MO of the peer RNC.
Mobile network code	**UtranNCell**.***Mnc***	Network plan (negotiation required)	This parameter specifies the MNC of the home RNC of the neighboring UTRAN cell. If this external cell works in RAN sharing mode, set this parameter to the MNC of the primary operator. If the peer NodeB is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **UCNOPERATOR** MO of the peer RNC.
UTRAN cell ID	**UtranNCell**.***CellId***	Network plan (negotiation required)	This parameter specifies the ID of the external UTRAN cell. It uniquely identifies a UTRAN cell within a NodeB. If the peer NodeB is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **UCELL** MO of the peer RNC.

The following table describes the parameters that must be set in the **InterRatPolicyCfgGroup** MO to configure an inter-RAT handover policy group.

Parameter Name	Parameter ID	Data Source	Setting Notes
InterRAT policy config group ID	**InterRatPolicyCfgGroup**.***InterRatPolicyCfgGroupId***	Network plan (negotiation not required)	This parameter specifies the ID of the inter-RAT handover policy group. If you want to set a different inter-RAT handover policy for services with a specific QCI, you must create a policy group. This parameter will be referenced by a standard or extended QCI. For details, see the **StandardQci** or **ExtendedQci** MO described later.

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Parameter Name	Parameter ID	Data Source	Setting Notes
UTRAN handover policy Configuration	**InterRatPolicyCfgGroup**.***UtranHoCfg***	Network plan (negotiation not required)	This parameter specifies the policy configuration for handovers to the UTRAN. Set this parameter based on the network plan. It is recommended that the default value be used if there is no special requirement for forbidding certain kind of handovers.
No handover flag	**InterRatPolicyCfgGroup**.***NoHoFlag***	Network plan (negotiation not required)	This parameter specifies whether to permit inter-RAT handovers. Set this parameter based on the network plan. It is recommended that the default value be used if there is no special requirement for forbidding certain kind of handovers.

The following table describes the parameters that must be set in the **StandardQci** MO to configure the inter-RAT handover policy group for a standardized QCI.

Parameter Name	Parameter ID	Data Source	Setting Notes
QoS Class Indication	**StandardQci**.***Qci***	Network plan (negotiation not required)	This parameter specifies the QCI of an EPS bearer. Set this parameter to the standard QCI with which services need to be configured with the inter-RAT handover policy.
InterRAT policy config group ID	**StandardQci**.***InterRatPolicyCfgGroupId***	Network plan (negotiation not required)	This parameter specifies the ID of the inter-RAT handover policy group. Set this parameter to the ID of an inter-RAT handover policy group that has been configured in an **InterRatPolicyCfgGroup** MO.

The following table describes the parameters that must be set in the **ExtendedQci** MO to configure the inter-RAT handover policy group for an extended QCI.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Extended QoS Class Identifier	**ExtendedQci**.***ExtendedQci***	Network plan (negotiation not required)	This parameter specifies the QCI of an EPS bearer. Set this parameter to the extended QCI with which services need to be configured with the inter-RAT handover policy.
InterRAT policy config group ID	**ExtendedQci**.***InterRatPolicyCfgGroupId***	Network plan (negotiation not required)	This parameter specifies the ID of the inter-RAT handover policy group. Set this parameter to the ID of an inter-RAT handover policy group that has been configured in an **InterRatPolicyCfgGroup** MO.

Scenario-Specific Data

The following table describes the parameter that must be set in the **ENodeBAlgoSwitch** MO to configure the mode for coverage-based inter-RAT handovers.

Parameter Name	Parameter ID	Data Source	Setting Notes
Handover Mode switch	**ENodeBAlgoSwitch**.***HoModeSwitch***	Network plan (negotiation not required)	Select a mode for each type of inter-RAT handover based on the network plan of the operator. Handover to UTRAN: ● UtranPsHoSwitch(UtranPsHoSwitch) ● UtranSrvccSwitch(UtranSrvccSwitch) ● UtranRedirectSwitch(UtranRedirectSwitch) Among the preceding switches, UtranRedirectSwitch(UtranRedirectSwitch) is basic to inter-RAT handover to UTRAN and must be turned on if the target RAT is UTRAN. Turn on or turn off other switches based on the network capability.

By default, the inter-RAT handover parameter group Group0 is used for services with all QCIs. To apply different inter-RAT handover parameter groups to services based on the QCI, the corresponding parameter group ID must be changed.

The following table describes the parameter that must be set in the **CellStandardQci** MO to configure parameter groups for services with standard QCIs.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Local cell ID	**CellStandardQci**.***LocalCellId***	Network plan (negotiation not required)	Ensure that this parameter has been set in a **Cell** MO based on the network plan.
QoS Class Indication	**CellStandardQci**.***Qci***	Network plan (negotiation not required)	Set this parameter to the index of the standard QCI for which the parameter group is configured.
InterRAT handover common group ID	**CellStandardQci**.***InterRatHoCommGroupId***	Network plan (negotiation not required)	Set this parameter to the index of the common inter-RAT handover parameter group.
InterRAT handover to UTRAN Group ID	**CellStandardQci**.***InterRatHoUtran-GroupId***	Network plan (negotiation not required)	Set this parameter to the index of the EUTRAN-to-UTRAN handover parameter group.
QCI priority for handover	**CellStandardQci**.***QciPriorityForHo***	Network plan (negotiation not required)	Set this parameter to the priority of the QCI.

The following table describes the parameters that must be set in the **CellExtendedQci** MO to configure parameter groups for services with extended QCIs.

Parameter Name	Parameter ID	Data Source	Setting Notes
Extended QCI	**CellExtendedQci**.***ExtendedQci***	Network plan (negotiation not required)	Set this parameter to the index of the extended QCI for which the parameter group is configured.
Local cell ID	**CellExtendedQci**.***LocalCellId***	Network plan (negotiation not required)	Ensure that this parameter has been set in a **Cell** MO based on the network plan.
InterRAT handover common group ID	**CellExtendedQci**.***InterRatHoCommGrou-pId***	Network plan (negotiation not required)	Set this parameter to the index of the common inter-RAT handover parameter group. For details about inter-RAT handover parameter group IDs and thresholds, see **8.1.7 Reconfiguration**.

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Parameter Name	Parameter ID	Data Source	Setting Notes
InterRAT handover to UTRAN Group ID	**CellExtendedQci**.***InterRatHoUtranGroupId***	Network plan (negotiation not required)	Set this parameter to the index of the EUTRAN-to-UTRAN handover parameter group.

The following table describes the parameter that must be set in the **UtranNCell** MO to configure blind handover priorities and measurement priorities for UTRAN neighboring cells.

Parameter Name	Parameter ID	Data Source	Setting Notes
Blind handover Priority	**UtranNCell**.***BlindHoPriority***	Network plan (negotiation not required)	A larger parameter value indicates a higher priority for the UTRAN neighboring cell to be selected during a blind handover. The value 0 indicates that blind handovers to the UTRAN neighboring cell are prohibited. It is recommended that this parameter be configured (with a value ranging from 1 to 32) for UTRAN neighboring cells whose coverages cover that of the serving cell. Set this parameter to a larger value for a neighboring cell that has wider overlapped coverage or lower frequency. Different cells can be configured with the same blind-handover priority.
Cell Measure Priority	**UtranNCell**.***CellMeasPriority***	Network plan (negotiation not required)	You are advised to set this parameter to **HIGH_PRIORITY(High Priority)** for neighboring cells with optimal network coverage and the maximum number of possible handovers, and to **LOW_PRIORITY(Low Priority)** for other neighboring cells.

The following table describes the parameter must be set in a **UtranNFreq** MO to configure the UTRAN frequency priority.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Connect frequency priority	**UtranNFreq.*ConnFreqPriority***	Network plan (negotiation not required)	This parameter specifies the frequency priority based on which the eNodeB selects a target frequency for blind redirection or contains a frequency in a measurement configuration. A larger value indicates a higher priority. If this priority is set to 0 for a frequency, this frequency is not selected as the target frequency for a blind redirection. You are advised to set this parameter for UTRAN frequencies with continuous coverage. If there are multiple UTRAN frequencies have continuous coverage, set this parameter to a larger value for a frequency in a lower frequency band. Set this parameter to 0 for a UTRAN frequency with non-continuous coverage. You must configure this parameter to a non-0 value for at least one UTRAN frequency to ensure that UEs that do not support UTRAN measurement can be handed over to the UTRAN in emergency conditions. Different frequencies can be configured with the same priority.

The following table describes the parameters must be set in a **CSFallBackBlindHoCfg** MO to configure RAT priorities for blind handovers. The **CSFallBackBlindHoCfg** MO takes effect for all inter-RAT handovers, rather than for only CSFB.

Parameter Name	Parameter ID	Data Source	Setting Notes
CN Operator ID	**CSFallBackBlind-HoCfg.*CnOperatorId***	Network plan (negotiation not required)	This parameter specifies the index of the operator. Ensure that this parameter has been set in a **CNOPERATOR** MO. Set this parameter for operators who need to be configured with priorities.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Highest priority InterRat	**CSFallBackBlind-HoCfg**.***InterRatHighest-Pri***	Network plan (negotiation not required)	This parameter specifies the highest-priority RAT for blind handovers. Setting this parameter to UTRAN ensures the optimal inter-RAT interoperability performance. If coverage holes exist in UTRAN, setting this parameter to GERAN to enhance the reliability of inter-RAT interoperability.
Second priority InterRat	**CSFallBackBlind-HoCfg**.***InterRatSecondPri***	Network plan (negotiation not required)	This parameter specifies the secondary-priority RAT for blind handovers. Set this parameter based on the value of **CSFallBackBlind-HoCfg**.*InterRatHighestPri*.
Lowest priority InterRat	**CSFallBackBlind-HoCfg**.***InterRatLowestPri***	Network plan (negotiation not required)	This parameter specifies the lowest-priority RAT for blind handovers. Set this parameter to the remaining available RAT or the RAT that do not support interoperability.

(Optional, required for only coverage-based inter-RAT handovers.) The following table describes the parameters must be set in a **CnOperatorHoCfg** MO to configure measurement offsets and priorities for coverage-based handovers to UTRAN or GERAN.

Parameter Name	Parameter ID	Data Source	Setting Notes
CN Operator ID	**CnOperatorHoCfg**.***CnOperatorId***	Network plan (negotiation not required)	This parameter specifies the index of the operator. Ensure that this parameter has been set in a **CnOperator** MO. Set this parameter for operators who need to be configured with priorities.

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Parameter Name	Parameter ID	Data Source	Setting Notes
UTRAN A2 Threshold RSRP Offset	**CnOperatorHoCfg**.***UtranA2ThdRsrpOffset***	Network plan (negotiation not required)	This parameter value plus the value of **InterRatHoCommGroup**.***InterRatHoA2ThdRsrp*** is the RSRP threshold for UTRAN measurement event A2. Set this parameter based on the actual conditions.

(Optional, required for only coverage-based inter-RAT handovers.) The following table describes the parameters must be set in a **UtranRanShare** MO to configure PLMN IDs of the operators that share a neighboring inter-RAT frequency.

This information is required to enable inter-RAT blind redirection triggered by no neighboring cells.

Parameter Name	Parameter ID	Data Source	Setting Notes
Local cell ID	**UtranRanShare**.***LocalCellId***	Network plan (negotiation not required)	Ensure that this parameter has been set in a **Cell** MO.
Downlink UARFCN	**UtranRanShare**.***UtranDlArfcn***	Network plan (negotiation not required)	This parameter specifies the DL UARFCN of the neighboring UTRAN cell. Add only the URTRAN frequencies for blind handovers triggered when there is no neighboring cells, and the UARFCN must have been set in a **UtranNFreq** MO.
Mobile country code	**UtranRanShare**.***Mcc***	Network plan (negotiation required)	This parameter specifies the MCC of the UARFCN for blind handovers triggered when there is no neighboring cells.
Mobile network code	**UtranRanShare**.***Mnc***	Network plan (negotiation required)	This parameter specifies the MNC of the UARFCN for blind handovers triggered when there is no neighboring cells.

8.1.5 Activation

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8.1.5.1 Using the CME to Perform Batch Configuration for Newly Deployed eNodeBs

Enter the values of related parameters in a summary data file, which also contains other data for the new eNodeBs to be deployed. Then, import the summary data file into the Configuration Management Express (CME) for batch configuration. For detailed instructions, see section "Creating eNodeBs in Batches" in the initial configuration guide for the eNodeB.

The summary data file may be a scenario-specific file provided by the CME or a customized file, depending on the following conditions:

- The managed objects (MOs) in **Table 8-1** are contained in a scenario-specific summary data file. In this situation, set the parameters in the MOs, and then verify and save the file.
- Some MOs in **Table 8-1** are not contained in a scenario-specific summary data file. In this situation, customize a summary data file to include the MOs before you can set the parameters.

Table 8-1 Parameters related to coverage-based inter-RAT handovers to UTRAN

MO	Sheet in the Summary Data File	Parameter Group	Remarks
ENodeBAlgoSwitch	User-defined sheet. **ENodeBAlgoSwitch** is recommended.	Handover Mode switch Frequency Layer Switch	This parameter must be customized on a list-type sheet of the template.
UtranNFreq	User-defined sheet. **UtranNFreq** is recommended.	Local cell identity, Downlink UARFCN, UTRAN version, UTRAN cell type indicator, Uplink UARFCN indicator, Uplink UARFCN, Reselection priority configure indicator, Cell reselection priority, PMAX(dBm), Frequency offset(dB), Minimum required quality level(dB), Minimum required RX level(dBm), UTRAN high priority threshold(2dB), UTRAN lower priority threshold(2dB)	This parameter must be customized on a list-type sheet of the template.

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MO	Sheet in the Summary Data File	Parameter Group	Remarks
UtranExternalCell	User-defined sheet. **UtranExternalCell** is recommended.	Mobile country code, Mobile network code, UTRAN cell identity, UTRAN cell type indicator, Downlink UARFCN, Uplink UARFCN configure indicator, Uplink UARFCN, RNC identity, Routing area code configure indicator, Routing area code, Primary scrambling code, Location area code, Cell name	This parameter must be customized on a list-type sheet of the template.
UtranExternalCellPlmn	User-defined sheet. **UtranExternalCellPlmn** is recommended.	UTRAN cell identity, Mobile country code, Mobile network code, Share mobile country code, Share mobile network code	This parameter must be customized on a list-type sheet of the template.
UtranNCell	User-defined sheet. **UtranNCell** is recommended.	Local cell identity, Mobile country code, Mobile network code, UTRAN cell identity, No handover indicator, No remove indicator, Blind handover priority, ANR flag, Local cell name, Neighbour cell name	This parameter must be customized on a list-type sheet of the template.

8.1.5.2 Using the CME to Perform Batch Configuration for Existing eNodeBs

Batch Activated

This feature can be batch activated using the CME. For detailed operations, see the following section in the CME product documentation or online help: Managing the CME > CME Guidelines > Enhanced Feature Management > Feature Operation and Maintenance.

Batch Reconfiguration

Batch reconfiguration using the CME is the recommended method to activate a feature on existing eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure is as follows:

Step 1 After creating a planned data area, choose **CME > Advanced > Customize Summary Data File** (U2000 client mode), or choose **Advanced > Customize Summary Data File** (CME client mode), to customize a summary data file for batch reconfiguration.

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NOTE

For context-sensitive help on a current task in the client, press **F1**.

Step 2 Choose **CME > LTE Application > Export Data > Export Base Station Bulk Configuration Data** (U2000 client mode), or choose **LTE Application > Export Data > Export Base Station Bulk Configuration Data** (CME client mode), to export the eNodeB data stored on the CME into the customized summary data file.

Step 3 In the summary data file, set the parameters in the MOs listed in **8.1.5.1 Using the CME to Perform Batch Configuration for Newly Deployed eNodeBs**, and close the file.

Step 4 Choose **CME > LTE Application > Import Data > Import Base Station Bulk Configuration Data** (U2000 client mode), or choose **LTE Application > Import Data > Import Base Station Bulk Configuration Data** (CME client mode), to import the summary data file into the CME.

Step 5 Choose **CME > Planned Area > Export Incremental Scripts** (U2000 client mode), or choose **Area Management > Planned Area > Export Incremental Scripts**, to export and activate the incremental scripts.

----End

8.1.5.3 Using the CME to Perform Single Configuration

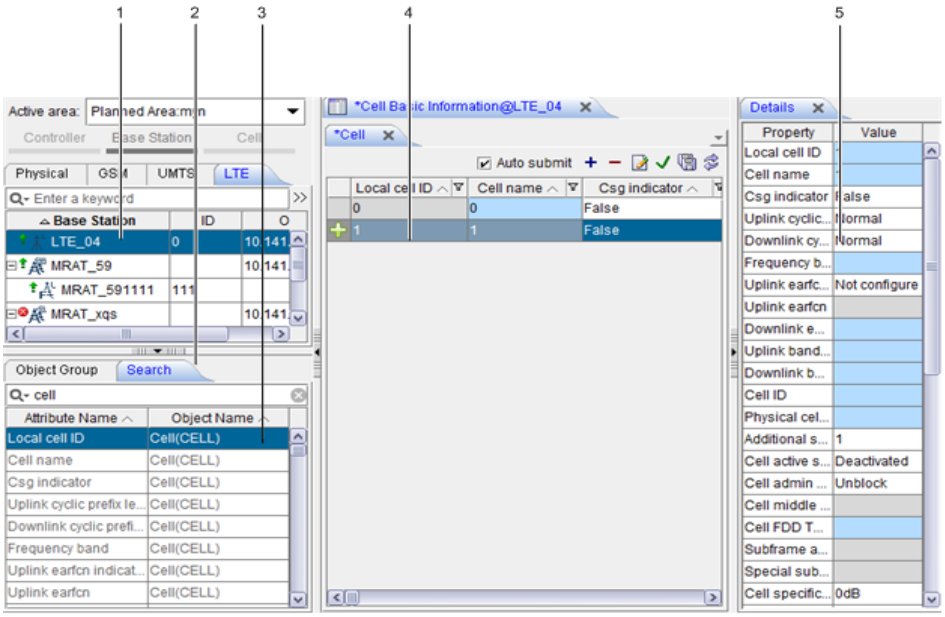
Using the CME to Perform Single Configuration

On the CME, set the parameters listed in the "Data Preparation" section for a single eNodeB. The procedure is as follows:

Step 1 In the planned data area, click **Base Station** in the upper left corner of the configuration window.

Step 2 In area 1 shown in **Figure 8-1**, select the eNodeB to which the MOs belong.

Figure 8-1 MO search and configuration window



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Step 3 On the **Search** tab page in area 2, enter an MO name, for example, **CELL**.

Step 4 In area 3, double-click the MO in the **Object Name** column. All parameters in this MO are displayed in area 4.

Step 5 Set the parameters in area 4 or 5.

Step 6 Choose **CME > Planned Area > Export Incremental Scripts** (U2000 client mode), or choose **Area Management > Planned Area > Export Incremental Scripts** (CME client mode), to export and activate the incremental scripts.

----End

8.1.5.4 Using MML Commands

Using MML Commands

Step 1 Run the **MOD ENODEBALGOSWITCH** command with the **UtranRedirectSwitch(UtranRedirectSwitch)** or **UtranPsHoSwitch(UtranPsHoSwitch)** check box selected under the **Handover Mode switch**.

Step 2 Run the following commands to add neighboring relationships with UTRAN cells:

1. **ADD UTRANNFREQ**
2. **ADD UTRANEXTERNALCELL**
3. (Optional) **ADD UTRANEXTERNALCELLPLMN**, which is used to configure the PLMN list for an external UTRAN cell when the neighboring UTRAN cell works in RAN sharing mode.
4. **ADD UTRANNCELL**

----End

MML Command Examples

//Activating coverage-based inter-RAT handover to UTRAN

```
MOD ENODEBALGOSWITCH: HoModeSwitch=UtranRedirectSwitch-1;
```

//Adding neighboring UTRAN cells

```
ADD UTRANNFREQ: LocalCellId=0, UtranDlArfcn=10650, UtranFddTddType=UTRAN_FDD,
UtranUlArfcnCfgInd=NOT_CFG, CellReselPriorityCfgInd=NOT_CFG;
ADD UTRANEXTERNALCELL: Mcc="460", Mnc="00", RncId=123, CellId=2,
UtranDlArfcn=10650, UtranUlArfcnCfgInd=NOT_CFG, UtranFddTddType=UTRAN_FDD,
RacCfgInd=CFG, Rac=130, PScrambCode=100, Lac=120;
ADD UTRANEXTERNALCELLPLMN: Mcc="460", Mnc="00", RncId=123, CellId=2,
ShareMcc="460", ShareMnc="20";
ADD UTRANNCELL: LocalCellId=0, Mcc="460", Mnc="00", RncId=123, CellId=2;
```

8.1.6 Activation Observation

The observation procedure is as follows:

Step 1 Trace messages over the Uu interface on the U2000 client after UEs access the network. Check for the A1- and A2-related measurement report information in the RRC_CONN_RECFG message.

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Step 2 Trace messages over the Uu interface while the UE is moving towards the cell edge. Check for the B1 or B2 measurement configuration in the RRC_CONN_RECFG message sent by the eNodeB after the eNodeB receives an A2-related measurement report from the UE.

Step 3 Trace messages over the Uu interface while the UE is moving towards an inter-frequency neighboring cell. Coverage-based inter-RAT handover is activated if either of the following situations occurs: After the eNodeB receives an event B1 or B2 measurement report from the UE, the eNodeB sends an RRC_MOBIL_FROM_EUTRA_CMD message or an RRC_CONN_REL message containing an inter-RAT frequency to the UE. After the eNodeB receives an event A2 measurement report for a blind handover, the UE does not support measurement of a specified RAT and the eNodeB sends an RRC_CONN_REL message containing an inter-RAT frequency to the UE.

----End

8.1.7 Reconfiguration

None

8.1.8 Deactivation

Using the CME to Perform Batch Configuration

Batch reconfiguration using the CME is the recommended method to deactivate a feature on eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure for feature deactivation is similar to that for feature activation described in **8.1.5.2 Using the CME to Perform Batch Configuration for Existing eNodeBs**. In the procedure, modify parameters according to **Table 8-2**.

Table 8-2 Parameters related to coverage-based inter-RAT handovers

MO	Sheet in the Summary Data File	Parameter Group	Setting Notes
ENodeBAlgoSwitch	User-defined sheet	Handover Mode switch	Clear the following check boxes: ● UtranPsHoSwitch ● UtranSrvccSwitch ● UtranRedirectSwitch

Using the CME to Perform Single Configuration

On the CME, set the parameters listed in each scenario for a single eNodeB. For detailed instructions, see **8.1.5.3 Using the CME to Perform Single Configuration** for feature activation.

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Using MML Commands

To deactivate coverage-based inter-RAT handover, clear the check boxes under the **Handover Mode switch** parameter listed in Table 8-2 by running the **MOD ENODEBALGOSWITCH** command.

MML Command Examples

//Deactivating coverage-based inter-RAT handover to UTRAN

```
MOD ENODEBALGOSWITCH:
HoModeSwitch=UtranPsHoSwitch-0&UtranSrvccSwitch-0&UtranRedirectSwitch-0;
```

8.1.9 Performance Monitoring

Parameter optimization for coverage-based inter-RAT handovers is similar to that for coverage-based inter-frequency handovers. In coverage-based inter-frequency handovers, event B1 measurement is performed. At present, an LTE system is usually deployed with other RAT systems at the same sites and the LTE system has the highest priority to carry services. Therefore, compared to inter-frequency handovers within the LTE system, inter-RAT handovers from the LTE system to other RAT systems have a lower event A2 threshold. This means that it is difficult to trigger a gap-assisted measurement for handovers from the LTE system to other RAT systems.

If inter-RAT handovers from an LTE system to a non-LTE system are not recommended, the corresponding algorithm needs to be disabled, and the frequency and neighboring cell information of this non-LTE system do not need to be configured in the LTE system. This prevents system performance deterioration.

Inter-RAT handovers are intended to ensure service continuity when users move around. How well these handovers serve this purpose reflects their performance. After activating inter-RAT handover, check the following KPIs to monitor and evaluate handover performance:

- KPIs indicating handover success rates: Inter-RAT Handover Out Success Rate
- KPI indicating the call drop rate: Abnormal Release Rate

If the handover success rates have not noticeably fallen and the call drop rate has not noticeably risen since inter-RAT handover was activated, this type of handover has been delivering good performance. Otherwise, this type of handover has been delivering poor performance.

Related counters are listed in Table 8-3 and Table 8-4.

Table 8-3 Counters related to coverage-based handover to UTRAN

Counter ID	Counter Name	Counter Description
1526729485	L.IRATHO.E2W.Coverage.PrepAttOut	Number of coverage-based handover preparation attempts from E-UTRAN to WCDMA network
1526729486	L.IRATHO.E2W.Coverage.ExecAttOut	Number of coverage-based handover execution attempts from E-UTRAN to WCDMA network

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Counter ID	Counter Name	Counter Description
1526729487	L.IRATHO.E2W.Coverage.ExecSuccOut	Number of successful coverage-based handover executions from E-UTRAN to WCDMA network

Table 8-4 Counter related to coverage-based redirection to UTRAN

Counter ID	Counter Name	Counter Description
1526729491	L.RRCRedirection.E2W.Coverage	Number of coverage-based redirections from E-UTRAN to WCDMA network

8.1.10 Parameter Optimization

Inter-RAT Handover Common Parameters

Common parameters related to inter-RAT handovers are contained in the **InterRatHoCommGroup** MO. The following table provides suggestions on how to reconfigure these parameters.

Parameter Name	Parameter ID	Data Source	Setting Notes
Local cell ID	**InterRatHoCommGroup.*LocalCellId***	Network plan (negotiation not required)	Ensure that this parameter has been set in a **Cell** MO based on the network plan.
InterRAT handover common group ID	**InterRatHoCommGroup.*InterRatHoCommGroupId***	Network plan (negotiation not required)	This parameter specifies the ID of a common parameter group related to inter-RAT handovers. If you want to set different inter-RAT handover parameters for services with a specific QCI, you must create a new parameter group, and reference the parameter group ID to the corresponding QCI. For detailed configuration, see **8.1.4 Data Preparation**.

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Parameter Name	Parameter ID	Data Source	Setting Notes
InterRAT A1A2 hysteresis	**InterRatHoCommGroup**.***InterRatHoA1A2Hyst***	Network plan (negotiation not required)	This parameter specifies the hysteresis for events A1 and A2 related to inter-RAT handover. This parameter is used to decrease frequent triggering of event evaluation caused by radio signal fluctuation. In this way, the probability of ping-pong handovers or handover decision errors is reduced. The value of this parameter has a negative correlation with the probability of ping-pong handovers or handover decision errors. The default value is recommended.
InterRAT A1A2 time to trigger	**InterRatHoCommGroup**.***InterRatHoA1A2TimeToTrig***	Network plan (negotiation not required)	This parameter specifies the time-to-trigger for inter-RAT measurement event A1 or A2. An appropriate setting of this parameter can effectively prevent unnecessary inter-RAT measurements and reduce the number of inter-RAT measurements. A larger value results in a smaller average number of inter-RAT measurements but a higher probability of service drops. The default value is recommended.
InterRAT A1 RSRP trigger threshold	**InterRatHoCommGroup**.***InterRatHoA1ThdRsrp***	Network plan (negotiation not required)	This parameter specifies the RSRP threshold for event A1 associated with inter-RAT handover. When the measured RSRP is above the threshold, the measurement report for event A1 is sent. A higher threshold makes it more difficult to trigger event A1 and stop inter-RAT measurement. A lower threshold makes it easier to trigger event A1 and stop inter-RAT measurement. The value of this parameter must be greater than or equal to (usually 4 dB greater than) the sum of **InterRatHoCommGroup**.*InterRatHoA1ThdRsrq* and threshold offset for inter-RAT measurement event A2 (set in the **CnOperatorHoCfg** MO).

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Parameter Name	Parameter ID	Data Source	Setting Notes
InterRAT A1 RSRQ trigger threshold	**InterRatHoCommGroup**.***InterRatHoA1ThdRsrq***	Network plan (negotiation not required)	This parameter specifies the RSRQ threshold for event A1 associated with inter-RAT handover. When the measured RSRQ is above the threshold, the measurement report for event A1 will be sent. A higher threshold makes it more difficult to trigger event A1 and stop inter-RAT measurement. A lower threshold makes it easier to trigger event A1 and stop inter-RAT measurement. The value of this parameter must be greater than or equal to (usually 2 dB greater than) the sum of the value of **InterRatHoCommGroup**.***InterRatHoA2ThdRsrq***.
InterRAT A2 RSRP trigger threshold	**InterRatHoCommGroup**.***InterRatHoA2ThdRsrp***	Network plan (negotiation not required)	This parameter specifies the RSRP threshold for event A2 associated with inter-RAT handover. When the measured RSRP is lower than the threshold, the measurement report for event A2 is sent. A higher threshold makes it easier to trigger event A2 and start inter-RAT measurement. A lower threshold makes it more difficult to trigger event A2 and start inter-RAT measurement. Set this parameter based on the actual conditions. Usually, set this parameter to the RSRP in the area where the UE throughput in the serving cell and that in the neighboring cell are similar.
InterRAT A2 RSRQ trigger threshold	**InterRatHoCommGroup**.***InterRatHoA2ThdRsrq***	Network plan (negotiation not required)	This parameter specifies the RSRQ threshold for event A2 associated with inter-RAT handover. When the measured RSRQ is lower than the threshold, the measurement report for event A2 is sent. A higher threshold makes it easier to trigger event A2 and start inter-RAT measurement. A lower threshold makes it more difficult to trigger event A2 and start inter-RAT measurement. Set this parameter based on the actual conditions. Usually, set this parameter to the RSRQ in the area where the UE throughput in the serving cell and that in the neighboring cell are similar.

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Parameter Groups Related to Inter-RAT Handovers to UTRAN

Parameter groups related to inter-RAT handovers to UTRAN are contained in the **InterRatHoUtranGroup** MO. The following table provides suggestions on how to reconfigure these parameters.

Parameter Name	Parameter ID	Data Source	Setting Notes
Local cell ID	**InterRatHoUtranGroup.*LocalCellId***	Network plan (negotiation not required)	Ensure that this parameter has been set in a **Cell** MO based on the network plan.
InterRAT handover to UTRAN Group ID	**InterRatHoUtranGroup.*InterRatHoUtranGroupId***	Network plan (negotiation not required)	This parameter specifies the ID of the parameter group related to inter-RAT handovers to UTRAN. If you want to set a different parameter group for services with a specific QCI, you must create a parameter group, and reference the parameter group ID to the corresponding QCI. For detailed configuration, see **8.1.4 Data Preparation**.
CoverageBased UTRAN ECN0 trigger threshold	**InterRatHoUtranGroup.*InterRatHoUtranB1ThdEcn0***	Network plan (negotiation not required)	This parameter specifies the Ec/No threshold for event B1 in a coverage-based, UL-quality-based, or distance-based handovers, or to SPID-based handovers to the HPLMN. A larger value of this parameter results in a lower probability of handovers to UTRAN, and a smaller value results in a higher probability. The default value is recommended.
CoverageBased UTRAN RSCP trigger threshold	**InterRatHoUtranGroup.*InterRatHoUtranB1ThdRscp***	Network plan (negotiation not required)	This parameter specifies the RSCP threshold for event B1 in a coverage-based, UL-quality-based, or distance-based handovers to UTRAN, or to SPID-based handovers to the HPLMN. A larger value of this parameter results in a lower probability of handovers to UTRAN, and a smaller value results in a higher probability. The default value is recommended.
UTRAN handover hysteresis	**InterRatHoUtranGroup.*InterRatHoUtranB1Hyst***	Network plan (negotiation not required)	This parameter specifies the hysteresis for event B1 related to inter-RAT handover to UTRAN. This parameter is used to prevent frequent triggering of event evaluation caused by radio signal fluctuation. In this way, the probability of ping-pong handovers or handover decision errors is reduced. The default value is recommended.

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Parameter Name	Parameter ID	Data Source	Setting Notes
UTRAN time to trigger	**InterRatHoUtranGroup**.***InterRatHoUtranB1TimeToTrig***	Network plan (negotiation not required)	This parameter specifies the time-to-trigger for event B1 related to inter-RAT handover to UTRAN. When detecting that the signal quality in at least one neighboring cell meets the entering condition, the UE does not send a measurement report to the eNodeB immediately. Instead, the UE sends a report only when the signal quality continuously meets the entering condition during the time-to-trigger period. If both UTRAN and GERAN neighboring cells are measured, you are advised to set the time-to-trigger for UTRAN neighboring cells to be smaller than that for GERAN neighboring cells, increasing the possibility of handovers to UTRAN neighboring cells. If both coverage-based inter-frequency handovers and coverage-based inter-RAT handovers are enabled, you are advised to set the time-to-trigger for the inter-RAT handover event be greater than that for the inter-frequency handover event, increasing the possibility of inter-frequency handovers.
Load Service Based UTRANB1 ECN0 threshold	**InterRatHoUtranGroup**.***LdSvBasedHoUtranB1ThdEcn0***	Network plan (negotiation not required)	This parameter specifies the Ec/No threshold for event B1 related to load- or service-based inter-RAT handover to UTRAN. A larger value of this parameter results in a lower probability of handovers to UTRAN, and a smaller value results in a higher probability. The default value is recommended.
Load Service Based UTRAN EventB1 RSCP trigger threshold	**InterRatHoUtranGroup**.***LdSvBasedHoUtranB1ThdRscp***	Network plan (negotiation not required)	This parameter specifies the RSCP threshold for event B1 related to load- or service-based inter-RAT handover to UTRAN. A larger value of this parameter results in a lower probability of handovers to UTRAN, and a smaller value results in a higher probability. The default value is recommended.

Inter-RAT Handover Parameters

Inter-RAT handover parameters are contained in the **InterRatHoComm** MO. The following table provides suggestions on how to reconfigure these parameters.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Max report cell number	**InterRatHoComm.*InterRatHoMaxRprtCell***	Network plan (negotiation not required)	This parameter specifies the maximum number of cells to be included in the measurement report. A smaller value of this parameter leads to less candidate target cells for handovers, which lowers the handover success rate but saves the air interface resources. A larger value leads to opposite effect. The default value is recommended.
Measurement report amount	**InterRatHoComm.*InterRatHoRprtAmount***	Network plan (negotiation not required)	This parameter specifies the number of periodical measurement reports to be sent after an inter-RAT handover event is triggered. It is used to prevent the impact of measurement report loss and internal processing failure on the handover. When the value is increased, the number of reports increases and the handover success rate rises to a certain extent. An excessively large number of reports, however, results in excessive signaling and hence causes a waste of radio interface resources. The default value is recommended.
Utran measurement trigger quantity	**InterRatHoComm.*InterRatHoUtranB1MeasQuan***	Network plan (negotiation not required)	This parameter specifies the quantity to be included in the measurements for handovers to UTRAN. The RSCP values are relatively stable, while the Ec/No values may vary with the network load. The value BOTH applies only to UEs complying with 3GPP Release 10. For UEs complying with 3GPP Release 8 or 9, the value BOTH takes the same effect as the value RSCP. The default value is recommended.

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Parameter Name	Parameter ID	Data Source	Setting Notes
UTRAN FDD measurement report interval	**InterRatHoComm.*InterRatHoUtranRprtInterval***	Network plan (negotiation not required)	This parameter specifies the interval at which periodic measurement reports are sent to the eNodeB after the event for inter-RAT handover to UTRAN is triggered. A smaller value of this parameter leads to more measurement reports over the air interface and more resources are consumed. A larger value of this parameter results in a larger handover delay, which lowers the handover success rate. The default value is recommended.
InterRat A1A2 measurement trigger quantity	**InterRatHoComm.*InterRatHoA1A2TrigQuan***	Network plan (negotiation not required)	This parameter specifies the quantity used to evaluate the triggering condition for the inter-RAT measurement event A1 or A2. The values are RSRP, RSRQ and BOTH. RSRP-based measurement results are stable, varying little with the load, and therefore there is little signal fluctuation. RSRQ-based measurement results vary with the load and are likely to reflect signal quality in real time. Set this parameter as required. By default, the reporting quantity for the inter-RAT measurement event A1 or A2 is BOTH, that is, both RSRP and RSRQ. It is recommended that the default value be used if the traffic load of LTE cells is not stable.
Inter RAT HO Trigger Event Type	**InterRatHoComm.*InterRatHoEventType***	Network plan (negotiation not required)	This parameter specifies the type of the inter-RAT handover event. This parameter applies only to coverage-based handovers. The default value is recommended.
Max Utran cell num in redirection	**InterRatHoComm.*CellInfoMaxUtranCellNum***	Network plan (negotiation not required)	This parameter specifies the maximum number of UTRAN cell system information messages that can be transmitted during a flash redirection procedure. The default value is recommended.

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8.1.11 Troubleshooting

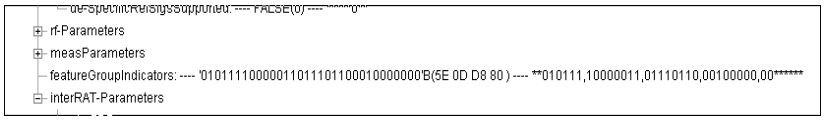
Fault 1: No Delivery of A1/A2-related Inter-RAT Measurement Configuration

Fault description: The eNodeB does not deliver the A1/A2-related inter-RAT measurement configuration to some UEs that have accessed the network.

Fault handling: Perform the following steps to determine whether the UEs support inter-RAT measurement or whether inter-RAT handover to the target system is enabled.

Step 1 Trace messages over the Uu interface on the U2000 client. Check for the target system and frequency band in the interRAT-Parameters IE of the RRC_UE_CAP_INFO message. If the interRAT-Parameters IE does not contain information about the target system and frequency band, the UE does not support inter-RAT measurement of the specified target system and frequency. **Figure 8-2** shows an example of an RRC_UE_CAP_INFO message.

Figure 8-2 RRC_UE_CAP_INFO message



Step 2 Run an MML command to check for the neighboring cell configurations. For example, if the target system is UTRAN, run the **LST UTRANNCELL** command to check whether the neighboring UTRAN cells are configured.

- If the neighboring UTRAN cells are not configured, configure the cells.
- If the neighboring UTRAN cells are configured, go to **Step 3**.

Step 3 Run the **LST ENODEBALGOSWITCH** command to check whether the corresponding handover mode is enabled.

- If the corresponding handover mode is disabled, enable it.
- If the corresponding handover mode is enabled, contact Huawei technical support.

----End

Fault 2: No Delivery of the B1/B2-related Measurement Configuration

Fault description After some UEs move to the cell edge, the eNodeB receives A2 measurement reports. However, the eNodeB does not deliver B1- or B2-related measurement configurations to the UEs.

Fault handling Perform the following steps to determine whether these UEs support B1- or B2-based inter-RAT measurement.

Step 1 Trace messages over the Uu interface on the U2000 client. Check whether the bits are 1 for target system in the featureGroupIndicators IE. For example, if the target system is UTRAN and the fifteenth and twenty-second bits are 1, the UE supports B1 and B2 measurements of the UTRAN. **Figure 8-3** shows an example of an RRC_UE_CAP_INFO message.

- If this IE indicates that the UE supports the target system and frequency band, go to **Step 2**.
- If this IE indicates that the UE does not support the target system and frequency band, the fault is caused due to UE capability.

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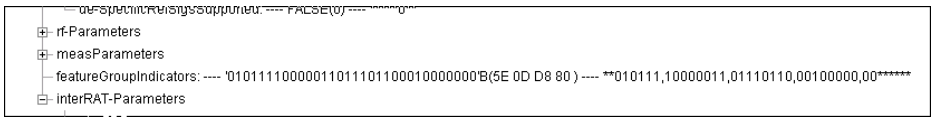
Figure 8-3 RRC_UE_CAP_INFO message



Step 2 Run an MML command to check for the neighboring cell configurations. For example, if the target system is UTRAN, run the **LST UTRANNCELL** command to check whether the neighboring UTRAN cells are configured.

- If the neighboring UTRAN cells are not configured, configure the cells.
- If the neighboring UTRAN cells are configured, go to **Step 3**.

Step 3 Run the **LST ENODEBALGOSWITCH** command to check whether the corresponding handover mode is enabled. If the corresponding handover mode is disabled, enable it. If the corresponding handover mode is enabled, contact Huawei technical support.

----End

Fault 3: No Delivery of a Redirection Command

Fault description After receiving A2 measurement reports for blind redirections from UEs that have moved to the cell edge, the eNodeB does not send redirection commands to these UEs.

Fault handling

Step 1 Run an MML command to check the blind handover priorities of inter-RAT neighboring cells. For example, if the target system is UTRAN, run the **LST UTRANNCELL** command to check whether the blind handover priorities configured for the neighboring UTRAN cells are in the range of 1 to 31.

- If the blind handover priorities configured for the neighboring UTRAN cells are not in this range, go to **Step 2**.
- If the blind handover priorities configured for the neighboring UTRAN cells are in this range, contact Huawei technical support.

Step 2 Run an MML command to change the blind handover priority to a value from 1 to 31. For example, if the target system is the UTRAN, run the **MOD UTRANNCELL** command.

----End

8.2 Service-based Inter-RAT Handover to UTRAN

8.2.1 When to Use Service-based Inter-RAT Handover to UTRAN

You are advised to enable service-based inter-RAT handover to UTRAN to divert some services to UTRAN in either of the following situations:

- In the early phase of LTE network construction, the E-UTRAN is incapable of providing continuous coverage.
- After the LTE network has stabilized, the E-UTRAN is heavily loaded with certain types of service.

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8.2.2 Required Information

Before you deploy service-based inter-RAT handover, collect information about the coverage and load of the target inter-RAT system. Ensure that the target inter-RAT system provides continuous coverage. If the target inter-RAT system is heavily loaded, service-based inter-RAT handover is not recommended. Service-based inter-RAT handover to a heavily loaded system affects user experience.

8.2.3 Deployment Requirements

Operating Environment

The feature LOFD-001019 PS Inter-RAT Mobility between E-UTRAN and UTRAN requires that the core-network equipment should support this feature. If the equipment is provided by Huawei, the version must be SAE1.2. If the core-network equipment is provided by another vendor, check with the vendor whether the equipment supports this feature.

Transmission Networking

None

License

To use service-based inter-RAT handovers, an operator must purchase and activate the licenses for the features listed in the following table.

Feature ID	Feature Name	License Control Item	NE	Sales Unit
LOFD-001019	PS Inter-RAT Mobility between E-UTRAN and UTRAN	PS Inter-RAT Mobility between E-UTRAN and UTRAN	eNodeB	per user in RRC connected mode
LOFD-001043	Service based inter-RAT handover to UTRAN	Service based inter-RAT handover to UTRAN	eNodeB	per user in RRC connected mode

8.2.4 Data Preparation

This section describes the data that you need to collect for setting parameters. Required data is data that you must collect for all scenarios. Collect scenario-specific data when necessary for a specific feature deployment scenario.

There are three types of data sources:

- Network plan (negotiation not required): parameter values planned and set by the operator
- Network plan (negotiation required): parameter values planned by the operator and negotiated with the EPC or peer transmission equipment

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- User-defined: parameter values set by users

Required Data

For details about the required data, see **8.1.4 Data Preparation**.

Scenario-specific Data

Service-based inter-RAT handover can be enabled only after coverage-based inter-RAT handover is enabled. This section describes the required parameters besides those for coverage-based inter-RAT handover. For details on the required parameters for coverage-based inter-RAT handover, see the **Scenario-Specific Data** part in **8.1.4 Data Preparation**.

The following table describes the parameter that must be set in the **ENodeBAlgoSwitch** MO to enable service-based inter-RAT handovers.

Parameter Name	Parameter ID	Data Source	Setting Notes
Handover Algo switch	**ENodeBAlgoSwitch.*HoAlgoSwitch***	Network plan (negotiation not required)	Turn on the **UtranServiceHoSwitch(UtranServiceHoSwitch)** switch based on the network plan of the operator.

The following table describes the parameters that must be set in the **ServiceIrHoCfgGroup** MO to configure a service-based inter-RAT handover policy.

Parameter Name	Parameter ID	Data Source	Setting Notes
CN Operator ID	**ServiceIrHoCfgGroup.*CnOperatorId***	Network plan (negotiation not required)	Ensure that this parameter has been set in a **CnOperator** MO based on the network plan.
Service-based inter-RAT handover policy group ID	**ServiceIrHoCfgGroup.*ServiceIrHoCfgGroupId***	Network plan (negotiation not required)	This parameter specifies the ID of the service-based inter-RAT handover policy group.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Inter-RAT handover state	**ServiceIrHoCfgGroup**.***InterRatHoState***	Network plan (negotiation not required)	This parameter specifies whether the service-based inter-RAT handover is forbidden, allowed, or required for services with the QCI. Parameter values 0, 1, and 2 indicate forbidden, allowed, and required, respectively. Set this parameter based on the application scenario.

The following tables describe the parameters that must be set to configure inter-RAT handover policy groups for services with standard or extended QCIs.

Parameters that must be set to configure parameter groups for services with standard QCIs are in the **CnOperatorStandardQci** MO.

Parameter Name	Parameter ID	Data Source	Setting Notes
CN Operator ID	**CnOperatorStandardQci**.***CnOperatorId***	Network plan (negotiation not required)	Ensure that this parameter has been set in a **CnOperator** MO based on the network plan.
QoS Class Indication	**CnOperatorStandardQci**.***Qci***	Network plan (negotiation not required)	This parameter specifies the index of the standard QCI for which the parameter group is configured.
Service-based inter-RAT handover policy group ID	**CnOperatorStandardQci**.***ServiceIrHoCfgGroupId***	Network plan (negotiation not required)	Set this parameter to the ID configured in **ServiceIrHoCfgGroup**.***ServiceIrHoCfgGroupId***.

Parameters that must be set to configure parameter groups for services with extended QCIs are in the **CnOperatorExtendedQci** MO.

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Parameter Name	Parameter ID	Data Source	Setting Notes
CN Operator ID	**CnOperatorExtendedQci**.***CnOperatorId***	Network plan (negotiation not required)	Ensure that this parameter has been set in a **CnOperator** MO based on the network plan.
QoS Class Indication	**CnOperatorExtendedQci**.***ExtendedQci***	Network plan (negotiation not required)	This parameter specifies the index of the extended QCI for which the parameter group is configured.
Service-based inter-RAT handover policy group ID	**CnOperatorExtendedQci**.***ServiceIrHoCfgGroupId***	Network plan (negotiation not required)	Set this parameter to the ID configured in **ServiceIrHoCfgGroup**.***ServiceIrHoCfgGroupId***.

8.2.5 Activation

Before activating service-based inter-RAT handover, ensure that coverage-based inter-RAT handover has been activated.

8.2.5.1 Using the CME to Perform Batch Configuration for Newly Deployed eNodeBs

Enter the values of related parameters in a summary data file, which also contains other data for the new eNodeBs to be deployed. Then, import the summary data file into the Configuration Management Express (CME) for batch configuration. For detailed instructions, see section "Creating eNodeBs in Batches" in the initial configuration guide for the eNodeB.

The summary data file may be a scenario-specific file provided by the CME or a customized file, depending on the following conditions:

- The managed objects (MOs) in **Table 8-5** are contained in a scenario-specific summary data file. In this situation, set the parameters in the MOs, and then verify and save the file.
- Some MOs in **Table 8-5** are not contained in a scenario-specific summary data file. In this situation, customize a summary data file to include the MOs before you can set the parameters.

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Table 8-5 Parameters related to service-based inter-RAT handovers to UTRAN

MO	Sheet in the Summary Data File	Parameter Group	Remarks
ENodeBAlgoSwitch	User-defined sheet. **ENodeBAlgoSwitch** is recommended.	Handover Algo switch, Handover Mode switch	This parameter must be customized on a list-type sheet of the template.
ServiceIrHoCfgGroup	User-defined sheet. **ServiceIrHoCfgGroup** is recommended.	CN Operator ID, Service-based inter-RAT handover policy group ID, Inter-RAT handover state	This parameter must be customized on a list-type sheet of the template.
UtranNFreq	User-defined sheet. **UtranNFreq** is recommended.	Local cell identity, Downlink UARFCN, UTRAN version, UTRAN cell type indicator, Uplink UARFCN indicator, Uplink UARFCN, Reselection priority configure indicator, Cell reselection priority, PMAX(dBm), Frequency offset(dB), Minimum required quality level(dB), Minimum required RX level(dBm), UTRAN high priority threshold(2dB), UTRAN lower priority threshold(2dB)	This parameter must be customized on a list-type sheet of the template.
UtranExternalCell	User-defined sheet. **UtranExternalCell** is recommended.	Mobile country code, Mobile network code, UTRAN cell identity, UTRAN cell type indicator, Downlink UARFCN, Uplink UARFCN configure indicator, Uplink UARFCN, RNC identity, Routing area code configure indicator, Routing area code, Primary scrambling code, Location area code, Cell name	This parameter must be customized on a list-type sheet of the template.

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MO	Sheet in the Summary Data File	Parameter Group	Remarks
UtranExternalCellPlmn	User-defined sheet. **UtranExternalCellPlmn** is recommended.	UTRAN cell identity, Mobile country code, Mobile network code, Share mobile country code, Share mobile network code	This parameter must be customized on a list-type sheet of the template.
UtranNCell	User-defined sheet. **UtranNCell** is recommended.	Local cell identity, Mobile country code, Mobile network code, UTRAN cell identity, No handover indicator, No remove indicator, Blind handover priority, ANR flag, Local cell name, Neighbour cell name	This parameter must be customized on a list-type sheet of the template.

8.2.5.2 Using the CME to Perform Batch Configuration for Existing eNodeBs

Batch Activated

This feature can be batch activated using the CME. For detailed operations, see the following section in the CME product documentation or online help: Managing the CME > CME Guidelines > Enhanced Feature Management > Feature Operation and Maintenance.

Batch Reconfiguration

Batch reconfiguration using the CME is the recommended method to activate a feature on existing eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure is as follows:

Step 1 After creating a planned data area, choose **CME > Advanced > Customize Summary Data File** (U2000 client mode), or choose **Advanced > Customize Summary Data File** (CME client mode), to customize a summary data file for batch reconfiguration.

NOTE

For context-sensitive help on a current task in the client, press **F1**.

Step 2 Choose **CME > LTE Application > Export Data > Export Base Station Bulk Configuration Data** (U2000 client mode), or choose **LTE Application > Export Data > Export Base Station Bulk Configuration Data** (CME client mode), to export the eNodeB data stored on the CME into the customized summary data file.

Step 3 In the summary data file, set the parameters in the MOs listed in **8.2.5.1 Using the CME to Perform Batch Configuration for Newly Deployed eNodeBs**, and close the file.

Step 4 Choose **CME > LTE Application > Import Data > Import Base Station Bulk Configuration Data** (U2000 client mode), or choose **LTE Application > Import Data >**

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Import Base Station Bulk Configuration Data (CME client mode), to import the summary data file into the CME.

Step 5 Choose **CME > Planned Area > Export Incremental Scripts** (U2000 client mode), or choose **Area Management > Planned Area > Export Incremental Scripts**, to export and activate the incremental scripts.

----End

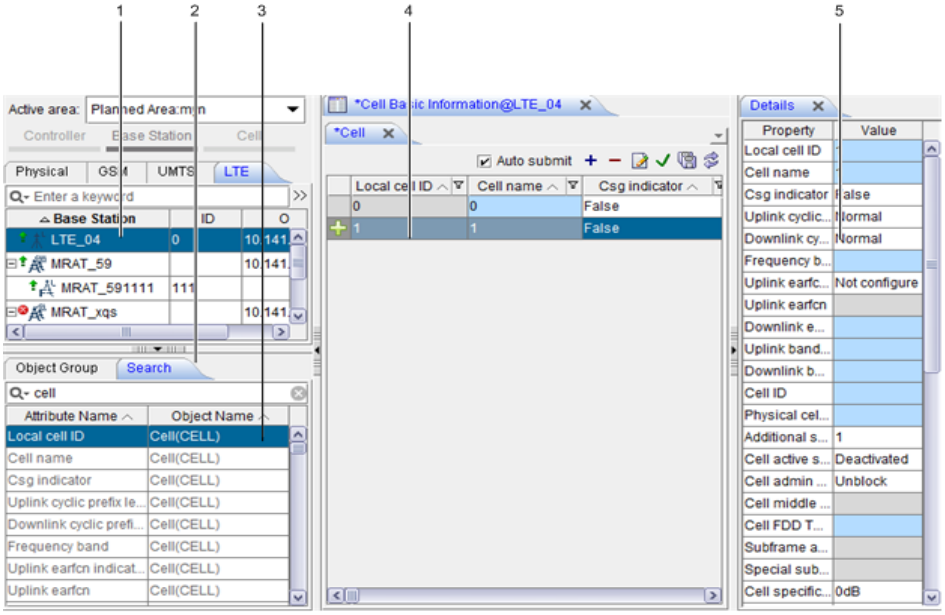
8.2.5.3 Using the CME to Perform Single Configuration

On the CME, set the parameters listed in the "Data Preparation" section for a single eNodeB. The procedure is as follows:

Step 1 In the planned data area, click **Base Station** in the upper left corner of the configuration window.

Step 2 In area 1 shown in **Figure 8-4**, select the eNodeB to which the MOs belong.

Figure 8-4 MO search and configuration window



Step 3 On the **Search** tab page in area 2, enter an MO name, for example, **CELL**.

Step 4 In area 3, double-click the MO in the **Object Name** column. All parameters in this MO are displayed in area 4.

Step 5 Set the parameters in area 4 or 5.

Step 6 Choose **CME > Planned Area > Export Incremental Scripts** (U2000 client mode), or choose **Area Management > Planned Area > Export Incremental Scripts** (CME client mode), to export and activate the incremental scripts.

----End

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8.2.5.4 Using MML Commands

Using MML Commands

Step 1 Run the **MOD ENODEBALGOSWITCH** command with the **UtranServiceHoSwitch(UtranServiceHoSwitch)** check box selected under the **Handover Algo switch**.

Step 2 Run the **MOD SERVICEIRHOCFGGROUP** command to configure a service-based inter-RAT handover policy group.

Step 3 Run the following commands to add the neighboring relationships with UTRAN cells:

1. **ADD UTRANNFREQ**
2. **ADD UTRANEXTERNALCELL**
3. (Optional) **ADD UTRANEXTERNALCELLPLMN**, which is used to configure the PLMN list for an external UTRAN cell when the neighboring UTRAN cell works in RAN sharing mode.
4. **ADD UTRANNCELL**

----End

MML Command Examples

//Activating service-based inter-RAT handover to UTRAN

```
MOD ENODEBALGOSWITCH: HoAlgoSwitch=UtranServiceHoSwitch-1;MOD
SERVICEIRHOCFGGROUP: CnOperatorId=0, ServiceIrHoCfgGroupId=0,
InterRatHoState=MUST_HO;
```

//Adding neighboring UTRAN cells

```
ADD UTRANNFREQ: LocalCellId=0, UtranDlArfcn=10650, UtranFddTddType=UTRAN_FDD,
UtranUlArfcnCfgInd=NOT_CFG, CellReselPriorityCfgInd=NOT_CFG;ADD
UTRANEXTERNALCELL: Mcc="460", Mnc="00", RncId=123, CellId=2, UtranDlArfcn=10650,
UtranUlArfcnCfgInd=NOT_CFG, UtranFddTddType=UTRAN_FDD, RacCfgInd=CFG, Rac=130,
PScrambCode=100, Lac=120;ADD UTRANEXTERNALCELLPLMN: Mcc="460", Mnc="00",
RncId=123, CellId=2, ShareMcc="460", ShareMnc="20";ADD UTRANNCELL: LocalCellId=0,
Mcc="460", Mnc="00", RncId=123, CellId=2;
```

8.2.6 Activation Observation

The observation procedure is as follows:

Step 1 Trace messages over the Uu interface for UEs running only voice services with QCI 1. Check the RRC_CONN_RECFG message. If the message contains B1 measurement information, you can infer that service-based inter-RAT handover is activated.

Step 2 If the eNodeB sends an RRC_MOBIL_FROM_EUTRA_CMD message after receiving the B1 measurement report from the UE, you can infer that service-based inter-RAT handover is triggered.

----End

8.2.7 Reconfiguration

None

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8.2.8 Deactivation

Using the CME to Perform Batch Configuration

Batch reconfiguration using the CME is the recommended method to deactivate a feature on eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure for feature deactivation is similar to that for feature activation described in **8.2.5.3 Using the CME to Perform Single Configuration**. In the procedure, modify parameters according to the table listed in each scenario.

Table 8-6 Parameters related to service-based inter-RAT handover

MO	Sheet in the Summary Data File	Parameter Group	Setting Notes
ENodeBAlgoSwitch	User-defined sheet	HoAlgoSwitch	Clear the following check boxes: UtranServiceHoSwitch, GeranServiceHoSwitch, Cdma1xRttServiceHoSwitch, CdmaHrpdServiceHoSwitch

Using the CME to Perform Single Configuration

On the CME, set the parameters listed in each scenario for a single eNodeB. For detailed instructions, see **8.2.5.3 Using the CME to Perform Single Configuration** for feature activation.

Using MML Commands

To deactivate service-based inter-RAT handover to UTRAN, turn off the corresponding switch by running the **MOD ENODEBALGOSWITCH** command.

MML Command Examples

//Deactivating service-based inter-RAT handover to UTRAN

```
MOD ENODEBALGOSWITCH: HoAlgoSwitch=UtranServiceHoSwitch-0;
```

8.2.9 Performance Monitoring

Performance monitoring for coverage-based inter-RAT handover is similar to that for inter-RAT mobility between E-UTRAN and UTRAN. For details, see **8.1.9 Performance Monitoring**.

Related counters are listed in **Table 8-7** and **Table 8-8**.

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Table 8-7 Counters related to service-based handover to UTRAN

Counter ID	Counter Name	Counter Description
1526730018	L.IRATHO.E2W.Service.PrepAttOut	Number of service-based handover preparation attempts from E-UTRAN to WCDMA network
1526730019	L.IRATHO.E2W.Service.ExecAttOut	Number of service-based handover execution attempts from E-UTRAN to WCDMA network
1526730020	L.IRATHO.E2W.Service.ExecSuccOut	Number of successful service-based handover executions from E-UTRAN to WCDMA network

Table 8-8 Counter related to service-based redirection to UTRAN

Counter ID	Counter Name	Counter Description
1526730024	L.RRCRedirection.E2W.Service	Number of service-based redirections from E-UTRAN to WCDMA network

8.2.10 Parameter Optimization

The reconfiguration of parameters that are not involved in this section is the same as the reconfiguration of parameters for coverage-based inter-RAT handovers. For details, see **8.1.10 Parameter Optimization**.

Parameter groups related to inter-RAT handovers to UTRAN are contained in the **InterRatHoUtranGroup** MO. The following table provides suggestions on how to reconfigure these parameters.

Parameter Name	Parameter ID	Data Source	Setting Notes
Name Load Service Based UTRANB1 ECN0 threshold	**InterRatHoUtranGroup.*LdSvBasedHoUtranB1ThdEcn0***	Network plan (negotiation not required)	This parameter specifies the Ec/No threshold for event B1 related to load- or service-based inter-RAT handover to UTRAN. A larger value of this parameter results in a lower probability of handovers to UTRAN, and a smaller value results in a higher probability. The default value is recommended.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Load Service Based UTRAN EventB1 RSCP trigger threshold	**InterRatHoUtranGroup.*LdSvBasedHoUtranB1ThdRscp***	Network plan (negotiation not required)	This parameter specifies the RSCP threshold for event B1 related to load- or service-based inter-RAT handover to UTRAN. A larger value of this parameter results in a lower probability of handovers to UTRAN, and a smaller value results in a higher probability. The default value is recommended.
UTRAN time to trigger	**InterRatHoUtranGroup.*InterRatHoUtranB1TimeToTrig***	Network plan (negotiation not required)	This parameter specifies the time-to-trigger for event B1 related to inter-RAT handover to UTRAN. When detecting that the signal quality in at least one neighboring cell meets the entering condition, the UE does not send a measurement report to the eNodeB immediately. Instead, the UE sends a report only when the signal quality continuously meets the entering condition during the time-to-trigger period. If both UTRAN and GERAN neighboring cells are measured, you are advised to set the time-to-trigger for UTRAN neighboring cells to be smaller than that for GERAN neighboring cells, increasing the possibility of handovers to UTRAN neighboring cells.

8.2.11 Troubleshooting

For details about troubleshooting, see **8.1.11 Troubleshooting**.

8.3 Distance-based Inter-RAT Handover to UTRAN

8.3.1 When to Use Distance-based Inter-RAT Handover to UTRAN

You are advised to enable distance-based inter-RAT handover when the E-UTRAN cannot cover the entire network and causes severe overshoot coverage for the cells of another RAT. The latter situation occurs if the coverage of the E-UTRAN overlaps with that of another RAT and the RF signals from an E-UTRAN cell travels distances significantly longer than the inter-site spacing specified in the network plan. If the cells of the other RAT are not configured as neighboring cells of the E-UTRAN cell, inter-frequency handovers cannot be triggered in time and call drops will occur. To reduce the probability of such call drops, you can enable distance-based inter-RAT handover.

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8.3.2 Required Information

Before deploying distance-based inter-RAT handover, you need to collect the normal inter-site spacing and the distance that the RF signals of the cell exerting cross-cell coverage can travel to properly set the TA threshold.

8.3.3 Deployment Requirements

Operating Environment

The feature LOFD-001019 PS Inter-RAT Mobility between E-UTRAN and UTRAN requires that the core-network equipment should support this feature. If the equipment is provided by Huawei, the version must be SAE1.2. If the core-network equipment is provided by another vendor, check with the vendor whether the equipment supports this feature.

Transmission Networking

None

License

To use distance-based inter-RAT handovers, an operator must purchase and activate the licenses for the features listed in the following table.

Feature ID	Feature Name	License Control Item	NE	Sales Unit
LOFD-001019	PS Inter-RAT Mobility between E-UTRAN and UTRAN	PS Inter-RAT Mobility between E-UTRAN and UTRAN	eNodeB	per user in RRC connected mode
LOFD-001072	Distance based Inter-RAT handover to UTRAN	Distance based Inter-RAT handover to UTRAN	eNodeB	per user in RRC connected mode

8.3.4 Data Preparation

This section describes the data that you need to collect for setting parameters. Required data is data that you must collect for all scenarios. Collect scenario-specific data when necessary for a specific feature deployment scenario.

There are three types of data sources:

- Network plan (negotiation not required): parameter values planned and set by the operator
- Network plan (negotiation required): parameter values planned by the operator and negotiated with the EPC or peer transmission equipment
- User-defined: parameter values set by users

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Required Data

For details about the required data, see **8.1.4 Data Preparation**.

Scenario-specific Data

Service-based inter-RAT handover can be enabled only after coverage-based inter-RAT handover is enabled. This section describes the required parameters besides those for coverage-based inter-RAT handover. For details on the required parameters for coverage-based inter-RAT handover, see the **Scenario-Specific Data** part in **8.1.4 Data Preparation**.

To enable distance-based inter-RAT handovers, the corresponding switch must be turned on and the measurement object type must be set to UTRAN or GERAN.

The following table describes the parameters that must be set in the **CellAlgoSwitch** MO to enable distance-based inter-RAT handovers.

Parameter Name	Parameter ID	Data Source	Setting Notes
Local cell ID	**CellAlgoSwitch**.***LocalCellId***	Network plan (negotiation not required)	Ensure that this parameter has been set in a **Cell** MO.
Distance-based Handover Switch	**CellAlgoSwitch**.***DistBasedHoSwitch***	Network plan (negotiation not required)	This parameter specifies whether to enable distance-based handovers. If this switch is turned on, distance-based handovers are allowed. If this switch is turned off, distance-based handovers to any cells are prohibited.

The following table describes the parameters that must be set in the **DistBasedHO** MO to set the measurement object type.

Parameter Name	Parameter ID	Data Source	Setting Notes
Local cell ID	**DistBasedHO**.***LocalCellId***	Network plan (negotiation not required)	Ensure that this parameter has been set in a **Cell** MO.
Distance-based Measurement Object Type	**DistBasedHO**.***DistBasedMeasObjType***	Network plan (negotiation not required)	This parameter specifies the measurement object type for distance-based handovers. If this parameter is set to **UTRAN** or **GERAN**, distanced-based handovers to neighboring UTRAN or GERAN cells are allowed.

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8.3.5 Activation

Before activating distance-based inter-RAT handover, ensure that coverage-based inter-RAT handover has been activated.

Using the CME to Perform Batch Configuration for Newly Deployed eNodeBs

Enter the values of related parameters in a summary data file, which also contains other data for the new eNodeBs to be deployed. Then, import the summary data file into the Configuration Management Express (CME) for batch configuration. For detailed instructions, see section "Creating eNodeBs in Batches" in the initial configuration guide for the eNodeB.

The summary data file may be a scenario-specific file provided by the CME or a customized file, depending on the following conditions:

- Related managed objects (MOs) are contained in a scenario-specific summary data file. In this situation, set the parameters in the MOs, and then verify and save the file.
- Some related MOs are not contained in a scenario-specific summary data file. In this situation, customize a summary data file to include the MOs before you can set the parameters.

Table 8-9 Parameters related to distance-based inter-RAT handovers to UTRAN

MO	Sheet in the Summary Data File	Parameter Group	Remarks
CellAlgoSwitch	User-defined sheet. **CellAlgoSwitch** is recommended.	Distance-based Handover Switch	This parameter must be customized on a list-type sheet of the template.
DistBasedHo	User-defined sheet. **DistBasedHo** is recommended.	Distance-based Measurement Object Type, Distance-based Handover Threshold	This parameter must be customized on a list-type sheet of the template.

Using the CME to Perform Batch Configuration for Existing eNodeBs

Batch reconfiguration using the CME is the recommended method to activate a feature on existing eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure is as follows:

Step 1 Choose **CME > Advanced > Customize Summary Data File** (U2000 client mode), or choose **Advanced > Customize Summary Data File** (CME client mode), to customize a summary data file for batch reconfiguration.

NOTE

For context-sensitive help on a current task in the client, press **F1**.

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Step 2 Choose **CME > LTE Application > Export Data > Export Base Station Bulk Configuration Data** (U2000 client mode), or choose **LTE Application > Export Data > Export Base Station Bulk Configuration Data** (CME client mode), to export the eNodeB data stored on the CME into the customized summary data file.

Step 3 In the summary data file, set the parameters in the MOs listed in **Table 8-9**, and close the file.

Step 4 Choose **CME > LTE Application > Import Data > Import Base Station Bulk Configuration Data** (U2000 client mode), or choose **LTE Application > Import Data > Import Base Station Bulk Configuration Data** (CME client mode), to import the summary data file into the CME.

Step 5 Choose **CME > Planned Area > Export Incremental Scripts** (U2000 client mode), or choose **Area Management > Planned Area > Export Incremental Scripts**, to export and activate the incremental scripts.

----End

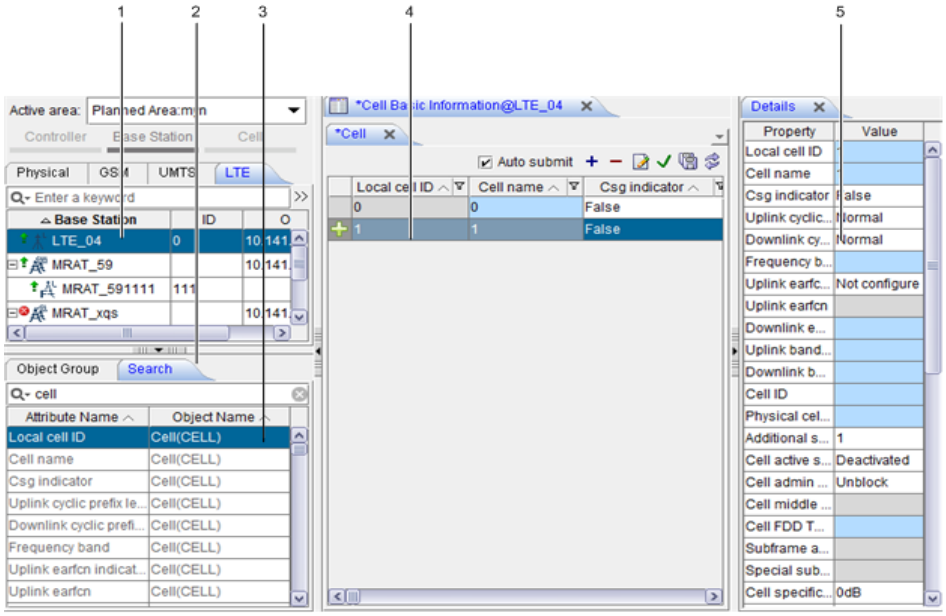
Using the CME to Perform Single Configuration

On the CME, set the parameters listed in the "Data Preparation" section for a single eNodeB. The procedure is as follows:

Step 1 In the planned data area, click **Base Station** in the upper left corner of the configuration window.

Step 2 In area 1 shown in **Figure 8-5**, select the eNodeB to which the MOs belong.

Figure 8-5 MO search and configuration window



Step 3 On the **Search** tab page in area 2, enter an MO name, for example, **CELL**.

Step 4 In area 3, double-click the MO in the **Object Name** column. All parameters in this MO are displayed in area 4.

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Step 5 Set the parameters in area 4 or 5.

Step 6 Choose **CME > Planned Area > Export Incremental Scripts** (U2000 client mode), or choose **Area Management > Planned Area > Export Incremental Scripts** (CME client mode), to export and activate the incremental scripts.

----End

Using MML Commands

Step 1 Run the **MOD CELLALGOSWITCH** command to enable distance-based inter-RAT handover.

Step 2 Run the **MOD DISTBASEDHO** command to set the measurement object type to UTRAN.

Step 3 Run the following commands to add neighbor relationships with UTRAN cells:

1. **ADD UTRANNFREQ**
2. **ADD UTRANEXTERNALCELL**
3. (Optional) **ADD UTRANEXTERNALCELLPLMN**, which is used to configure the PLMN list for an external UTRAN cell when the neighboring UTRAN cell works in RAN sharing mode.
4. **ADD UTRANNCELL**

----End

MML Command Examples

//Activating distance-based inter-RAT handover to UTRAN

```
MOD CELLALGOSWITCH: LocalCellId=0, DistBasedHoSwitch=ON;MOD DISTBASEDHO:
LocalCellId=0, DistBasedMeasObjType=UTRAN-1;
```

//Adding neighboring UTRAN cells

```
ADD UTRANNFREQ: LocalCellId=0, UtranDlArfcn=10650, UtranFddTddType=UTRAN_FDD,
UtranUlArfcnCfgInd=NOT_CFG, CellReselPriorityCfgInd=NOT_CFG;ADD
UTRANEXTERNALCELL: Mcc="460", Mnc="00", RncId=123, CellId=2, UtranDlArfcn=10650,
UtranUlArfcnCfgInd=NOT_CFG, UtranFddTddType=UTRAN_FDD, RacCfgInd=CFG, Rac=130,
PScrambCode=100, Lac=120;ADD UTRANEXTERNALCELLPLMN: Mcc="460", Mnc="00",
RncId=123, CellId=2, ShareMcc="460", ShareMnc="20";ADD UTRANNCELL: LocalCellId=0,
Mcc="460", Mnc="00", RncId=123, CellId=2;
```

8.3.6 Activation Observation

The observation procedure is as follows:

Step 1 Trace messages over the Uu interface on the U2000 client after the UE accesses the network at the center of an E-UTRAN cell that exerts overshoot coverage to other cells. Move the UE toward the cell edge. During the movement, coverage-based A2 measurements are not triggered due to overshoot coverage, but the threshold for distance-based handover is reached. Check the RRC_CONN_RECFG message. If the message contains the distance-based inter-RAT measurement configuration, you can infer that distance-based inter-RAT handover is activated.

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Step 2 If the eNodeB sends an RRC_CONN_RECFG message after receiving the B1 measurement report from the UE for a distance-based inter-RAT handover, you can infer that distance-based inter-RAT handover is triggered.

----End

8.3.7 Reconfiguration

None

8.3.8 Deactivation

Using the CME to Perform Batch Configuration

Batch reconfiguration using the CME is the recommended method to deactivate a feature on eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure for feature deactivation is similar to that for feature activation described in **Table 8-10**. In the procedure, modify parameters according to the table listed in each scenario.

Table 8-10 Parameters related to distance-based inter-RAT handovers

MO	Sheet in the Summary Data File	Parameter Group	Setting Notes
CellAlgoSwitch	User-defined sheet	Distance-based Handover Switch	Set the switch to **OFF(Off)**.

Using the CME to Perform Single Configuration

On the CME, set the parameters listed in each scenario for a single eNodeB. For detailed instructions, see **Using the CME to Perform Single Configuration** for feature activation.

Using MML Commands

To deactivate distance-based handover to UTRAN, run the **MOD DISTBASEDHO** command, and clear the corresponding check box under the **Distance-based Measurement Object Type** parameter.

MML Command Examples

//Deactivating distance-based inter-RAT handover to UTRAN

```
MOD DISTBASEDHO: LocalCellId=0, DistBasedMeasObjType=UTRAN-0;
```

8.3.9 Performance Monitoring

Performance monitoring for distance-based inter-RAT handover is similar to that for inter-RAT mobility between E-UTRAN and UTRAN. For details, see **8.1.9 Performance Monitoring**.

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Related counters are listed in **Table 8-11** and **Table 8-12**.

Table 8-11 Counters related to distance-based handover to UTRAN

Counter ID	Counter Name	Counter Description
1526730034	L.IRATHO.E2W.Distance.PrepAttOut	Number of distance-based handover preparation attempts from E-UTRAN to WCDMA network
1526730035	L.IRATHO.E2W.Distance.ExecAttOut	Number of distance-based handover execution attempts from E-UTRAN to WCDMA network
1526730036	L.IRATHO.E2W.Distance.ExecSuccOut	Number of successful distance-based handover executions from E-UTRAN to WCDMA network

Table 8-12 Counter related to distance-based redirection to UTRAN

Counter ID	Counter Name	Counter Description
1526730040	L.RRCRedirection.E2W.Distance	Number of distance-based redirections from E-UTRAN to WCDMA network

8.3.10 Parameter Optimization

The reconfiguration of parameters that are not involved in this section is the same as the reconfiguration of parameters for coverage-based inter-RAT handovers. For details, see **8.1.10 Parameter Optimization**.

Distance-Based Handover Threshold

Distance-based handover threshold is contained in the **DistBasedHO** MO. The following table provides suggestions on how to reconfigure this parameter.

Parameter Name	Parameter ID	Data Source	Setting Notes
Distance-based Handover Threshold	**DistBasedHO.*DistBasedHOThd***	Network plan (negotiation not required)	This parameter specifies the threshold for distance-based handovers. It is recommended that you set this parameter based on the planned distance between sites.

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Parameter Group Related to Inter-RAT Handover to UTRAN

The parameter group related to inter-RAT handovers to UTRAN are contained in the **InterRatHoUtranGroup** MO. The following table provides suggestions on how to reconfigure these parameters.

Parameter Name	Parameter ID	Data Source	Setting Notes
CoverageBased UTRAN ECN0 trigger threshold	**InterRatHoUtranGroup**.***InterRatHoUtranB1ThdEcn0***	Network plan (negotiation not required)	This parameter specifies the Ec/No threshold for event B1 related to coverage-based, UL-quality-based, or distance-based handover, or to SPID-based handover to the HPLMN. A larger value of this parameter results in a lower probability of handovers to UTRAN, and a smaller value results in a higher probability. The default value is recommended.
CoverageBased UTRAN RSCP trigger threshold	**InterRatHoUtranGroup**.***InterRatHoUtranB1ThdRscp***	Network plan (negotiation not required)	This parameter specifies the RSCP threshold for event B1 related to coverage-based, UL-quality-based, or distance-based handover, or to SPID-based handover to the HPLMN. A larger value of this parameter results in a lower probability of handovers to UTRAN, and a smaller value results in a higher probability. The default value is recommended.
UTRAN time to trigger	**InterRatHoUtranGroup**.***InterRatHoUtranB1TimeToTrig***	Network plan (negotiation not required)	This parameter specifies the time-to-trigger for event B1 related to inter-RAT handover to UTRAN. When detecting that the signal quality in at least one neighboring cell meets the entering condition, the UE does not send a measurement report to the eNodeB immediately. Instead, the UE sends a report only when the signal quality continuously meets the entering condition during the time-to-trigger period. If both UTRAN and GERAN neighboring cells are measured, you are advised to set the time-to-trigger for UTRAN neighboring cells to be smaller than that for GERAN neighboring cells, increasing the possibility of handovers to UTRAN neighboring cells. If both distance-based inter-frequency handovers and distance-based inter-RAT handovers are enabled, you are advised to set the time-to-trigger for the inter-RAT handover event be greater than that for the inter-frequency handover event, increasing the possibility of inter-frequency handovers.

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8.3.11 Troubleshooting

For details about troubleshooting, see **8.1.11 Troubleshooting**.

8.4 UL-Quality-based Inter-RAT Handover to UTRAN

8.4.1 When to Use UL-Quality-based Inter-RAT Handover to UTRAN

Coverage-based inter-RAT handovers are triggered based on DL signal quality. However, there may be a huge imbalance between the DL and UL signal quality. The UL signal quality may be relatively poor while the DL signal quality is fairly satisfactory. If this occurs, prompt triggering of inter-RAT handovers cannot be ensured. To handle this problem, you can enable UL-quality-based inter-RAT handover. It is good practice to enable UL-quality-based inter-RAT handover after the LTE network has stabilized. In the early phase of network deployment, however, enable this type of handover if it is required in the network plan.

If both UL-quality-based inter-frequency handover and UL-quality-based inter-RAT handover are available, the former has higher priority.

8.4.2 Required Information

Before deploying UL-quality-based inter-RAT handover, you need to collect the causes of the imbalance between the UL and DL signal quality as well as the possible magnitudes of the imbalance. In addition, you need to collect the requirements of the local network plan for the UL data rate at the cell edge.

8.4.3 Deployment Requirements

Operating Environment

The feature LOFD-001019 PS Inter-RAT Mobility between E-UTRAN and UTRAN requires that the core-network equipment should support this feature. If the equipment is provided by Huawei, the version must be SAE1.2. If the core-network equipment is provided by another vendor, check with the vendor whether the equipment supports this feature.

Transmission Networking

None

License

To use UL-quality-based inter-RAT handovers, an operator must purchase and activate the license for the feature listed in the following table.

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Feature ID	Feature Name	License Control Item	NE	Sales Unit
LOFD-001019	PS Inter-RAT Mobility between E-UTRAN and UTRAN	PS Inter-RAT Mobility between E-UTRAN and UTRAN	eNodeB	per user in RRC connected mode

8.4.4 Data Preparation

This section describes the data that you need to collect for setting parameters. Required data is data that you must collect for all scenarios. Collect scenario-specific data when necessary for a specific feature deployment scenario.

There are three types of data sources:

- Network plan (negotiation not required): parameter values planned and set by the operator
- Network plan (negotiation required): parameter values planned by the operator and negotiated with the EPC or peer transmission equipment
- User-defined: parameter values set by users

Required Data

For details about the required data, see **8.1.4 Data Preparation**.

Scenario-specific Data

Service-based inter-RAT handover can be enabled only after coverage-based inter-RAT handover is enabled. This section describes the required parameters besides those for coverage-based inter-RAT handover. For details on the required parameters for coverage-based inter-RAT handover, see the **Scenario-Specific Data** part in **8.1.4 Data Preparation**.

The following table describes the parameter that must be set in the **ENodeBAlgoSwitch** MO to enable UL-quality-based inter-RAT handover.

Parameter Name	Parameter ID	Data Source	Setting Notes
Handover Algo switch	**ENodeBAlgoSwitch.*HoAlgoSwitch***	Network plan (negotiation not required)	To enable UL-quality-based inter-RAT handover, select the **UlQualityInterRATHoSwitch** check box under this parameter.

8.4.5 Activation

Before activating UL-quality-based inter-RAT handover, ensure that coverage-based inter-RAT handover has been activated. Currently, the target RAT cannot be specified for an UL-quality-based inter-RAT handover.

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Using the CME to Perform Batch Configuration for Newly Deployed eNodeBs

Enter the values of related parameters in a summary data file, which also contains other data for the new eNodeBs to be deployed. Then, import the summary data file into the Configuration Management Express (CME) for batch configuration. For detailed instructions, see section "Creating eNodeBs in Batches" in the initial configuration guide for the eNodeB.

The summary data file may be a scenario-specific file provided by the CME or a customized file, depending on the following conditions:

- Related managed objects (MOs) are contained in a scenario-specific summary data file. In this situation, set the parameters in the MOs, and then verify and save the file.
- Some related MOs are not contained in a scenario-specific summary data file. In this situation, customize a summary data file to include the MOs before you can set the parameters.

Table 8-13 Parameters related to UL-quality-based inter-RAT handovers to UTRAN

MO	Sheet in the Summary Data File	Parameter Group	Remarks
ENodeBAlgoSwitch	User-defined sheet. **ENodeBAlgoSwitch** is recommended.	Handover Algo switch, Handover Mode switch	This parameter must be customized on a list-type sheet of the template.

Using the CME to Perform Batch Configuration for Existing eNodeBs

Batch reconfiguration using the CME is the recommended method to activate a feature on existing eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure is as follows:

Step 1 Choose **CME > Advanced > Customize Summary Data File** (U2000 client mode), or choose **Advanced > Customize Summary Data File** (CME client mode), to customize a summary data file for batch reconfiguration.

NOTE

For context-sensitive help on a current task in the client, press **F1**.

Step 2 Choose **CME > LTE Application > Export Data > Export Base Station Bulk Configuration Data** (U2000 client mode), or choose **LTE Application > Export Data > Export Base Station Bulk Configuration Data** (CME client mode), to export the eNodeB data stored on the CME into the customized summary data file.

Step 3 In the summary data file, set the parameters in the MOs listed in **Table 8-13**, and close the file.

Step 4 Choose **CME > LTE Application > Import Data > Import Base Station Bulk Configuration Data** (U2000 client mode), or choose **LTE Application > Import Data >**

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Import Base Station Bulk Configuration Data (CME client mode), to import the summary data file into the CME.

Step 5 Choose **CME > Planned Area > Export Incremental Scripts** (U2000 client mode), or choose **Area Management > Planned Area > Export Incremental Scripts**, to export and activate the incremental scripts.

----End

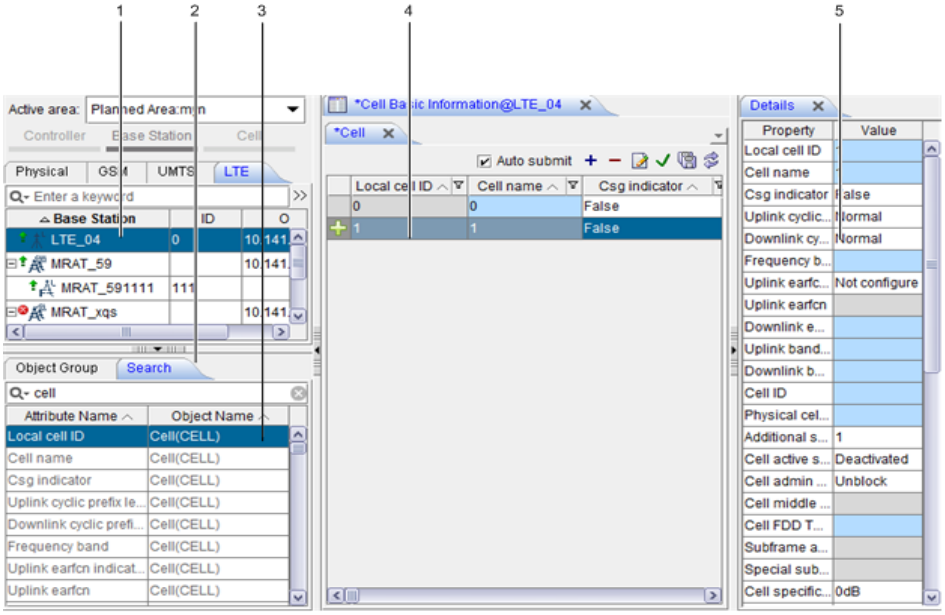
Using the CME to Perform Single Configuration

On the CME, set the parameters listed in the "Data Preparation" section for a single eNodeB. The procedure is as follows:

Step 1 In the planned data area, click **Base Station** in the upper left corner of the configuration window.

Step 2 In area 1 shown in **Figure 8-6**, select the eNodeB to which the MOs belong.

Figure 8-6 MO search and configuration window



Step 3 On the **Search** tab page in area 2, enter an MO name, for example, **CELL**.

Step 4 In area 3, double-click the MO in the **Object Name** column. All parameters in this MO are displayed in area 4.

Step 5 Set the parameters in area 4 or 5.

Step 6 Choose **CME > Planned Area > Export Incremental Scripts** (U2000 client mode), or choose **Area Management > Planned Area > Export Incremental Scripts** (CME client mode), to export and activate the incremental scripts.

----End

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Using MML Commands

Step 1 Run the **MOD ENODEBALGOSWITCH** command to enable uplink-quality-based handover and configure inter-RAT handover to UTRAN.

Step 2 Run the following commands to add neighbor relationships with UTRAN cells:

1. **ADD UTRANNFREQ**
2. **ADD UTRANEXTERNALCELL**
3. (Optional) **ADD UTRANEXTERNALCELLPLMN**, which is used to configure the PLMN list for an external UTRAN cell when the neighboring UTRAN cell works in RAN sharing mode.
4. **ADD UTRANNCELL**

----End

MML Command Examples

//Activating UL-quality-based inter-RAT handover to UTRAN

```
MOD ENODEBALGOSWITCH: HoAlgoSwitch=UlQualityInterRATHoSwitch-1;
```

//Adding neighboring UTRAN cells

```
ADD UTRANNFREQ: LocalCellId=0, UtranDlArfcn=10650, UtranFddTddType=UTRAN_FDD,
UtranUlArfcnCfgInd=NOT_CFG, CellReselPriorityCfgInd=NOT_CFG;ADD
UTRANEXTERNALCELL: Mcc="460", Mnc="00", RncId=123, CellId=2, UtranDlArfcn=10650,
UtranUlArfcnCfgInd=NOT_CFG, UtranFddTddType=UTRAN_FDD, RacCfgInd=CFG, Rac=130,
PScrambCode=100, Lac=120;ADD UTRANEXTERNALCELLPLMN: Mcc="460", Mnc="00",
RncId=123, CellId=2, ShareMcc="460", ShareMnc="20";ADD UTRANNCELL: LocalCellId=0,
Mcc="460", Mnc="00", RncId=123, CellId=2;
```

8.4.6 Activation Observation

The observation procedure is as follows:

Step 1 Use a UE to access the network at the center of a cell in the 2600 MHz band. Then, move the UE from the cell center to the cell edge. During this process, the uplink quality of the UE becomes poor.

Step 2 Observe the user-level measurement result on the U2000 (choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**, and observe the value of **BLER Monitoring** in **User Performance Monitoring**.)

- When the user uplink IBLER decreases gradually, observe the messages traced over the Uu interface. If the RRC_CONN_RECFG message from the eNodeB contains measurement control information for the inter-RAT handover and the MOBIL_FROM_EUTRA_CMD message received by the eNodeB contains information of the target cell for an inter-RAT handover, you can infer that UL-quality-based inter-RAT handover has been activated.
- When the user uplink IBLER decreases suddenly to lower than 99%, observe the messages traced over the Uu interface. If the RRC_CONN_REL message from the eNodeB contains target inter RAT frequency or frequency group information, you can infer that UL-quality-based inter-RAT handover has been activated.

----End

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8.4.7 Reconfiguration

None

8.4.8 Deactivation

Using the CME to Perform Batch Configuration

Batch reconfiguration using the CME is the recommended method to deactivate a feature on eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure for feature deactivation is similar to that for feature activation described in **Table 8-14**. In the procedure, modify parameters according to the table listed in each scenario.

Table 8-14 Parameters related to UL-Quality-based inter-RAT handovers

MO	Sheet in the Summary Data File	Parameter Group	Setting Notes
ENodeBAlgoSwitch	User-defined sheet	HoAlgoSwitch	Clear the **UlQualityInterRATHoSwitch** check box under this parameter.

Using the CME to Perform Single Configuration

On the CME, set the parameters listed in each scenario for a single eNodeB. For detailed instructions, see **Using the CME to Perform Single Configuration** for feature activation.

Using MML Commands

To deactivate UL-quality-based inter-RAT handover, turn off the corresponding switch by running the **MOD ENODEBALGOSWITCH** command.

MML Command Examples

//Deactivating UL-quality-based inter-RAT handover

```
MOD ENODEBALGOSWITCH: HoAlgoSwitch=UlQualityInterRATHoSwitch-0;
```

8.4.9 Performance Monitoring

Performance monitoring for UL-quality-based inter-RAT handover is similar to that for inter-RAT mobility between E-UTRAN and UTRAN. For details, see **8.1.9 Performance Monitoring**.

Related counters are listed in **Table 8-15** and **Table 8-16**.

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Table 8-15 Counters related to UL-quality-based handover to UTRAN

Counter ID	Counter Name	Counter Description
1526730026	L.IRATHO.E2W.ULquality.PrepAttOut	Number of UL-quality-based handover preparation attempts from E-UTRAN to WCDMA network
1526730027	L.IRATHO.E2W.ULquality.ExecAttOut	Number of UL-quality-based handover execution attempts from E-UTRAN to WCDMA network
1526730028	L.IRATHO.E2W.ULquality.ExecSuccOut	Number of successful UL-quality-based handover executions from E-UTRAN to WCDMA network

Table 8-16 Counters related to UL-quality-based redirection to UTRAN

Counter ID	Counter Name	Counter Description
1526730032	L.RRCRedirection.E2W.Ulquality	Number of UL-quality-based redirections from E-UTRAN to WCDMA network

8.4.10 Parameter Optimization

The reconfiguration of parameters that are not involved in this section is the same as the reconfiguration of parameters for coverage-based inter-RAT handovers. For details, see **8.1.10 Parameter Optimization**.

Parameter groups related to inter-RAT handovers to UTRAN are contained in the **InterRatHoUtranGroup** MO. The following table provides suggestions on how to reconfigure these parameters.

Parameter Name	Parameter ID	Data Source	Setting Notes
CoverageBased UTRAN ECN0 trigger threshold	**InterRatHoUtranGroup.*InterRatHoUtranB1ThdEcn0***	Network plan (negotiation not required)	This parameter specifies the Ec/No threshold for event B1 in a coverage-based, UL-quality-based, or distance-based handovers, or to SPID-based handovers to the HPLMN. A larger value of this parameter results in a lower probability of handovers to UTRAN, and a smaller value results in a higher probability. The default value is recommended.

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Parameter Name	Parameter ID	Data Source	Setting Notes
CoverageBased UTRAN RSCP trigger threshold	**InterRatHoUtranGroup.*InterRatHoUtranB1ThdRscp***	Network plan (negotiation not required)	This parameter specifies the RSCP threshold for event B1 in a coverage-based, UL-quality-based, or distance-based handovers to UTRAN, or to SPID-based handovers to the HPLMN. A larger value of this parameter results in a lower probability of handovers to UTRAN, and a smaller value results in a higher probability. The default value is recommended.
UTRAN time to trigger	**InterRatHoUtranGroup.*InterRatHoUtranB1TimeToTrig***	Network plan (negotiation not required)	This parameter specifies the time-to-trigger for event B1 related to inter-RAT handover to UTRAN. When detecting that the signal quality in at least one neighboring cell meets the entering condition, the UE does not send a measurement report to the eNodeB immediately. Instead, the UE sends a report only when the signal quality continuously meets the entering condition during the time-to-trigger period. If both UTRAN and GERAN neighboring cells are measured, you are advised to set the time-to-trigger for UTRAN neighboring cells to be smaller than that for GERAN neighboring cells, increasing the possibility of handovers to UTRAN neighboring cells. If both UL-quality-based inter-frequency handovers and UL-quality-based inter-RAT handovers are enabled, you are advised to set the time-to-trigger for the inter-RAT handover event be greater than that for the inter-frequency handover event, increasing the possibility of inter-frequency handovers.

8.4.11 Troubleshooting

For details about troubleshooting, see **8.1.11 Troubleshooting**.

8.5 E-UTRAN to UTRAN CS/PS Steering

8.5.1 When to Use E-UTRAN to UTRAN CS/PS Steering

You are advised to enable E-UTRAN to UTRAN CS/PS steering on the eNodeB if the following conditions are met:

- Inter-RAT interoperability between E-UTRAN and UTRAN is enabled.
- The operator has multiple UTRAN frequencies and these frequencies are planned for CS/PS steering.

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8.5.2 Required Information

Before deploying E-UTRAN to UTRAN CS/PS steering, collect information about frequencies used to carry CS and PS services in UTRAN.

8.5.3 Deployment Requirements

Operating Environment

Both coverage-based inter-RAT handover to UTRAN and CSFB to UTRAN are enabled. For details, see **8.1.3 Deployment Requirements** and *CS Fallback*.

Transmission Networking

None

License

To use E-UTRAN to UTRAN CS/PS steering, an operator must purchase and activate the license for the feature listed in the following table.

Feature ID	Feature Name	License Control Item	NE	Sales Unit
LOFD-001078	E-UTRAN to UTRAN CS/PS Steering	E-UTRAN to UTRAN CS/PS Steering	eNodeB	per user in RRC connected mode

In addition, this feature requires the features LOFD-001019 PS Inter-RAT Mobility between E-UTRAN and UTRAN and LOFD-001033 CS Fallback to UTRAN. For details about license requirements, see **8.1.3 Deployment Requirements** and *CS Fallback*.

8.5.4 Data Preparation

This section describes the data that you need to collect for setting parameters. Required data is data that you must collect for all scenarios. Collect scenario-specific data when necessary for a specific feature deployment scenario.

There are three types of data sources:

- Network plan (negotiation not required): parameter values planned and set by the operator
- Network plan (negotiation required): parameter values planned by the operator and negotiated with the EPC or peer transmission equipment
- User-defined: parameter values set by users

Required Data

For details about required data for coverage-based inter-RAT handover to UTRAN, see **8.1.4 Data Preparation**.

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Scenario-specific Data

UL-quality-based inter-RAT handover to GERAN can be enabled only after coverage-based inter-RAT handover is enabled. This section describes the required parameters besides those for coverage-based inter-RAT handover. For details on the required parameters for coverage-based inter-RAT handover, see the **Scenario-Specific Data** part in 8.1.4 Data Preparation.

For details about data preparation of CSFB to UTRAN, see *CS Fallback*.

The following table describes the parameter that must be set in the **ENodeBAlgoSwitch** MO to enable E-UTRAN to UTRAN CS/PS steering.

Parameter Name	Parameter ID	Data Source	Setting Notes
Frequency Layer Switch	**ENodeBAlgoSwitch**.***FreqLayerSwtich***	Network plan (negotiation not required)	Select one or both of the following options based on the actual network deployment: ● UtranFreqLayerMeasS-witch(UtranFreqLayerMeasS-witch) ● UtranFreqLayerBlind-Switch(UtranFreqLayerBlind-Switch)

The following table describes the parameters that must be set in the **UtranNFreq** MO to set CS and PS service priorities for UTRAN frequencies.

Parameter Name	Parameter ID	Data Source	Setting Notes
PS service priority	**UtranNFreq**.***PsPriority***	Network plan (negotiation not required)	Set this parameter based on the network plan about CS and PS service priorities. Do not set this parameter to **Priority_0** for UTRAN frequencies for PS services. The priority must be consistent with that in the live network. If the UTRAN frequencies carry PS services have the same priority in the live network, set this parameter to Priority_16 for all UTRAN frequencies carry PS services. Set this parameter to Priority_0 for all UTRAN frequencies carry CS services. Different frequencies can be configured with the same priority.

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Parameter Name	Parameter ID	Data Source	Setting Notes
CS service priority	**UtranNFreq**.***CsPriority***	Network plan (negotiation not required)	Set this parameter based on the network plan about CS and PS service priorities. Do not set this parameter to Priority_0 for UTRAN frequencies for CS services. The priority must be consistent with that in the live network. If the UTRAN frequencies carry CS services have the same priority in the live network, set this parameter to **Priority_16** for all UTRAN frequencies carry CS services. Set this parameter to **Priority_0** for all UTRAN frequencies carry PS services. Different frequencies can be configured with the same priority.

8.5.5 Activation

Before enabling E-UTRAN to UTRAN CS/PS steering, ensure that coverage-based inter-RAT handover to UTRAN and CSFB to UTRAN have been enabled. For details, see **8.1.5 Activation** and *CS Fallback*.

Using the CME to Perform Batch Configuration for Newly Deployed eNodeBs

Enter the values of related parameters in a summary data file, which also contains other data for the new eNodeBs to be deployed. Then, import the summary data file into the Configuration Management Express (CME) for batch configuration. For detailed instructions, see section "Creating eNodeBs in Batches" in the initial configuration guide for the eNodeB.

Then, import the summary data file as shown in **Table 8-17** into the Configuration Management Express (CME) for batch configuration.

The summary data file in **Table 8-17** may be a scenario-specific file provided by the CME or a customized file, depending on the following conditions:

- The managed objects (MOs) in **Table 8-17** are contained in a scenario-specific summary data file. In this situation, set the parameters in the MOs, and then verify and save the file.
- Some MOs in **Table 8-17** are not contained in a scenario-specific summary data file. In this situation, customize a summary data file to include the MOs before you can set the parameters.

Related MOs and parameters in each scenario are listed in **Table 8-17**.

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Table 8-17 Parameters related to E-UTRAN to UTRAN CS/PS steering

MO	Sheet in the Summary Data File	Parameter Group	Remarks
ENodeBAlgoSwitch	User-defined sheet. **ENodeBAlgoSwitch** is recommended.	Frequency Layer Switch	This parameter must be customized on a list-type sheet of the template.
UtranNFreq	User-defined sheet. **UtranNFreq** is recommended.	PS service priority, CS service priority	This parameter must be customized on a list-type sheet of the template.

Using the CME to Perform Batch Configuration for Existing eNodeBs

Batch reconfiguration using the CME is the recommended method to activate a feature on existing eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure is as follows:

Step 1 Choose CME > Advanced > Customize Summary Data File (U2000 client mode), or choose Advanced > Customize Summary Data File (CME client mode), to customize a summary data file for batch reconfiguration.

NOTE

For context-sensitive help on a current task in the client, press F1.

Step 2 Choose CME > LTE Application > Export Data > Export Base Station Bulk Configuration Data (U2000 client mode), or choose LTE Application > Export Data > Export Base Station Bulk Configuration Data (CME client mode), to export the eNodeB data stored on the CME into the customized summary data file.

Step 3 In the summary data file, set the parameters in the MOs listed in **Using the CME to Perform Batch Configuration for Newly Deployed eNodeBs** and close the file.

Step 4 Choose CME > LTE Application > Import Data > Import Base Station Bulk Configuration Data (U2000 client mode), or choose LTE Application > Import Data > Import Base Station Bulk Configuration Data (CME client mode), to import the summary data file into the CME.

Step 5 Choose CME > Planned Area > Export Incremental Scripts (U2000 client mode), or choose Area Management > Planned Area > Export Incremental Scripts (CME client mode), to export and activate the incremental scripts.

----End

Using the CME to Perform Single Configuration

On the CME, set the parameters listed in the "Data Preparation" section for a single eNodeB. The procedure is as follows:

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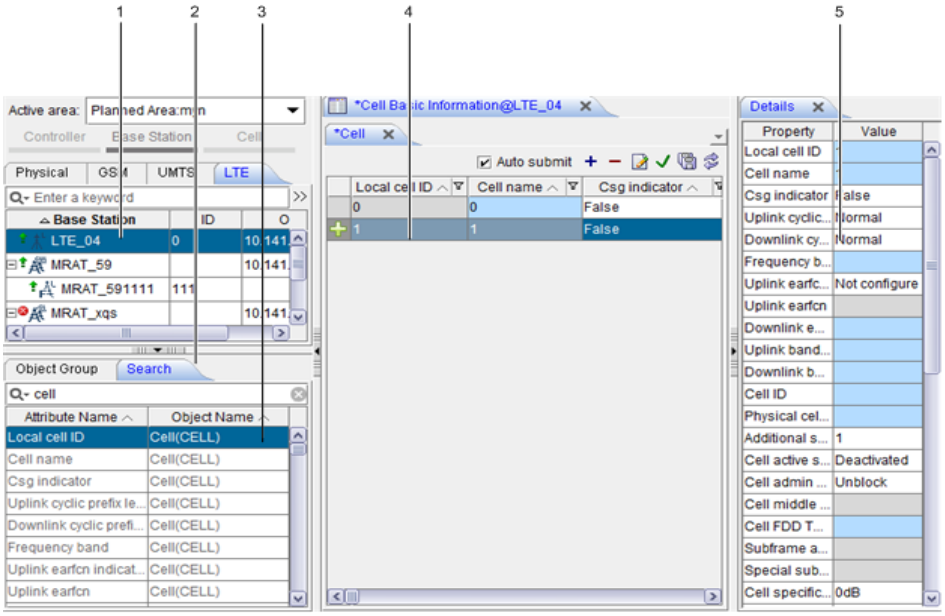
Step 1 In the planned data area, click **Base Station** in the upper left corner of the configuration window.

Step 2 In area 1 shown in **Figure 8-7**, select the eNodeB to which the MOs belong.

Figure 8-7 MO search and configuration window



Step 3 On the **Search** tab page in area 2, enter an MO name, for example, **CELL**.

Step 4 In area 3, double-click the MO in the **Object Name** column. All parameters in this MO are displayed in area 4.

Step 5 Set the parameters in area 4 or 5.

Step 6 Choose CME > Planned Area > Export Incremental Scripts (U2000 client mode), or choose Area Management > Planned Area > Export Incremental Scripts (CME client mode), to export and activate the incremental scripts.

----End

Using MML Commands

- Measurement for E-UTRAN to UTRAN CS/PS steering

Step 1 Run the **MOD ENODEBALGOSWITCH** command to enable the measurement for E-UTRAN to UTRAN CS/PS steering.

Step 2 Run the **MOD UTRANNFREQ** command to set CS and PS service priorities for UTRAN frequencies.

----End

- Blind handover for E-UTRAN to UTRAN CS/PS steering

Step 1 Run the **MOD ENODEBALGOSWITCH** command to enable blind handover for E-UTRAN to UTRAN CS/PS steering.

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Step 2 Run the **MOD UTRANNFREQ** command to set CS and PS service priorities for UTRAN frequencies.

----End

MML Command Examples

- Measurement for E-UTRAN to UTRAN CS/PS steering

//Enabling the measurement for E-UTRAN to UTRAN CS/PS steering

```
MOD ENODEBALGOSWITCH: FreqLayerSwtich=UtranFreqLayerMeasSwitch-1;
```

//Setting CS and PS service priorities for UTRAN frequencies

```
MOD UTRANNFREQ: LocalCellId=0, UtranDlArfcn=10650, CsPriority= Priority_16;
MOD UTRANNFREQ: LocalCellId=0, UtranDlArfcn=10650, PsPriority= Priority_16;
```

- Blind handover for E-UTRAN to UTRAN CS/PS steering

//Enabling blind handover for E-UTRAN to UTRAN CS/PS steering

```
MOD ENODEBALGOSWITCH: FreqLayerSwtich=UtranFreqLayerBlindSwitch-1;
```

//Setting CS and PS service priorities for UTRAN frequencies

```
MOD UTRANNFREQ: LocalCellId=0, UtranDlArfcn=10650, CsPriority= Priority_16;
MOD UTRANNFREQ: LocalCellId=0, UtranDlArfcn=10650, PsPriority= Priority_16;
```

8.5.6 Activation Observation

The observation procedure is as follows:

Step 1 Trace messages over the Uu interface on the U2000 client after UEs access the network.

Step 2 Trace messages over the Uu interface while the UE is moving towards the cell edge.

After the eNodeB receives an A2-related measurement report from the UE, check the RRC_CONN_RECFG message. If the message contains the UTRAN frequency with the highest PS service priority, you can infer that E-UTRAN to UTRAN CS/PS steering has been activated.

Step 3 Trace messages over the Uu interface after the UE moves towards the cell edge.

After the eNodeB receives a B1- or B2-related measurement report from the UE, check the RRC_MOBIL_FROM_EUTRA_CMD message or the RRC_CONN_REL message. If the RRC_MOBIL_FROM_EUTRA_CMD message contains the cell under the UTRAN frequency with the highest PS service priority or if the RRC_CONN_REL message contains the UTRAN frequency with the highest PS service priority, you can infer that E-UTRAN to UTRAN CS/PS steering has been triggered.

NOTE

For details about activation observations for CS service priority configuration, see *CS Fallback*.

----End

8.5.7 Reconfiguration

None

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8.5.8 Deactivation

Using the CME to Perform Batch Configuration

Batch reconfiguration using the CME is the recommended method to deactivate a feature on eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure for feature deactivation is similar to that for feature activation described in **Table 8-18**. In the procedure, modify parameters according to the table listed in each scenario.

Table 8-18 Parameters related to E-UTRAN to UTRAN CS/PS steering

MO	Sheet in the Summary Data File	Parameter Group	Setting Notes
ENodeBAlgoSwitch	User-defined sheet	FreqLayerSwtich	Clear the following check boxes: ● UtranFreqLayerMeasSwitch ● UtranFreqLayerBlindSwitch

Using the CME to Perform Single Configuration

On the CME, set the parameters listed in each scenario for a single eNodeB. For detailed instructions, see **Using the CME to Perform Single Configuration** for feature activation.

Using MML Commands

- Disabling the measurement for E-UTRAN to UTRAN CS/PS steering

Run the **MOD ENODEBALGOSWITCH** command with the **UtranFreqLayerMeasSwitch(UtranFreqLayerMeasSwitch)** check box cleared under the **Frequency Layer Switch** parameter.

- Disabling blind handover for E-UTRAN to UTRAN CS/PS steering

Run the **MOD ENODEBALGOSWITCH** command with the **UtranFreqLayerBlindSwitch(UtranFreqLayerBlindSwitch)** check box cleared under the **Frequency Layer Switch** parameter.

MML Command Examples

- //Disabling the measurement for E-UTRAN to UTRAN CS/PS steering

```
MOD ENODEBALGOSWITCH: FreqLayerSwtich=UtranFreqLayerMeasSwitch-0;
```

- //Disabling blind handover for E-UTRAN to UTRAN CS/PS steering

```
MOD ENODEBALGOSWITCH: FreqLayerSwtich=UtranFreqLayerBlindSwitch-0;
```

8.5.9 Performance Monitoring

Performance monitoring for E-UTRAN to UTRAN CS/PS steering is similar to that for inter-RAT mobility between E-UTRAN and UTRAN. For details, see **8.1.9 Performance Monitoring**.

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8.5.10 Parameter Optimization

For details about parameter optimization, see **8.1.10 Parameter Optimization**.

8.5.11 Troubleshooting

For details about troubleshooting, see **8.1.11 Troubleshooting**.

8.6 Separate Mobility Policies to UTRAN for Multi PLMN

8.6.1 When to Use Separate Mobility Policies to UTRAN for Multi PLMN

You are advised to enable Separate Mobility Policies to UTRAN for Multi PLMN when the interoperability between E-UTRAN and UTRAN is enabled and one of the following conditions is met:

- Some cells in the operators' UTRAN support PS handovers while some cells do not.
- Some cells in the operators' UTRAN support the RIM procedure while some cells do not.

8.6.2 Required Information

Before deploying Separate Mobility Policies to UTRAN for Multi PLMN, collect information about whether the UTRAN supports PS handovers and RIM procedures. The information is required during eNodeB configuration.

8.6.3 Deployment Requirements

Operating Environment

Either coverage-based inter-RAT handover to UTRAN or CSFB to UTRAN are enabled. For details, see **8.1.3 Deployment Requirements** and *CS Fallback*.

Transmission Networking

None

License

The operator has purchased and activated the license for the feature listed in the following table.

Feature ID	Feature Name	License Control Item	NE	Sales Unit
LOFD-070216	Separate Mobility Policies to UTRAN for Multi PLMN	Separate Mobility Policies to UTRAN for Multi PLMN	eNodeB	per user in RRC connected mode

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This feature requires LOFD-001019 PS Inter-RAT Mobility between E-UTRAN and UTRAN and LOFD-001033 CS Fallback to UTRAN. For details about license requirements, see **8.1.3 Deployment Requirements** and *CS Fallback*.

8.6.4 Data Preparation

This section describes the data that you need to collect for setting parameters. Required data is data that you must collect for all scenarios. Collect scenario-specific data when necessary for a specific feature deployment scenario.

There are three types of data sources:

- Network plan (negotiation not required): parameter values planned and set by the operator
- Network plan (negotiation required): parameter values planned by the operator and negotiated with the EPC or peer transmission equipment
- User-defined: parameter values set by users

Required Data

This feature can be enabled only after coverage-based inter-RAT handover to UTRAN is enabled. For details about required data for coverage-based inter-RAT handover to UTRAN, see **8.1.4 Data Preparation**.

Scenario-specific Data

UL-quality-based inter-RAT handover to GERAN can be enabled only after coverage-based inter-RAT handover is enabled. This section describes the required parameters besides those for coverage-based inter-RAT handover. For details on the required parameters for coverage-based inter-RAT handover, see the **Scenario-specific Data** part in **8.1.4 Data Preparation**.

For details about data preparation of CSFB to UTRAN, see *CS Fallback*.

The following table describes the parameters that must be set in the **ENodeBAlgoSwitch** MO to enable Separate Mobility Policies to UTRAN for Multi PLMN.

Parameter Name	Parameter ID	Data Source	Setting Notes
Multi-Operator Control Switch	**ENodeBAlgoSwitch.*MultiOpCtrlSwitch***	Network plan (negotiation not required)	Turn on this switch when some cells in the network support PS handovers or the RIM procedure.

The following table describes the parameters that must be set in the **UtranNetworkCapCfg** MO to configure the UTRAN capability.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Mobile country code	**UtranNetworkCapCfg**.***Mcc***	Network plan (negotiation required)	This parameter specifies the MCC of the home NodeB of the neighboring UTRAN cell. If this neighboring cell works in RAN sharing mode, set this parameter to the MCC of the primary operator. If the peer NodeB is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **URNCBASIC** MO of the Huawei NodeB.
Mobile network code	**UtranNetworkCapCfg**.***Mnc***	Network plan (negotiation required)	This parameter specifies the MNC of the home NodeB of the neighboring UTRAN cell. If this neighboring UTRAN cell works in eRAN sharing mode, set this parameter to the MNC of the primary operator. If the peer NodeB is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **URNCBASIC** MO of the peer NodeB.
Rnc Id	**UtranNetworkCapCfg**.***RncId***	Network plan (negotiation not required)	This parameter specifies the RNC ID of the neighboring UTRAN cell. The value range of an RNC ID is (0-4095, 4294967295). 0-4095 indicates a specific RNC on the network. 4294967295 indicates all RNCs under the PLMN. That is, the policy applies to all RNCs under the PLMN. If the peer NodeB is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **URNCBASIC** MO of the peer NodeB.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Network Capability Configuration	**UtranNetworkCapCfg**.***NetworkCapCfg***	Network plan (negotiation not required)	If an MME, SGSN, or RNC of an operator does not support PS handovers or the RIM procedure, the corresponding bit area must be configured. If this parameter is not set, PS handovers or the RIM procedure is supported by default. ● **PsHoCapCfg** indicates whether PS handovers are supported. If this check box is selected, PS handovers are supported. If this check box is not selected, PS handovers are not supported. ● **SiByRimCapCfg** affects the RIM procedure that is based on core networks. eCoordinator-based RIM procedure is not affected by this parameter. – When **SiByRimCapCfg** is set to **OFF**, obtaining SI through the RIM procedure based on core networks is not supported. – When **SiByRimCapCfg** is set to **ON**, obtaining SI through the RIM procedure based on core networks is supported.

8.6.5 Activation

Before enabling Separate Mobility Policies to UTRAN for Multi PLMN, ensure that coverage-based inter-RAT handover to UTRAN or CSFB to UTRAN has been enabled. For details, see **8.1.5 Activation** and *CS Fallback*.

Using the CME to Perform Batch Configuration for Newly Deployed eNodeBs

Enter the values of related parameters in a summary data file, which also contains other data for the new eNodeBs to be deployed. Then, import the summary data file into the Configuration Management Express (CME) for batch configuration. For detailed instructions, see section "Creating eNodeBs in Batches" in the initial configuration guide for the eNodeB.

The summary data file may be a scenario-specific file provided by the CME or a customized file, depending on the following conditions:

- Related managed objects (MOs) are contained in a scenario-specific summary data file. In this situation, set the parameters in the MOs, and then verify and save the file.
- Some related MOs are not contained in a scenario-specific summary data file. In this situation, customize a summary data file to include the MOs before you can set the parameters.

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Table 8-19 Parameters related to Separate Mobility Policies to UTRAN for Multi PLMN

MO	Sheet in the Summary Data File	Parameter Group	Remarks
ENodeBAlgoSwitch	User-defined sheet. **ENodeBAlgoSwitch** is recommended.	Multi-Operator Control Switch	This parameter must be customized on a list-type sheet of the template.
NetworkCapCfg	User-defined sheet. **NetworkCapCfg** is recommended.	Mobile country code, Mobile network code, Rnc Id, Network Capability Configuration	This parameter must be customized on a list-type sheet of the template.

Using the CME to Perform Batch Configuration for Existing eNodeBs

Batch reconfiguration using the CME is the recommended method to activate a feature on existing eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure is as follows:

Step 1 Choose **CME > Advanced > Customize Summary Data File** (U2000 client mode), or choose **Advanced > Customize Summary Data File** (CME client mode), to customize a summary data file for batch reconfiguration.

NOTE

For context-sensitive help on a current task in the client, press **F1**.

Step 2 Choose **CME > LTE Application > Export Data > Export Base Station Bulk Configuration Data** (U2000 client mode), or choose **LTE Application > Export Data > Export Base Station Bulk Configuration Data** (CME client mode), to export the eNodeB data stored on the CME into the customized summary data file.

Step 3 In the summary data file, set the parameters in the MOs listed in Table 8-19, and close the file.

Step 4 Choose **CME > LTE Application > Import Data > Import Base Station Bulk Configuration Data** (U2000 client mode), or choose **LTE Application > Import Data > Import Base Station Bulk Configuration Data** (CME client mode), to import the summary data file into the CME.

Step 5 Choose **CME > Planned Area > Export Incremental Scripts** (U2000 client mode), or choose **Area Management > Planned Area > Export Incremental Scripts**, to export and activate the incremental scripts.

----End

Using the CME to Perform Single Configuration

On the CME, set the parameters listed in the "Data Preparation" section for a single eNodeB. The procedure is as follows:

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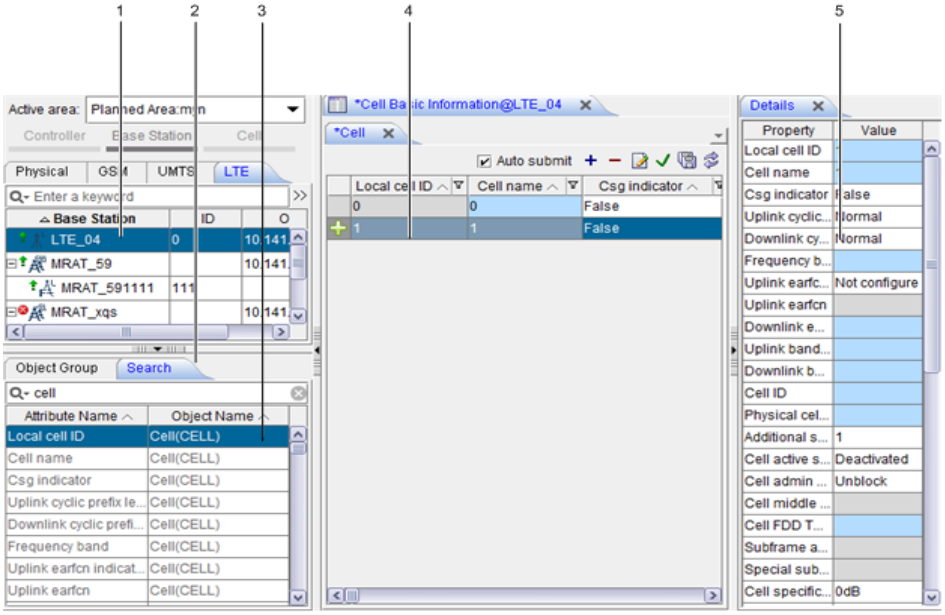
Step 1 In the planned data area, click **Base Station** in the upper left corner of the configuration window.

Step 2 In area 1 shown in **Figure 8-8**, select the eNodeB to which the MOs belong.

Figure 8-8 MO search and configuration window



Step 3 On the **Search** tab page in area 2, enter an MO name, for example, **CELL**.

Step 4 In area 3, double-click the MO in the **Object Name** column. All parameters in this MO are displayed in area 4.

Step 5 Set the parameters in area 4 or 5.

Step 6 Choose **CME > Planned Area > Export Incremental Scripts** (U2000 client mode), or choose **Area Management > Planned Area > Export Incremental Scripts** (CME client mode), to export and activate the incremental scripts.

----End

Using MML Commands

Step 1 Run the **MOD ENODEBALGOSWITCH** command to turn on the multi-PLMN control switch.

Step 2 Run the **MOD NETWORKCAPCFG** command to configure the UTRAN capability.

----End

MML Command Examples

//Turning on the multi-PLMN control switch

```
MOD ENODEBALGOSWITCH: MultiOpCtrlSwitch=UtranSepOpMobilitySwitch-1;
```

//Configuring the UTRAN capability

```
MOD UTRANNETWORKCAPCFG: Mcc="460", Mnc="32", RncId=0, NetworkCapCfg=SiByRimCapCfg;
```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8.6.6 Activation Observation

The activation procedure is as follows:

Step 1 Trace messages over the Uu interface on the U2000 client after UEs access the network.

Step 2 Trace messages over the Uu interface while the UE is moving towards the cell edge. After the eNodeB receives a B1- or B2-related measurement report from the UE:

- If the neighboring UTRAN cell with the strongest signal quality does not support PS handovers and the RIM procedure and the RRC_CONN_REL message traced over the Uu interface carries the frequency of that cell, you can infer that the PS handover supporting capability identification function has been triggered.
- If flash CS fallback to UTRAN is enabled and the RRC_CONN_REL message does not carry system information for the cells using the target frequency, you can infer that the RIM supporting capability identification function has been triggered.

----End

8.6.7 Reconfiguration

None

8.6.8 Deactivation

Using the CME to Perform Batch Configuration

Batch reconfiguration using the CME is the recommended method to deactivate a feature on eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure for feature deactivation is similar to that for feature activation described in **Table 8-20**. In the procedure, modify parameters according to the table listed in each scenario.

Table 8-20 Parameters related to Separate Mobility Policies to UTRAN for Multi PLMN

MO	Sheet in the Summary Data File	Parameter Group	Setting Notes
ENodeBAlgoSwitch	User-defined sheet	Multi-Operator Control Switch	Clear the **MultiOpCtrlSwitch** check box under this parameter.

Using the CME to Perform Single Configuration

On the CME, set the parameters listed in each scenario for a single eNodeB. For detailed instructions, see **Using the CME to Perform Single Configuration** for feature activation.

Using MML Commands

To deactivate Separate Mobility Policies to UTRAN for Multi PLMN, run the **MOD ENODEBALGOSWITCH** command and turn off the corresponding switch under **MultiOpCtrlSwitch**.

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MML Command Examples

//Deactivating Separate Mobility Policies to UTRAN for Multi PLMN

```
MOD ENODEBALGOSWITCH: MultiOpCtrlSwitch=UtranSepOpMobilitySwitch-0;
```

8.6.9 Performance Monitoring

Performance monitoring for Separate Mobility Policies to UTRAN for Multi PLMN is similar to that for inter-RAT mobility between E-UTRAN and UTRAN. For details, see **8.1.9 Performance Monitoring**.

8.6.10 Parameter Optimization

For details about parameter optimization, see **8.1.10 Parameter Optimization**.

8.6.11 Troubleshooting

For details about troubleshooting, see **8.1.11 Troubleshooting**.

8.7 RIM Based LTE Target Cell Selection

8.7.1 When to Use RIM Based LTE Target Cell Selection

It is recommended that RIM Based LTE Target Cell Selection be enabled when the interoperability between E-UTRAN and UTRAN is enabled and one of the following conditions is met:

- Service-Based PS Service Redirection from UMTS to LTE is activated.
- Service-Based PS Handover from UMTS to LTE is activated.
- Load Based PS Redirection from UMTS to LTE is activated.
- Load Based PS Handover from UMTS to LTE is activated.

8.7.2 Required Information

None

8.7.3 Requirements

Operating Environment

The core networks (MME or SGSN) must support the RIM procedure when the RIM procedure is based on core networks.

If the RIM procedure is performed through the eCoordinator, the RNC/BSC, eNodeB, and eCoordinator must all be provided by Huawei and with the switch for supporting the RIM procedure through eCoordinator turned on.

At least one of the following features is activated: Service-Based PS Service Redirection from UMTS to LTE, Service-Based PS Handover from UMTS to LTE, Load Based PS Redirection from UMTS to LTE, and Load Based PS Handover from UMTS to LTE.

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Transmission Networking

None

License

The operator has purchased and activated the license for the feature listed in the following table.

Feature ID	Feature Name	License Control Item	NE	Sales Unit
LOFD-070203	RIM Based LTE Target Cell Selection	RIM Based LTE Target Cell Selection	eNodeB	per cell

8.7.4 Data Preparation

This section describes the data that you need to collect for setting parameters. Required data is data that you must collect for all scenarios. Collect scenario-specific data when necessary for a specific feature deployment scenario.

There are three types of data sources:

- Network plan (negotiation not required): parameter values planned and set by the operator
- Network plan (negotiation required): parameter values planned by the operator and negotiated with the EPC or peer transmission equipment
- User-defined: parameter values set by users

Required Data

The following table describes the parameter that must be set in the **GlobalProcSwitch** MO to configure the E-UTRAN load transfer switch.

Parameter Name	Parameter ID	Data Source	Setting Notes
EUTRAN Load Transfer Switch	**GlobalProcSwitch.*EutranLoadTransSwitch***	Network plan (negotiation not required)	Indicates whether to send E-UTRAN cell load information. **RimUtranSwitch** specifies whether to transfer E-UTRAN cell load information to the UTRAN through the RIM procedure. This bit applies only to LTE FDD. If this switch is turned on, the E-UTRAN transfers E-UTRAN cell load information to the UTRAN and notifies the UTRAN of any cell load changes through the RIM procedure.

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Scenario-specific Data

The following table describes the parameters that must be set in the **ENodeBAlgoSwitch** MO to specify whether the RIM procedure is initiated by the eCoordinator.

Parameter Name	Parameter ID	Data Source	Setting Notes
Support RIM by eCoordinator Switch	**ENodeBAlgoSwitch.*RimOnEcoSwitch***	Network plan (negotiation not required)	● When this parameter is set to **OFF(Off)**, the RIM procedure is performed through core networks. ● When this parameter is set to **ON(On)**, the RIM procedure is performed through the eCoordinator.

8.7.5 Activation

Using the CME to Perform Batch Configuration for Newly Deployed eNodeBs

Enter the values of related parameters in a summary data file, which also contains other data for the new eNodeBs to be deployed. For detailed instructions, see section "Creating eNodeBs in Batches" in the initial configuration guide for the eNodeB.

Then, import the summary data file as shown in **Table 8-21** into the Configuration Management Express (CME) for batch configuration.

The summary data file in **Table 8-21** may be a scenario-specific file provided by the CME or a customized file, depending on the following conditions:

- The managed objects (MOs) in **Table 8-21** are contained in a scenario-specific summary data file. In this situation, set the parameters in the MOs, and then verify and save the file.
- Some MOs in **Table 8-21** are not contained in a scenario-specific summary data file. In this situation, customize a summary data file to include the MOs before you can set the parameters.

Table 8-21 Parameters related to RIM Based LTE Target Cell Selection

MO	Sheet in the Summary Data File	Parameter Group	Remarks
GLOBALPROCSWITCH	User-defined sheet. **GLOBALPROCSWITCH** is recommended.	EUTRAN Load Transfer Switch	This parameter must be customized on a list-type sheet of the template.

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MO	Sheet in the Summary Data File	Parameter Group	Remarks
ENODEBALGOSWITCH	User-defined sheet. **ENODEBALGOSWITCH** is recommended.	Support RIM by eCoordinator Switch	This parameter must be customized on a list-type sheet of the template.

Using the CME to Perform Batch Configuration for Existing eNodeBs

Batch reconfiguration using the CME is the recommended method to activate a feature on existing eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure is as follows:

Step 1 Choose CME > Advanced > Customize Summary Data File (U2000 client mode), or choose Advanced > Customize Summary Data File (CME client mode), to customize a summary data file for batch reconfiguration.

NOTE

For context-sensitive help on a current task in the client, press F1.

Step 2 Choose CME > LTE Application > Export Data > Export Base Station Bulk Configuration Data (U2000 client mode), or choose LTE Application > Export Data > Export Base Station Bulk Configuration Data (CME client mode), to export the eNodeB data stored on the CME into the customized summary data file.

Step 3 In the summary data file, set the parameters in the MOs listed in **Using the CME to Perform Batch Configuration for Newly Deployed eNodeBs** and close the file.

Step 4 Choose CME > LTE Application > Import Data > Import Base Station Bulk Configuration Data (U2000 client mode), or choose LTE Application > Import Data > Import Base Station Bulk Configuration Data (CME client mode), to import the summary data file into the CME.

Step 5 Choose CME > Planned Area > Export Incremental Scripts (U2000 client mode), or choose Area Management > Planned Area > Export Incremental Scripts (CME client mode), to export and activate the incremental scripts.

----End

Using the CME to Perform Single Configuration

On the CME, set the parameters listed in the "Data Preparation" section for a single eNodeB. The procedure is as follows:

Step 1 In the planned data area, click **Base Station** in the upper left corner of the configuration window.

Step 2 In area 1 shown in **Figure 8-9**, select the eNodeB to which the MOs belong.

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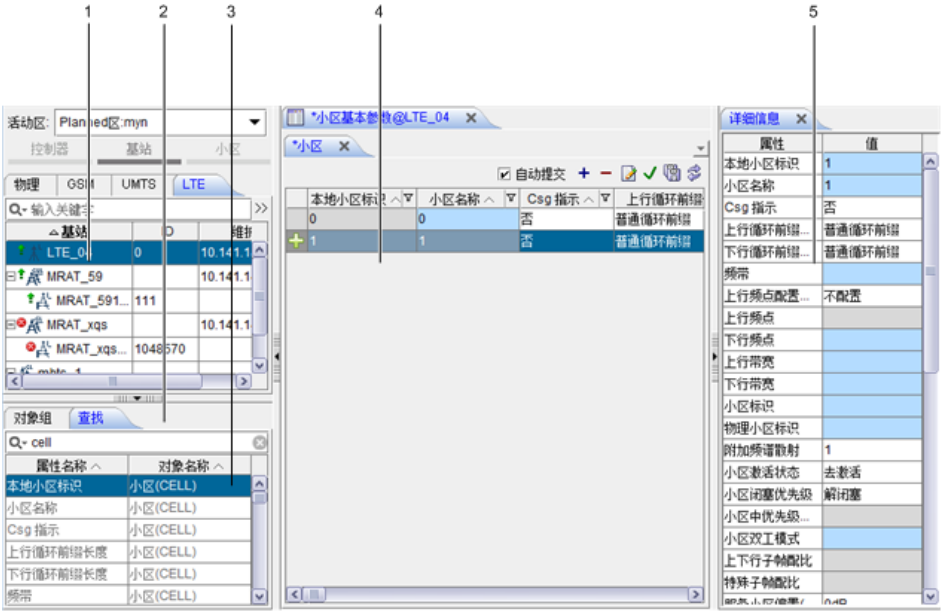
Figure 8-9 MO search and configuration window



Step 3 On the **Search** tab page in area 2, enter an MO name, for example, **CELL**.

Step 4 In area 3, double-click the MO in the **Object Name** column. All parameters in this MO are displayed in area 4.

Step 5 Set the parameters in area 4 or 5.

Step 6 Choose CME > Planned Area > Export Incremental Scripts (U2000 client mode), or choose Area Management > Planned Area > Export Incremental Scripts (CME client mode), to export and activate the incremental scripts.

----End

Using MML Commands

Run the **MOD GLOBALPROCSWITCH** command and clear the **RimUtranSwitch** check box under the **EutranLoadTransSwitch** parameter.

- If the RIM procedure is based on core networks, run the **MOD ENODEBALGOSWITCH** command to turn off **RimOnEcoSwitch**.
- If the RIM procedure is based on the eCoordinator, run the **MOD ENODEBALGOSWITCH** command to turn on **RimOnEcoSwitch**.

MML Command Examples

//Activating RIM Based LTE Target Cell Selection

```
MOD GLOBALPROCSWITCH: EutranLoadTransSwitch=RimUtranSwitch-1;
```

- If the RIM procedure is based on core networks:
  ```
  MOD ENODEBALGOSWITCH: RimOnEcoSwitch=OFF;
  ```
- If the RIM procedure is based on the eCoordinator:
  ```
  MOD ENODEBALGOSWITCH: RimOnEcoSwitch=ON;
  ```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8.7.6 Activation Observation

Table 8-22 lists the counters in the RIM messages, which are used to observe whether RIM Based LTE Target Cell Selection is enabled.

Table 8-22 RIM-related counters

Counter ID	Counter Name	Description
1526729475	L.RIM.Load.W2E.Req	Number of Times the eNodeB Receives Load Information Requests from WCDMA Network
1526729476	L.RIM.Load.W2E.Resp	Number of Times the eNodeB Sends Load Information Responses to WCDMA Network
1526729477	L.RIM.Load.W2E.Update	Number of Times the eNodeB Sends Load Information Updates to WCDMA Network
1526729478	L.RIM.Load.W2E.Update.ACK	Number of times the eNodeB receives load information update acknowledgments from WCDMA network

- If counters 1526729475 and 1526729476 are not 0, RIM Based LTE Target Cell Selection is enabled.
- If counters 1526729477 and 1526729478 are not 0, E-UTRAN cell load status has changed and the E-UTRAN informs the UTRAN of the changes through the RIM procedure.

8.7.7 Reconfiguration

None

8.7.8 Deactivation

Using the CME to Perform Batch Configuration

Batch reconfiguration using the CME is the recommended method to deactivate a feature on eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure for feature deactivation is similar to that for feature activation described in **Using the CME to Perform Batch Configuration for Existing eNodeBs**. In the procedure, modify parameters according to the table listed in each scenario.

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Table 8-23 RIM Based LTE Target Cell Selection

MO	Sheet in the Summary Data File	Parameter Group	Setting Notes
ENodeBAlgoSwitch	User-defined sheet	HoAlgoSwitch	Clear the **UlQualityInterRATHoSwitch** check box under this parameter.

Using the CME to Perform Single Configuration

On the CME, set the parameters listed in each scenario for a single eNodeB. For detailed instructions, see **Using the CME to Perform Single Configuration** for feature activation.

Using MML Commands

Run the **MOD GLOBALPROCSWITCH** command and clear the **RimUtranSwitch** check box under the **EutranLoadTransSwitch** parameter.

MML Command Examples

//Deactivating RIM Based LTE Target Cell Selection

```
MOD ENODEBALGOSWITCH: EutranLoadTransSwitch=RimUtranSwitch-0;
```

8.7.9 Performance Monitoring

Table 8-24 lists the counters for monitoring the performance of RIM Based LTE Target Cell Selection.

Table 8-24 Counters for monitoring the performance of RIM Based LTE Target Cell Selection

Counter ID	Counter Name	Description
1526729475	L.RIM.Load.W2E.Req	Number of Times the eNodeB Receives Load Information Requests from WCDMA Network
1526729476	L.RIM.Load.W2E.Resp	Number of Times the eNodeB Sends Load Information Responses to WCDMA Network
1526729477	L.RIM.Load.W2E.Update	Number of Times the eNodeB Sends Load Information Updates to WCDMA Network

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Counter ID	Counter Name	Description
1526729478	L.RIM.Load.W2E.Update.ACK	Number of times the eNodeB receives load information update acknowledgments from WCDMA network

The percentage of eNodeB initially responding to the RIM message = L.RIM.Load.W2E.Resp/L.RIM.Load.W2E.Req. If the value is not 100%, the possible causes are as follows:

- The response message is incorrectly parsed.
- The requested target LTE cell is unavailable.
- The requested target LTE cell is abnormal.

eNodeB load update success rate = L.RIM.Load.W2E.Update.ACK/ L.RIM.Load.W2E.Update. If the value is not 100%, the possible causes are as follows:

- Data transmission is abnormal.
- The core networks or the eCoordinator works abnormally.
- The RNC works abnormally.

8.7.10 Parameter Optimization

If an LTE cell load is high, decrease the value of **CellMLB**.***InterRatMlbUeNumThd*** to increase the probability of LTE cell transition to the congested state, which reduces the incoming handovers of UMTS UEs.

If an LTE cell load is light, increase the value of **CellMLB**.***InterRatMlbUeNumThd*** to decrease the probability of LTE cell transition to the congested state, which increases the incoming handovers of UMTS UEs.

8.7.11 Troubleshooting

Fault Description

Traffic statistics are 0.

Traffic statistics include the following counters:

L.RIM.Load.W2E.Req (Number of Times the eNodeB Receives Load Information Requests from WCDMA Network), L.RIM.Load.W2E.Resp (Number of Times the eNodeB Sends Load Information Responses to WCDMA Network), L.RIM.Load.W2E.Update (Number of Times the eNodeB Sends Load Information Updates to WCDMA Network), and L.RIM.Load.W2E.Update.ACK (Number of times the eNodeB receives load information update acknowledgments from WCDMA network)

Trouble Handling

If the RIM procedure is based on core networks, perform the following steps to troubleshoot:

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Step 1 On the eNodeB side, check whether the switch controlling the RIM Based LTE Target Cell Selection feature is turned on:

- If the switch is turned off, turn on the switch.
- If the switch is turned on, check whether the license is valid and go to **Step 2**.

Step 2 On the RNC side, check whether the switch controlling the RIM Based LTE Target Cell Selection feature is turned on: If the peer RNC is a Huawei device, see *Interoperability Between UMTS and LTE* for details.

- If the switch is turned off, turn on the switch.
- If the switch is turned on, go to **Step 3**.

Step 3 Check whether the core network supports the RIM procedure.

- If the core network does not support the RIM procedure, enable this function on the core network side.
- If the core network supports the RIM procedure, go to **Step 4**

Step 4 On the eNodeB side, check whether the switch controlling the RIM Based LTE Target Cell Selection feature is changed from **ON** to **OFF** and then to **ON**.

- If yes, turn on the switch controlling the RIM Based LTE Target Cell Selection feature on the RNC side.
- Otherwise, contact Huawei technical support.

----End

If the RIM procedure is based on the eCoordinator, perform the following steps to troubleshoot:

Step 1 On the eNodeB side, check whether the switch controlling the RIM Based LTE Target Cell Selection feature is turned on:

- If the switch is turned off, turn on the switch.
- If the switch is turned on, check whether the license is valid and go to **Step 2**.

Step 2 On the eNodeB, check whether **RimOnEcoSwitch** is turned on.

- If the switch is turned off, turn on the switch.
- If the switch is turned on, go to **Step 3**.

Step 3 On the RNC side, check whether the switch controlling the RIM Based LTE Target Cell Selection feature is turned on: If the peer RNC is a Huawei device, see *Interoperability Between UMTS and LTE* for details.

- If the switch is turned off, turn on the switch.
- If the switch is turned on, go to **Step 4**.

Step 4 On the RNC, check whether **RimOnEcoSwitch** is turned on.

- If the switch is turned off, turn on the switch.
- If the switch is turned on, go to **Step 5**.

Step 5 Check whether **RimOnEcoSwitch** is turned on.

- If the switch is turned off, turn on the switch.
- If the switch is turned on, contact Huawei technical support.

----End

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8.8 PS Inter-RAT Mobility Between E-UTRAN and GERAN

8.8.1 When to Use PS Inter-RAT Mobility Between E-UTRAN and GERAN

Mobility management is fundamental to service continuity. If the E-UTRAN cannot cover the entire network and therefore requires other RATs, such as UTRAN or GERAN to supplement coverage, you are advised to enable coverage-based inter-RAT handover.

Coverage-based inter-RAT handover is the basis for other types of inter-RAT handover. Before enabling another type of inter-RAT handover, you need to enable coverage-based inter-RAT handover.

Event B1 or event B2 triggers coverage-based inter-RAT handovers. If there are no special requirements for the serving cell signal quality, event B2 may not be used.

8.8.2 Required Information

Inter-RAT handovers have requirements on networks and UEs. Before enabling an inter-RAT handover policy, you need to check whether the networks support inter-RAT operations and whether the UE supports inter-RAT measurements and handovers.

- If the devices in the networks do not support inter-RAT operations, disable the inter-RAT PS handover policies.
- If the devices in the networks do not support SRVCC to inter-RAT systems, disable the inter-RAT SRVCC policies.
- If the devices in the networks do not support CCO to GERAN, disable the CCO/NACC policies to GERAN.
- If blind handovers are used to implement inter-RAT handovers, you need to collect the information about the neighboring cells to which blind handovers will be allowed as well as their blind-handover priorities ranging from 1 to 32. These priorities have a positive correlation with the values. Note that the value 0 indicates that blind handovers to the neighboring cell are prohibited.
- If an operator requires flash redirection or NACC, ensure that the equipment on the network side supports RIM. If the equipment does not support RIM, flash redirection cannot be used.

The information to be collected before deploying coverage-based inter-RAT handover is the same as that before deploying coverage-based inter-frequency handover.

Verify that the inter-RAT networks provide continuous coverage in inter-RAT networking scenarios where eNodeBs often share sites with base stations of other RATs. Since the LTE network often provides higher data rates and has a lower threshold for inter-RAT measurement event A2, it is recommended that LTE networks preferentially carry services.

You need to determine whether event B1 or event B2 is used for inter-RAT measurements. If event B2 is configured, event B2 is used for inter-RAT measurements. If event B1 is configured and the UE support event B1, event B1 is used for inter-RAT measurements. If event B1 is configured but the UE only supports event B2, event B2 is used for inter-RAT measurements.

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8.8.3 Requirements

Operating Environment

The feature LOFD-001020 PS Inter-RAT Mobility between E-UTRAN and GERAN requires that the core-network equipment should support this feature. If the core-network equipment is provided by Huawei, the equipment version must be PS11.0. If the core-network equipment is provided by another vendor, check with the vendor whether the equipment supports this feature.

Transmission Networking

None

License

To use coverage-based inter-RAT handovers, an operator must purchase and activate the licenses for the features listed in the following table.

Feature ID	Feature Name	License Control Item	NE	Sales Unit
LOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN	PS Inter-RAT Mobility between E-UTRAN and GERAN	eNodeB	per user in RRC connected mode

8.8.4 Data Preparation

This section describes the data that you need to collect for setting parameters. Required data is data that you must collect for all scenarios. Collect scenario-specific data when necessary for a specific feature deployment scenario.

There are three types of data sources:

- Network plan (negotiation not required): parameter values planned and set by the operator
- Network plan (negotiation required): parameter values planned by the operator and negotiated with the EPC or peer transmission equipment
- User-defined: parameter values set by users

Required Data

This section describes the data required for configuring neighboring GERAN cells.

NOTE

During coverage-based inter-RAT handovers, the eNodeB can perform a blind handover when there is no neighboring cells. To enable this function, you can configure only UTRAN neighboring frequencies, GERAN neighboring frequency groups, or GERAN BCCH neighboring frequencies.

To enable other inter-RAT handover features, you must configure required data described in this section.

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The following table describes the parameters that must be set in the **GeranNfreqGroup** MO to configure a group of neighboring GERAN carrier frequencies.

Parameter Name	Parameter ID	Data Source	Setting Notes
Local cell ID	**GeranNfreqGroup**.***LocalCellId***	Network plan (negotiation not required)	Ensure that this parameter has been set in a **Cell** MO based on the network plan.
BCCH group ID	**GeranNfreqGroup**.***BcchGroupId***	Network plan (negotiation required)	This parameter specifies the ID of the GERAN carrier frequency group.
GERAN version	**GeranNfreqGroup**.***GeranVersion***	Network plan (negotiation required)	This parameter specifies the RAT supported by the GERAN carrier frequency group. If the peer BTS is provided by Huawei, this parameter must have the same value as the corresponding parameter in the **GCELLGPRS** MO of the peer BSC.
Starting ARFCN	**GeranNfreqGroup**.***StartingArfcn***	Network plan (negotiation required)	This parameter specifies the first ARFCN in the GERAN carrier frequency group. If the peer BTS is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **GCELL** MO of the peer BSC.
Band indicator	**GeranNfreqGroup**.***BandIndicator***	Network plan (negotiation required)	This parameter specifies the GSM band containing the GERAN carrier frequency group. If the peer BTS is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **GCELL** MO of the peer BSC.

The following table describes the parameters that must be set in the **GeranNfreqGroupArfcn** MO to configure a neighboring frequency in a GERAN carrier frequency group.

Parameter Name	Parameter ID	Data Source	Setting Notes
Local cell ID	**GeranNfreqGroupArfcn**.***LocalCellId***	Network plan (negotiation not required)	Ensure that this parameter has been set in a **Cell** MO based on the network plan.

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Parameter Name	Parameter ID	Data Source	Setting Notes
BCCH group ID	**GeranNfreqGroupArfcn**.***BcchGroupId***	Network plan (negotiation required)	This parameter specifies the ID of the GERAN carrier frequency group. Ensure that this parameter has been set in a **GERANNFREQGROUP** MO.
GERAN ARFCN	**GeranNfreqGroupArfcn**.***GeranArfcn***	Network plan (negotiation required)	This parameter specifies the ARFCN of the GERAN BCCH. If the peer BTS is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **GCELL** MO of the peer BSC. You are advised to configure frequencies on different bands in different frequency groups.

The following table describes the parameters that must be set in the **GeranExternalCell** MO to configure an external GERAN cell.

Parameter Name	Parameter ID	Data Source	Setting Notes
Mobile country code	**GeranExternalCell**.***Mcc***	Network plan (negotiation required)	This parameter specifies the MCC of the home BTS of the external GERAN cell. If this external cell works in RAN sharing mode, set this parameter to the MCC of the primary operator. If the peer BTS is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **GCNOPERATOR** MO of the peer BSC.
Mobile network code	**GeranExternalCell**.***Mnc***	Network plan (negotiation required)	This parameter specifies the MNC of the home BTS of the external GERAN cell. If this external cell works in RAN sharing mode, set this parameter to the MNC of the primary operator. If the peer BTS is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **GCNOPERATOR** MO of the peer BSC.
GERAN cell ID	**GeranExternalCell**.***GeranCellId***	Network plan (negotiation required)	This parameter specifies the ID of the external GERAN cell. If the peer BTS is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **GCELL** MO of the peer BSC.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Location area code	**GeranExternalCell**.***Lac***	Network plan (negotiation required)	This parameter specifies the LAC of the external GERAN cell. It uniquely identifies a GERAN cell within a PLMN. If the peer BTS is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **GCELL** MO of the peer BSC.
Routing area code configure indicator	**GeranExternalCell**.***RacCfgInd***	Network plan (negotiation required)	This parameter specifies whether to set the routing area code (RAC) for an external GERAN cell. Set this parameter to **CFG** if the neighboring GERAN cell supports PS services.
Routing area code	**GeranExternalCell**.***Rac***	Network plan (negotiation required)	This parameter specifies the RAC. If **RacCfgInd** is set to **CFG**, this parameter is mandatory. If the peer BTS is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **GCELLGPRS** MO of the peer BSC.
Band indicator	**GeranExternalCell**.***BandIndicator***	Network plan (negotiation required)	This parameter specifies the GSM band in which the external GERAN cell operates. If the peer BTS is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **GCELL** MO of the peer BSC.
GERAN ARFCN	**GeranExternalCell**.***GeranArfcn***	Network plan (negotiation required)	This parameter specifies the ARFCN of the GERAN BCCH. If the peer BTS is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **GCELL** MO of the peer BSC.
Network color code	**GeranExternalCell**.***NetworkColourCode***	Network plan (negotiation required)	This parameter specifies the NCC of the GERAN. It uniquely identifies a network within a country. If the peer BTS is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **GCELL** MO of the peer BSC.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Base station color code	**GeranExternalCell**.***BaseStationColourCode***	Network plan (negotiation required)	This parameter specifies the BCC of the home BTS of the external GERAN cell. It is used by the UE to discriminate between cells on the same frequency. If the peer BTS is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **GCELL** MO of the peer BSC.
DTM indication	**GeranExternalCell**.***DtmInd***	Network plan (negotiation required)	This parameter specifies whether the external GERAN cell supports dual service mode. If the cell supports dual service mode, both CS and PS services can be handed over to the cell. Otherwise, only CS services or PS services can be handed over to the cell. If the peer BTS is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **GCELL** MO of the peer BSC.

(Optional) The following table describes the parameters that must be set in the **GeranExternalCellPlmn** MO to configure an additional PLMN ID (representing a secondary operator) for an external GERAN cell.

Parameter Name	Parameter ID	Data Source	Setting Notes
GERAN cell ID	**GeranExternalCellPlmn**.***GeranCellId***	Network plan (negotiation required)	This parameter specifies the ID of the external GERAN cell. It uniquely identifies a GERAN cell within a BTS. If the peer BTS is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **GCELL** MO of the peer BSC.
Location area code	**GeranExternalCellPlmn**.***Lac***	Network plan (negotiation required)	This parameter specifies the LAC of the external GERAN cell. It uniquely identifies a GERAN cell within a PLMN. If the peer BTS is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **GCELL** MO of the peer BSC.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Mobile country code	**GeranExternalCellPlmn**.***Mcc***	Network plan (negotiation required)	This parameter specifies the MCC of the home BTS of the external GERAN cell. If this external cell works in RAN sharing mode, set this parameter to the MCC of the primary operator. If the peer BTS is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **GCNOPERATOR** MO of the peer BSC.
Mobile network code	**GeranExternalCellPlmn**.***Mnc***	Network plan (negotiation required)	This parameter specifies the MNC of the home BTS of the external GERAN cell. If this external cell works in RAN sharing mode, set this parameter to the MNC of the primary operator. If the peer BTS is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **GCNOPERATOR** MO of the peer BSC.
Share mobile country code	**GeranExternalCellPlmn**.***ShareMcc***	Network plan (negotiation required)	This parameter specifies the MCC of the secondary operator. If the peer BTS is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **GCNOPERATOR** MO of the peer BSC.
Share mobile network code	**GeranExternalCellPlmn**.***ShareMnc***	Network plan (negotiation required)	This parameter specifies the MNC of the secondary operator. If the peer BTS is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **GCNOPERATOR** MO of the peer BSC.

The following table describes the parameters that must be set in the **GERANNCELL** MO to configure the neighbor relationship with a GERAN cell.

Parameter Name	Parameter ID	Data Source	Setting Notes
Local cell ID	**GeranNcell**.***LocalCellId***	Network plan (negotiation not required)	Ensure that this parameter has been set in a **Cell** MO.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Mobile country code	**GeranNcell.*Mcc***	Network plan (negotiation required)	This parameter specifies the MCC of the home BTS of the neighboring GERAN cell. If this external cell works in RAN sharing mode, set this parameter to the MCC of the primary operator. If the peer BTS is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **GCNOPERATOR** MO of the peer BSC.
Mobile network code	**GeranNcell.*Mnc***	Network plan (negotiation required)	This parameter specifies the MNC of the home BTS of the neighboring GERAN cell. If this neighboring cell works in RAN sharing mode, set this parameter to the MNC of the primary operator. If the peer BTS is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **GCNOPERATOR** MO of the peer BSC.
Location area code	**GeranNcell.*Lac***	Network plan (negotiation required)	This parameter specifies the LAC of the neighboring GERAN cell. It uniquely identifies a GERAN cell within a PLMN. If the peer BTS is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **GCELL** MO of the peer BSC.
GERAN cell ID	**GeranNcell.*GeranCellId***	Network plan (negotiation required)	This parameter specifies the ID of the neighboring GERAN cell. It uniquely identifies a GERAN cell within a BTS. If the peer BTS is provided by Huawei, set this parameter to the same value as the corresponding parameter in the **GCELL** MO of the peer BSC.

The following table describes the parameters that must be set in the **InterRatPolicyCfgGroup** MO to configure an inter-RAT handover policy group.

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Parameter Name	Parameter ID	Data Source	Setting Notes
InterRAT policy config group ID	**InterRatPolicyCfgGroup**.***InterRatPolicyCfgGroupId***	Network plan (negotiation not required)	This parameter specifies the ID of the inter-RAT handover policy group. If you want to set a different inter-RAT handover policy for services with a specific QCI, you must create a policy group. This parameter will be referenced by a standard or extended QCI. For details, see the **StandardQci** or **ExtendedQci** MO described later.
GERAN GSM handover policy Configuration	**InterRatPolicyCfgGroup**.***GeranGsmHoCfg***	Network plan (negotiation not required)	This parameter specifies the policy configuration for handovers to GSM cells. Set this parameter based on the network plan. It is recommended that the default value be used if there is no special requirement for forbidding certain kind of handovers.
GERAN GPRS EDGE handover policy Configuration	**InterRatPolicyCfgGroup**.***GeranGprsEdgeHoCfg***	Network plan (negotiation not required)	This parameter specifies the policy configuration for handovers to GPRS/EDGE cells. Set this parameter based on the network plan. It is recommended that the default value be used if there is no special requirement for forbidding certain kind of handovers.
No handover flag	**InterRatPolicyCfgGroup**.***NoHoFlag***	Network plan (negotiation not required)	This parameter specifies whether to permit inter-RAT handovers. Set this parameter based on the network plan. It is recommended that the default value be used if there is no special requirement for forbidding certain kind of handovers.

The following table describes the parameters that must be set in the **StandardQci** MO to configure the inter-RAT handover policy group for a standardized QCI.

Parameter Name	Parameter ID	Data Source	Setting Notes
QoS Class Indication	**StandardQci**.***Qci***	Network plan (negotiation not required)	This parameter specifies the QCI of an EPS bearer. Set this parameter to the standard QCI with which services need to be configured with the inter-RAT handover policy.

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Parameter Name	Parameter ID	Data Source	Setting Notes
InterRAT policy config group ID	**StandardQci**.***InterRatPolicyCfgGroupId***	Network plan (negotiation not required)	This parameter specifies the ID of the inter-RAT handover policy group. Set this parameter to the ID of an inter-RAT handover policy group that has been configured in an **InterRatPolicyCfgGroup** MO.

The following table describes the parameters that must be set in the **ExtendedQci** MO to configure the inter-RAT handover policy group for an extended QCI.

Parameter Name	Parameter ID	Data Source	Setting Notes
Extended QoS Class Identifier	**ExtendedQci**.***ExtendedQci***	Network plan (negotiation not required)	This parameter specifies the QCI of an evolved packet system (EPS) bearer. Set this parameter to the extended QCI with which services need to be configured with the inter-RAT handover policy.
InterRAT policy config group ID	**ExtendedQci**.***InterRatPolicyCfgGroupId***	Network plan (negotiation not required)	This parameter specifies the ID of a parameter group related to inter-RAT handover. Set this parameter to the ID of an inter-RAT handover policy group that has been configured in an **InterRatPolicyCfgGroup** MO.

Scenario-specific Data

The following table describes the parameters that must be set in the **ENodeBAlgoSwitch** MO to configure the mode for coverage-based inter-RAT handovers.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Handover Mode switch	**ENodeBAlgoSwitch**.***HoModeSwitch***	Network plan (negotiation not required)	Select a mode for each type of inter-RAT handover based on the network plan of the operator. Handover to GERAN: ● GeranPsHoSwitch ● GeranNaccSwitch ● GeranCcoSwitch ● GeranSrvccSwitch ● GeranRedirectSwitch(GeranRedirectSwitch) Among the preceding switches, **GeranRedirectSwitch**(GeranRedirectSwitch) is basic to inter-RAT handover. This redirection switch must be turned on. Other inter-RAT handover switches can be turned on based on the network capability.

By default, the inter-RAT handover parameter group Group0 is used for services with all QCIs. To apply different inter-RAT handover parameter groups to services based on the QCI, the corresponding parameter group ID must be changed.

The following table describes the parameter that must be set in the **CellStandardQci** MO to configure parameter groups for services with standard QCIs.

Parameter Name	Parameter ID	Data Source	Setting Notes
Local cell ID	**CellStandardQci**.***LocalCellId***	Network plan (negotiation not required)	Ensure that this parameter has been set in a **Cell** MO based on the network plan.
QoS Class Indication	**CellStandardQci**.***Qci***	Network plan (negotiation not required)	Set this parameter to the index of the standard QCI for which the parameter group is configured.
InterRAT handover common group ID	**CellStandardQci**.***InterRatHoCommGroupId***	Network plan (negotiation not required)	Set this parameter to the index of the common inter-RAT handover parameter group.
InterRAT handover to GERAN Group ID	**CellStandardQci**.***InterRatHoGeranGroupId***	Network plan (negotiation not required)	Set this parameter to the index of the EUTRAN-to-GERAN handover parameter group.

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Parameter Name	Parameter ID	Data Source	Setting Notes
QCIPriorityFor Handover	**CellStandardQci**. ***QciPriorityForHo***	Network plan (negotiation not required)	Set this parameter to the priority of the corresponding QCI.

The following table describes the parameter that must be set in the **CellExtendedQci** MO to configure parameter groups for services with extended QCIs.

Parameter Name	Parameter ID	Data Source	Setting Notes
Extended QCI	**CellExtended Qci**.***Extended Qci***	Network plan (negotiation not required)	Set this parameter to the index of the extended QCI for which the parameter group is configured.
Local cell ID	**CellExtended Qci**.***LocalCell Id***	Network plan (negotiation not required)	Ensure that this parameter has been set in a **Cell** MO based on the network plan.
InterRAT handover common group ID	**CellExtended Qci**.***InterRat HoCommGro upId***	Network plan (negotiation not required)	Set this parameter to the index of the common inter-RAT handover parameter group.
InterRAT handover to GERAN Group ID	**CellExtended Qci**.***InterRat HoGeranGro upId***	Network plan (negotiation not required)	Set this parameter to the index of the EUTRAN-to-GERAN handover parameter group.

The following table describes the parameter that must be set in the **GeranNcell** MO to configure blind handover priorities and measurement priorities for GERAN neighboring cells.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Blind handover priority	**GeranNcell**.***BlindHoPriority***	Network plan (negotiation not required)	A larger parameter value indicates a higher priority for the GERAN neighboring cell to be selected during a blind handover. The value 0 indicates that blind handovers to the GERAN neighboring cell are prohibited. It is recommended that this parameter be set to a value ranging from 1 to 32 for GERAN neighboring cells whose coverages cover that of the serving cell. Set this parameter to a larger value for a neighboring cell that has wider overlapped coverage or a lower frequency. Different cells can be configured with the same blind-handover priority.

The following table describes the parameter must be set in a **GeranNfreqGroup** MO to configure the GERAN frequency priority.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Connect frequency priority	**GeranNfreqGroup**.***ConnFreqPriority***	Network plan (negotiation not required)	This parameter specifies the frequency priority based on which the eNodeB selects a target frequency for blind redirection or contains a frequency in a measurement configuration. A larger value indicates a higher priority. If this priority is set to 0 for a frequency, this frequency is not selected as the target frequency for a blind redirection. You are advised to set this parameter for GERAN frequency groups with continuous coverage. If there are multiple frequency groups have continuous coverage, set this parameter to a larger value for a frequency group in a lower frequency band. Set this parameter to 0 for a GETRAN frequency group with non-continuous coverage. You must configure this parameter to 0 for at least one GERAN frequency group to ensure that UEs that do not support GERAN measurement can be handed over to the GERAN in emergency conditions. Different frequencies can be configured with the same priority.

The following table describes the parameters must be set in a **CSFallBackBlindHoCfg** MO to configure RAT priorities for blind handovers. The **CSFallBackBlindHoCfg** MO takes effect for all inter-RAT handovers, rather than for only CSFB.

Parameter Name	Parameter ID	Data Source	Setting Notes
CN Operator ID	**CSFallBackBlind-HoCfg**.***CnOperatorId***	Network plan (negotiation not required)	This parameter specifies the index of the operator. Ensure that this parameter has been set in a **CnOperator** MO. Set this parameter for operators who need to be configured with priorities.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Highest priority InterRat	**CSFallBackBlind-HoCfg**.***InterRatHighest-Pri***	Network plan (negotiation not required)	This parameter specifies the highest-priority RAT for blind handovers. Setting this parameter to UTRAN ensures the optimal inter-RAT interoperability performance. If coverage holes exist in UTRAN, setting this parameter to GERAN to enhance the reliability of inter-RAT interoperability.
Second priority InterRat	**CSFallBackBlind-HoCfg**.***InterRatSecondPri***	Network plan (negotiation not required)	This parameter specifies the secondary-priority RAT for blind handovers. Set this parameter based on the value of **CSFallBackBlind-HoCfg**.***InterRatHighestPri***.
Lowest priority InterRat	**CSFallBackBlind-HoCfg**.***InterRatLowestPri***	Network plan (negotiation not required)	This parameter specifies the lowest-priority RAT for blind handovers. Set this parameter to the remaining available RAT or the RAT that do not support interoperability.

(Optional, required for only coverage-based inter-RAT handovers.) The following table describes the parameters must be set in a **CnOperatorHoCfg** MO to configure measurement offsets and priorities for coverage-based handovers to UTRAN or GERAN.

Parameter Name	Parameter ID	Data Source	Setting Notes
CN Operator ID	**CnOperatorHoCfg**.***CnOperatorId***	Network plan (negotiation not required)	This parameter specifies the index of the operator. Ensure that this parameter has been set in a **CnOperator** MO. Set this parameter for operators who need to be configured with priorities.

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Parameter Name	Parameter ID	Data Source	Setting Notes
GERAN A2 Threshold RSRP Offset	**CnOperatorHoCfg**.***GeranA2ThdRsrpOffset***	Network plan (negotiation not required)	This parameter value plus the value of **InterRatHoCommGroup**.***InterRatHoA2ThdRsrp*** is the RSRP threshold for GERAN measurement event A2 Set this parameter based on the actual conditions.

(Optional, required for only coverage-based inter-RAT handovers.) The following table describes the parameters must be set in a **GeranRanShare** MO to configure PLMN IDs of the operators that share a neighboring inter-RAT frequency. This information is required to enable inter-RAT blind redirection triggered by no neighboring cells.

Parameter Name	Parameter ID	Data Source	Setting Notes
Local cell ID	**GeranRanShare**.***LocalCellId***	Network plan (negotiation not required)	Ensure that this parameter has been set in a **Cell** MO.
BCCH group ID	**GeranRanShare**.***BcchGroupId***	Network plan (negotiation not required)	This parameter specifies the index of the GERAN carrier frequency group. It uniquely identifies a GERAN carrier frequency group. Add only the GERAN frequency groups for blind handovers triggered when there is no neighboring cells, and the frequency groups must have been set in a **GeranNfreqGroup** MO.
Mobile country code	**GeranRanShare**.***Mcc***	Network plan (negotiation required)	This parameter specifies the MCC of the GERAN frequency group for blind handovers triggered when there is no neighboring cells.
Mobile network code	**GeranRanShare**.***Mnc***	Network plan (negotiation required)	This parameter specifies the MNC of the GERAN frequency group for blind handovers triggered when there is no neighboring cells.

8.8.5 Activation

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8.8.5.1 Using the CME to Perform Batch Configuration for Newly Deployed eNodeBs

Enter the values of related parameters in a summary data file, which also contains other data for the new eNodeBs to be deployed. For detailed instructions, see section "Creating eNodeBs in Batches" in the initial configuration guide for the eNodeB.

The summary data file may be a scenario-specific file provided by the CME or a customized file, depending on the following conditions:

- Related managed objects (MOs) are contained in a scenario-specific summary data file. In this situation, set the parameters in the MOs, and then verify and save the file.
- Some related MOs are not contained in a scenario-specific summary data file. In this situation, customize a summary data file to include the MOs before you can set the parameters.

Table 8-25 Parameters related to coverage-based inter-RAT handovers to GERAN

MO	Sheet in the Summary Data File	Parameter Group	Remarks
ENodeBAlgoSwitch	User-defined sheet. **ENodeBAlgoSwitch** is recommended.	Handover Mode switch	This parameter must be customized on a list-type sheet of the template.
GeranNfreqGroup	User-defined sheet. **GeranNfreqGroup** is recommended.	Local cell identity, BCCH group identity, GERAN version, Starting ARFCN, Band indicator, Cell reselection priority configure indicator, Cell reselection priority, PMAX configure indicator, PMAX(dBm), Minimum required RX level(dBm), High priority threshold(2dB), Lower priority threshold(2dB), Frequency offset(dB), NCC monitoring permitted	This parameter must be customized on a list-type sheet of the template.
GeranNfreqGroupArfcn	User-defined sheet. **GeranNfreqGroupArfcn** is recommended.	Local cell identity, BCCH group identity, GERAN ARFCN	This parameter must be customized on a list-type sheet of the template.

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MO	Sheet in the Summary Data File	Parameter Group	Remarks
GeranExternalCell	User-defined sheet. **GeranExternalCell** is recommended.	Mobile country code, Mobile network code, GERAN cell identity, Location area code, Routing area code configure indicator, Routing area code, Band indicator, GERAN ARFCN, Network colour code, Base station colour code, DTM indication, Cell name	This parameter must be customized on a list-type sheet of the template.
GeranExternalCellPlmn	User-defined sheet. **GeranExternalCellPlmn** is recommended.	GERAN cell identity, Location area code, Mobile country code, Mobile network code, Share mobile country code, Share mobile network code	This parameter must be customized on a list-type sheet of the template.
GeranNcell	User-defined sheet. **GeranNcell** is recommended.	Local cell identity, Mobile country code, Mobile network code, Location area code, GERAN cell identity, No remove indicator, No handover indicator, Blind handover priority, ANR flag/Local cell name, Neighbour cell name	This parameter must be customized on a list-type sheet of the template.

8.8.5.2 Using the CME to Perform Batch Configuration for Existing eNodeBs

Batch Activated

This feature can be batch activated using the CME. For detailed operations, see the following section in the CME product documentation or online help: Managing the CME > CME Guidelines > Enhanced Feature Management > Feature Operation and Maintenance.

Batch Reconfiguration

Batch reconfiguration using the CME is the recommended method to activate a feature on existing eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure is as follows:

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Step 1 After creating a planned data area, choose **CME > Advanced > Customize Summary Data File** (U2000 client mode), or choose **Advanced > Customize Summary Data File** (CME client mode), to customize a summary data file for batch reconfiguration.

NOTE

For context-sensitive help on a current task in the client, press **F1**.

Step 2 Choose **CME > LTE Application > Export Data > Export Base Station Bulk Configuration Data** (U2000 client mode), or choose **LTE Application > Export Data > Export Base Station Bulk Configuration Data** (CME client mode), to export the eNodeB data stored on the CME into the customized summary data file.

Step 3 In the summary data file, set the parameters in the MOs listed in **8.8.5.1 Using the CME to Perform Batch Configuration for Newly Deployed eNodeBs**, and close the file.

Step 4 Choose **CME > LTE Application > Import Data > Import Base Station Bulk Configuration Data** (U2000 client mode), or choose **LTE Application > Import Data > Import Base Station Bulk Configuration Data** (CME client mode), to import the summary data file into the CME.

Step 5 Choose **CME > Planned Area > Export Incremental Scripts** (U2000 client mode), or choose **Area Management > Planned Area > Export Incremental Scripts**, to export and activate the incremental scripts.

----End

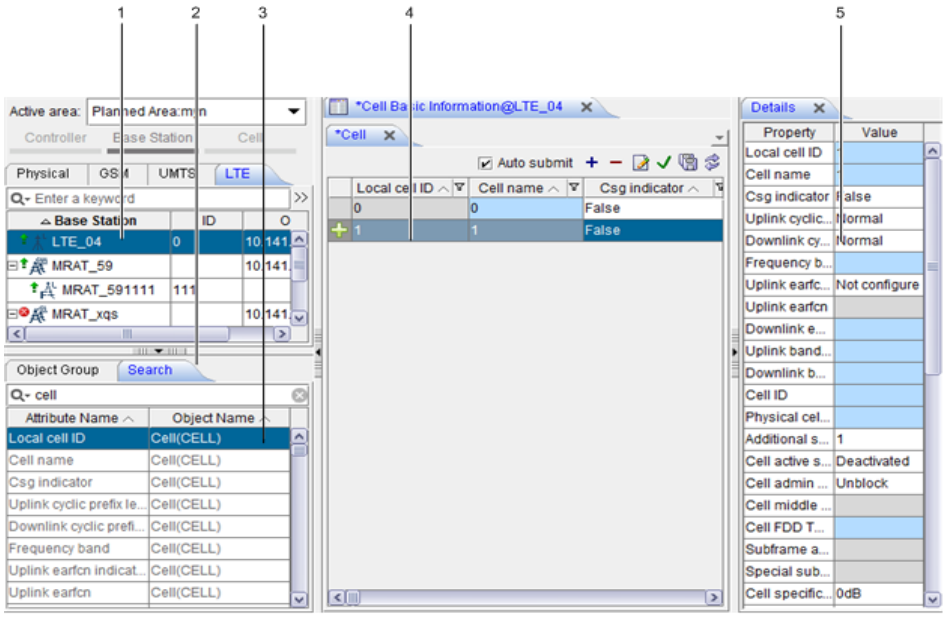
8.8.5.3 Using the CME to Perform Single Configuration

On the CME, set the parameters listed in the "Data Preparation" section for a single eNodeB. The procedure is as follows:

Step 1 In the planned data area, click **Base Station** in the upper left corner of the configuration window.

Step 2 In area 1 shown in **Figure 8-10**, select the eNodeB to which the MOs belong.

Figure 8-10 MO search and configuration window



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Step 3 On the **Search** tab page in area 2, enter an MO name, for example, **CELL**.

Step 4 In area 3, double-click the MO in the **Object Name** column. All parameters in this MO are displayed in area 4.

Step 5 Set the parameters in area 4 or 5.

Step 6 Choose **CME > Planned Area > Export Incremental Scripts** (U2000 client mode), or choose **Area Management > Planned Area > Export Incremental Scripts** (CME client mode), to export and activate the incremental scripts.

----End

8.8.5.4 Using MML Commands

Using MML Commands

Step 1 Run the **MOD ENODEBALGOSWITCH** command with the **GeranRedirectSwitch(GeranRedirectSwitch)** or **GeranPsHoSwitch(GeranPsHoSwitch)** check box selected under the **Handover Mode switch**.

Step 2 Run the following commands to add neighboring relationships with GERAN cells:

1. **ADD GERANNFREQGROUP**
2. **ADD GERANNFREQGROUPARFCN**
3. **ADD GERANEXTERNALCELL**
4. (Optional) **ADD GERANEXTERNALCELLPLMN**, which is used to configure the PLMN list for an external GERAN cell when the neighboring GERAN cell works in RAN sharing mode.
5. **ADD GERANNCELL**

----End

MML Command Examples

//Activating coverage-based inter-RAT handover to GERAN

```
MOD ENODEBALGOSWITCH: HoModeSwitch=GeranRedirectSwitch-1;
```

//Adding neighboring GERAN cells

```
ADD GERANNFREQGROUP: LocalCellId=0, BcchGroupId=0, GeranVersion=GSM,
StartingArfcn=600, BandIndicator=GSM_dcs1800, CellReselPriorityCfgInd=NOT_CFG,
PmaxGeranCfgInd=NOT_CFG;
ADD GERANNFREQGROUPARFCN: LocalCellId=0, BcchGroupId=0, GeranArfcn=601;
ADD GERANEXTERNALCELL: Mcc="460", Mnc="00", GeranCellId=1, Lac=5, RacCfgInd=CFG,
Rac=0, BandIndicator=GSM_dcs1800, GeranArfcn=601, NetworkColourCode=1,
BaseStationColourCode=1;
ADD GERANEXTERNALCELLPLMN: GeranCellId=1, Lac=1, Mcc="460", Mnc="00",
ShareMcc="460", ShareMnc="20";5.   ADD GERANNCELL: LocalCellId=0, Mcc="460",
Mnc="00", Lac=1, GeranCellId=1;
```

8.8.6 Activation Observation

The activation observation procedure for coverage-based inter-RAT handover to GERAN is the same as that for coverage-based inter-RAT handover to UTRAN. For details, see **8.1.6 Activation Observation**.

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8.8.7 Reconfiguration

None

8.8.8 Deactivation

Using the CME to Perform Batch Configuration

Batch reconfiguration using the CME is the recommended method to deactivate a feature on eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure for feature deactivation is similar to that for feature activation described in **8.8.5.2 Using the CME to Perform Batch Configuration for Existing eNodeBs**. In the procedure, modify parameters according to **Table 8-26**.

Table 8-26 Coverage-based Inter-RAT Handover

MO	Sheet in the Summary Data File	Parameter Group	Setting Notes
ENodeBAlgoSwitch	User-defined sheet	Handover Mode switch	Clear the following check boxes: ● GeranPsHoSwitch ● GeranSrvccSwitch ● GeranRedirectSwitch ● GeranCcoSwitch ● GeranNaccSwitch

Using the CME to Perform Single Configuration

On the CME, set the parameters listed in each scenario for a single eNodeB. For detailed instructions, see **8.8.5.3 Using the CME to Perform Single Configuration** for feature activation.

Using MML Commands

To deactivate this feature, clear the corresponding check boxes under the **Handover Mode switch** parameter by running the **MOD ENODEBALGOSWITCH** command.

MML Command Examples

//Deactivating coverage-based inter-RAT handover to GERAN

```
MOD ENODEBALGOSWITCH:
HoModeSwitch=GeranPsHoSwitch-0&GeranCcoSwitch-0&GeranSrvccSwitch-0&GeranRedirectSw
itch-0;
```


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8.8.9 Performance Monitoring

Performance monitoring for coverage-based E-UTRAN to GERAN inter-RAT handover is similar to that for inter-RAT mobility between E-UTRAN and UTRAN. For details, see **8.1.9 Performance Monitoring**.

Related counters are listed in **Table 8-27** and **Table 8-28**.

Table 8-27 Counters related to coverage-based handover to GERAN

Counter ID	Counter Name	Description
1526729488	L.IRATHO.E2G.Coverage.PrepAttOut	Number of coverage-based handover preparation attempts from E-UTRAN to GERAN
1526729489	L.IRATHO.E2G.Coverage.ExecAttOut	Number of coverage-based handover execution attempts from E-UTRAN to GERAN
1526729490	L.IRATHO.E2G.Coverage.ExecSuccOut	Number of successful coverage-based handover executions from E-UTRAN to GERAN

Table 8-28 Counter related to coverage-based redirection to GERAN

Counter ID	Counter Name	Description
1526729492	L.RRCRedirection.E2G.Coverage	Number of coverage-based redirections from E-UTRAN to GERAN

8.8.10 Parameter Optimization

The optimization for E-UTRAN to GERAN inter-RAT handover parameter groups are the same as that for E-UTRAN to GERAN inter-RAT handovers. For details, see **8.1.10 Parameter Optimization**.

Parameter Groups Related to Inter-RAT Handovers to GERAN

Parameter groups related to inter-RAT handovers to GERAN are contained in the **InterRatHoGeranGroup** MO. The following table provides suggestions on how to reconfigure these parameters.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Local cell ID	**InterRatHoGeranGroup**.***LocalCellId***	Network plan (negotiation not required)	Ensure that this parameter has been set in a **Cell** MO based on the network plan.
InterRAT handover to GERAN Group ID	**InterRatHoGeranGroup**.***InterRatHoGeranGroupId***	Network plan (negotiation not required)	This parameter specifies the ID of the parameter group related to inter-RAT handovers to GERAN. If you want to set a different parameter group for services with a specific QCI, you must create a parameter group, and reference the parameter group ID to the corresponding QCI.
GERAN handover hysteresis	**InterRatHoGeranGroup**.***InterRatHoGeranB1Hyst***	Network plan (negotiation not required)	This parameter specifies the hysteresis for event B1 related to inter-RAT handover to GERAN. This parameter is used to prevent frequent triggering of event evaluation caused by radio signal fluctuation. In this way, the probability of ping-pong handovers or handover decision errors is reduced. The default value is recommended.
CoverageBased GERAN trigger threshold	**InterRatHoGeranGroup**.***InterRatHoGeranB1Thd***	Network plan (negotiation not required)	This parameter specifies the RSSI threshold for event B1 in a coverage-based, UL-quality-based, or a distance-based handover to GERAN. A larger value of this parameter results in a lower probability of handovers to GERAN, and a smaller value results in a higher probability. The default value is recommended.

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Parameter Name	Parameter ID	Data Source	Setting Notes
GERAN time to trigger	**InterRatHoGeranGroup.*InterRatHoGeranB1TimeToTrig***	Network plan (negotiation not required)	This parameter specifies the time-to-trigger for event B1 related to inter-RAT handover to GERAN. When detecting that the signal quality in at least one neighboring cell meets the entering condition, the UE does not send a measurement report to the eNodeB immediately. Instead, the UE sends a report only when the signal quality continuously meets the entering condition during the time-to-trigger period. If both UTRAN and GERAN neighboring cells are measured, you are advised to set the time-to-trigger for GERAN neighboring cells to be greater than that for UTRAN neighboring cells, increasing the possibility of handovers to UTRAN neighboring cells. If both coverage-based inter-frequency handovers and coverage-based inter-RAT handovers are enabled, you are advised to set the time-to-trigger for the inter-RAT handover event be greater than that for the inter-frequency handover event, increasing the possibility of inter-frequency handovers.
Load Service Based Geran EventB1 trigger threshold	**InterRatHoGeranGroup.*LdSvBasedHoGeranB1Thd***	Network plan (negotiation not required)	This parameter specifies the RSSI threshold for event B1 related to load- or service-based inter-RAT handover to GERAN. A larger value of this parameter results in a lower probability of handovers to UTRAN, and a smaller value results in a higher probability. The default value is recommended.

Inter-RAT Handover Parameters

Inter-RAT handover parameters are contained in the **InterRatHoComm** MO. The following table provides suggestions on how to reconfigure these parameters.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Max report cell number	**InterRatHoComm.*InterRatHoMaxRprtCell***	Network plan (negotiation not required)	This parameter specifies the maximum number of cells to be included in the measurement report. A smaller value of this parameter leads to less candidate target cells for handovers, which lowers the handover success rate but saves the air interface resources. A larger value leads to opposite effect. It is recommended that the default value be used.
Measurement report amount	**InterRatHoComm.*InterRatHoRprtAmount***	Network plan (negotiation not required)	This parameter specifies the number of periodical measurement reports to be sent after an inter-RAT handover event is triggered. It is used to prevent the impact of measurement report loss and internal processing failure on the handover. When the value is increased, the number of reports increases and the handover success rate rises to a certain extent. An excessively large number of reports, however, results in excessive signaling and hence causes a waste of radio interface resources. The default value is recommended.
Geran measurement report interval	**InterRatHoComm.*InterRatHoGeranRprtInterval***	Network plan (negotiation not required)	This parameter specifies the interval at which periodic measurement reports are sent to the eNodeB after the event for inter-RAT handover to GERAN is triggered. A smaller value of this parameter leads to more measurement reports over the air interface and more resources are consumed. A larger value of this parameter results in a larger handover delay, which lowers the handover success rate. The default value is recommended.

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Parameter Name	Parameter ID	Data Source	Setting Notes
InterRat A1A2 measurement trigger quantity	**InterRatHoComm.*InterRatHoA1A2TrigQuan***	Network plan (negotiation not required)	This parameter specifies the quantity used to evaluate the triggering condition for the inter-RAT measurement event A1 or A2. The values are RSRP, RSRQ and BOTH. RSRP-based measurement results are stable, varying little with the load, and therefore there is little signal fluctuation. RSRQ-based measurement results vary with the load and are likely to reflect signal quality in real time. Set this parameter as required. By default, the reporting quantity for the inter-RAT measurement event A1 or A2 is BOTH, that is, both RSRP and RSRQ. It is recommended that the default value be used if the traffic load of LTE cells is not stable.
Inter RAT HO Trigger Event Type	**InterRatHoComm.*InterRatHoEventType***	Network plan (negotiation not required)	This parameter specifies the type of the inter-RAT handover event. This parameter applies only to coverage-based handovers. The default value is recommended.
Max Geran cell num in redirection	**InterRatHoComm.*CellInfoMaxGeranCellNum***	Network plan (negotiation not required)	This parameter specifies the maximum number of GERAN cell system information messages that can be transmitted during a flash redirection procedure. The default value is recommended.

8.8.11 Troubleshooting

For details about troubleshooting, see **8.1.11 Troubleshooting**.

8.9 Service-based Inter-RAT Handover to GERAN

8.9.1 When to Use Service-based Inter-RAT Handover to GERAN

You are advised to enable service-based inter-RAT handover to divert some services to another RAT in either of the following situations:

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- In the early phase of LTE network construction, the E-UTRAN is incapable of providing continuous coverage.
- After the LTE network has stabilized, the E-UTRAN is heavily loaded with certain types of service.

8.9.2 Required Information

Before you deploy service-based inter-RAT handover, collect information about the coverage and load of the target inter-RAT system. Ensure that the target inter-RAT system provides continuous coverage. If the target inter-RAT system is heavily loaded, service-based inter-RAT handover is not recommended. Service-based inter-RAT handover to a heavily loaded system affects user experience.

8.9.3 Deployment Requirements

Operating Environment

The feature LOFD-001020 PS Inter-RAT Mobility between E-UTRAN and GERAN requires that the core-network equipment should support this feature. If the core-network equipment is provided by Huawei, the equipment version must be PS11.0. If the core-network equipment is provided by another vendor, check with the vendor whether the equipment supports this feature.

Transmission Networking

None

License

To use service-based inter-RAT handover to GERAN, an operator must purchase and activate the licenses for the features listed in the following table.

Feature ID	Feature Name	License Control Item	NE	Sales Unit
LOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN	PS Inter-RAT Mobility between E-UTRAN and GERAN	eNodeB	per user in RRC connected mode
LOFD-001046	Service based inter-RAT handover to GERAN	Service based inter-RAT handover to GERAN	eNodeB	per user in RRC connected mode

8.9.4 Data Preparation

This section describes the data that you need to collect for setting parameters. Required data is data that you must collect for all scenarios. Collect scenario-specific data when necessary for a specific feature deployment scenario.

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There are three types of data sources:

- Network plan (negotiation not required): parameter values planned and set by the operator
- Network plan (negotiation required): parameter values planned by the operator and negotiated with the EPC or peer transmission equipment
- User-defined: parameter values set by users

Required Data

For details about the required data, see **8.8.4 Data Preparation**.

Scenario-specific Data

Service-based inter-RAT handover can be enabled only after coverage-based inter-RAT handover is enabled. This section describes the required parameters besides those for coverage-based inter-RAT handover. For details on the required parameters for coverage-based inter-RAT handover, see the **Scenario-specific Data** part in **8.8.4 Data Preparation**.

The following table describes the parameter that must be set in the **ENodeBAlgoSwitch** MO to enable service-based inter-RAT handover to GERAN.

Parameter Name	Parameter ID	Data Source	Setting Notes
Handover Algo switch	**ENodeBAlgoSwitch.*HoAlgoSwitch***	Network plan (negotiation not required)	Turn on the **GeranServiceHoSwitch(GeranServiceHoSwitch)** switch based on the network plan of the operator.

The handover policy for service-based inter-RAT handover to GERAN is the same as that for service-based inter-RAT handover to UTRAN. For detailed configuration, see **8.2.4 Data Preparation**.

8.9.5 Activation

Before activating service-based inter-RAT handover, ensure that coverage-based inter-RAT handover has been activated.

8.9.5.1 Using the CME to Perform Batch Configuration for Newly Deployed eNodeBs

Enter the values of related parameters in a summary data file, which also contains other data for the new eNodeBs to be deployed. Then, import the summary data file into the Configuration Management Express (CME) for batch configuration. For detailed instructions, see section "Creating eNodeBs in Batches" in the initial configuration guide for the eNodeB.

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The summary data file may be a scenario-specific file provided by the CME or a customized file, depending on the following conditions:

- Related managed objects (MOs) are contained in a scenario-specific summary data file. In this situation, set the parameters in the MOs, and then verify and save the file.
- Some related MOs are not contained in a scenario-specific summary data file. In this situation, customize a summary data file to include the MOs before you can set the parameters.

Table 8-29 Parameters related to service-based inter-RAT handovers to GERAN

MO	Sheet in the Summary Data File	Parameter Group	Remarks
ENodeBAlgoSwitch	User-defined sheet. **ENodeBAlgoSwitch** is recommended.	Handover Algo switch, Handover Mode switch	This parameter must be customized on a list-type sheet of the template.
ServiceIrHoCfgGroup	User-defined sheet. **ServiceIrHoCfgGroup** is recommended.	CN Operator ID, Service-based inter-RAT handover policy group ID, Inter-RAT handover state	This parameter must be customized on a list-type sheet of the template.
GeranNfreqGroup	User-defined sheet. **GeranNfreqGroup** is recommended.	Local cell identity, BCCH group identity, GERAN version, Starting ARFCN, Band indicator, Cell reselection priority configure indicator, Cell reselection priority, PMAX configure indicator, PMAX(dBm), Minimum required RX level(dBm), High priority threshold(2dB), Lower priority threshold(2dB), Frequency offset(dB), NCC monitoring permitted	This parameter must be customized on a list-type sheet of the template.
GeranNfreqGroupArfcn	User-defined sheet. **GeranNfreqGroupArfcn** is recommended.	Local cell identity, BCCH group identity, GERAN ARFCN	This parameter must be customized on a list-type sheet of the template.

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MO	Sheet in the Summary Data File	Parameter Group	Remarks
GeranExternalCell	User-defined sheet. **GeranExternalCell** is recommended.	Mobile country code, Mobile network code, GERAN cell identity, Location area code, Routing area code configure indicator, Routing area code, Band indicator, GERAN ARFCN, Network colour code, Base station colour code, DTM indication, Cell name	This parameter must be customized on a list-type sheet of the template.
GeranExternalCellPlmn	User-defined sheet. **GeranExternalCellPlmn** is recommended.	GERAN cell identity, Location area code, Mobile country code, Mobile network code, Share mobile country code, Share mobile network code	This parameter must be customized on a list-type sheet of the template.
GeranNcell	User-defined sheet. **GeranNcell** is recommended.	Local cell identity, Mobile country code, Mobile network code, Location area code, GERAN cell identity, No remove indicator, No handover indicator, Blind handover priority, ANR flag/Local cell name, Neighbour cell name	This parameter must be customized on a list-type sheet of the template.

8.9.5.2 Using the CME to Perform Batch Configuration for Existing eNodeBs

Batch Activated

This feature can be batch activated using the CME. For detailed operations, see the following section in the CME product documentation or online help: Managing the CME > CME Guidelines > Enhanced Feature Management > Feature Operation and Maintenance.

Batch Reconfiguration

Batch reconfiguration using the CME is the recommended method to activate a feature on existing eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure is as follows:

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Step 1 After creating a planned data area, choose **CME > Advanced > Customize Summary Data File** (U2000 client mode), or choose **Advanced > Customize Summary Data File** (CME client mode), to customize a summary data file for batch reconfiguration.

NOTE

For context-sensitive help on a current task in the client, press **F1**.

Step 2 Choose **CME > LTE Application > Export Data > Export Base Station Bulk Configuration Data** (U2000 client mode), or choose **LTE Application > Export Data > Export Base Station Bulk Configuration Data** (CME client mode), to export the eNodeB data stored on the CME into the customized summary data file.

Step 3 In the summary data file, set the parameters in the MOs listed in **8.9.5.1 Using the CME to Perform Batch Configuration for Newly Deployed eNodeBs**, and close the file.

Step 4 Choose **CME > LTE Application > Import Data > Import Base Station Bulk Configuration Data** (U2000 client mode), or choose **LTE Application > Import Data > Import Base Station Bulk Configuration Data** (CME client mode), to import the summary data file into the CME.

Step 5 Choose **CME > Planned Area > Export Incremental Scripts** (U2000 client mode), or choose **Area Management > Planned Area > Export Incremental Scripts**, to export and activate the incremental scripts.

----End

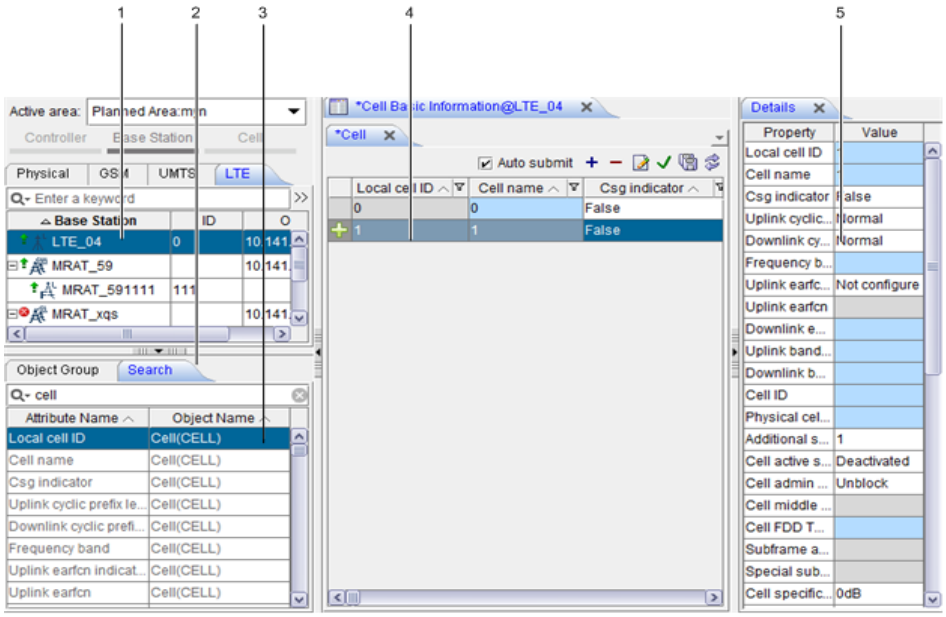
8.9.5.3 Using the CME to Perform Single Configuration

On the CME, set the parameters listed in the "Data Preparation" section for a single eNodeB. The procedure is as follows:

Step 1 In the planned data area, click **Base Station** in the upper left corner of the configuration window.

Step 2 In area 1 shown in **Figure 8-11**, select the eNodeB to which the MOs belong.

Figure 8-11 MO search and configuration window



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Step 3 On the **Search** tab page in area 2, enter an MO name, for example, **CELL**.

Step 4 In area 3, double-click the MO in the **Object Name** column. All parameters in this MO are displayed in area 4.

Step 5 Set the parameters in area 4 or 5.

Step 6 Choose **CME > Planned Area > Export Incremental Scripts** (U2000 client mode), or choose **Area Management > Planned Area > Export Incremental Scripts** (CME client mode), to export and activate the incremental scripts.

----End

8.9.5.4 Using MML Commands

Using MML Commands

Step 1 Run the **MOD ENODEBALGOSWITCH** command with the **GeranServiceHoSwitch(GeranServiceHoSwitch)** check box selected under the **Handover Algo switch**.

Step 2 Run the **MOD CNOPERATORSTANDARDQCI** command to configure a service-based inter-RAT handover policy group.

Step 3 Run the following commands to add neighbor relationships with GERAN cells:

1. **ADD GERANNFREQGROUP**
2. **ADD GERANNFREQGROUPARFCN**
3. **ADD GERANEXTERNALCELL**
4. (Optional) **ADD GERANEXTERNALCELLPLMN**, which is used to configure the PLMN list for an external GERAN cell when the neighboring GERAN cell works in RAN sharing mode.
5. **ADD GERANNCELL**

----End

MML Command Examples

//Activating service-based inter-RAT handover to GERAN

```
MOD ENODEBALGOSWITCH: HoAlgoSwitch=GeranServiceHoSwitch-1;MOD
SERVICEIRHOCFGGROUP: CnOperatorId=0, ServiceIrHoCfgGroupId=0,
InterRatHoState=MUST_HO;
```

//Adding neighboring relationships with GERAN cells

```
ADD GERANNFREQGROUP: LocalCellId=0, BcchGroupId=0, GeranVersion=GSM,
StartingArfcn=600, BandIndicator=GSM_dcs1800, CellReselPriorityCfgInd=NOT_CFG,
PmaxGeranCfgInd=NOT_CFG;ADD GERANNFREQGROUPARFCN: LocalCellId=0, BcchGroupId=0,
GeranArfcn=601;ADD GERANEXTERNALCELL: Mcc="460", Mnc="00", GeranCellId=1, Lac=5,
RacCfgInd=CFG, Rac=0, BandIndicator=GSM_dcs1800, GeranArfcn=601,
NetworkColourCode=1, BaseStationColourCode=1;ADD GERANEXTERNALCELLPLMN:
GeranCellId=1, Lac=1, Mcc="460", Mnc="00", ShareMcc="460", ShareMnc="20";5.ADD
GERANNCELL: LocalCellId=0, Mcc="460", Mnc="00", Lac=1, GeranCellId=1;ADD
GERANNCELL: LocalCellId=0, Mcc="460", Mnc="00", Lac=1, GeranCellId=1;
```


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8.9.6 Activation Observation

The activation observation procedure for service-based inter-RAT handover to GERAN is the same as that for service-based inter-RAT handover to UTRAN. For details, see **8.2.6 Activation Observation**.

8.9.7 Reconfiguration

None

8.9.8 Deactivation

Using the CME to Perform Batch Configuration

Batch reconfiguration using the CME is the recommended method to deactivate a feature on eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure for feature deactivation is similar to that for feature activation described in **8.9.5.2 Using the CME to Perform Batch Configuration for Existing eNodeBs**. In the procedure, modify parameters according to the table listed in each scenario.

Table 8-30 Service-based inter-RAT handover

MO	Sheet in the Summary Data File	Parameter Group	Setting Notes
ENodeBAlgoSwitch	User-defined sheet	HoAlgoSwitch	Clear the following check boxes: UtranServiceHoSwitch, GeranServiceHoSwitch, Cdma1xRttServiceHoSwitch, CdmaHrpdServiceHoSwitch

Using the CME to Perform Single Configuration

On the CME, set the parameters listed in each scenario for a single eNodeB. For detailed instructions, see **8.9.5.3 Using the CME to Perform Single Configuration** for feature activation.

Using MML Commands

To deactivate service-based inter-RAT handover to GERAN, turn off the corresponding switch by running the **MOD ENODEBALGOSWITCH** command.

MML Command Examples

//Deactivating service-based inter-RAT handover to GERAN

```
MOD ENODEBALGOSWITCH: HoAlgoSwitch=GeranServiceHoSwitch-0;
```


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8.9.9 Performance Monitoring

Performance monitoring for Separate Mobility Policies to UTRAN for Multi PLMN is similar to that for inter-RAT mobility between E-UTRAN and UTRAN. For details, see **8.1.9 Performance Monitoring**.

Related counters are listed in **Table 8-31** and **Table 8-32**.

Table 8-31 Counters related to service-based handover to GERAN

Counter ID	Counter Name	Counter Description
1526730021	L.IRATHO.E2G.Service.PrepAttOut	Number of service-based handover preparation attempts from E-UTRAN to GERAN
1526730022	L.IRATHO.E2G.Service.ExecAttOut	Number of service-based handover execution attempts from E-UTRAN to GERAN
1526730023	L.IRATHO.E2G.Service.ExecSuccOut	Number of successful service-based handover executions from E-UTRAN to GERAN

Table 8-32 Counters related to service-based redirection to GERAN

Counter ID	Counter Name	Counter Description
1526730025	L.RRCRedirection.E2G.Service	Number of service-based redirections from E-UTRAN to GERAN

8.9.10 Parameter Optimization

The reconfiguration of parameters that are not involved in this section is the same as the reconfiguration of parameters for coverage-based inter-RAT handovers. For details, see **8.8.10 Parameter Optimization**.

Parameter groups related to inter-RAT handover to GERAN are contained in the **InterRatHoGeranGroup** MO. The following table provides suggestions on how to reconfigure these parameters.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Load Service Based Geran EventB1 trigger threshold	**InterRatHoGeranGroup.*LdSvBasedHoGeranB1Thd***	Network plan (negotiation not required)	This parameter specifies the RSSI threshold for event B1 related to load- or service-based inter-RAT handover to GERAN. A larger value of this parameter results in a lower probability of handovers to GERAN, and a smaller value results in a higher probability. The default value is recommended.
GERAN time to trigger	**InterRatHoGeranGroup.*InterRatHoGeranB1TimeToTrig***	Network plan (negotiation not required)	This parameter specifies the time-to-trigger for event B1 related to inter-RAT handover to GERAN. When detecting that the signal quality in at least one neighboring cell meets the entering condition, the UE does not send a measurement report to the eNodeB immediately. Instead, the UE sends a report only when the signal quality continuously meets the entering condition during the time-to-trigger period. If both UTRAN and GERAN neighboring cells are measured, you are advised to set the time-to-trigger for GERAN neighboring cells to be greater than that for UTRAN neighboring cells, increasing the possibility of handovers to UTRAN neighboring cells.

8.9.11 Troubleshooting

For details about troubleshooting, see **8.1.11 Troubleshooting**.

8.10 Distance-based Inter-RAT Handover to GERAN

8.10.1 When to Use Distance based Inter-RAT handover to GERAN

You are advised to enable distance-based inter-RAT handover when the E-UTRAN cannot cover the entire network and causes severe overshoot coverage for the cells of another RAT. The latter situation occurs if the coverage of the E-UTRAN overlaps with that of another RAT and the RF signals from an E-UTRAN cell travels distances significantly longer than the inter-site spacing specified in the network plan. If the cells of the other RAT are not configured as neighboring cells of the E-UTRAN cell, inter-frequency handovers cannot be triggered in time and call drops will occur. To reduce the probability of such call drops, you can enable distance-based inter-RAT handover.

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8.10.2 Required Information

Information required for distance-based inter-RAT handover to GERAN is the same as information required for distance-based inter-RAT handover to UTRAN. For details, see **8.3.2 Required Information**.

8.10.3 Deployment Requirements

Operating Environment

The feature LOFD-001020 PS Inter-RAT Mobility between E-UTRAN and GERAN requires that the core-network equipment should support this feature. If the core-network equipment is provided by Huawei, the equipment version must be PS11.0. If the core-network equipment is provided by another vendor, check with the vendor whether the equipment supports this feature.

Transmission Networking

None

License

To use distance-based inter-RAT handovers, an operator must purchase and activate the licenses for the features listed in the following table.

Feature ID	Feature Name	License Control Item	NE	Sales Unit
LOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN	PS Inter-RAT Mobility between E-UTRAN and GERAN	eNodeB	per user in RRC connected mode
LOFD-001073	Distance based Inter-RAT handover to GERAN	Distance based Inter-RAT handover to GERAN	eNodeB	per user in RRC connected mode

8.10.4 Data Preparation

This section describes the data that you need to collect for setting parameters. Required data is data that you must collect for all scenarios. Collect scenario-specific data when necessary for a specific feature deployment scenario.

There are three types of data sources:

- Network plan (negotiation not required): parameter values planned and set by the operator
- Network plan (negotiation required): parameter values planned by the operator and negotiated with the EPC or peer transmission equipment

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- User-defined: parameter values set by users

Required Data

For details about the required data, see **8.8.4 Data Preparation**.

Scenario-specific Data

Distance-based inter-RAT handover can be enabled only after coverage-based inter-RAT handover is enabled. This section describes the required parameters besides those for coverage-based inter-RAT handover. For details on the required parameters for coverage-based inter-RAT handover, see the **Scenario-specific Data** part in **8.8.4 Data Preparation**.

To enable distance-based inter-RAT handover, the corresponding switch must be turned on and the measurement object type must be set to UTRAN or GERAN.

The following table describes the parameters that must be set in the **CellAlgoSwitch** MO to enable distance-based inter-RAT handovers.

Parameter Name	Parameter ID	Data Source	Setting Notes
Local cell ID	**CellAlgoSwitch**.***LocalCellId***	Network plan (negotiation not required)	Ensure that this parameter has been set in a **Cell** MO.
Distance-based Handover Switch	**CellAlgoSwitch**.***DistBasedHoSwitch***	Network plan (negotiation not required)	This parameter specifies whether to enable distance-based handovers. If this switch is turned on, distance-based handovers are allowed. If this switch is turned off, distance-based handovers to any cells are prohibited.

The following table describes the parameters that must be set in the **DistBasedHO** MO to set the measurement object type.

Parameter Name	Parameter ID	Data Source	Setting Notes
Local cell ID	**DistBasedHO**.***LocalCellId***	Network plan (negotiation not required)	Ensure that this parameter has been set in a **Cell** MO.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Distance-based Measurement Object Type	**DistBasedHO**.***DistBasedMeasObjType***	Network plan (negotiation not required)	This parameter specifies the measurement object type for distance-based handovers. If this parameter is set to UTRAN or GERAN, distanced-based handovers to inter-frequency neighboring cells are allowed.

8.10.5 Activation

Before activating distance-based inter-RAT handover, ensure that coverage-based inter-RAT handover has been activated.

Using the CME to Perform Batch Configuration for Newly Deployed eNodeBs

Enter the values of related parameters in a summary data file, which also contains other data for the new eNodeBs to be deployed. Then, import the summary data file into the Configuration Management Express (CME) for batch configuration. For detailed instructions, see section "Creating eNodeBs in Batches" in the initial configuration guide for the eNodeB.

The summary data file may be a scenario-specific file provided by the CME or a customized file, depending on the following conditions:

- Related managed objects (MOs) are contained in a scenario-specific summary data file. In this situation, set the parameters in the MOs, and then verify and save the file.
- Some related MOs are not contained in a scenario-specific summary data file. In this situation, customize a summary data file to include the MOs before you can set the parameters.

Table 8-33 Parameters related to distance-based inter-RAT handovers to GERAN

MO	Sheet in the Summary Data File	Parameter Group	Remarks
CellAlgoSwitch	User-defined sheet. **CellAlgoSwitch** is recommended.	Distance-based Handover Switch	This parameter must be customized on a list-type sheet of the template.
DistBasedHo	User-defined sheet. **DistBasedHo** is recommended.	Distance-based Measurement Object Type, Distance-based Handover Threshold	This parameter must be customized on a list-type sheet of the template.

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Using the CME to Perform Batch Configuration for Existing eNodeBs

Batch reconfiguration using the CME is the recommended method to activate a feature on existing eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure is as follows:

Step 1 Choose **CME > Advanced > Customize Summary Data File** (U2000 client mode), or choose **Advanced > Customize Summary Data File** (CME client mode), to customize a summary data file for batch reconfiguration.

NOTE

For context-sensitive help on a current task in the client, press **F1**.

Step 2 Choose **CME > LTE Application > Export Data > Export Base Station Bulk Configuration Data** (U2000 client mode), or choose **LTE Application > Export Data > Export Base Station Bulk Configuration Data** (CME client mode), to export the eNodeB data stored on the CME into the customized summary data file.

Step 3 In the summary data file, set the parameters in the MOs listed in **Table 8-33**, and close the file.

Step 4 Choose **CME > LTE Application > Import Data > Import Base Station Bulk Configuration Data** (U2000 client mode), or choose **LTE Application > Import Data > Import Base Station Bulk Configuration Data** (CME client mode), to import the summary data file into the CME.

Step 5 Choose **CME > Planned Area > Export Incremental Scripts** (U2000 client mode), or choose **Area Management > Planned Area > Export Incremental Scripts**, to export and activate the incremental scripts.

----End

Using the CME to Perform Single Configuration

On the CME, set the parameters listed in the "Data Preparation" section for a single eNodeB. The procedure is as follows:

Step 1 In the planned data area, click **Base Station** in the upper left corner of the configuration window.

Step 2 In area 1 shown in **Figure 8-12**, select the eNodeB to which the MOs belong.

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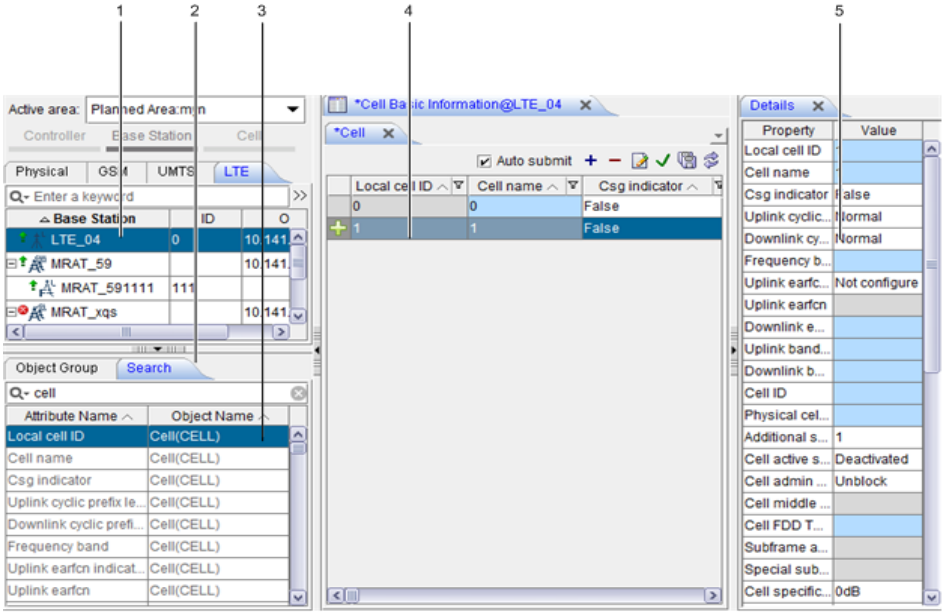
Figure 8-12 MO search and configuration window



Step 3 On the **Search** tab page in area 2, enter an MO name, for example, **CELL**.

Step 4 In area 3, double-click the MO in the **Object Name** column. All parameters in this MO are displayed in area 4.

Step 5 Set the parameters in area 4 or 5.

Step 6 Choose **CME > Planned Area > Export Incremental Scripts** (U2000 client mode), or choose **Area Management > Planned Area > Export Incremental Scripts** (CME client mode), to export and activate the incremental scripts.

----End

Using MML Commands

Step 1 Run the **MOD CELLALGOSWITCH** command to enable distance-based inter-RAT handover.

Step 2 Run the **MOD DISTBASEDHO** command to set the measurement object type to GERAN.

Step 3 Run the following commands to add neighbor relationships with GERAN cells:

1. **ADD GERANNFREQGROUP**
2. **ADD GERANNFREQGROUPARFCN**
3. **ADD GERANEXTERNALCELL**
4. (Optional) **ADD GERANEXTERNALCELLPLMN**, which is used to configure the PLMN list for an external GERAN cell when the neighboring GERAN cell works in RAN sharing mode.
5. **ADD GERANNCELL**

----End

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MML Command Examples

//Activating distance-based inter-RAT handover to GERAN

```
MOD CELLALGOSWITCH: LocalCellId=0, DistBasedHoSwitch=ON;MOD DISTBASEDHO:
LocalCellId=0, DistBasedMeasObjType=GERAN-1;
```

//Add neighboring relationships with GERAN cells

```
ADD GERANNFREQGROUP: LocalCellId=0, BcchGroupId=0, GeranVersion=GSM,
StartingArfcn=600, BandIndicator=GSM_dcs1800, CellReselPriorityCfgInd=NOT_CFG,
PmaxGeranCfgInd=NOT_CFG;ADD GERANNFREQGROUPARFCN: LocalCellId=0, BcchGroupId=0,
GeranArfcn=601;ADD GERANEXTERNALCELL: Mcc="460", Mnc="00", GeranCellId=1, Lac=5,
RacCfgInd=CFG, Rac=0, BandIndicator=GSM_dcs1800, GeranArfcn=601,
NetworkColourCode=1, BaseStationColourCode=1;ADD GERANEXTERNALCELLPLMN:
GeranCellId=1, Lac=1, Mcc="460", Mnc="00", ShareMcc="460", ShareMnc="20";5.ADD
GERANNCELL: LocalCellId=0, Mcc="460", Mnc="00", Lac=1, GeranCellId=1;ADD
GERANNCELL: LocalCellId=0, Mcc="460", Mnc="00", Lac=1, GeranCellId=1;
```

8.10.6 Activation Observation

The activation observation procedure for distance-based inter-RAT handover to GERAN is the same as that for distance-based inter-RAT handover to UTRAN. For details, see **8.3.6 Activation Observation**.

8.10.7 Reconfiguration

None

8.10.8 Deactivation

Using the CME to Perform Batch Configuration

Batch reconfiguration using the CME is the recommended method to deactivate a feature on eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure for feature deactivation is similar to that for feature activation described in **Using the CME to Perform Batch Configuration for Existing eNodeBs**. In the procedure, modify parameters according to the table listed in each scenario.

Table 8-34 Parameters related to distance-based inter-RAT handovers

MO	Sheet in the Summary Data File	Parameter Group	Setting Notes
CellAlgoSwitch	User-defined sheet	Distance-based Handover Switch	Set the switch to **OFF(Off)**.

Using the CME to Perform Single Configuration

On the CME, set the parameters listed in each scenario for a single eNodeB. For detailed instructions, see **Using the CME to Perform Single Configuration** for feature activation.

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Using MML Commands

To deactivate distance-based inter-RAT handover to GERAN, run the **MOD DISTBASEDHO** command and set the measurement object type to GERAN.

MML Command Examples

//Deactivating distance-based inter-RAT handover to GERAN

```
MOD DISTBASEDHO: LocalCellId=0, DistBasedMeasObjType=GERAN-0;
```

8.10.9 Performance Monitoring

Performance monitoring for distance-based inter-RAT handover to GERAN is similar to that for inter-RAT mobility between E-UTRAN and UTRAN. For details, see **8.1.9 Performance Monitoring**.

Related counters are listed in **Table 8-35** and **Table 8-36**.

Table 8-35 Counters related to distance-based handover to GERAN

Counter ID	Counter Name	Description
1526730037	L.IRATHO.E2G.Distance .PrepAttOut	Number of distance-based handover preparation attempts from E-UTRAN to GERAN
1526730038	L.IRATHO.E2G.Distance .ExecAttOut	Number of distance-based handover execution attempts from E-UTRAN to GERAN
1526730039	L.IRATHO.E2G.Distance .ExecSuccOut	Number of successful distance-based handover executions from E-UTRAN to GERAN

Table 8-36 Counter related to distance-based redirection to GERAN

Counter ID	Counter Name	Description
1526730041	L.RRCRedirection.E2G.D istance	Number of distance-based redirections from E-UTRAN to GERAN

8.10.10 Parameter Optimization

The reconfiguration of parameters that are not involved in this section is the same as the reconfiguration of parameters for coverage-based inter-RAT handovers. For details, see **8.8.10 Parameter Optimization**.

Distance-Based Handover Threshold

Distance-based handover threshold is contained in the **DistBasedHO** MO. The following table provides suggestions on how to reconfigure this parameter.

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Parameter Name	Parameter ID	Data Source	Setting Notes
Distance-based Handover Threshold	**DistBasedHO**.***DistBasedHOThd***	Network plan (negotiation not required)	This parameter specifies the threshold for distance-based handovers. It is recommended that you set this parameter based on the planned distance between sites.

Parameter Groups Related to Inter-RAT Handovers to GERAN

Parameter groups related to inter-RAT handovers to GERAN are contained in the **InterRatHoGeranGroup** MO. The following table provides suggestions on how to reconfigure these parameters.

Parameter Name	Parameter ID	Data Source	Setting Notes
CoverageBased GERAN trigger threshold	**InterRatHoGeranGroup**.***InterRatHoGeranB1Thd***	Network plan (negotiation not required)	This parameter specifies the RSSI threshold for event B1 related to coverage-based, UL-quality-based, or distance-based handover to GERAN. A larger value of this parameter results in a lower probability of handovers to GERAN, and a smaller value results in a higher probability. The default value is recommended.

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Parameter Name	Parameter ID	Data Source	Setting Notes
GERAN time to trigger	**InterRatHoGeranGroup.*InterRatHoGeranB1TimeToTrig***	Network plan (negotiation not required)	This parameter specifies the time-to-trigger for event B1 related to inter-RAT handover to GERAN. When detecting that the signal quality in at least one neighboring cell meets the entering condition, the UE does not send a measurement report to the eNodeB immediately. Instead, the UE sends a report only when the signal quality continuously meets the entering condition during the time-to-trigger period. If both UTRAN and GERAN neighboring cells are measured, you are advised to set the time-to-trigger for GERAN neighboring cells to be greater than that for UTRAN neighboring cells, increasing the possibility of handovers to UTRAN neighboring cells. If both distance-based inter-frequency handovers and distance-based inter-RAT handovers are enabled, you are advised to set the time-to-trigger for the inter-RAT handover event be greater than that for the inter-frequency handover event, increasing the possibility of inter-frequency handovers.

8.10.11 Troubleshooting

For details about troubleshooting, see **8.1.11 Troubleshooting**.

8.11 UL-Quality-based Inter-RAT Handover to GERAN

8.11.1 When to Use UL-Quality-based Inter-RAT Handover to GERAN

Coverage-based inter-RAT handovers are triggered based on DL signal quality. However, there may be a huge imbalance between the DL and UL signal quality. The UL signal quality may be relatively poor while the DL signal quality is fairly satisfactory. If this occurs, prompt triggering of inter-RAT handovers cannot be ensured. To handle this problem, you can enable UL-quality-based inter-RAT handover. It is good practice to enable UL-quality-based inter-RAT handover after the LTE network has stabilized. In the early phase of network deployment, however, enable this type of handover if it is required in the network plan.

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If both UL-quality-based inter-frequency handover and UL-quality-based inter-RAT handover are available, the former has higher priority.

8.11.2 Required Information

Information required for UL-quality-based inter-RAT handover to GERAN is the same as information required for UL-quality-based inter-RAT handover to UTRAN. For details, see **8.1.10 Parameter Optimization**.

8.11.3 Deployment Requirements

Operating Environment

The feature LOFD-001020 PS Inter-RAT Mobility between E-UTRAN and GERAN requires that the core-network equipment should support this feature. If the core-network equipment is provided by Huawei, the equipment version must be PS11.0. If the core-network equipment is provided by another vendor, check with the vendor whether the equipment supports this feature.

Transmission Networking

None

License

To use UL-quality-based inter-RAT handovers, an operator must purchase and activate the license for the feature listed in the following table.

Feature ID	Feature Name	License Control Item	NE	Sales Unit
LOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN	PS Inter-RAT Mobility between E-UTRAN and GERAN	eNodeB	per user in RRC connected mode

8.11.4 Data Preparation

This section describes the data that you need to collect for setting parameters. Required data is data that you must collect for all scenarios. Collect scenario-specific data when necessary for a specific feature deployment scenario.

There are three types of data sources:

- Network plan (negotiation not required): parameter values planned and set by the operator
- Network plan (negotiation required): parameter values planned by the operator and negotiated with the EPC or peer transmission equipment
- User-defined: parameter values set by users

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Required Data

For details about the required data, see **8.8.4 Data Preparation**.

Scenario-specific Data

UL-quality-based inter-RAT handover to GERAN can be enabled only after coverage-based inter-RAT handover is enabled. This section describes the required parameters besides those for coverage-based inter-RAT handover. For details on the required parameters for coverage-based inter-RAT handover, see the **Scenario-specific Data** part in **8.8.4 Data Preparation**.

The following table describes the parameter that must be set in the **ENodeBAlgoSwitch** MO to enable UL-quality-based inter-RAT handover.

Parameter Name	Parameter ID	Data Source	Setting Notes
Handover Algo switch	**ENodeBAlgoSwitch.*HoAlgoSwitch***	Network plan (negotiation not required)	To enable UL-quality-based inter-RAT handover, select the **UlQualityInterRATHoSwitch(UlQualityInterRATHoSwitch)** check box under this parameter.

8.11.5 Activation

Before activating UL-quality-based inter-RAT handover, ensure that coverage-based inter-RAT handover has been activated. Currently, the target RAT cannot be specified for an UL-quality-based inter-RAT handover.

Using the CME to Perform Batch Configuration for Newly Deployed eNodeBs

Enter the values of related parameters in a summary data file, which also contains other data for the new eNodeBs to be deployed. Then, import the summary data file into the Configuration Management Express (CME) for batch configuration. For detailed instructions, see section "Creating eNodeBs in Batches" in the initial configuration guide for the eNodeB.

The summary data file may be a scenario-specific file provided by the CME or a customized file, depending on the following conditions:

- Related managed objects (MOs) are contained in a scenario-specific summary data file. In this situation, set the parameters in the MOs, and then verify and save the file.
- Some related MOs are not contained in a scenario-specific summary data file. In this situation, customize a summary data file to include the MOs before you can set the parameters.

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Table 8-37 Parameters related to UL-quality-based inter-RAT handovers to GERAN

MO	Sheet in the Summary Data File	Parameter Group	Remarks
ENodeBAlgoSwitch	User-defined sheet. **ENodeBAlgoSwitch** is recommended.	Handover Algo switch, Handover Mode switch	This parameter must be customized on a list-type sheet of the template.

Using the CME to Perform Batch Configuration for Existing eNodeBs

Batch reconfiguration using the CME is the recommended method to activate a feature on existing eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure is as follows:

Step 1 Choose **CME > Advanced > Customize Summary Data File** (U2000 client mode), or choose **Advanced > Customize Summary Data File** (CME client mode), to customize a summary data file for batch reconfiguration.

NOTE

For context-sensitive help on a current task in the client, press **F1**.

Step 2 Choose **CME > LTE Application > Export Data > Export Base Station Bulk Configuration Data** (U2000 client mode), or choose **LTE Application > Export Data > Export Base Station Bulk Configuration Data** (CME client mode), to export the eNodeB data stored on the CME into the customized summary data file.

Step 3 In the summary data file, set the parameters in the MOs listed in **Table 8-37**, and close the file.

Step 4 Choose **CME > LTE Application > Import Data > Import Base Station Bulk Configuration Data** (U2000 client mode), or choose **LTE Application > Import Data > Import Base Station Bulk Configuration Data** (CME client mode), to import the summary data file into the CME.

Step 5 Choose **CME > Planned Area > Export Incremental Scripts** (U2000 client mode), or choose **Area Management > Planned Area > Export Incremental Scripts**, to export and activate the incremental scripts.

----End

Using the CME to Perform Single Configuration

On the CME, set the parameters listed in the "Data Preparation" section for a single eNodeB. The procedure is as follows:

Step 1 In the planned data area, click **Base Station** in the upper left corner of the configuration window.

Step 2 In area 1 shown in **Figure 8-13**, select the eNodeB to which the MOs belong.

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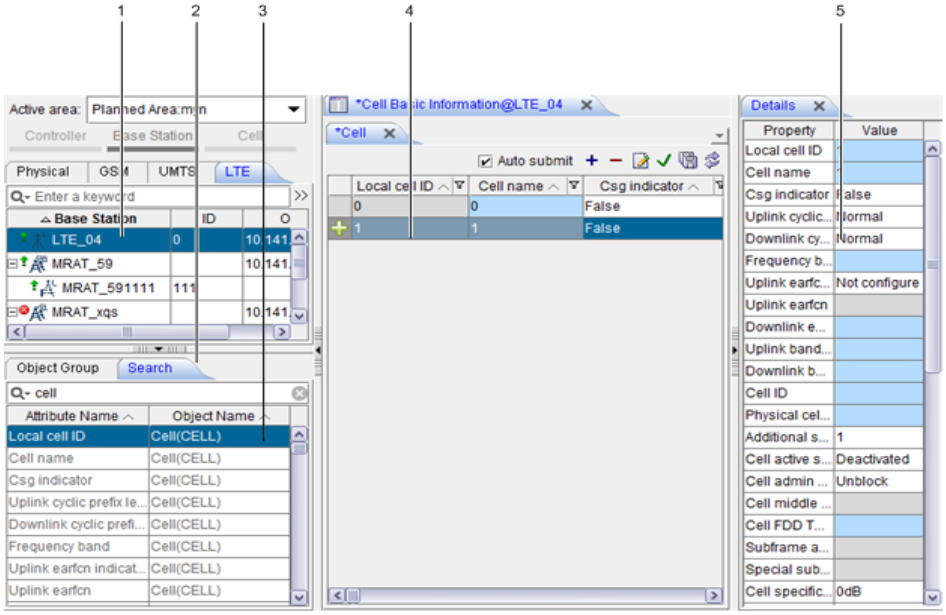
Figure 8-13 MO search and configuration window



Step 3 On the **Search** tab page in area 2, enter an MO name, for example, **CELL**.

Step 4 In area 3, double-click the MO in the **Object Name** column. All parameters in this MO are displayed in area 4.

Step 5 Set the parameters in area 4 or 5.

Step 6 Choose **CME > Planned Area > Export Incremental Scripts** (U2000 client mode), or choose **Area Management > Planned Area > Export Incremental Scripts** (CME client mode), to export and activate the incremental scripts.

----End

Using MML Commands

Step 1 Run the **MOD ENODEBALGOSWITCH** command to enable uplink-quality-based handover and configure inter-RAT handover to GERAN.

Step 2 Run the following commands to add neighboring relationships with GERAN cells:

1. **ADD GERANNFREQGROUP**
2. **ADD GERANNFREQGROUPARFCN**
3. **ADD GERANEXTERNALCELL**
4. (Optional) **ADD GERANEXTERNALCELLPLMN**, which is used to configure the PLMN list for an external GERAN cell when the neighboring GERAN cell works in RAN sharing mode.
5. **ADD GERANNCELL**

----End

MML Command Examples

//Activating UL-quality-based inter-RAT handover to GERAN

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```
MOD ENODEBALGOSWITCH: HoAlgoSwitch=UlQualityInterRATHoSwitch-1;
```

//Adding neighboring GERAN cells

```
ADD GERANNFREQGROUP: LocalCellId=0, BcchGroupId=0, GeranVersion=GSM,
StartingArfcn=600, BandIndicator=GSM_dcs1800, CellReselPriorityCfgInd=NOT_CFG,
PmaxGeranCfgInd=NOT_CFG;ADD GERANNFREQGROUPARFCN: LocalCellId=0, BcchGroupId=0,
GeranArfcn=601;ADD GERANEXTERNALCELL: Mcc="460", Mnc="00", GeranCellId=1, Lac=5,
RacCfgInd=CFG, Rac=0, BandIndicator=GSM_dcs1800, GeranArfcn=601,
NetworkColourCode=1, BaseStationColourCode=1;ADD GERANEXTERNALCELLPLMN:
GeranCellId=1, Lac=1, Mcc="460", Mnc="00", ShareMcc="460", ShareMnc="20";5.ADD
GERANNCELL: LocalCellId=0, Mcc="460", Mnc="00", Lac=1, GeranCellId=1;ADD
GERANNCELL: LocalCellId=0, Mcc="460", Mnc="00", Lac=1, GeranCellId=1;
```

8.11.6 Activation Observation

The activation observation procedure for UL-quality-based inter-RAT handover to GERAN is the same as that for UL-quality-based inter-RAT handover to UTRAN. For details, see **8.4.6 Activation Observation**.

8.11.7 Reconfiguration

None

8.11.8 Deactivation

Using the CME to Perform Batch Configuration

Batch reconfiguration using the CME is the recommended method to deactivate a feature on eNodeBs. This method reconfigures all data, except neighbor relationships, for multiple eNodeBs in a single procedure. The procedure for feature deactivation is similar to that for feature activation described in **Using the CME to Perform Batch Configuration for Existing eNodeBs**. In the procedure, modify parameters according to the table listed in each scenario.

Table 8-38 UL-quality-based inter-RAT handover

MO	Sheet in the Summary Data File	Parameter Group	Setting Notes
ENodeBAlgoSwitch	User-defined sheet	HoAlgoSwitch	Clear the **UlQualityInterRATHoSwitch** check box under this parameter.

Using the CME to Perform Single Configuration

On the CME, set the parameters listed in each scenario for a single eNodeB. For detailed instructions, see **Using the CME to Perform Single Configuration** for feature activation.

Using MML Commands

To deactivate UL-quality-based inter-RAT handover, turn off the corresponding switch by running the **MOD ENODEBALGOSWITCH** command.

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MML Command Examples

//Deactivating UL-quality-based inter-RAT handover

```
MOD ENODEBALGOSWITCH: HoAlgoSwitch=UlQualityInterRATHoSwitch-0;
```

8.11.9 Performance Monitoring

Performance monitoring for UL-quality-based inter-RAT handover is similar to that for inter-RAT mobility between E-UTRAN and UTRAN. For details, see **8.1.9 Performance Monitoring**.

Related counters are listed in **Table 8-39** and **Table 8-40**.

Table 8-39 Counters related to UL-quality-based handover to GERAN

Counter ID	Counter Name	Counter Description
1526730029	L.IRATHO.E2G.ULquality.PrepAttOut	Number of UL-quality-based handover preparation attempts from E-UTRAN to GERAN
1526730030	L.IRATHO.E2G.ULquality.ExecAttOut	Number of UL-quality-based handover execution attempts from E-UTRAN to GERAN
1526730031	L.IRATHO.E2G.ULquality.ExecSuccOut	Number of successful UL-quality-based handover executions from E-UTRAN to GERAN

Table 8-40 Counters related to UL-quality-based redirection to GERAN

Counter ID	Counter Name	Counter Description
1526730033	L.RRCRedirection.E2G.Ulquality	Number of UL-quality-based redirections from E-UTRAN to GERAN

8.11.10 Parameter Optimization

The reconfiguration of parameters that are not involved in this section is the same as the reconfiguration of parameters for coverage-based inter-RAT handovers. For details, see **8.8.10 Parameter Optimization**.

Parameter groups related to inter-RAT handovers to GERAN are contained in the **InterRatHoGeranGroup** MO. The following table provides suggestions on how to reconfigure these parameters.

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Parameter Name	Parameter ID	Data Source	Setting Notes
CoverageBased GERAN trigger threshold	**InterRatHoGeranGroup.*InterRatHoGeranB1Thd***	Network plan (negotiation not required)	This parameter specifies the RSSI threshold for event B1 related to coverage-based, UL-quality-based, or distance-based handover to GERAN. A larger value of this parameter results in a lower probability of handovers to GERAN, and a smaller value results in a higher probability. The default value is recommended.
GERAN time to trigger	**InterRatHoGeranGroup.*InterRatHoGeranB1TimeToTrig***	Network plan (negotiation not required)	This parameter specifies the time-to-trigger for event B1 related to inter-RAT handover to GERAN. When detecting that the signal quality in at least one neighboring cell meets the entering condition, the UE does not send a measurement report to the eNodeB immediately. Instead, the UE sends a report only when the signal quality continuously meets the entering condition during the time-to-trigger period. If both UTRAN and GERAN neighboring cells are measured, you are advised to set the time-to-trigger for GERAN neighboring cells to be greater than that for UTRAN neighboring cells, increasing the possibility of handovers to UTRAN neighboring cells. If both UL-quality-based inter-frequency handovers and UL-quality-based inter-RAT handovers are enabled, you are advised to set the time-to-trigger for the inter-RAT handover event be greater than that for the inter-frequency handover event, increasing the possibility of inter-frequency handovers.

8.11.11 Troubleshooting

For details about troubleshooting, see **8.1.11 Troubleshooting**.

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9 Parameters

Table 9-1 Parameters

MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
ENodeBAlgoSwitch	HighLoadNetOptSwitch	MOD ENODEBALGOSWITCH LST ENODEBALGOSWITCH	LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001043 / TDLOFD-001043 LOFD-001046 / TDLOFD-001046 LOFD-001072 / TDLOFD-001072 LOFD-001073 / TDLOFD-001073	PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN Service based Inter-RAT handover to UTRAN Service based Inter-RAT handover to GERAN	Meaning: Indicates whether to enable network performance optimization when the network is heavily loaded. This parameter includes the following switch: SPECSIGRETRANSOPTSWITCH: Indicates whether to optimize special signaling retransmission. If this switch is on, the eNodeB adaptively configures the maximum number of special signaling retransmissions that can be initiated by the RLC layer based on the number of UEs in the cell. If this switch is off, the eNodeB does not adaptively configure the maximum number of special signaling retransmissions based on the number of UEs in the cell. GUI Value Range: SPECSIGRETRANSOPTS-WITCH(SPECSIGRETRANSOPTSWITCH) Unit: None Actual Value Range: SPECSIGRETRANSOPTS-WITCH Default Value: SPECSIGRETRANSOPTS-WITCH:Off

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
			LOFD-001078	Distance based Inter-RAT handover to UTRAN Distance based Inter-RAT handover to GERAN E-UTRAN to UTRAN CS/PS Steering	

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
ENodeBAlgoSwitch	HoModeSwitch	MOD ENODEBALGOSWITCH LST ENODEBALGOSWITCH	LOFD-001022 / TDLOFD-001022 LOFD-001023 / TDLOFD-001023 LOFD-001033 / TDLOFD-001033 LOFD-001034 / TDLOFD-001034 LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001021 / TDLOFD-001021 TDLOFD-001052 TDLOFD-001088 TDLOFD-001043	SRVCC to UTRAN SRVCC to GERAN CS Fallback to UTRAN CS Fallback to GERAN PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000 Flash CS Fallback to UTRAN CS Fallback	Meaning: Indicates the switches corresponding to the inputs based on which the eNodeB determines handover policies. EutranVoipCapSwitch: UtranVoipCapSwitch: If this switch is turned on, UTRAN supports VoIP. If this switch is turned off, UTRAN does not support VoIP. GeranVoipCapSwitch: This switch will be removed in later versions. In this version, the setting of this switch is still synchronized between the U2000 and the eNodeB, but it is no longer used internally. Therefore, you are advised to avoid using this switch. Cdma1xRttVoipCapSwitch: If this switch is turned on, CDMA2000 1xRTT supports VoIP. If this switch is turned off, CDMA2000 1xRTT does not support VoIP. UtranPsHoSwitch: If this switch is turned on, UTRAN supports PS handovers. If this switch is turned off, UTRAN does not support PS handovers. GeranPsHoSwitch: If this switch is turned on, GERAN supports PS handovers. If this switch is turned off, GERAN does not support PS handovers. CdmaHrpdNonOtpimisedHoSwitch: If this switch is turned on, non-optimized handovers to CDMA2000 HRPD are enabled. If this switch is turned off, non-optimized handovers to CDMA2000 HRPD are disabled. CdmaHrpdOptimisedHoSwitch: If this switch is turned on, optimized handovers to CDMA2000 HRPD are enabled. If this switch is turned off, optimized handovers to CDMA2000 HRPD are disabled. GeranNaccSwitch: This switch does not take effect if GeranCcoSwitch is disabled. If this switch is turned on, the GERAN supports network assisted cell change (NACC). If this switch is turned off, the GERAN does not support NACC. GeranCcoSwitch: If this switch is turned on, the GERAN supports cell change order (CCO). If this switch is turned off, the GERAN does not support CCO. UtranSrvccSwitch: If this switch is turned on, the UTRAN supports single radio voice call continuity (SRVCC). If this switch is turned off, the UTRAN does not support SRVCC. GeranSrvccSwitch: If this switch is turned on, the GERAN supports SRVCC. If this switch is turned off, the GERAN does not support SRVCC. Cdma1xRttSrvccSwitch: If this switch is turned on, the CDMA2000 1xRTT supports SRVCC. If this switch is turned off, the CDMA2000 1xRTT does not support SRVCC. UtranRedirectSwitch: If this switch is turned on, redirection to UTRAN is enabled. If this switch is turned off, redirection to UTRAN is disabled. GeranRedirectSwitch: If this switch is turned

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
			TDLOFD-001072 TDLOFD-001046 TDLOFD-001073	Steering to UTRAN Service based Inter-RAT handover to UTRAN Distance based Inter-RAT handover to UTRAN Service based Inter-RAT handover to GERAN Distance based Inter-RAT handover to GERAN	on, redirection to GERAN is enabled. If this switch is turned off, redirection to GERAN is disabled. CdmaHrpdRedirectSwitch: If this switch is turned on, redirection to CDMA2000 HRPD is enabled. If this switch is turned off, redirection to CDMA2000 HRPD is disabled. Cdma1xRttRedirectSwitch: If this switch is turned on, redirection to CDMA2000 1xRTT is enabled. If this switch is turned off, redirection to CDMA2000 1xRTT is disabled. BlindHoSwitch: If this switch is turned on, blind handovers for CSFB are enabled. If this switch is turned off, blind handovers for CSFB are disabled. LcsSrvccSwitch: If this switch is turned on, an SRVCC procedure is triggered when a UE receives a CSFB instruction during a VoIP service. If this switch is turned off, an SRVCC procedure is not triggered when a UE receives a CSFB instruction during a VoIP service. AutoGapSwitch: If this switch is turned on and UEs support automatic measurement gap configurations on the target frequency, the eNodeB does not deliver gap configurations to UEs. If this switch is turned off, the eNodeB delivers gap configurations to UEs during all inter-frequency and inter-RAT measurements. UeVoipOnHspaCapSwitch: If this switch is turned on and the eNodeB attempts to hand over UEs using voice services to UTRAN, the eNodeB checks UE capabilities when determining whether PS handover is applied. UEs must support voiceOverPS-HS-UTRA-FDD-r9 if the target UTRAN cell works in FDD mode or voiceOverPS-HS-UTRA-TDD128-r9 if the target UTRAN cell works in TDD mode. If this switch is turned off, the eNodeB does not check UE capabilities when handing over UEs to UTRAN based on PS handovers. UtranFddB1CapSwitch: If this switch is turned on, the setting of bit 41 of FGI specifying the UE capability of event B1 measurement on FDD UTRAN cells must be considered. If this switch is turned off, the setting of bit 41 of FGI does not need to be considered. GUI Value Range: EutranVoipCapSwitch(EutranVoipCapSwitch), UtranVoipCapSwitch(UtranVoipCapSwitch), GeranVoipCapSwitch(GeranVoipCapSwitch), Cdma1xRttVoipCapSwitch(Cdma1xRttVoipCapSwitch), UtranPsHoSwitch(UtranPsHoSwitch), GeranPsHoSwitch(GeranPsHoSwitch), CdmaHrpdNonOtpimisedHoSwitch(CdmaHrpdNonOtpimisedHoSwitch), CdmaHrpdOptimisedHoSwitch(CdmaHrpdOptimised-

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
					HoSwitch), GeranNaccSwitch(GeranNaccSwitch), GeranCcoSwitch(GeranCcoSwitch), UtranSrvccSwitch(UtranSrvccSwitch), GeranSrvccSwitch(GeranSrvccSwitch), Cdma1xRttSrvccSwitch(Cdma1xRttSrvccSwitch), UtranRedirectSwitch(UtranRedirectSwitch), GeranRedirectSwitch(GeranRedirectSwitch), CdmaHrpdRedirectSwitch(CdmaHrpdRedirectS-witch), Cdma1xRttRedirectSwitch(Cdma1xRttRedirectSwitch), BlindHoSwitch(BlindHoSwitch), LcsSrvccSwitch(LcsSrvccSwitch), AutoGapSwitch(AutoGapSwitch), UeVoipOnHspaCapSwitch(UeVoipOnHspaCapS-witch), UtranFddB1CapSwitch(UtranFddB1CapSwitch) Unit: None Actual Value Range: EutranVoipCapSwitch, UtranVoipCapSwitch, GeranVoipCapSwitch, Cdma1xRttVoipCapSwitch, UtranPsHoSwitch, GeranPsHoSwitch, CdmaHrpdNonOtpimisedHoS-witch, CdmaHrpdOptimisedHoSwitch, GeranNaccSwitch, GeranCcoSwitch, UtranSrvccSwitch, GeranSrvccSwitch, Cdma1xRttSrvccSwitch, UtranRedirectSwitch, GeranRedirectSwitch, CdmaHrpdRedirectSwitch, Cdma1xRttRedirectSwitch, BlindHoSwitch, LcsSrvccSwitch, AutoGapSwitch, UeVoipOnHspaCapSwitch, UtranFddB1CapSwitch Default Value: EutranVoipCapSwitch:On, UtranVoipCapSwitch:Off, GeranVoipCapSwitch:Off, Cdma1xRttVoipCapSwitch:Off, UtranPsHoSwitch:Off, GeranPsHoSwitch:Off, CdmaHrpdNonOtpimisedHoSwitch:Off, CdmaHrpdOptimisedHoSwitch:Off, GeranNaccSwitch:Off, GeranCcoSwitch:Off, UtranSrvccSwitch:Off, GeranSrvccSwitch:Off, Cdma1xRttSrvccSwitch:Off, UtranRedirectSwitch:Off, GeranRedirectSwitch:Off, CdmaHrpdRedirectSwitch:Off, Cdma1xRttRedirectSwitch:Off, BlindHoSwitch:Off, LcsSrvccSwitch:Off, AutoGapSwitch:Off, UeVoipOnHspaCapSwitch:Off, UtranFddB1CapSwitch:Off

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
InterRatHoComm	InterRatHoA1A2TrigQuan	MOD INTERRATHOCOMM LST INTERRATHOCOMM	LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001021 / TDLOFD-001021 TDLOFD-001022 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000 SRVCC to UTRAN SRVCC to GERAN	Meaning: Indicates the triggering quantity for the inter-RAT measurement event A1 or A2 report. The quantity can be RSRP, RSRQ, or both RSRP and RSRQ. The measured RSRP values are stable, varying little with the load, and therefore there is little signal fluctuation. The measured RSRQ values vary with the load and are likely to reflect the signal quality of the cell in real time. For details, see 3GPP TS 36.331. GUI Value Range: RSRP, RSRQ, BOTH Unit: None Actual Value Range: RSRP, RSRQ, BOTH Default Value: RSRP

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
InterRatHoCommGroup	InterRatHoA1A2Hyst	ADD INTERRATHOCOMMGROUP MOD INTERRATHOCOMMGROUP LST INTERRATHOCOMMGROUP	LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001021 / TDLOFD-001021 TDLOFD-001022 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000 SRVCC to UTRAN SRVCC to GERAN	Meaning: Indicates the hysteresis for inter-RAT measurement events A1 and A2. This parameter is used to prevent frequent triggering of event evaluation caused by radio signal fluctuation. In this way, the probability of ping-pong handovers or handover decision errors is reduced. A larger value of this parameter results in a lower probability. GUI Value Range: 0~30 Unit: 0.5dB Actual Value Range: 0~15 Default Value: 2

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
InterRatHoCommGroup	InterRatHoA2ThdRsrp	ADD INTERRATHOCOMMGROUP MOD INTERRATHOCOMMGROUP LST INTERRATHOCOMMGROUP	LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001021 / TDLOFD-001021 TDLOFD-001022 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000 SRVCC to UTRAN SRVCC to GERAN	Meaning: Indicates the RSRP threshold for inter-RAT measurement event A2. When the measured RSRP value of the serving cell is below this threshold, an event A2 report will be sent. GUI Value Range: -141~-43 Unit: dBm Actual Value Range: -141~-43 Default Value: -115

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
InterRatHoCommGroup	InterRatHoA2ThdRsrq	ADD INTERRATHOCOMMGROUP MOD INTERRATHOCOMMGROUP LST INTERRATHOCOMMGROUP	LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001021 / TDLOFD-001021 TDLOFD-001022 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000 SRVCC to UTRAN SRVCC to GERAN	Meaning: Indicates the RSRQ threshold for inter-RAT measurement event A2. When the measured RSRQ value of the serving cell is below this threshold, an event A2 report will be sent. GUI Value Range: -40~-6 Unit: 0.5dB Actual Value Range: -20~-3 Default Value: -24

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
InterRatHoCommGroup	InterRatHoA1A2TimeToTrig	ADD INTERRATHOCOMMGROUP MOD INTERRATHOCOMMGROUP LST INTERRATHOCOMMGROUP	LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001021 / TDLOFD-001021 TDLOFD-001022 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000 SRVCC to UTRAN SRVCC to GERAN	Meaning: Indicates the time-to-trigger for inter-RAT measurement event A1 or A2. When detecting that the signal quality in the serving cell meets the entering condition, the UE does not send a measurement report to the eNodeB immediately. Instead, the UE sends a report only when the signal quality continuously meets the entering condition during the time-to-trigger. This parameter helps decrease the number of occasionally triggered event reports, the average number of handovers, and the number of wrong handovers. In summary, it helps prevent unnecessary handovers. GUI Value Range: 0ms, 40ms, 64ms, 80ms, 100ms, 128ms, 160ms, 256ms, 320ms, 480ms, 512ms, 640ms, 1024ms, 1280ms, 2560ms, 5120ms Unit: ms Actual Value Range: 0ms, 40ms, 64ms, 80ms, 100ms, 128ms, 160ms, 256ms, 320ms, 480ms, 512ms, 640ms, 1024ms, 1280ms, 2560ms, 5120ms Default Value: 640ms

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
ENodeBAlgoSwitch	HoAlgoSwitch	MOD ENODEBALGOSWITCH LST ENODEBALGOSWITCH	LBFD-00201801 / TDLBFD-00201801 LBFD-00201802 / TDLBFD-00201802 LBFD-00201804 / TDLBFD-00201804 LBFD-00201805 / TDLBFD-00201805 LOFD-001033 / TDLOFD-001033 LOFD-001034 / TDLOFD-001034 LOFD-001035 / TDLOFD-001035 LOFD-001052 / TDLOFD-001052 LOFD-001053 /	Coverage Based Intra-frequency Handover Distance Based Inter-frequency Handover Service Based Inter-frequency Handover CS Fallback to UTRAN CS Fallback to GERAN CS Fallback to CDMA2000 1xRTT Flash CS Fallback to UTRAN Flash CS Fallback to GERAN CS Fallback	Meaning: Indicates the switches used to enable or disable handover algorithms. The switches are described as follows: IntraFreqCoverHoSwitch: If this switch is turned on, coverage-based intra-frequency handovers are enabled to ensure service continuity. If this switch is turned off, coverage-based intra-frequency handovers are disabled. InterFreqCoverHoSwitch: If this switch is turned on, coverage-based inter-frequency handovers are enabled to ensure service continuity. If this switch is turned off, coverage-based inter-frequency handovers are disabled. UtranCsfbSwitch: If this switch is turned on, CSFB to UTRAN is enabled and UEs can fall back to UTRAN. If this switch is turned off, CSFB to UTRAN is disabled. GeranCsfbSwitch: If this switch is turned on, CSFB to GERAN is enabled and UEs can fall back to GERAN. If this switch is turned off, CSFB to GERAN is disabled. Cdma1xRttCsfbSwitch: If this switch is turned on, CSFB to CDMA2000 1xRTT is enabled and UEs can fall back to CDMA2000 1xRTT. If this switch is turned off, CSFB to CDMA2000 1xRTT is disabled. UtranServiceHoSwitch: If this switch is turned on, service-based handovers to UTRAN are enabled and UEs using a specific type of services can be handed over to UTRAN. If this switch is turned off, service-based handovers to UTRAN are disabled. GeranServiceHoSwitch: If this switch is turned on, service-based handovers to GERAN are enabled and UEs using a specific type of services can be handed over to GERAN. If this switch is turned off, service-based handovers to GERAN are disabled. CdmaHrpdServiceHoSwitch: If this switch is turned on, service-based handovers to CDMA2000 HRPD cells are enabled and UEs using a specific type of services can be handed over to CDMA2000 HRPD cells. If this switch is turned off, service-based handovers to CDMA2000 HRPD cells are disabled. Cdma1xRttServiceHoSwitch: If this switch is turned on, service-based handovers to CDMA2000 1xRTT

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
			TDLOFD-001053 LOFD-001088 / TDLOFD-001088 LOFD-001089 / TDLOFD-001089 LOFD-001090 / TDLOFD-001090 LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001043 / TDLOFD-001043 LOFD-001046 / TDLOFD-001046 LOFD-001072 / TDLOFD-001072 LOFD-001073 / TDLOF	Steering to UTRAN CS Fallback Steering to GERAN Enhanced CS Fallback to CDMA2000 1xRTT PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN Service based inter-RAT handover to UTRAN Service based inter-RAT handover to GERAN Distance based	are enabled and UEs using a specific type of services can be handed over to CDMA2000 1xRTT. If this switch is turned off, service-based handovers to CDMA2000 1xRTT are disabled. UlQualityInterRATHoSwitch: If this switch is turned on, UL-quality-based inter-RAT handovers are enabled and UEs can be handed over to inter-RAT cells to ensure service continuity when the UL signal quality is poor. If this switch is turned off, UL-quality-based inter-RAT handovers are disabled. InterPlmnHoSwitch: If this switch is turned on, inter-PLMN handovers are enabled and UEs can be handed over to cells in other PLMNs. If this switch is turned off, inter-PLMN handovers are disabled. UtranFlashCsfbSwitch: This switch takes effect only when UtranCsfbSwitch is turned on. If UtranFlashCsfbSwitch is turned on, flash CSFB to UTRAN is enabled and the eNodeB sends system information of candidate target UTRAN cells to UEs during redirection. If UtranFlashCsfbSwitch is turned off, flash CSFB to UTRAN is disabled. GeranFlashCsfbSwitch: This switch takes effect only when GeranCsfbSwitch is turned on. If GeranFlashCsfbSwitch is turned on, flash CSFB to GERAN is enabled and the eNodeB sends system information of candidate target GERAN cells to UEs during redirection. If GeranFlashCsfbSwitch is turned off, flash CSFB to GERAN is disabled. ServiceBasedInterFreqHoSwitch: If this switch is turned on, service-based inter-frequency handovers are enabled and UEs using a specific type of services can be handed over to inter-frequency cells. If this switch is turned off, service-based inter-frequency handovers are disabled. UlQualityInterFreqHoSwitch: If this switch is turned on, UL-quality-based inter-frequency handovers are enabled and UEs can be handed over to inter-frequency cells to ensure service continuity when the UL signal quality is poor. If this switch is turned off, UL-quality-based inter-frequency handovers are disabled. CsfbAdaptiveBlindHoSwitch: This switch takes effect only when BlindHoSwitch is enabled. If CsfbAdaptiveBlindHoSwitch is turned on, adaptive

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
			D-001073 TDLBFD-002018 TDLOFD-001022 TDLOFD-070228 LOFD-070202 / TDLOFD-070202	inter-RAT handover to UTRAN Distance based inter-RAT handover to GERAN Mobility Management Coverage Based Inter-frequency Handover SRVCC to UTRAN Service-Request Based Inter-frequency Handover Ultra-Flash CSFB	blind handovers for CSFB are enabled and appropriate handover mechanisms are selected for UEs based on their locations. If CsfbAdaptiveBlindHoSwitch is turned off, adaptive blind handovers for CSFB are disabled. UtranCsfbSteeringSwitch: If this switch is turned on, CSFB steering to UTRAN is enabled and CSFB policies for UEs in idle mode can be configured. If this switch is turned off, CSFB steering to UTRAN is disabled. GeranCsfbSteeringSwitch: If this switch is turned on, CSFB steering to GERAN is enabled and CSFB policies for UEs in idle mode can be configured. If this switch is turned off, CSFB steering to GERAN is disabled. CSFBLoadInfoSwitch: If this switch is turned on, load-based CSFB is enabled and a target cell for CSFB is selected based on loads of candidate target cells. If this switch is turned off, load-based CSFB is disabled. Cdma1XrttEcsfbSwitch: If this switch is turned on, eCSFB to CDMA2000 1xRTT is enabled and UEs can fall back to CDMA2000 1xRTT through handovers. If this switch is turned off, eCSFB to CDMA2000 1xRTT is disabled. EmcBlindHoA1Switch: If this switch is turned on, blind handover event A1 measurements are enabled. If a blind handover event measurement conflicts with a handover procedure, an emergency blind handover can be triggered after the handover procedure is complete. If this switch is turned off, blind handover event A1 measurements are disabled. If a blind handover event measurement conflicts with a handover procedure, an emergency blind handover cannot be triggered. EmcInterFreqBlindHoSwitch: If this switch is turned on, the eNodeB preferentially performs an inter-frequency blind handover when an emergency blind handover is triggered. If this switch is turned off, the eNodeB only performs an inter-RAT blind handover when an emergency blind handover is triggered. EPlmnSwitch: Indicates whether handovers to neighboring cells under the equivalent PLMNs (EPLMNs) are allowed. When inter-PLMN handovers

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
					are allowed, handovers to neighboring cells under the EPLMNs are allowed if this switch is turned on, and not allowed if this switch is turned off. The EPLMNs are delivered by the MME to the UE. ServiceReqInterFreqHoSwitch: If this parameter is set to ON, the service request-based inter-frequency handover algorithm is enabled, handing UEs of a specific service to a frequency in LTE mode. If this parameter is set to OFF, the service request-based inter-frequency handover algorithm is disabled. This parameter is dedicated to LTE TDD cells. VoipHoControlSwitch: Indicates whether the eNodeB filters out target cells that do not support VoIP services when processing intra-RAT handovers for VoIP services. UtranFlashCsfbSwitch: This switch takes effect only when UtranCsfbSwitch is turned on. GeranFlashCsfbSwitch: This switch takes effect only when GeranCsfbSwitch is turned on. CsfbAdaptiveBlindHoSwitch: This switch takes effect only when BlindHoSwitch is turned on. GeranUltraFlashCsfbSwitch: If this switch is turned on, UEs can fall back to GERAN by performing ultra-flash CSFB. If this switch is turned off, ultra-flash CSFB to GERAN is disabled. This switch is dedicated to LTE TDD cells. GUI Value Range: IntraFreqCoverHoSwitch(IntraFreqCoverHoSwitch), InterFreqCoverHoSwitch(InterFreqCoverHoSwitch), UtranCsfbSwitch(UtranCsfbSwitch), GeranCsfbSwitch(GeranCsfbSwitch), Cdma1xRttCsfbSwitch(Cdma20001xRttCsfbSwitch), UtranServiceHoSwitch(UtranServiceHoSwitch), GeranServiceHoSwitch(GeranServiceHoSwitch), CdmaHrpdServiceHoSwitch(Cdma2000HrpdServiceHoSwitch), Cdma1xRttServiceHoSwitch(Cdma20001xRttServiceHoSwitch), UlQualityInterRATHoSwitch(UlQualityInterRATHoSwitch), InterPlmnHoSwitch(InterPlmnHoSwitch), UtranFlashCsfbSwitch(UtranFlashCsfbSwitch), GeranFlashCsfbSwitch(GeranFlashCsfbSwitch), ServiceBasedInterFreqHoSwitch(ServiceBasedInterFreqHoSwitch), UlQualityInterFreqHoSwitch(UlQualityInterFreqHoSwitch), CsfbAdaptiveBlindHoSwitch(CsfbAdaptiveBlind-

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
					HoSwitch), UtranCsfbSteeringSwitch(UtranCsfbSteeringSwitch), GeranCsfbSteeringSwitch(GeranCsfbSteeringSwitch), CSFBLoadInfoSwitch(CSFBLoadInfoSwitch), Cdma1XrttEcsfbSwitch(Cdma1XrttEcsfbSwitch), EmcBlindHoA1Switch(EmcBlindHoA1Switch), EmcInterFreqBlindHoSwitch(EmcInterFreqBlindHoSwitch), EPlmnSwitch(EPlmnSwitch), ServiceReqInterFreqHoSwitch(ServiceReqInterFreqHoSwitch), VoipHoControlSwitch(VoipHoControlSwitch), UtranUltraFlashCsfbSwitch(UtranUltraFlashCsfbSwitch), GeranUltraFlashCsfbSwitch(GeranUltraFlashCsfbSwitch) Unit: None Actual Value Range: IntraFreqCoverHoSwitch, InterFreqCoverHoSwitch, UtranCsfbSwitch, GeranCsfbSwitch, Cdma1xRttCsfbSwitch, UtranServiceHoSwitch, GeranServiceHoSwitch, CdmaHrpdServiceHoSwitch, Cdma1xRttServiceHoSwitch, UlQualityInterRATHoSwitch, InterPlmnHoSwitch, UtranFlashCsfbSwitch, GeranFlashCsfbSwitch, ServiceBasedInterFreqHoSwitch, UlQualityInterFreqHoSwitch, CsfbAdaptiveBlindHoSwitch, UtranCsfbSteeringSwitch, GeranCsfbSteeringSwitch, CSFBLoadInfoSwitch, Cdma1XrttEcsfbSwitch, EmcBlindHoA1Switch, EmcInterFreqBlindHoSwitch, EPlmnSwitch, ServiceReqInterFreqHoSwitch, VoipHoControlSwitch, UtranUltraFlashCsfbSwitch, GeranUltraFlashCsfbSwitch Default Value: IntraFreqCoverHoSwitch:On, InterFreqCoverHoSwitch:On, UtranCsfbSwitch:Off, GeranCsfbSwitch:Off, Cdma1xRttCsfbSwitch:Off, UtranServiceHoSwitch:Off, GeranServiceHoSwitch:Off, CdmaHrpdServiceHoSwitch:Off, Cdma1xRttServiceHoSwitch:Off, UlQualityInterRATHoSwitch:Off, InterPlmnHoSwitch:Off, UtranFlashCsfbSwitch:Off, GeranFlashCsfbSwitch:Off, ServiceBasedInterFreqHoSwitch:Off, UlQualityInterFreqHoSwitch:Off, CsfbAdaptiveBlindHoSwitch:Off, UtranCsfbSteeringSwitch:Off, GeranCsfbSteeringSwitch:Off, CSFBLoadInfoSwitch:Off, Cdma1XrttEcsfbSwitch:Off, EmcBlindHoA1Switch:Off, EmcInterFreqBlindHoSwitch:Off, EPlmnSwitch:Off, ServiceReqInterFreqHoSwitch:Off, VoipHoControlSwitch:Off,

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
					UtranUltraFlashCsfbSwitch:Off, GeranUltraFlashCsfbSwitch:Off
CnOperatorHoCfg	UtranA2ThdRsrpOffset	MOD CNOPERATORHOCFG LST CNOPERATORHOCFG	LOFD-001019 / TDLOFD-001019	PS Inter-RAT Mobility between E-UTRAN and UTRAN	Meaning: Indicates the offset of the RSRP threshold for UTRAN inter-RAT handover event A2 relative to the RSRP threshold for inter-RAT handover event A2 (which is specified by InterRatHoA2ThdRsrp). If the value of UtranA2ThdRsrpOffset and that of GeranA2ThdRsrpOffset are identical (excluding the value -100), UTRAN measurement and GERAN measurement are not prioritized, and the RSRP threshold delivered by the eNodeB is the sum of the value of this parameter and the RSRP threshold for inter-RAT measurement event A2. If both UtranA2ThdRsrpOffset and GeranA2ThdRsrpOffset are set to -100, UTRAN measurement and GERAN measurement are not prioritized, and the RSRP threshold delivered by the eNodeB is the RSRP threshold for inter-RAT measurement event A2. If only UtranA2ThdRsrpOffset is set to -100, UTRAN measurement is not performed and the RSRP threshold delivered by the eNodeB is the sum of the value of GeranA2ThdRsrpOffset and the RSRP threshold for inter-RAT measurement event A2. GUI Value Range: -100,-20~20 Unit: dBm Actual Value Range: -100,-20~20 Default Value: 0

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CnOperatorHoCfg	GeranA2ThdRsrpOffset	MOD CNOPERATORHOCFG LST CNOPERATORHOCFG	LOFD-001020 / TDLOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN	Meaning: Indicates the offset of the RSRP threshold for GERAN inter-RAT handover event A2 relative to the RSRP threshold for inter-RAT handover event A2 (which is specified by InterRatHoA2ThdRsrp). If the value of UtranA2ThdRsrpOffset and that of GeranA2ThdRsrpOffset are identical (excluding the value -100), UTRAN measurement and GERAN measurement are not prioritized, and the RSRP threshold delivered by the eNodeB is the sum of the value of this parameter and the RSRP threshold for inter-RAT measurement event A2. If both UtranA2ThdRsrpOffset and GeranA2ThdRsrpOffset are set to -100, UTRAN measurement and GERAN measurement are not prioritized, and the RSRP threshold delivered by the eNodeB is the RSRP threshold for inter-RAT measurement event A2. If only GeranA2ThdRsrpOffset is set to -100, GERAN measurement is not performed and the RSRP threshold delivered by the eNodeB is the sum of the value of UtranA2ThdRsrpOffset and the RSRP threshold for inter-RAT measurement event A2. GUI Value Range: -100,-20~20 Unit: dBm Actual Value Range: -100,-20~20 Default Value: 0

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
InterRatHoCommGroup	InterRatHoA1ThdRsrp	ADD INTERRATHOCOMMGROUP MOD INTERRATHOCOMMGROUP LST INTERRATHOCOMMGROUP	LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001021 / TDLOFD-001021 TDLOFD-001022 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000 SRVCC to UTRAN SRVCC to GERAN	Meaning: Indicates the RSRP threshold for inter-RAT measurement event A1. When the measured RSRP value of the serving cell exceeds this threshold, an event A1 report will be sent. GUI Value Range: -141~-43 Unit: dBm Actual Value Range: -141~-43 Default Value: -111

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
InterRatHoCommGroup	InterRatHoA1ThdRsrq	ADD INTERRATHOCOMMGROUP MOD INTERRATHOCOMMGROUP LST INTERRATHOCOMMGROUP	LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001021 / TDLOFD-001021 TDLOFD-001022 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000 SRVCC to UTRAN SRVCC to GERAN	Meaning: Indicates the RSRQ threshold for inter-RAT measurement event A1. When the measured RSRQ value of the serving cell exceeds this threshold, an event A1 report will be sent. GUI Value Range: -40~-6 Unit: 0.5dB Actual Value Range: -20~-3 Default Value: -20

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranNFreq	UtranDlArfcn	ADD UTRANNFREQ LST UTRANNFREQ MOD UTRANNFREQ RMV UTRANNFREQ	LBFD-00201803 / TDLBFD-00201803 LOFD-001019 / TDLOFD-001019 LBFD-002009 / TDLBFD-002009 TDLOFD-001078	Cell Selection and Re-selection PS Inter-RAT Mobility between E-UTRAN and UTRAN Broadcast of system information E-UTRAN to UTRAN CS/PS steering	Meaning: Indicates the DL UARFCN of the neighboring cell on the UTRAN frequency. The UARFCN range in each FDD frequency band is shown as follows: Band 1: Normal UARFCNs: [10562-10838] Special UARFCNs: none Band 2: Normal UARFCNs: [9662-9938] Special UARFCNs: (412,437,462,487,512,537,562,587,612,637,662,687) Band 3: Normal UARFCNs: [1162-1513] Special UARFCNs: none Band 4: Normal UARFCNs: [1537-1738] Special UARFCNs: (1887, 1912, 1937, 1962, 1987, 2012, 2037, 2062, 2087) Band 5: Normal UARFCNs: [4357-4458] Special UARFCNs: (1007, 1012, 1032, 1037, 1062, 1087) Band 6: Normal UARFCNs: [4387-4413] Special UARFCNs: (1037, 1062) Band 7: Normal UARFCNs: [2237-2563] Special UARFCNs: (2587, 2612, 2637, 2662, 2687, 2712, 2737, 2762, 2787, 2812, 2837, 2862, 2887, 2912) Band 8: Normal UARFCNs: [2937-3088] Special UARFCNs: none

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
					Band 9: Normal UARFCNs: [9237-9387] Special UARFCNs: none The UARFCN range in each TDD frequency band is shown as follows: Band 1: Normal UARFCNs: [9500-9600] and [10050-10125] Special UARFCNs: none Band 2: Normal UARFCNs: [9250-9550] and [9650-9950] Special UARFCNs: none Band 3: Normal UARFCNs: [9550-9650] Special UARFCNs: none Band 4: Normal UARFCNs: [12850-13100] Special UARFCNs: (2112, 2137, 2162, 2187, 2212, 2237, 2262, 2287, 2312, 2337) Band 5: Normal UARFCNs: [11500-12000] Special UARFCNs: none Band 6: Normal UARFCNs: [9400-9600] Special UARFCNs: none For details, see 3GPP TS 25.104 and 3GPP TS 25.105. GUI Value Range: 0~16383 Unit: None Actual Value Range: 0~16383 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranNFreq	OffsetFreq	ADD UTRANNFREQ MOD UTRANNFREQ LST UTRANNFREQ	LOFD-001019 / TDLOFD-001019 TDLOFD-001022 TDLOFD-001033 TDLOFD-001052	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN Flash CS Fallback to UTRAN	Meaning: Indicates the frequency offset of the cell on the UTRAN frequency. It determines the probability of triggering measurement reports for events B1 and B2. For details, see 3GPP TS 36.331. GUI Value Range: -15~15 Unit: dB Actual Value Range: -15~15 Default Value: 0
HoMeasComm	UtranFilterCoeffRscp	MOD HOMEASCOMM LST HOMEASCOMM	LOFD-001019 / TDLOFD-001019 TDLOFD-001022 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN	Meaning: Indicates the filtering coefficient for RSCP measurements on UTRAN. A larger value of this parameter indicates a stronger smoothing effect and higher resistance to fast fading, but it may weaken the tracing capability towards varying signals. For details, see 3GPP TS 36.331. GUI Value Range: FC0, FC1, FC2, FC3, FC4, FC5, FC6, FC7, FC8, FC9, FC11, FC13, FC15, FC17, FC19 Unit: None Actual Value Range: FC0, FC1, FC2, FC3, FC4, FC5, FC6, FC7, FC8, FC9, FC11, FC13, FC15, FC17, FC19 Default Value: FC6

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
HoMeasComm	UtranFilterCoeffEcn0	MOD HOMEASCOMM LST HOMEASCOMM	LOFD-001019 / TDLOFD-001019 TDLOFD-001022 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN	Meaning: Indicates the upper-layer filtering coefficient for Ec/No measurements on UTRAN. A larger value of this parameter indicates a stronger smoothing effect and higher resistance to fast fading, but it may weaken the tracing capability towards varying signals. For details, see 3GPP TS 36.331. GUI Value Range: FC0, FC1, FC2, FC3, FC4, FC5, FC6, FC7, FC8, FC9, FC11, FC13, FC15, FC17, FC19 Unit: None Actual Value Range: FC0, FC1, FC2, FC3, FC4, FC5, FC6, FC7, FC8, FC9, FC11, FC13, FC15, FC17, FC19 Default Value: FC6

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
HoMeasComm	GapPatternType	MOD HOMEASCOMM LST HOMEASCOMM	LBFD-00201802 / TDLBFD-00201802 LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001021 / TDLOFD-001021 TDLBFD-002018 TDLBFD-00201804 TDLBFD-00201805 TDLOFD-001022 TDLOFD-001023 TDLOFD-001033 TDLOFD-001034 TDLOFD-001035	Coverage Based Inter-frequency Handover PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000 Mobility Management Distance Based Inter-frequency Handover Service Based Inter-frequency	Meaning: Indicates the measurement gap pattern. The eNodeB defines measurement gaps for UEs in connected mode. During the measurement gaps, a UE measures the signal quality of the cells on a specific frequency. There are two gap patterns: pattern 1 and pattern 2. In pattern 1, the gap width is 6 ms and the repetition period is 40 ms. In pattern 2, the gap width is 6 ms and the repetition period is 80 ms. For details, see 3GPP TS 36.331. Pattern 1 and pattern 2 correspond to Gap Pattern Id 0 and Gap Pattern Id 1 in 3GPP TS 36.331, respectively. GUI Value Range: GAP_PATTERN_TYPE_1, GAP_PATTERN_TYPE_2 Unit: None Actual Value Range: GAP_PATTERN_TYPE_1, GAP_PATTERN_TYPE_2 Default Value: GAP_PATTERN_TYPE_1

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
			TDLOFD-001043 TDLOFD-001072 TDLOFD-001046 TDLOFD-001073	Handover SRVCC to UTRAN SRVCC to GERAN CS Fallback to UTRAN CS Fallback to GERAN CS Fallback to CDMA2000 1xRTT Service based Inter-RAT handover to UTRAN Distance based Inter-RAT handover to UTRAN Service based Inter-RAT handover to GERAN Distance based Inter-	

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
				RAT handover to GERAN	

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranNFreq	ConnFreqPriority	ADD UTRANNFREQ MOD UTRANNFREQ LST UTRANNFREQ	LOFD-001019 / TDLOFD-001019 TDLBFD-00201803 TDLOFD-001022 TDLOFD-001033 TDLOFD-001052 TDLOFD-001043 TDLOFD-001072 TDLOFD-001078	PS Inter-RAT Mobility between E-UTRAN and UTRAN Cell Selection and Re-selection SRVCC to UTRAN CS Fallback to UTRAN Flash CS Fallback to UTRAN Service based Inter-RAT handover to UTRAN Distance based Inter-RAT handover to UTRAN E-UTRAN to UTRAN CS/PS steering	Meaning: Indicates the frequency priority based on which the eNodeB selects a target frequency for blind redirection or contains a frequency in a measurement configuration. If a blind redirection is triggered and the target neighboring cell is not specified, the eNodeB selects a target frequency based on this priority. If a measurement configuration is to be delivered, the eNodeB preferentially delivers a frequency with the highest priority. If this priority is set to 0 for a frequency, this frequency is not selected as the target frequency for a blind redirection. A larger value indicates a higher priority. GUI Value Range: 0~8 Unit: None Actual Value Range: 0~8 Default Value: 0

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranNCell	CellMeasPriority	ADD UTRANNCELL MOD UTRANNCELL LST UTRANNCELL	LOFD-001019 TDLOFD-001022 TDLOFD-001033 TDLOFD-001052 TDLOFD-001019 TDLOFD-001043 TDLOFD-001072 TDLOFD-001078	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN Flash CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN Service based Inter-RAT handover to UTRAN Distance based Inter-RAT handover to UTRAN E-UTRAN to UTRAN	Meaning: Indicates the priority of measurement on the neighboring UTRAN cell. The eNodeB preferentially contains the information about a neighboring cell with this priority set to HIGH_PRIORITY while delivering a measurement configuration. GUI Value Range: LOW_PRIORITY(Low Priority), HIGH_PRIORITY(High Priority) Unit: None Actual Value Range: LOW_PRIORITY, HIGH_PRIORITY Default Value: LOW_PRIORITY(Low Priority)

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
				CS/PS steering	
InterRatHoComm	InterRatHoEventType	MOD INTERRATHOCOMM LST INTERRATHOCOMM	LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001021 / TDLOFD-001021 TDLOFD-001022 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000 SRVCC to UTRAN SRVCC to GERAN	Meaning: Indicates the type of the inter-RAT handover event. The event type can be B1 or B2. This parameter applies only to coverage-based handovers. For details, see 3GPP TS 36.331. GUI Value Range: EventB1(EventB1), EventB2(EventB2) Unit: None Actual Value Range: EventB1, EventB2 Default Value: EventB1(EventB1)

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
InterRatHoComm	InterRatHoUtranB1MeasQuan	MOD INTERRATHOCOMM LST INTERRATHOCOMM	LOFD-001019 / TDLOFD-001019 LOFD-001022 / TDLOFD-001022 LOFD-001033 / TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN	Meaning: Indicates the quantity to be included in the measurements for handovers to UTRAN. For details, see 3GPP TS 36.331. This parameter is dedicated to UTRAN FDD. The RSCP values are relatively stable, while the Ec/No values may vary with the network load. The value BOTH applies only to UEs complying with 3GPP Release 10. For UEs complying with 3GPP Release 8 or 9, the value BOTH takes the same effect as the value RSCP. GUI Value Range: RSCP, ECN0, BOTH Unit: None Actual Value Range: RSCP, ECN0, BOTH Default Value: ECN0
InterRatHoUtranGroup	InterRatHoUtranB1Hyst	ADD INTERRATHOUTRANGROUP MOD INTERRATHOUTRANGROUP LST INTERRATHOUTRANGROUP	LOFD-001019 / TDLOFD-001019 TDLOFD-001022	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN	Meaning: Indicates the hysteresis for event B1 related to inter-RAT handover to UTRAN. This parameter is used to prevent frequent triggering of event evaluation caused by radio signal fluctuation. In this way, the probability of ping-pong handovers or handover decision errors is reduced. A larger value of this parameter results in a lower probability of ping-pong handovers or handover decision errors. GUI Value Range: 0~30 Unit: 0.5dB Actual Value Range: 0~15 Default Value: 0
InterRatHoUtranGroup	InterRatHoUtranB1ThdRscp	ADD INTERRATHOUTRANGROUP MOD INTERRATHOUTRANGROUP LST INTERRATHOUTRANGROUP	LOFD-001019 / TDLOFD-001019 TDLOFD-001022	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN	Meaning: Indicates the RSCP threshold for event B1 related to coverage-based inter-RAT handover to UTRAN. This parameter specifies the requirement for RSCP of the target UTRAN cell. When the measurement value exceeds this threshold, a measurement report will be sent. GUI Value Range: -120~-25 Unit: dBm Actual Value Range: -120~-25 Default Value: -103

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
InterRatHoUtranGroup	InterRatHoUtranB1ThdEcn0	ADD INTERRATHOUTRANGROUP MOD INTERRATHOUTRANGROUP LST INTERRATHOUTRANGROUP	LOFD-001019 / TDLOFD-001019 TDLOFD-001022	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN	Meaning: Indicates the Ec/N0 threshold for event B1 related to coverage-based inter-RAT handover to UTRAN. This parameter specifies the requirement for Ec/N0 of the target UTRAN cell. For a cell with large signal fading variance, set this parameter to a large value to prevent unnecessary handovers. For a cell with small signal fading variance, set this parameter to a small value to ensure timely handovers. A large value of this parameter results in a low probability of handover to the UTRAN cell, and a small value leads to a high probability. When the measurement value exceeds this threshold, a measurement report will be sent. GUI Value Range: -48~0 Unit: 0.5dB Actual Value Range: -24~0 Default Value: -24
InterRatHoUtranGroup	InterRatHoUtranB1TimeToTrig	ADD INTERRATHOUTRANGROUP MOD INTERRATHOUTRANGROUP LST INTERRATHOUTRANGROUP	LOFD-001019 / TDLOFD-001019 TDLOFD-001022	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN	Meaning: Indicates the time-to-trigger for event B1 related to inter-RAT handover to UTRAN. When detecting that the signal quality in at least one neighboring cell meets the entering condition, the UE does not send a measurement report to the eNodeB immediately. Instead, the UE sends a report only when the signal quality continuously meets the entering condition during the time-to-trigger. This parameter helps decrease the number of occasionally triggered event reports, the average number of handovers and the number of wrong handovers, preventing unnecessary handovers. GUI Value Range: 0ms, 40ms, 64ms, 80ms, 100ms, 128ms, 160ms, 256ms, 320ms, 480ms, 512ms, 640ms, 1024ms, 1280ms, 2560ms, 5120ms Unit: ms Actual Value Range: 0ms, 40ms, 64ms, 80ms, 100ms, 128ms, 160ms, 256ms, 320ms, 480ms, 512ms, 640ms, 1024ms, 1280ms, 2560ms, 5120ms Default Value: 320ms

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
InterRatHoComm	InterRatHoMaxRprtCell	MOD INTERRATHOCOMM LST INTERRATHOCOMM	LOFD-001090 / TDLOFD-001090 LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001021 / TDLOFD-001021 TDLBFD-001022 TDLBFD-001023 TDLBFD-001043 TDLBFD-001072 TDLBFD-001046 TDLBFD-001073	Enhanced CS Fallback to CDMA2000 1xRTT PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000 SRVCC to UTRAN SRVCC to GERAN Service based Inter-RAT handover to UTRAN Distance based	Meaning: Indicates the maximum number of cells to be included in the measurement report after an inter-RAT handover event or a periodic inter-RAT measurement for ANR purposes is triggered. For details, see 3GPP TS 36.331. GUI Value Range: 1~8 Unit: None Actual Value Range: 1~8 Default Value: 4

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
				Inter-RAT handover to UTRAN Service based Inter-RAT handover to GERAN Distance based Inter-RAT handover to GERAN	

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
InterRatHoComm	InterRatHoRprtAmount	MOD INTERRATHOCOMM LST INTERRATHOCOMM	LOFD-001090 / TDLOFD-001090 LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001021 / TDLOFD-001021 TDLBFD-001022 TDLBFD-001023 TDLBFD-001043 TDLBFD-001072 TDLBFD-001046 TDLBFD-001073	Enhanced CS Fallback to CDMA2000 1xRTT PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000 SRVCC to UTRAN SRVCC to GERAN Service based Inter-RAT handover to UTRAN Distance based	Meaning: Indicates the number of periodical measurement reports to be sent after an inter-RAT handover event is triggered. It is used to prevent the impact of measurement report loss and internal processing failure on the handover. For details, see 3GPP TS 36.331. GUI Value Range: r1(1), r2(2), r4(4), r8(8), r16(16), r32(32), r64(64), Infinity(Infinity) Unit: None Actual Value Range: r1, r2, r4, r8, r16, r32, r64, Infinity Default Value: Infinity(Infinity)

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
				Inter-RAT handover to UTRAN Service based Inter-RAT handover to GERAN Distance based Inter-RAT handover to GERAN	
InterRatHoComm	InterRatHoUtranRprtInterval	MOD INTERRATHOCOMM LST INTERRATHOCOMM	LOFD-001019 / TDLOFD-001019 TDLOFD-001022 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN	Meaning: Indicates the interval at which periodical measurement reports are sent after the event for the handover to UTRAN is triggered. For details, see 3GPP TS 36.331. GUI Value Range: 120ms, 240ms, 480ms, 640ms, 1024ms, 2048ms, 5120ms, 10240ms, 1min, 6min, 12min, 30min, 60min Unit: None Actual Value Range: 120ms, 240ms, 480ms, 640ms, 1024ms, 2048ms, 5120ms, 10240ms, 1min, 6min, 12min, 30min, 60min Default Value: 480ms

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CSFallBackBlindHoCfg	InterRatHighestPri	MOD CSFALLBACKBLINDHOCFG LST CSFALLBACKBLINDHOCFG	LOFD-001033 / TDLOFD-001033 LOFD-001034 / TDLOFD-001034 LOFD-001035 / TDLOFD-001035 LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001021 TDLOFD-001052 TDLOFD-001053 TDLOFD-001090 TDLOFD-001043 TDLOFD-001072 TDLOFD-001046	CS Fallback to UTRAN CS Fallback to GERAN CS Fallback to CDMA2000 1xRTT PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000 Flash CS Fallback to UTRAN Flash CS Fallback	Meaning: Indicates the highest-priority RAT for handovers. It is UTRAN by default. If this parameter is set to UTRAN, GERAN, or CDMA2000, the highest-priority RAT is UTRAN, GERAN, or CDMA2000, respectively. GUI Value Range: UTRAN, GERAN, CDMA2000 Unit: None Actual Value Range: UTRAN, GERAN, CDMA2000 Default Value: UTRAN

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
			TDLOFD-001073	to GERAN Enhanced CS Fallback to CDMA2000 1xRTT Service based Inter-RAT handover to UTRAN Distance based Inter-RAT handover to UTRAN Service based Inter-RAT handover to GERAN Distance based Inter-RAT handover to GERAN	

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CSFallBackBlindHoCfg	InterRatSecondPri	MOD CSFALLBACKBLINDHOCFG LST CSFALLBACKBLINDHOCFG	LOFD-001033 / TDLOFD-001033 LOFD-001034 / TDLOFD-001034 LOFD-001035 / TDLOFD-001035 LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001021 TDLOFD-001052 TDLOFD-001053 TDLOFD-001090 TDLOFD-001043 TDLOFD-001072 TDLOFD-001046	CS Fallback to UTRAN CS Fallback to GERAN CS Fallback to CDMA2000 1xRTT PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000 Flash CS Fallback to UTRAN Flash CS Fallback	Meaning: Indicates the medium-priority RAT for handovers. It is GERAN by default. If this parameter is set to UTRAN, GERAN, or CDMA2000, the medium-priority RAT is UTRAN, GERAN, or CDMA2000, respectively. If this parameter is set to NULL, no medium-priority RAT is specified and only the highest-priority RAT can be selected for handovers. GUI Value Range: UTRAN, GERAN, CDMA2000, NULL Unit: None Actual Value Range: UTRAN, GERAN, CDMA2000, NULL Default Value: GERAN

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
			TDLOFD-001073	to GERAN Enhanced CS Fallback to CDMA2000 1xRTT Service based Inter-RAT handover to UTRAN Distance based Inter-RAT handover to UTRAN Service based Inter-RAT handover to GERAN Distance based Inter-RAT handover to GERAN	

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CSFallBackBlindHoCfg	InterRatLowestPri	MOD CSFALLBACKBLINDHOCFG LST CSFALLBACKBLINDHOCFG	LOFD-001033 / TDLOFD-001033 LOFD-001034 / TDLOFD-001034 LOFD-001035 / TDLOFD-001035 LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001021 TDLOFD-001052 TDLOFD-001053 TDLOFD-001090 TDLOFD-001043 TDLOFD-001072 TDLOFD-001046	CS Fallback to UTRAN CS Fallback to GERAN CS Fallback to CDMA2000 1xRTT PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000 Flash CS Fallback to UTRAN Flash CS Fallback	Meaning: Indicates the lowest-priority RAT for handovers. It is CDMA2000 by default. If this parameter is set to UTRAN, GERAN, or CDMA2000, the lowest-priority RAT is UTRAN, GERAN, or CDMA2000, respectively. If this parameter is set to NULL, no lowest-priority RAT is specified and only the highest- or medium-priority RAT can be selected for handovers. GUI Value Range: UTRAN, GERAN, CDMA2000, NULL Unit: None Actual Value Range: UTRAN, GERAN, CDMA2000, NULL Default Value: CDMA2000

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
			TDLOFD-001073	to GERAN Enhanced CS Fallback to CDMA2000 1xRTT Service based Inter-RAT handover to UTRAN Distance based Inter-RAT handover to UTRAN Service based Inter-RAT handover to GERAN Distance based Inter-RAT handover to GERAN	

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranNCell	BlindHoPriority	ADD UTRANNCELL MOD UTRANNCELL LST UTRANNCELL	LOFD-001019 TDLOFD-001022 TDLOFD-001033 TDLOFD-001052 TDLOFD-001019 TDLOFD-001043 TDLOFD-001072 TDLOFD-001078	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN Flash CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN Service based Inter-RAT handover to UTRAN Distance based Inter-RAT handover to UTRAN E-UTRAN to UTRAN	Meaning: Indicates the priority of the neighboring cell during blind handovers. Blind handover is a process in which the eNodeB instructs a UE to hand over to a specified neighboring cell. There are 32 priorities altogether. The priority has a positive correlation with the value of this parameter. Note that the value 0 indicates that blind handovers to the neighboring cell are not allowed. GUI Value Range: 0~32 Unit: None Actual Value Range: 0~32 Default Value: 0

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
				CS/PS steering	
ServiceIrHoCfgGroup	InterRatHoState	ADD SERVICEIRHOCFGGROUP MOD SERVICEIRHOCFGGROUP LST SERVICEIRHOCFGGROUP	LOFD-001043 / TDLOFD-001043 LOFD-001046 / TDLOFD-001046	Service based inter-RAT handover to UTRAN Service based inter-RAT handover to GERAN	Meaning: Indicates whether service-based inter-RAT handovers are required, allowed, or not allowed for a QCI. GUI Value Range: NO_HO, PERMIT_HO, MUST_HO Unit: None Actual Value Range: NO_HO, PERMIT_HO, MUST_HO Default Value: NO_HO
CnOperatorStandardQci	Qci	LST CNOPERATORSTANDARDQCI MOD CNOPERATORSTANDARDQCI	None	None	Meaning: Indicates the QoS Class Identifier (QCI) of an EPS bearer. Different QCIs represent different QoS specifications such as the packet delay budget, packet error loss rate, and resource type (whether the service is a GBR service or not). For details, see Table 6.1.7 in 3GPP TS 23.203. GUI Value Range: QCI1(QCI 1), QCI2(QCI 2), QCI3(QCI 3), QCI4(QCI 4), QCI5(QCI 5), QCI6(QCI 6), QCI7(QCI 7), QCI8(QCI 8), QCI9(QCI 9) Unit: None Actual Value Range: QCI1, QCI2, QCI3, QCI4, QCI5, QCI6, QCI7, QCI8, QCI9 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CnOperatorExtendedQci	ExtendedQci	ADD CNOPERATOREXTENDEDQCI LST CNOPERATOREXTENDEDQCI MOD CNOPERATOREXTENDEDQCI RMV CNOPERATOREXTENDEDQCI	LBFD-002032	Extended-QCI	Meaning: Indicates the extended QoS Class Identifier (QCI), which is required by the operator for user differentiation. GUI Value Range: 10~254 Unit: None Actual Value Range: 10~254 Default Value: None
InterRatHoUtranGroup	LdSvBasedHoUtranB1ThdRscp	ADD INTERRATHOUTRANGROUP MOD INTERRATHOUTRANGROUP LST INTERRATHOUTRANGROUP	LOFD-001043 TDLOFD-001022	Service based inter-RAT handover to UTRAN SRVCC to UTRAN	Meaning: Indicates the RSCP threshold for event B1 related to load- or service-based inter-RAT handover to UTRAN. This parameter specifies the requirement for RSCP of the target UTRAN cell. When the measurement value exceeds this threshold, a measurement report will be sent. GUI Value Range: -120~-25 Unit: dBm Actual Value Range: -120~-25 Default Value: -101

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
InterRatHoUtranGroup	LdSvBasedHoUtranB1ThdEcn0	ADD INTERRATHOUTRANGROUP MOD INTERRATHOUTRANGROUP LST INTERRATHOUTRANGROUP	LOFD-001043 TDLOFD-001022	Service based inter-RAT handover to UTRAN SRVCC to UTRAN	Meaning: Indicates the Ec/N0 threshold for event B1 related to load- or service-based inter-RAT handover to UTRAN. This parameter specifies the requirement for Ec/N0 of the target UTRAN cell. When the measurement value exceeds this threshold, a measurement report may be sent. For a cell with large signal fading variance, set this parameter to a large value to prevent unnecessary handovers. For a cell with small signal fading variance, set this parameter to a small value to ensure timely handovers. GUI Value Range: -48~0 Unit: 0.5dB Actual Value Range: -24~0 Default Value: -18
CellAlgoSwitch	DistBasedHoSwitch	MOD CELLALGOSWITCH LST CELLALGOSWITCH	LBFD-00201804 / TDLBFD-00201804 LOFD-001072 / TDLOFD-001072 LOFD-001073 / TDLOFD-001073	Distance Based Inter-frequency Handover Distance based Inter-RAT handover to UTRAN Distance based Inter-RAT handover to GERAN	Meaning: Indicates the switch used to enable or disable distance-based handovers. If this switch is turned on, distance-based handovers are allowed. If this switch is turned off, distance-based handovers are prohibited. GUI Value Range: OFF(Off), ON(On) Unit: None Actual Value Range: OFF, ON Default Value: OFF(Off)

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
DistBasedHO	DistBasedMeasObjType	MOD DISTBASEDHO LST DISTBASEDHO	LOFD-001072 / TDLOFD-001072 LOFD-001073 / TDLOFD-001073 LBFD-00201804 / TDLBFD-00201804	Distance based Inter-RAT handover to UTRAN Distance based Inter-RAT handover to GERAN Distance Based Inter-frequency Handover	Meaning: Indicates the measurement object type for distance-based handovers. The measurement object type can be E-UTRAN, UTRAN, or GERAN. GUI Value Range: EUTRAN, GERAN, UTRAN Unit: None Actual Value Range: EUTRAN, GERAN, UTRAN Default Value: EUTRAN:Off, GERAN:Off, UTRAN:Off
DistBasedHO	DistBasedHOThd	MOD DISTBASEDHO LST DISTBASEDHO	LBFD-00201804 / TDLBFD-00201804 LOFD-001072 / TDLOFD-001072 LOFD-001073 / TDLOFD-001073	Distance Based Inter-frequency Handover Distance based Inter-RAT handover to UTRAN Distance based Inter-RAT handover to GERAN	Meaning: Indicates the threshold for distance-based handovers. If the distance between a UE and the cell center is greater than this threshold, a distance-based handover is triggered for this UE. GUI Value Range: 1~1000 Unit: 100m Actual Value Range: 100~100000 Default Value: 10

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
ENodeBAlgoSwitch	FreqLayerSwtich	MOD ENODEBALGOSWITCH LST ENODEBALGOSWITCH	LOFD-001087 LOFD-001078 / TDLOFD-001078 TDLOFD-001022 TDLOFD-001033 TDLOFD-001052 TDLOFD-001088	SRVCC Flexible Steering to UTRAN E-UTRAN to UTRAN CS/PS Steering SRVCC to UTRAN CS Fallback to UTRAN Flash CS Fallback to UTRAN CS Fallback Steering to UTRAN	Meaning: This parameter includes the following three switches: UtranFreqLayerMeasSwitch, UtranFreqLayerBlindSwitch, and UtranSrvccSwitch. UtranSrvccSwitch takes effect only when UtranFreqLayerMeasSwitch is turned on. If UtranFreqLayerMeasSwitch is turned on, the UTRAN hierarchy-based measurement algorithm takes effect for measurements related to coverage-based and CSFB-triggered handovers from EUTRAN to UTRAN. If UtranFreqLayerBlindSwitch is turned on, the UTRAN hierarchy-based blind-handover algorithm takes effect for coverage-based and CSFB-triggered blind handovers from EUTRAN to UTRAN. If UtranSrvccSwitch is turned on, the UTRAN SRVCC hierarchy-based measurement algorithm takes effect for SRVCC-triggered handovers from EUTRAN to UTRAN. GUI Value Range: UtranFreqLayerMeasSwitch(UtranFreqLayerMeasSwitch), UtranFreqLayerBlindSwitch(UtranFreqLayerBlindSwitch), UtranSrvccSteeringSwitch(UtranSrvccSteeringSwitch) Unit: None Actual Value Range: UtranFreqLayerMeasSwitch, UtranFreqLayerBlindSwitch, UtranSrvccSteeringSwitch Default Value: UtranFreqLayerMeasSwitch:Off, UtranFreqLayerBlindSwitch:Off, UtranSrvccSteeringSwitch:Off

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranNFreq	PsPriority	ADD UTRANNFREQ MOD UTRANNFREQ LST UTRANNFREQ	LOFD-001019 / TDLOFD-001019 TDLOFD-001078	PS Inter-RAT Mobility between E-UTRAN and UTRAN E-UTRAN to UTRAN CS/PS steering	Meaning: Indicates the priority for the neighboring UTRAN frequency to carry PS services. In coverage-based inter-RAT PS handovers, if UtranFreqLayerMeasSwitch is turned on, the eNodeB determines the UTRAN frequency to be delivered to a UE based on PS service priorities. The eNodeB delivers the UTRAN frequency with the highest PS priority to the UE by default.In blind handovers to UTRAN, the target cell for a blind handover is selected based on PS service priorities. The cell on the UTRAN frequency with the highest PS priority is selected by default.When this parameter is set to Priority_0 for a UTRAN frequency, the UTRAN frequency is not included in PS service priority arrangement. GUI Value Range: Priority_0(Priority 0), Priority_1(Priority 1), Priority_2(Priority 2), Priority_3(Priority 3), Priority_4(Priority 4), Priority_5(Priority 5), Priority_6(Priority 6), Priority_7(Priority 7), Priority_8(Priority 8), Priority_9(Priority 9), Priority_10(Priority 10), Priority_11(Priority 11), Priority_12(Priority 12), Priority_13(Priority 13), Priority_14(Priority 14), Priority_15(Priority 15), Priority_16(Priority 16) Unit: None Actual Value Range: Priority_0, Priority_1, Priority_2, Priority_3, Priority_4, Priority_5, Priority_6, Priority_7, Priority_8, Priority_9, Priority_10, Priority_11, Priority_12, Priority_13, Priority_14, Priority_15, Priority_16 Default Value: Priority_2(Priority 2)

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
ENodeBAlgoSwitch	HoSignalingOptSwitch	MOD ENODEBALGOSWITCH LST ENODEBALGOSWITCH	LBFD-00201802 / TDLBFD-00201802 LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020	Coverage Based Inter-frequency Handover PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN	Meaning: Indicates the switches used to enable or disable handover signaling optimization. ReduceInvalidA1A2RptSigSwitch: Indicates whether to optimize the sequence in delivering A1/A2 measurement configurations during coverage- or frequency-priority-based handovers. If this option is selected: (1) In coverage-based handovers, the eNodeB delivers A2 measurement configurations to UEs first, and then delivers A1 measurement configurations to UEs after receiving event A2 reports, reducing signaling overheads in reporting unnecessary event A1. (2) In frequency-priority-based handovers, the eNodeB delivers A1 measurement configurations to UEs first, and then delivers A2 measurement configurations to UEs after receiving event A1 report, reducing signaling overheads in reporting unnecessary event A2. MultiQciHoExecOptSwitch: Indicates whether the handover decision is triggered based on the admission result of all QCIs in SPID-based handovers to HPLMN and frequency-priority-based handovers. If this option is selected, the handover decision is triggered only when all QCIs are successfully admitted. If this option is not selected, the handover decision is triggered when any of the QCIs is successfully admitted. ReduceInvalidFreqPriHoSwitch: Indicates whether to avoid repeated switches between the frequency-priority-based intra-eNodeB handover algorithm and the load-based intra-eNodeB handover algorithm. If this option is selected, the eNodeB evaluates whether to perform a frequency-priority-based intra-eNodeB handover based on the settings of InterFreqMlbSwitch of the MlbAlgoSwitch parameter and the load status on the target cell. If the target cell is heavily loaded, the eNodeB does not trigger the handover. If this option is not selected, the eNodeB evaluates whether to perform a frequency-priority-based intra-eNodeB handover without checking the settings of InterFreqMlbSwitch of the MlbAlgoSwitch parameter on the target cell. GUI Value Range: ReduceInvalidA1A2RptSigSwitch(ReduceInvalidA1A2RptSigSwitch), MultiQciHoExecOptSwitch(MultiQciHoExecOptSwitch), ReduceInvalidFreqPriHoSwitch(ReduceInvalidFreqPriHoSwitch) Unit: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
					Actual Value Range: ReduceInvalidA1A2RptSigSwitch, MultiQciHoExecOptSwitch, ReduceInvalidFreqPri-HoSwitch Default Value: ReduceInvalidA1A2RptSigSwitch:Off, MultiQciHoExecOptSwitch:Off, ReduceInvalidFreq-PriHoSwitch:Off
InterRatHoComm	CellInfoMaxUtranCellNum	MOD INTERRATHOCOMM LST INTERRATHOCOMM	LOFD-001019 / TDLOFD-001019 TDLOFD-001052	PS Inter-RAT Mobility between E-UTRAN and UTRAN Flash CS Fallback to UTRAN	Meaning: Indicates the maximum number of UTRAN cell system information messages that can be transmitted during a flash redirection procedure. GUI Value Range: 1~16 Unit: None Actual Value Range: 1~16 Default Value: 8

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
ENodeBAlgoSwitch	RedirectSwitch	MOD ENODEBALGOSWITCH LST ENODEBALGOSWITCH	LBFD-002023 / TDLBFD-002023 LBFD-002024 / TDLBFD-002024 LOFD-00102901 / TDLOFD-00102901 LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001021 / TDLOFD-001021	Admission Control Congestion Control Radio/transport resource pre-emption PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000	Meaning: Indicates the switch used to enable or disable the redirection algorithm. This switch does not control redirection or fast redirection for CSFB. GeranFlashRedirectSwitch: If this switch and GeranRedirectSwitch are turned on, fast redirection to GERAN can be performed. UranFlashRedirectSwitch: If this switch and UtranRedirectSwitch are turned on, fast redirection to UTRAN can be performed. LoadBasedSwitch: If this switch is turned on, load based redirection can be performed. GUI Value Range: LoadBasedSwitch, GeranFlashRedirectSwitch, UranFlashRedirectSwitch Unit: None Actual Value Range: LoadBasedSwitch, GeranFlashRedirectSwitch, UranFlashRedirectSwitch Default Value: LoadBasedSwitch:Off, GeranFlashRedirectSwitch:Off, UranFlashRedirectSwitch:Off

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
InterRatPolicyCfgGroup	UtranHoCfg	ADD INTERRATPOLICYCFGGROUP MOD INTERRATPOLICYCFGGROUP LST INTERRATPOLICYCFGGROUP	LOFD-001022 / TDLOFD-001022 LOFD-001019 / TDLOFD-001019	SRVCC to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN	Meaning: Indicates the policy of handovers to UTRAN. PS_HO: indicates whether to allow PS handovers to UTRAN. SRVCC: indicates whether to allow transfers to UTRAN in SRVCC mode. REDIRECTION: indicates whether to allow redirection to UTRAN. GUI Value Range: PS_HO, SRVCC, REDIRECTION Unit: None Actual Value Range: PS_HO, SRVCC, REDIRECTION Default Value: PS_HO:On, SRVCC:Off, REDIRECTION:Off

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
InterRatPolicyCfgGroup	InterRatPolicyCfgGroupId	ADD INTERRATPOLICYCFGGROUP LST INTERRATPOLICYCFGGROUP MOD INTERRATPOLICYCFGGROUP RMV INTERRATPOLICYCFGGROUP	LOFD-001022 / TDLOFD-001022 LOFD-001023 / TDLOFD-001023 LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001021 / TDLOFD-001021	SRVCC to UTRAN SRVCC to GERAN PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000	Meaning: Indicates the group ID of the parameters related to the inter-RAT handover with a specific QCI. GUI Value Range: 0~39 Unit: None Actual Value Range: 0~39 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
Standard Qci	InterRatPolicyCfgGroupId	MOD STANDARDQCI LST STANDARDQCI	LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001021 / TDLOFD-001021 TDLOFD-001022 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000 SRVCC to UTRAN SRVCC to GERAN	Meaning: Indicates the ID of a parameter group related to inter-RAT handover. GUI Value Range: 0~39 Unit: None Actual Value Range: 0~39 Default Value: 0

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
ExtendedQci	InterRatPolicyCfgGroupId	ADD EXTENDEDQCI MOD EXTENDEDQCI LST EXTENDEDQCI	LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001021 / TDLOFD-001021 TDLOFD-001022 TDLOFD-001023 TDLOFD-001035	PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000 SRVCC to UTRAN SRVCC to GERAN CS Fallback to CDMA2000 1xRTT	Meaning: Indicates the group ID of the parameters related to the inter-RAT handover with the extended QCI. GUI Value Range: 0~39 Unit: None Actual Value Range: 0~39 Default Value: 0

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
ENodeBAlgoSwitch	MultiOpCtrlSwitch	MOD ENODEBALGOSWITCH LST ENODEBALGOSWITCH	LOFD-001022 LOFD-001087 LOFD-001033 LOFD-001052 LOFD-001068 LOFD-001088 LOFD-001019 LOFD-001043 LOFD-001072 LOFD-001078	SRVCC to UTRAN SRVCC Flexible Steering to UTRAN CS Fallback to UTRAN Flash CS Fallback to UTRAN CS Fallback with LAI to UTRAN CS Fallback Steering to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN Service based inter-RAT Distance based Inter-RAT handover to UTRAN	Meaning: Indicates the switch used to control whether operators can adopt different policies. This parameter is a bit-filed-type parameter. By specifying the bit fields under this parameter, operators can adopt different policies on the corresponding RAT. UtranSepOpMobilitySwitch is a switch used to control whether operators can adopt different mobility policies on their UTRANs. If this switch is turned on, operators can adopt different policies (for example, PS handover capability and RIM-based system information reading capability) on their UTRANs. If this switch is turned off, operators cannot adopt different policies on their UTRANs. GUI Value Range: UtranSepOpMobility-Switch(UtranSepOpMobilitySwitch) Unit: None Actual Value Range: UtranSepOpMobilitySwitch Default Value: UtranSepOpMobilitySwitch:Off

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
				E-UTRAN to UTRAN CS/PS Steering	

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranNetworkCapCfg	NetworkCapCfg	ADD UTRANNETWORKCAPCFG MOD UTRANNETWORKCAPCFG LST UTRANNETWORKCAPCFG	LOFD-001022 / TDLOFD-001022 LOFD-001033 / TDLOFD-001033 LOFD-001052 / TDLOFD-001052 LOFD-001068 / TDLOFD-001068 LOFD-001088 / TDLOFD-001088 LOFD-001019 / TDLOFD-001019 LOFD-001043 / TDLOFD-001043 LOFD-001072 / TDLOFD-001072 LOFD-070202 / TDLOFD-070202	SRVCC to UTRAN CS Fallback to UTRAN Flash CS Fallback to UTRAN CS Fallback with LAI to UTRAN CS Fallback Steering to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN Service based inter-RAT handover to UTRAN Distance based Inter-RAT handover to UTRAN	Meaning: Indicates the UTRAN capabilities for an operator including PS handover capability, capability of obtaining system information (SI) of the UTRAN through RAN Information Management (RIM) procedures, VoIP capability, and Ultra-Flash CSFB capability. If the MME, SGSN, or RNC of the operator does not support PS handovers, RIM procedures, VoIP, or Ultra-Flash CSFB, set this parameter to indicate the incapabilities. If this parameter is not set, UTRAN capabilities are supported by default. PsHoCapCfg indicates whether PS handovers are supported. If this option is selected, PS handovers are supported. If this option is cleared, PS handovers are not supported. SiByRimCapCfg indicates whether the capability of obtaining SI of the UTRAN through RIM procedures is supported. If the option is selected, obtaining SI of the UTRAN through RIM procedures is supported. If the option is cleared, obtaining SI of the UTRAN through RIM procedures is not supported. VoipCapCfg indicates the VoIP capability of the UTRAN. If this option is selected, the VoIP is supported. VoIP services established in the LTE network can be transferred to the UTRAN using PS handovers so that UEs can perform voice services in the UTRAN. If this option is cleared, VoIP is not supported. VoIP services established in the LTE network cannot be transferred to the UTRAN using PS handovers. UltraFlashCsfbCapCfg indicates the SRVCC capability. If this option is selected, SRVCC is supported and voice services can continue in the UTRAN by SRVCC. If this option is not selected, SRVCC is not supported. GUI Value Range: PsHoCapCfg(PsHoCapCfg), SiByRimCapCfg(SiByRimCapCfg), VoipCapCfg(VoipCapCfg), UltraFlashCsfbCapCfg(UltraFlashCsfbCapCfg) Unit: None Actual Value Range: PsHoCapCfg, SiByRimCapCfg, VoipCapCfg, UltraFlashCsfbCapCfg Default Value: PsHoCapCfg:off, SiByRimCapCfg:off, VoipCapCfg:on, UltraFlashCsfbCapCfg:off

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
				Ultra-Flash CSFB	
UtranNetworkCapCfg	RncId	ADD UTRANNETWORKCAPCFG LST UTRANNETWORKCAPCFG MOD UTRANNETWORKCAPCFG RMV UTRANNETWORKCAPCFG	None	None	Meaning: Indicates the RNC ID of the UTRAN cell. The 28-bit UTRAN cell ID is comprised of the RNC cell ID and the RNC ID that are represented by the least significant 16 bits and the most significant 12 bits, respectively. The formula for calculating the UTRAN cell ID is as follows: UTRAN cell ID = RNC ID x 65536 + RNC cell ID. The value range of RNC ID is (0~4095, 4294967295). 0~4095 indicates the actual RNC ID on the network. 4294967295 indicates the special ID of all RNCs under the same PLMN. GUI Value Range: 0~4095,4294967295 Unit: None Actual Value Range: 0~4095,4294967295 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
HoMeasComm	OptHoPreFailPunishTimer	MOD HOMEASCOMM LST HOMEASCOMM	LBFD-00201805 / TDLBFD-00201805 LOFD-001043 / TDLOFD-001043 LOFD-001046 / TDLOFD-001046 LOFD-00105401 / TDLOFD-00105401	Service Based Inter-frequency Handover Service based inter-RAT handover to UTRAN Service based inter-RAT handover to GERAN Camp & Handover Based on SPID	Meaning: Indicates the punish time of handover preparation failures in optimized handovers, including frequency-priority-based handovers, service-based handovers, and SPID-based handovers back to the HPLMN. GUI Value Range: 0~20 Unit: 30s Actual Value Range: 0~600 Default Value: 0

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
GlobalProcSwitch	EutranLoadTransSwitch	MOD GLOBALPROCSWITCH LST GLOBALPROCSWITCH	LOFD-070203 SEFD-090205 SEFD-090206	RIM Based LTE Target Cell Selection Intelligent WiFi Detection and Selection-LTE FDD Intelligent WiFi Detection and Selection-LTE TDD	Meaning: Indicates whether to send E-UTRAN cell load information. The RimUtranSwitch option controls whether to send E-UTRAN cell load information through RAN information management (RIM) procedures to the UTRAN. This option applies only to LTE FDD. If the RimUtranSwitch option is selected, the eNodeB sends cell load information through RIM procedures to the UTRAN and notifies the UTRAN of cell load changes through RIM procedures. The EcoSwitch option controls whether to send cell load information to the eCoordinator. This option applies to both LTE FDD and LTE TDD. If the EcoSwitch option is selected, the eNodeB sends load information to the eCoordinator and notifies the eCoordinator of cell load changes through RIM procedures. GUI Value Range: RimUtranSwitch(RimUtranSwitch), EcoSwitch(EcoSwitch) Unit: None Actual Value Range: RimUtranSwitch, EcoSwitch Default Value: RimUtranSwitch:off, EcoSwitch:off

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
ENodeBAlgoSwitch	RimOnEcoSwitch	MOD ENODEBALGOSWITCH LST ENODEBALGOSWITCH	MRFD-090211 LOFD-001052/TDLOFD-001052 LOFD-001019/TDLOFD-001019 LOFD-001044/TDLOFD-001044 LOFD-001033/TDLOFD-001033	eCoordinator based RIM process optimization Flash CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN Inter-RAT Load Sharing to UTRAN(based on UMTS cell load information) CS Fallback to UTRAN (based on UMTS cell load information)	Meaning: Indicates whether the RAN information management (RIM) procedure is initiated by the eCoordinator. If this parameter is set to ON, the RIM procedure is initiated by the eCoordinator. If this parameter is set to OFF, the RIM procedure is initiated by the core network. GUI Value Range: OFF(Off), ON(On) Unit: None Actual Value Range: OFF, ON Default Value: OFF(Off)

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CellMLB	InterRatMlbUeNumThd	MOD CELLMLB LST CELLMLB	LOFD-001044 / TDLOFD-001044 LOFD-001045 / TDLOFD-001045	Inter-RAT Load Sharing to UTRAN Inter-RAT Load Sharing to GERAN	Meaning: Indicates the threshold of the number of synchronized UEs for triggering inter-RAT load sharing. Inter-RAT load sharing is triggered when the PRB usage of the cell is greater than or equal to the sum of the InterRatMlbThd and LoadOffset parameter values and the number of synchronized UEs in the cell is greater than or equal to the sum of the InterRatMlbUeNumThd and InterRatMlbUeNumOffset parameter values. Inter-RAT load sharing stops if the number of synchronized UEs in the cell is less than the InterRatMlbUeNumThd parameter value or the PRB usage of the cell is less than the InterRatMlbThd parameter value. The actual threshold of the number of synchronized UEs equals the percentage value multiplied by 1000.For example, parameter value 1 indicates that the threshold for the number of synchronized UEs is 10 (1000 x 1%), parameter value 2 indicates threshold 20, and parameter value 10 indicates threshold 100. Note that if the parameter value is 100 or 99, the threshold for the number of synchronized UEs is 1 or 2, respectively. GUI Value Range: 1~100 Unit: % Actual Value Range: 1~100 Default Value: 15

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CellMLB	InterRatMlbUeNumOffset	MOD CELLMLB LST CELLMLB	LOFD-001044 / TDLOFD-001044 LOFD-001045 / TDLOFD-001045 LOFD-070215 / TDLOFD-070215	Inter-RAT Load Sharing to UTRAN Inter-RAT Load Sharing to GERAN Intra-LTE User Number Load Balancing	Meaning: Indicates the offset of the number of synchronized UEs for triggering inter-RAT load sharing. Inter-RAT load sharing is triggered when the PRB usage of the cell is greater than or equal to the sum of the InterRatMlbThd and LoadOffset parameter values and the number of synchronized UEs in the cell is greater than or equal to the sum of the InterRatMlbUeNumThd and InterRatMlbUeNumOffset parameter values. Inter-RAT load sharing to transfer to-be-released UEs is triggered if the number of synchronized UEs is greater than or equal to the sum of the InterRATIdleMlbUeNumThd and InterRatMlbUeNumOffset parameter values and the PRB usage of the cell is greater than or equal to the sum of the InterRatMlbThd and LoadOffset parameter values. GUI Value Range: 0~10000 Unit: None Actual Value Range: 0~10000 Default Value: 0
GeranNfreqGroup	Starting Arfcn	ADD GERANNFREQGROUP MOD GERANNFREQGROUP LST GERANNFREQGROUP	LBFD-0201803 / TDLBFD-00201803 LOFD-001020 / TDLOFD-001020	Cell Selection and Re-selection PS Inter-RAT Mobility between E-UTRAN and GERAN	Meaning: Indicates the first ARFCN in the GERAN carrier frequency group. If the GERAN ARFCN is not between 512 and 810, one ARFCN can only in one frequency band. For details, see 3GPP TS 45.005. GUI Value Range: 0~1023 Unit: None Actual Value Range: 0~1023 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
GeranNfreqGroup	BandIndicator	ADD GERANNFREQGROUP MOD GERANNFREQGROUP LST GERANNFREQGROUP	LBFD-00201803 / TDLBFD-00201803 LOFD-001020 / TDLOFD-001020	Cell Selection and Re-selection PS Inter-RAT Mobility between E-UTRAN and GERAN	Meaning: Indicates the GERAN band indicator. If the GERAN ARFCN is between 512 and 810, this parameter indicates whether the GERAN ARFCN is in the 1800 MHz band or the 1900 MHz band. If the GERAN ARFCN is not in either band, this parameter does not take effect and can be set to either value and one ARFCN can only in one frequency band. For details, see 3GPP TS 45.005. GUI Value Range: GSM_dcs1800, GSM_pcs1900 Unit: None Actual Value Range: GSM_dcs1800, GSM_pcs1900 Default Value: None
GeranNfreqGroupArfcn	GeranArfcn	ADD GERANNFREQGROUPARFCN LST GERANNFREQGROUPARFCN RMV GERANNFREQGROUPARFCN	LBFD-00201803 / TDLBFD-00201803 LOFD-001020 / TDLOFD-001020 LBFD-002009 / TDLBFD-002009	Cell Selection and Re-selection PS Inter-RAT Mobility between E-UTRAN and GERAN Broadcast of system information	Meaning: Specifies the ARFCN of the GERAN BCCH. If the GERAN ARFCN value is not between 512 and 810, one ARFCN can only in one frequency band. For details about this parameter, see 3GPP TS 45.005. GUI Value Range: 0~1023 Unit: None Actual Value Range: 0~1023 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
GeranNfreqGroup	OffsetFreq	ADD GERANNFREQGROUP MOD GERANNFREQGROUP LST GERANNFREQGROUP	LOFD-001020 / TDLOFD-001020 TDLOFD-001023 TDLOFD-001034 TDLOFD-001053	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN Flash CS Fallback to GERAN	Meaning: Indicates the frequency offset of the group of neighboring GERAN carrier frequencies. This parameter is used in the decision of the UE to send measurement reports for inter-RAT handovers. For details, see 3GPP TS 36.331. GUI Value Range: -15~15 Unit: dB Actual Value Range: -15~15 Default Value: 0
HoMeasComm	GeranFilterCoeff	MOD HOMEASCOMM LST HOMEASCOMM	LOFD-001020 / TDLOFD-001020 TDLOFD-001023 TDLOFD-001034 TDLOFD-001053	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN Flash CS Fallback to GERAN	Meaning: Indicates the L3 filtering coefficient used for the measurement quantity in inter-RAT measurements of GERAN. A great value of this parameter indicates a strong smoothing effect and a high anti-fast-fading capability, but a low signal change tracing capability. For details, see 3GPP TS 36.331. GUI Value Range: FC0, FC1, FC2, FC3, FC4, FC5, FC6, FC7, FC8, FC9, FC11, FC13, FC15, FC17, FC19 Unit: None Actual Value Range: FC0, FC1, FC2, FC3, FC4, FC5, FC6, FC7, FC8, FC9, FC11, FC13, FC15, FC17, FC19 Default Value: FC6

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
GeranNfreqGroup	ConnFreqPriority	ADD GERANNFREQGROUP MOD GERANNFREQGROUP LST GERANNFREQGROUP	LOFD-001020 / TDLOFD-001020 TDLOFD-001023 TDLOFD-001034 TDLOFD-001053	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN Flash CS Fallback to GERAN	Meaning: Indicates the frequency group priority based on which the eNodeB selects a target frequency group for blind redirection or delivers a frequency group in measurement configuration messages. If a blind redirection is triggered and the target neighboring cell is not specified, the eNodeB selects a target frequency group based on the setting of this parameter. If a measurement configuration is to be delivered, the eNodeB preferentially delivers the frequency group with the highest priority. If this parameter is set to 0 for a frequency group, this frequency group is not selected as the target frequency group for a blind redirection. A larger value indicates a higher priority. GUI Value Range: 0~8 Unit: None Actual Value Range: 0~8 Default Value: 0
InterRatHoGeranGroup	InterRatHoGeranB1Hyst	ADD INTERRATHOGERANGROUP MOD INTERRATHOGERANGROUP LST INTERRATHOGERANGROUP	LOFD-001020 / TDLOFD-001020 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN	Meaning: Indicates the hysteresis for event B1 related to inter-RAT handover to GERAN. This parameter is used to prevent frequent triggering of event evaluation caused by radio signal fluctuation. In this way, the probability of ping-pong handovers or handover decision errors is reduced. A larger value of this parameter results in a lower probability of ping-pong handovers or handover decision errors. GUI Value Range: 0~30 Unit: 0.5dB Actual Value Range: 0~15 Default Value: 2

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
InterRatHoGeranGroup	InterRatHoGeranB1Thd	ADD INTERRATHOGERANGROUP MOD INTERRATHOGERANGROUP LST INTERRATHOGERANGROUP	LOFD-001020 / TDLOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN	Meaning: Indicates the RSSI threshold for event B1 related to coverage-based inter-RAT handover to GERAN. A UE sends a measurement report related to event B1 to the eNodeB when the RSSI in at least one GERAN cell exceeds this threshold and other triggering conditions are met. For details, see 3GPP TS 36.331. GUI Value Range: -110~-48 Unit: dBm Actual Value Range: -110~-48 Default Value: -100
InterRatHoGeranGroup	InterRatHoGeranB1TimeToTrig	ADD INTERRATHOGERANGROUP MOD INTERRATHOGERANGROUP LST INTERRATHOGERANGROUP	LOFD-001020 / TDLOFD-001020 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN	Meaning: Indicates the time-to-trigger for event B1 related to inter-RAT handover to GERAN. When detecting that the signal quality in at least one neighboring cell meets the entering condition, the UE does not send a measurement report to the eNodeB immediately. Instead, the UE sends a report only when the signal quality continuously meets the entering condition during the time-to-trigger. This parameter helps decrease the number of occasionally triggered event reports, the average number of handovers and the number of wrong handovers, preventing unnecessary handovers. GUI Value Range: 0ms, 40ms, 64ms, 80ms, 100ms, 128ms, 160ms, 256ms, 320ms, 480ms, 512ms, 640ms, 1024ms, 1280ms, 2560ms, 5120ms Unit: ms Actual Value Range: 0ms, 40ms, 64ms, 80ms, 100ms, 128ms, 160ms, 256ms, 320ms, 480ms, 512ms, 640ms, 1024ms, 1280ms, 2560ms, 5120ms Default Value: 640ms

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
InterRatHoComm	InterRatHoGeranRprtInterval	MOD INTERRATHOCOMM LST INTERRATHOCOMM	LOFD-001020 / TDLOFD-001020 LOFD-001021 TDLOFD-001023 TDLOFD-001033 TDLOFD-001034 TDLOFD-001035 TDLOFD-001019 TDLOFD-001046 TDLOFD-001073	PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000 SRVCC to GERAN CS Fallback to UTRAN CS Fallback to GERAN CS Fallback to CDMA2000 1xRTT PS Inter-RAT Mobility between E-UTRAN and UTRAN Service based Inter-RAT handove	Meaning: Indicates the interval at which periodic measurement reports are sent to the eNodeB after the event for inter-RAT handover to GERAN is triggered. For details, see 3GPP TS 36.331. GUI Value Range: 120ms, 240ms, 480ms, 640ms, 1024ms, 2048ms, 5120ms, 10240ms, 1min, 6min, 12min, 30min, 60min Unit: None Actual Value Range: 120ms, 240ms, 480ms, 640ms, 1024ms, 2048ms, 5120ms, 10240ms, 1min, 6min, 12min, 30min, 60min Default Value: 480ms

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
				r to GERAN Distance based Inter-RAT handover to GERAN	
GeranNcell	BlindHoPriority	ADD GERANNCELL MOD GERANNCELL LST GERANNCELL	LOFD-001020 / TDLOFD-001020 TDLOFD-001023 TDLOFD-001034 TDLOFD-001053	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN Flash CS Fallback to GERAN	Meaning: Indicates the priority of the neighboring cell during blind handovers. Blind handover is a process in which the eNodeB instructs a UE to hand over to a specified neighboring cell. There are 32 priorities altogether. The priority has a positive correlation with the value of this parameter. Note that the value 0 indicates that blind handovers to the neighboring cell are not allowed. GUI Value Range: 0~32 Unit: None Actual Value Range: 0~32 Default Value: 0
InterRatHoGeranGroup	LdSvBasedHoGeranB1Thd	ADD INTERRATHOGERANGROUP MOD INTERRATHOGERANGROUP LST INTERRATHOGERANGROUP	LOFD-001046 / TDLOFD-001046	Service based inter-RAT handover to GERAN	Meaning: Indicates the RSSI threshold for event B1 related to load- or service-based inter-RAT handover to GERAN. When the measured RSSI value exceeds this threshold, a measurement report will be sent. GUI Value Range: -110~-48 Unit: dBm Actual Value Range: -110~-48 Default Value: -98

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
InterRatHoComm	CellInfoMaxGeranCellNum	MOD INTERRATHOCOMM LST INTERRATHOCOMM	LOFD-001020 / TDLOFD-001020 TDLOFD-001053	PS Inter-RAT Mobility between E-UTRAN and GERAN Flash CS Fallback to GERAN	Meaning: Indicates the maximum number of GERAN cell system information messages that can be transmitted during a flash redirection procedure. GUI Value Range: 1~32 Unit: None Actual Value Range: 1~32 Default Value: 8
InterRatPolicyCfgGroup	GeranGsmHoCfg	ADD INTERRATPOLICYCFGGROUP MOD INTERRATPOLICYCFGGROUP LST INTERRATPOLICYCFGGROUP	LOFD-001023 / TDLOFD-001023 LOFD-001020 / TDLOFD-001020	SRVCC to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN	Meaning: Indicates the switch used to enable or disable the single radio voice call continuity (SRVCC) handover to GERAN GSM. If this switch is set to On, the UE can perform the SRVCC handover to the GERAN GSM cell. If this switch is set to Off, the UE cannot perform the SRVCC handover to the GERAN GSM cell. GUI Value Range: SRVCC Unit: None Actual Value Range: SRVCC Default Value: SRVCC:On

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
InterRatPolicyCfgGroup	GeranGprsEdgeHoCfg	ADD INTERRATPOLICYCFGGROUP MOD INTERRATPOLICYCFGGROUP LST INTERRATPOLICYCFGGROUP	TDLOFD-001023 LOFD-001020 / TDLOFD-001020	SRVCC to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN	Meaning: Indicates the policy of handovers to GPRS/EDGE. PS_HO: Indicates whether PS handovers to neighboring GPRS/EDGE cells are allowed. NACC_HO: Indicates whether NACC handovers to neighboring GPRS/EDGE cells are allowed. CCO_HO: Indicates whether CCO handovers to neighboring GPRS/EDGE cells are allowed. SRVCC: Indicates whether SRVCC handovers to neighboring GPRS/EDGE cells are allowed. REDIRECTION: Indicates whether redirections to neighboring GPRS/EDGE cells are allowed. GUI Value Range: PS_HO, NACC_HO, CCO_HO, SRVCC, REDIRECTION Unit: None Actual Value Range: PS_HO, NACC_HO, CCO_HO, SRVCC, REDIRECTION Default Value: PS_HO:On, NACC_HO:Off, CCO_HO:Off, SRVCC:Off, REDIRECTION:Off

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
eNBRsvdPara	RsvdSwPara1	MOD ENBRSVDPARA LST ENBRSVDPARA	None	None	Meaning: Indicates reserved 32-bit switch parameter 1 that is reserved for future requirements. Note on parameter replacement: Reserved parameters are temporarily used in patch versions and will be replaced with new parameters. For example, the ID of a new parameter can signify the parameter function. Therefore, avoid using this parameter. GUI Value Range: RsvdSwPara1_bit1(ReservedSwitchParameter1_bit1), RsvdSwPara1_bit2(ReservedSwitchParameter1_bit2), RsvdSwPara1_bit3(ReservedSwitchParameter1_bit3), RsvdSwPara1_bit4(ReservedSwitchParameter1_bit4), RsvdSwPara1_bit5(ReservedSwitchParameter1_bit5), RsvdSwPara1_bit6(ReservedSwitchParameter1_bit6), RsvdSwPara1_bit7(ReservedSwitchParameter1_bit7), RsvdSwPara1_bit8(ReservedSwitchParameter1_bit8), RsvdSwPara1_bit9(ReservedSwitchParameter1_bit9), RsvdSwPara1_bit10(ReservedSwitchParameter1_bit10), RsvdSwPara1_bit11(ReservedSwitchParameter1_bit11), RsvdSwPara1_bit12(ReservedSwitchParameter1_bit12), RsvdSwPara1_bit13(ReservedSwitchParameter1_bit13), RsvdSwPara1_bit14(ReservedSwitchParameter1_bit14), RsvdSwPara1_bit15(ReservedSwitchParameter1_bit15), RsvdSwPara1_bit16(ReservedSwitchParameter1_bit16), RsvdSwPara1_bit17(ReservedSwitchParameter1_bit17), RsvdSwPara1_bit18(ReservedSwitchParameter1_bit18), RsvdSwPara1_bit19(ReservedSwitchParameter1_bit19), RsvdSwPara1_bit20(ReservedSwitchParameter1_bit20), RsvdSwPara1_bit21(ReservedSwitchParameter1_bit21), RsvdSwPara1_bit22(ReservedSwitchParameter1_bit22), RsvdSwPara1_bit23(ReservedSwitchParameter1_bit23), RsvdSwPara1_bit24(ReservedSwitchParame-

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
					ter1_bit24), RsvdSwPara1_bit25(ReservedSwitchParameter1_bit25), RsvdSwPara1_bit26(ReservedSwitchParameter1_bit26), RsvdSwPara1_bit27(ReservedSwitchParameter1_bit27), RsvdSwPara1_bit28(ReservedSwitchParameter1_bit28), RsvdSwPara1_bit29(ReservedSwitchParameter1_bit29), RsvdSwPara1_bit30(ReservedSwitchParameter1_bit30), RsvdSwPara1_bit31(ReservedSwitchParameter1_bit31), RsvdSwPara1_bit32(ReservedSwitchParameter1_bit32) Unit: None Actual Value Range: RsvdSwPara1_bit1, RsvdSwPara1_bit2, RsvdSwPara1_bit3, RsvdSwPara1_bit4, RsvdSwPara1_bit5, RsvdSwPara1_bit6, RsvdSwPara1_bit7, RsvdSwPara1_bit8, RsvdSwPara1_bit9, RsvdSwPara1_bit10, RsvdSwPara1_bit11, RsvdSwPara1_bit12, RsvdSwPara1_bit13, RsvdSwPara1_bit14, RsvdSwPara1_bit15, RsvdSwPara1_bit16, RsvdSwPara1_bit17, RsvdSwPara1_bit18, RsvdSwPara1_bit19, RsvdSwPara1_bit20, RsvdSwPara1_bit21, RsvdSwPara1_bit22, RsvdSwPara1_bit23, RsvdSwPara1_bit24, RsvdSwPara1_bit25, RsvdSwPara1_bit26, RsvdSwPara1_bit27, RsvdSwPara1_bit28, RsvdSwPara1_bit29, RsvdSwPara1_bit30, RsvdSwPara1_bit31, RsvdSwPara1_bit32 Default Value: RsvdSwPara1_bit1:Off, RsvdSwPara1_bit2:Off, RsvdSwPara1_bit3:Off, RsvdSwPara1_bit4:Off, RsvdSwPara1_bit5:Off, RsvdSwPara1_bit6:Off, RsvdSwPara1_bit7:Off, RsvdSwPara1_bit8:Off, RsvdSwPara1_bit9:Off, RsvdSwPara1_bit10:Off, RsvdSwPara1_bit11:Off, RsvdSwPara1_bit12:Off, RsvdSwPara1_bit13:Off, RsvdSwPara1_bit14:Off, RsvdSwPara1_bit15:Off, RsvdSwPara1_bit16:Off, RsvdSwPara1_bit17:Off, RsvdSwPara1_bit18:Off, RsvdSwPara1_bit19:Off, RsvdSwPara1_bit20:Off, RsvdSwPara1_bit21:Off, RsvdSwPara1_bit22:Off, RsvdSwPara1_bit23:Off, RsvdSwPara1_bit24:Off, RsvdSwPara1_bit25:Off,

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
					RsvdSwPara1_bit26:Off, RsvdSwPara1_bit27:Off, RsvdSwPara1_bit28:Off, RsvdSwPara1_bit29:Off, RsvdSwPara1_bit30:Off, RsvdSwPara1_bit31:Off, RsvdSwPara1_bit32:Off
UtranNFreq	LocalCellId	ADD UTRANNFREQ LST UTRANNFREQ MOD UTRANNFREQ RMV UTRANNFREQ	None	None	Meaning: Indicates the cell ID of the local cell. It uniquely identifies a cell within an eNodeB. GUI Value Range: 0~255 Unit: None Actual Value Range: 0~255 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranExternalCell	Mcc	ADD UTRANEXTERNALCELL ADD UTRANEXTERNALCELLBAND LST UTRANEXTERNALCELL LST UTRANEXTERNALCELLBAND MOD UTRANEXTERNALCELL RMV UTRANEXTERNALCELL RMV UTRANEXTERNALCELLBAND	LOFD-001019 / TDLOFD-001019	PS Inter-RAT Mobility between E-UTRAN and UTRAN	Meaning: Indicates the mobile country code of the external UTRAN cell. The PLMN consists of the MCC and the MNC. The MCC consists of three digits. The MNC consists of two to three digits. For example, MCC = 123, MNC = 45, PLMN = 12345. GUI Value Range: 3 characters Unit: None Actual Value Range: 000~999 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranExternalCell	Mnc	ADD UTRANEXTERNALCELL ADD UTRANEXTERNALCELLBAND LST UTRANEXTERNALCELL LST UTRANEXTERNALCELLBAND MOD UTRANEXTERNALCELL RMV UTRANEXTERNALCELL RMV UTRANEXTERNALCELLBAND	LOFD-001019 / TDLOFD-001019	PS Inter-RAT Mobility between E-UTRAN and UTRAN	Meaning: Indicates the mobile network code of the external UTRAN cell. The PLMN consists of the MCC and the MNC. The MCC consists of three digits. The MNC consists of two to three digits. For example, MCC = 123, MNC = 45, PLMN = 12345. GUI Value Range: 2~3 characters Unit: None Actual Value Range: 00~99,000~999 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranExternalCell	CellId	ADD UTRANEXTERNALCELL ADD UTRANEXTERNALCELLBAND LST UTRANEXTERNALCELL LST UTRANEXTERNALCELLBAND MOD UTRANEXTERNALCELL RMV UTRANEXTERNALCELL RMV UTRANEXTERNALCELLBAND	LOFD-001019 / TDLOFD-001019	PS Inter-RAT Mobility between E-UTRAN and UTRAN	Meaning: Indicates the RNC cell ID of an external UTRAN cell. It uniquely identifies a cell within an RNC. The 28-bit UTRAN cell ID is comprised of the RNC cell ID and the RNC ID that are represented by the least significant 16 bits and the most significant 12 bits, respectively. The formula for calculating the UTRAN cell ID is as follows: UTRAN cell ID = RNC ID x 65536 + RNC cell ID. For details, see 3GPP TS 25.401. GUI Value Range: 0~65535 Unit: None Actual Value Range: 0~65535 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranExternalCell	UtranDlArfcn	ADD UTRANEXTERNALCELL MOD UTRANEXTERNALCELL LST UTRANEXTERNALCELL	LOFD-001019 / TDLOFD-001019	PS Inter-RAT Mobility between E-UTRAN and UTRAN	Meaning: Indicates the DL UARFCN of the external UTRAN cell. The UARFCN range in each FDD frequency band is shown as follows: Band 1: Normal UARFCNs: [10562-10838] Special UARFCNs: none Band 2: Normal UARFCNs: [9662-9938] Special UARFCNs: (412,437,462,487,512,537,562,587,612,637,662,687) Band 3: Normal UARFCNs: [1162-1513] Special UARFCNs: none Band 4: Normal UARFCNs: [1537-1738] Special UARFCNs: (1887, 1912, 1937, 1962, 1987, 2012, 2037, 2062, 2087) Band 5: Normal UARFCNs: [4357-4458] Special UARFCNs: (1007, 1012, 1032, 1037, 1062, 1087) Band 6: Normal UARFCNs: [4387-4413] Special UARFCNs: (1037, 1062) Band 7: Normal UARFCNs: [2237-2563] Special UARFCNs: (2587, 2612, 2637, 2662, 2687, 2712, 2737, 2762, 2787, 2812, 2837, 2862, 2887, 2912) Band 8: Normal UARFCNs: [2937-3088] Special UARFCNs: none

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
					Band 9: Normal UARFCNs: [9237-9387] Special UARFCNs: none The UARFCN range in each TDD frequency band is shown as follows: Band 1: Normal UARFCNs: [9500-9600] and [10050-10125] Special UARFCNs: none Band 2: Normal UARFCNs: [9250-9550] and [9650-9950] Special UARFCNs: none Band 3: Normal UARFCNs: [9550-9650] Special UARFCNs: none Band 4: Normal UARFCNs: [12850-13100] Special UARFCNs: (2112, 2137, 2162, 2187, 2212, 2237, 2262, 2287, 2312, 2337) Band 5: Normal UARFCNs: [11500-12000] Special UARFCNs: none Band 6: Normal UARFCNs: [9400-9600] Special UARFCNs: none For details, see 3GPP TS 25.104 and 3GPP TS 25.105. GUI Value Range: 0~16383 Unit: None Actual Value Range: 0~16383 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranExternalCell	RncId	ADD UTRANEXTERNALCELL ADD UTRANEXTERNALCELLBAND LST UTRANEXTERNALCELL LST UTRANEXTERNALCELLBAND MOD UTRANEXTERNALCELL RMV UTRANEXTERNALCELL RMV UTRANEXTERNALCELLBAND	LOFD-001052 / TDLOFD-001052	Flash CS Fallback to UTRAN	Meaning: Indicates the RNC ID of the external UTRAN cell. The 28-bit UTRAN cell ID is comprised of the RNC cell ID and the RNC ID that are represented by the least significant 16 bits and the most significant 12 bits, respectively. The formula for calculating the UTRAN cell ID is as follows: UTRAN cell ID = RNC ID x 65536 + RNC cell ID. For details, see 3GPP TS 25.401. GUI Value Range: 0~4095 Unit: None Actual Value Range: 0~4095 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranExternalCell	RacCfgInd	ADD UTRANEXTERNALCELL MOD UTRANEXTERNALCELL LST UTRANEXTERNALCELL	LOFD-001019 / TDLOFD-001019	PS Inter-RAT Mobility between E-UTRAN and UTRAN	Meaning: Indicates whether to set the routing area code (RAC) of the external UTRAN cell. This parameter must be set to CFG(Configure) if the external UTRAN cell supports packet switched (PS) services. GUI Value Range: NOT_CFG(Not configure), CFG(Configure) Unit: None Actual Value Range: NOT_CFG, CFG Default Value: NOT_CFG(Not configure)

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranExternalCell	Rac	ADD UTRANEXTERNALCELL MOD UTRANEXTERNALCELL LST UTRANEXTERNALCELL	LOFD-001019 / TDLOFD-001019 LOFD-001034 / TDLOFD-001034 LOFD-001052 / TDLOFD-001052 TDLOFD-001033 TDLOFD-001043 TDLOFD-001072 TDLOFD-001078	PS Inter-RAT Mobility between E-UTRAN and UTRAN CS Fallback to GERAN Flash CS Fallback to UTRAN CS Fallback to UTRAN Service based Inter-RAT handover to UTRAN Distance based Inter-RAT handover to UTRAN E-UTRAN to UTRAN CS/PS steering	Meaning: Indicates the routing area code. GUI Value Range: 0~255 Unit: None Actual Value Range: 0~255 Default Value: 0

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranExternalCell	PScrambCode	ADD UTRANEXTERNALCELL MOD UTRANEXTERNALCELL LST UTRANEXTERNALCELL	LOFD-001019 / TDLOFD-001019	PS Inter-RAT Mobility between E-UTRAN and UTRAN	Meaning: Indicates the primary DL scrambles of the external UTRAN cell. For details, see 3GPP TS 25.331. GUI Value Range: 0~511 Unit: None Actual Value Range: 0~511 Default Value: None
UtranExternalCell	Lac	ADD UTRANEXTERNALCELL MOD UTRANEXTERNALCELL LST UTRANEXTERNALCELL	LOFD-001022 / TDLOFD-001022 LOFD-001068 / TDLOFD-001068 TDLOFD-001078	SRVCC to UTRAN CS Fallback with LAI to UTRAN E-UTRAN to UTRAN CS/PS steering	Meaning: Indicates the LAC of the external UTRAN cell. GUI Value Range: 1~65533,65535 Unit: None Actual Value Range: 1~65533,65535 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranExternalCellPlmn	CellId	ADD UTRANEXTERNALCELLPLMN LST UTRANEXTERNALCELLPLMN RMV UTRANEXTERNALCELLPLMN	LOFD-001036 / TDLOFD-001036 LOFD-001037 / TDLOFD-001037	RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier	Meaning: Indicates the RNC cell ID of an external UTRAN cell. It uniquely identifies a cell within an RNC. The 28-bit UTRAN cell ID is comprised of the RNC cell ID and the RNC ID that are represented by the least significant 16 bits and the most significant 12 bits, respectively. The formula for calculating the UTRAN cell ID is as follows: UTRAN cell ID = RNC ID x 65536 + RNC cell ID. For details, see 3GPP TS 25.401. GUI Value Range: 0~65535 Unit: None Actual Value Range: 0~65535 Default Value: None
UtranExternalCellPlmn	Mcc	ADD UTRANEXTERNALCELLPLMN LST UTRANEXTERNALCELLPLMN RMV UTRANEXTERNALCELLPLMN	LOFD-001036 / TDLOFD-001036 LOFD-001037 / TDLOFD-001037	RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier	Meaning: Indicates the mobile country code of the external UTRAN cell. A PLMN ID is comprised of an MCC and an MNC. The MCC consists of three digits. The MNC consists of two to three digits. For example, if the MCC is 123 and the MNC is 45, then the PLMN ID is 12345. GUI Value Range: 3 characters Unit: None Actual Value Range: 000~999 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranExternalCellPlmn	Mnc	ADD UTRANEXTERNALCELLPLMN LST UTRANEXTERNALCELLPLMN RMV UTRANEXTERNALCELLPLMN	LOFD-001036 / TDLOFD-001036 LOFD-001037 / TDLOFD-001037	RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier	Meaning: Indicates the mobile network code of the external UTRAN cell. A PLMN ID is comprised of an MCC and an MNC. The MCC consists of three digits. The MNC consists of two to three digits. For example, if the MCC is 123 and the MNC is 45, then the PLMN ID is 12345. GUI Value Range: 2~3 characters Unit: None Actual Value Range: 00~99,000~999 Default Value: None
UtranExternalCellPlmn	ShareMcc	ADD UTRANEXTERNALCELLPLMN LST UTRANEXTERNALCELLPLMN RMV UTRANEXTERNALCELLPLMN	LOFD-001036 / TDLOFD-001036 LOFD-001037 / TDLOFD-001037	RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier	Meaning: Indicates the mobile country code of the external UTRAN cell that is shared among multiple operators. A PLMN ID is comprised of an MCC and an MNC. The MCC consists of three digits. The MNC consists of two to three digits. For example, if the MCC is 123 and the MNC is 45, then the PLMN ID is 12345. GUI Value Range: 3 characters Unit: None Actual Value Range: 000~999 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranExternalCellPlmn	ShareMnc	ADD UTRANEXTERNALCELLPLMN LST UTRANEXTERNALCELLPLMN RMV UTRANEXTERNALCELLPLMN	LOFD-001036 / TDLOFD-001036 LOFD-001037 / TDLOFD-001037	RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier	Meaning: Indicates the mobile network code of the external UTRAN cell that is shared among multiple operators. A PLMN ID is comprised of an MCC and an MNC. The MCC consists of three digits. The MNC consists of two to three digits. For example, if the MCC is 123 and the MNC is 45, then the PLMN ID is 12345. GUI Value Range: 2~3 characters Unit: None Actual Value Range: 00~99,000~999 Default Value: None
UtranNCell	LocalCellId	ADD UTRANNCELL DSP UTRANRIMINFO DSP UTRANRIMLOADINFO LST UTRANNCELL MOD UTRANNCELL RMV UTRANNCELL	None	None	Meaning: Indicates the cell ID of the local cell. It uniquely identifies a cell within an eNodeB. GUI Value Range: 0~255 Unit: None Actual Value Range: 0~255 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranNCell	Mcc	ADD UTRANNCELL LST UTRANNCELL MOD UTRANNCELL RMV UTRANNCELL DSP UTRANRIMINFO DSP UTRANRIMLOADINFO	LOFD-001019 / TDLOFD-001019	PS Inter-RAT Mobility between E-UTRAN and UTRAN	Meaning: Indicates the mobile country code of the neighboring UTRAN cell. The PLMN consists of the MCC and the MNC. The MCC consists of three digits. The MNC consists of two to three digits. For example, MCC = 123, MNC = 45, PLMN = 12345. GUI Value Range: 3 characters Unit: None Actual Value Range: 000~999 Default Value: None
UtranNCell	Mnc	ADD UTRANNCELL LST UTRANNCELL MOD UTRANNCELL RMV UTRANNCELL DSP UTRANRIMINFO DSP UTRANRIMLOADINFO	LOFD-001019 / TDLOFD-001019	PS Inter-RAT Mobility between E-UTRAN and UTRAN	Meaning: Indicates the mobile network code of the neighboring UTRAN cell. The PLMN consists of the MCC and the MNC. The MCC consists of three digits. The MNC consists of two to three digits. For example, MCC = 123, MNC = 45, PLMN = 12345. GUI Value Range: 2~3 characters Unit: None Actual Value Range: 00~99,000~999 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranNCell	CellId	ADD UTRANNCELL LST UTRANNCELL MOD UTRANNCELL RMV UTRANNCELL DSP UTRANRIMINFO DSP UTRANRIMLOADINFO	LOFD-001052 / TDLOFD-001052 LOFD-001019 / TDLOFD-001019	Flash CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN	Meaning: Indicates the RNC cell ID of an external UTRAN cell. It uniquely identifies a cell within an RNC. The 28-bit UTRAN cell ID is comprised of the RNC cell ID and the RNC ID that are represented by the least significant 16 bits and the most significant 12 bits, respectively. The formula for calculating the UTRAN cell ID is as follows: UTRAN cell ID = RNC ID x 65536 + RNC cell ID. For details, see 3GPP TS 25.401. GUI Value Range: 0~65535 Unit: None Actual Value Range: 0~65535 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
InterRatPolicyCfgGroup	NoHoFlag	ADD INTERRATPOLICYCFGGROUP MOD INTERRATPOLICYCFGGROUP LST INTERRATPOLICYCFGGROUP	LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001021 / TDLOFD-001021 TDLBFD-00201801 TDLOFD-001022 TDLOFD-001023 TDLOFD-001033 TDLOFD-001034	PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000 Coverage Based Intra-frequency Handover SRVCC to UTRAN SRVCC to GERAN CS Fallback to UTRAN CS Fallback to GERAN	Meaning: Indicates whether to permit or forbid the inter-RAT handover under the QCI. GUI Value Range: PERMIT_HO_ENUM(Permit Ho), FORBID_HO_ENUM(Forbid Ho) Unit: None Actual Value Range: PERMIT_HO_ENUM, FORBID_HO_ENUM Default Value: PERMIT_HO_ENUM(Permit Ho)

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
StandardQci	Qci	LST STANDARDQCI MOD STANDARDQCI	LOFD-00101502 / TDLOFD-00101502 TDLBFD-002025 TDLOFD-001015	Dynamic Scheduling Basic Scheduling Enhanced Scheduling	Meaning: Indicates the QoS Class Identifier (QCI) of an EPS bearer. Different QCIs represent different QoS specifications such as the packet delay budget, packet error loss rate, and resource type (whether the service is a GBR service or not). For details, see Table 6.1.7 in 3GPP TS 23.203. GUI Value Range: QCI1(QCI 1), QCI2(QCI 2), QCI3(QCI 3), QCI4(QCI 4), QCI5(QCI 5), QCI6(QCI 6), QCI7(QCI 7), QCI8(QCI 8), QCI9(QCI 9) Unit: None Actual Value Range: QCI1, QCI2, QCI3, QCI4, QCI5, QCI6, QCI7, QCI8, QCI9 Default Value: None
ExtendedQci	ExtendedQci	ADD EXTENDEDQCI LST EXTENDEDQCI MOD EXTENDEDQCI RMV EXTENDEDQCI	LBFD-002032 LOFD-00101502 / TDLOFD-00101502	Extended-QCI Dynamic Scheduling	Meaning: Indicates the extended QoS Class Identifier (QCI), which is required by the operator for user differentiation. GUI Value Range: 10~254 Unit: None Actual Value Range: 10~254 Default Value: None
CellStandardQci	LocalCellId	LST CELLSTANDARDQCI MOD CELLSTANDARDQCI	None	None	Meaning: Indicates the local ID of the cell. It uniquely identifies a cell within a BS. GUI Value Range: 0~255 Unit: None Actual Value Range: 0~255 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CellStandardQci	Qci	LST CELLSTANDARDQCI MOD CELLSTANDARDQCI	LBFD-002025 / TDLBFD-002025 LOFD-0101502 / TDLOFD-00101502	Basic Scheduling Dynamic Scheduling	Meaning: Indicates the QoS class identifier (QCI) of an evolved packet system (EPS) bearer. Different QCIs indicate different QoS requirements, such as the packet delay budget, packet error loss rate, and resource type. For details, see Table 6.1.7 in 3GPP TS 23.203. GUI Value Range: QCI1(QCI 1), QCI2(QCI 2), QCI3(QCI 3), QCI4(QCI 4), QCI5(QCI 5), QCI6(QCI 6), QCI7(QCI 7), QCI8(QCI 8), QCI9(QCI 9) Unit: None Actual Value Range: QCI1, QCI2, QCI3, QCI4, QCI5, QCI6, QCI7, QCI8, QCI9 Default Value: None
CellStandardQci	InterRatHoCommGroupId	MOD CELLSTANDARDQCI LST CELLSTANDARDQCI	LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001021 / TDLOFD-001021 TDLOFD-001022 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000 SRVCC to UTRAN SRVCC to GERAN	Meaning: Indicates the ID of a common parameter group related to inter-RAT handover. GUI Value Range: 0~9 Unit: None Actual Value Range: 0~9 Default Value: 0

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CellStandardQci	InterRatHoUtranGroupId	MOD CELLSTANDARDQCI LST CELLSTANDARDQCI	LOFD-001019 / TDLOFD-001019 TDLOFD-001022	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN	Meaning: Indicates the ID of a parameter group related to inter-RAT handover to UTRAN. GUI Value Range: 0~9 Unit: None Actual Value Range: 0~9 Default Value: 0

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CellStandardQci	QciPriorityForHo	MOD CELLSTANDARDQCI LST CELLSTANDARDQCI	LBFD-00201801 LBFD-00201802 LBFD-00201804 LBFD-00201805 LOFD-001072 LOFD-001073	Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover Distance Based Inter-frequency Handover Service Based Inter-frequency Handover Distance based Inter-RAT handover to UTRAN Distance based Inter-RAT handover to GERAN	Meaning: Indicates the QCI priority used by each QCI-based handover algorithm. A larger value of this parameter indicates a lower priority. This parameter applies to handover parameter selection based on QCI priorities, target frequency selection in service-based inter-frequency handovers, and inter-RAT handover policy selection. For a service that involves all the preceding selection, the setting of this parameter determines the handover parameter, target frequency, and handover policy corresponding to a QCI. GUI Value Range: 1~9 Unit: None Actual Value Range: 1~9 Default Value: 9

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CellExtendedQci	ExtendedQci	ADD CELLEXTENDEDQCI LST CELLEXTENDEDQCI MOD CELLEXTENDEDQCI RMV CELLEXTENDEDQCI	LBFD-002032	Extended-QCI	Meaning: Indicates the extended QoS Class Identifier (QCI), which is required by the operator for service differentiation. GUI Value Range: 10~254 Unit: None Actual Value Range: 10~254 Default Value: None
CellExtendedQci	LocalCellId	ADD CELLEXTENDEDQCI LST CELLEXTENDEDQCI MOD CELLEXTENDEDQCI RMV CELLEXTENDEDQCI	None	None	Meaning: Indicates the local ID of the cell. It uniquely identifies a cell within a BS. GUI Value Range: 0~255 Unit: None Actual Value Range: 0~255 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CellExtendedQci	InterRatHoCommGroupId	ADD CELLEXTENDEDQCI MOD CELLEXTENDEDQCI LST CELLEXTENDEDQCI	LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001021 / TDLOFD-001021	PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000	Meaning: Indicates the ID of a group of common parameters related to inter-RAT handover. GUI Value Range: 0~9 Unit: None Actual Value Range: 0~9 Default Value: 0
CellExtendedQci	InterRatHoUtranGroupId	ADD CELLEXTENDEDQCI MOD CELLEXTENDEDQCI LST CELLEXTENDEDQCI	LOFD-001019 / TDLOFD-001019	PS Inter-RAT Mobility between E-UTRAN and UTRAN	Meaning: Indicates the ID of a parameter group related to inter-RAT handover to UTRAN. GUI Value Range: 0~9 Unit: None Actual Value Range: 0~9 Default Value: 0

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CSFallBackBlindHoCfg	CnOperatorId	LST CSFALLBACKBLINDHOCFG MOD CSFALLBACKBLINDHOCFG	LOFD-001033 / TDLOFD-001033 LOFD-001034 / TDLOFD-001034 TDLOFD-001052 TDLOFD-001053 TDLOFD-001035 TDLOFD-001090	CS Fallback to UTRAN CS Fallback to GERAN Flash CS Fallback to UTRAN Flash CS Fallback to GERAN CS Fallback to CDMA2000 1xRTT Enhanced CS Fallback to CDMA2000 1xRTT	Meaning: Indicates the index of the operator. GUI Value Range: 0~5 Unit: None Actual Value Range: 0~5 Default Value: None
CnOperatorHoCfg	CnOperatorId	LST CNOPERATORHOCFG MOD CNOPERATORHOCFG	None	None	Meaning: Indicates the ID of the operator. GUI Value Range: 0~5 Unit: None Actual Value Range: 0~5 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranRanShare	LocalCellId	ADD UTRANRANSHARE LST UTRANRANSHARE MOD UTRANRANSHARE RMV UTRANRANSHARE	None	None	Meaning: Indicates the cell identity of the local cell. It uniquely identifies a cell within an eNodeB. GUI Value Range: 0~255 Unit: None Actual Value Range: 0~255 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranRanShare	UtranDlArfcn	ADD UTRANRANSHARE LST UTRANRANSHARE MOD UTRANRANSHARE RMV UTRANRANSHARE	LOFD-001036 / TDLOFD-001036 TDLOFD-001112 LOFD-070206	RAN Sharing with Common Carrier MOCN Flexible Priority Based Camping Hybrid RAN Sharing	Meaning: Indicates the DL UARFCN of the neighboring cell operating on the UTRAN frequency. The UARFCN range in each FDD frequency band is shown as follows: Band 1: Normal UARFCNs: [10562-10838] Special UARFCNs: none Band 2: Normal UARFCNs: [9662-9938] Special UARFCNs: (412, 437, 462, 487, 512, 537, 562, 587, 612, 637, 662, 687) Band 3: Normal UARFCNs: [1162-1513] Special UARFCNs: none Band 4: Normal UARFCNs: [1537-1738] Special UARFCNs: (1887, 1912, 1937, 1962, 1987, 2012, 2037, 2062, 2087) Band 5: Normal UARFCNs: [4357-4458] Special UARFCNs: (1007, 1012, 1032, 1037, 1062, 1087) Band 6: Normal UARFCNs: [4387-4413] Special UARFCNs: (1037, 1062) Band 7: Normal UARFCNs: [2237-2563] Special UARFCNs: (2587, 2612, 2637, 2662, 2687, 2712, 2737, 2762, 2787, 2812, 2837, 2862, 2887, 2912) Band 8: Normal UARFCNs: [2937-3088]

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
					Special UARFCNs: none Band 9: Normal UARFCNs: [9237-9387] Special UARFCNs: none The UARFCN range in each TDD frequency band is shown as follows: Band 1: Normal UARFCNs: [9500-9600] and [10050-10125] Special UARFCNs: none Band 2: Normal UARFCNs: [9250-9550] and [9650-9950] Special UARFCNs: none Band 3: Normal UARFCNs: [9550-9650] Special UARFCNs: none Band 4: Normal UARFCNs: [12850-13100] Special UARFCNs: (2112, 2137, 2162, 2187, 2212, 2237, 2262, 2287, 2312, 2337) Band 5: Normal UARFCNs: [11500-12000] Special UARFCNs: none Band 6: Normal UARFCNs: [9400-9600] Special UARFCNs: none For details, see 3GPP TS 25.104 and 3GPP TS 25.105. GUI Value Range: 0~16383 Unit: None Actual Value Range: 0~16383 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranRanShare	Mcc	ADD UTRANRANSHARE LST UTRANRANSHARE MOD UTRANRANSHARE RMV UTRANRANSHARE	LOFD-001036 / TDLOFD-001036 TDLOFD-001112 LOFD-070206	RAN Sharing with Common Carrier MOCN Flexible Priority Based Camping Hybrid RAN Sharing	Meaning: Indicates the mobile country code (MCC) of the neighboring UTRAN cell. A public land mobile network (PLMN) ID is comprised of an MCC and a mobile network code (MNC). The MCC consists of three digits. The MNC consists of two to three digits. For example, if the MCC is 123 and the MNC is 45, then the PLMN ID is 12345. GUI Value Range: 3 characters Unit: None Actual Value Range: 000~999 Default Value: None
UtranRanShare	Mnc	ADD UTRANRANSHARE LST UTRANRANSHARE MOD UTRANRANSHARE RMV UTRANRANSHARE	LOFD-001036 / TDLOFD-001036 TDLOFD-001112 LOFD-070206	RAN Sharing with Common Carrier MOCN Flexible Priority Based Camping Hybrid RAN Sharing	Meaning: Indicates the mobile network code (MNC) of the neighboring UTRAN cell. A public land mobile network (PLMN) ID is comprised of a mobile country code (MCC) and an MNC. The MCC consists of three digits. The MNC consists of two to three digits. For example, if the MCC is 123 and the MNC is 45, then the PLMN ID is 12345. GUI Value Range: 2~3 characters Unit: None Actual Value Range: 00~99,000~999 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
InterRatHoCommGroup	LocalCellId	ADD INTERRATHOCOMMGROUP LST INTERRATHOCOMMGROUP MOD INTERRATHOCOMMGROUP RMV INTERRATHOCOMMGROUP	None	None	Meaning: Indicates the local ID of the cell. It uniquely identifies a cell within a BS. GUI Value Range: 0~255 Unit: None Actual Value Range: 0~255 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
InterRatHoCommGroup	InterRatHoCommGroupId	ADD INTERRATHOCOMMGROUP LST INTERRATHOCOMMGROUP MOD INTERRATHOCOMMGROUP RMV INTERRATHOCOMMGROUP	LOFD-001019 / TDLOFD-001019 LOFD-001020 / TDLOFD-001020 LOFD-001021 / TDLOFD-001021 TDLOFD-001022 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000 SRVCC to UTRAN SRVCC to GERAN	Meaning: Indicates the ID of the common parameter group related to inter-RAT handover. GUI Value Range: 0~9 Unit: None Actual Value Range: 0~9 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
InterRatHoUtranGroup	LocalCellId	ADD INTERRATHOUTRANGROUP LST INTERRATHOUTRANGROUP MOD INTERRATHOUTRANGROUP RMV INTERRATHOUTRANGROUP	None	None	Meaning: Indicates the local ID of the cell. It uniquely identifies a cell within a BS. GUI Value Range: 0~255 Unit: None Actual Value Range: 0~255 Default Value: None
InterRatHoUtranGroup	InterRatHoUtranGroupId	ADD INTERRATHOUTRANGROUP LST INTERRATHOUTRANGROUP MOD INTERRATHOUTRANGROUP RMV INTERRATHOUTRANGROUP	LOFD-001019 / TDLOFD-001019 TDLOFD-001022	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN	Meaning: Indicates the ID of the parameter group related to inter-RAT handover to UTRAN. GUI Value Range: 0~9 Unit: None Actual Value Range: 0~9 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
ServiceIrHoCfgGroup	CnOperatorId	ADD SERVICEIRHOCFGGROUP LST SERVICEIRHOCFGGROUP MOD SERVICEIRHOCFGGROUP RMV SERVICEIRHOCFGGROUP	LOFD-001043 / TDLOFD-001043 LOFD-001046 / TDLOFD-001046	Service based inter-RAT handover to UTRAN Service based inter-RAT handover to GERAN	Meaning: Indicates the index of the operator. GUI Value Range: 0~5 Unit: None Actual Value Range: 0~5 Default Value: None
ServiceIrHoCfgGroup	ServiceIrHoCfgGroupId	ADD SERVICEIRHOCFGGROUP LST SERVICEIRHOCFGGROUP MOD SERVICEIRHOCFGGROUP RMV SERVICEIRHOCFGGROUP	LOFD-001043 / TDLOFD-001043 LOFD-001046 / TDLOFD-001046	Service based inter-RAT handover to UTRAN Service based inter-RAT handover to GERAN	Meaning: Indicates the ID of the service-based inter-RAT handover policy group. GUI Value Range: 0~9 Unit: None Actual Value Range: 0~9 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CnOperatorStandardQci	CnOperatorId	LST CNOPERATORSTANDARDQCI MOD CNOPERATORSTANDARDQCI	None	None	Meaning: Indicates the index of the operator. GUI Value Range: 0~5 Unit: None Actual Value Range: 0~5 Default Value: None
CnOperatorStandardQci	ServiceIrHoCfgGroupId	MOD CNOPERATORSTANDARDQCI LST CNOPERATORSTANDARDQCI	LOFD-001043 / TDLOFD-001043 LOFD-001046 / TDLOFD-001046	Service based inter-RAT handover to UTRAN Service based inter-RAT handover to GERAN	Meaning: Indicates the ID of the service-based inter-RAT handover policy group. GUI Value Range: 0~9 Unit: None Actual Value Range: 0~9 Default Value: 0

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CnOperatorExtendedQci	CnOperatorId	ADD CNOPERATOREXTENDEDQCI LST CNOPERATOREXTENDEDQCI MOD CNOPERATOREXTENDEDQCI RMV CNOPERATOREXTENDEDQCI	LBFD-002032	Extended-QCI	Meaning: Indicates the index of the operator. GUI Value Range: 0~5 Unit: None Actual Value Range: 0~5 Default Value: None
CnOperatorExtendedQci	ServiceIrHoCfgGroupId	ADD CNOPERATOREXTENDEDQCI MOD CNOPERATOREXTENDEDQCI LST CNOPERATOREXTENDEDQCI	LOFD-001043 / TDLOFD-001043 LOFD-001046 / TDLOFD-001046	Service based inter-RAT handover to UTRAN Service based inter-RAT handover to GERAN	Meaning: Indicates the ID of the service-based inter-RAT handover policy group. GUI Value Range: 0~9 Unit: None Actual Value Range: 0~9 Default Value: 0

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CellAlgoSwitch	LocalCellId	LST CELLALGOSWITCH MOD CELLALGOSWITCH	None	None	Meaning: Indicates the local ID of the cell. It uniquely identifies a cell within a BS. GUI Value Range: 0~255 Unit: None Actual Value Range: 0~255 Default Value: None
DistBasedHO	LocalCellId	LST DISTBASEDHO MOD DISTBASEDHO	None	None	Meaning: Indicates the local identity of the cell. It uniquely identifies a cell within an eNodeB. GUI Value Range: 0~255 Unit: None Actual Value Range: 0~255 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranNFreq	CsPriority	ADD UTRANNFREQ MOD UTRANNFREQ LST UTRANNFREQ	LOFD-001078 / TDLOFD-001078 TDLOFD-001033 TDLOFD-001052 TDLOFD-001088	E-UTRAN to UTRAN CS/PS Steering CS Fallback to UTRAN Flash CS Fallback to UTRAN CS Fallback Steering to UTRAN	Meaning: Indicates the priority for a neighboring UTRAN frequency to carry CS services. In coverage-based inter-RAT SRVCC or CSFB to UTRAN, if UtranFreqLayerMeasSwitch is turned on, the eNodeB determines the UTRAN frequency to be delivered to a UE based on CS service priorities. The eNodeB delivers the UTRAN frequency with the highest CS priority to the UE by default.In blind handovers for CSFB to UTRAN, if UtranFreqLayerBlindSwitch is turned on, the target cell for a blind handover is selected based on CS service priorities. The cell on the UTRAN frequency with the highest CS priority is selected by default.When this parameter is set to Priority_0 for a UTRAN frequency, the UTRAN frequency is not included in CS service priority arrangement. GUI Value Range: Priority_0(Priority 0), Priority_1(Priority 1), Priority_2(Priority 2), Priority_3(Priority 3), Priority_4(Priority 4), Priority_5(Priority 5), Priority_6(Priority 6), Priority_7(Priority 7), Priority_8(Priority 8), Priority_9(Priority 9), Priority_10(Priority 10), Priority_11(Priority 11), Priority_12(Priority 12), Priority_13(Priority 13), Priority_14(Priority 14), Priority_15(Priority 15), Priority_16(Priority 16) Unit: None Actual Value Range: Priority_0, Priority_1, Priority_2, Priority_3, Priority_4, Priority_5, Priority_6, Priority_7, Priority_8, Priority_9, Priority_10, Priority_11, Priority_12, Priority_13, Priority_14, Priority_15, Priority_16 Default Value: Priority_2(Priority 2)

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
UtranNetworkCapCfg	Mcc	ADD UTRANNETWORKCAPCFG LST UTRANNETWORKCAPCFG MOD UTRANNETWORKCAPCFG RMV UTRANNETWORKCAPCFG	None	None	Meaning: Indicates the mobile country code (MCC) of a target radio network controller (RNC). A public land mobile network (PLMN) ID is comprised of an MCC and a mobile network code (MNC). The MCC consists of three digits. The MNC consists of two or three digits. For example, if the MCC is 123 and the MNC is 45, the PLMN ID is 12345. GUI Value Range: 3 characters Unit: None Actual Value Range: 000~999 Default Value: None
UtranNetworkCapCfg	Mnc	ADD UTRANNETWORKCAPCFG LST UTRANNETWORKCAPCFG MOD UTRANNETWORKCAPCFG RMV UTRANNETWORKCAPCFG	None	None	Meaning: Indicates the mobile network code (MNC) of a target radio network controller (RNC). A public land mobile network (PLMN) ID is comprised of an MCC and a mobile network code (MNC). The MCC consists of three digits. The MNC consists of two or three digits. For example, if the MCC is 123 and the MNC is 45, the PLMN ID is 12345. GUI Value Range: 2~3 characters Unit: None Actual Value Range: 00~99,000~999 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
GeranNfreqGroup	LocalCellId	ADD GERANNFREQGROUP LST GERANNFREQGROUP MOD GERANNFREQGROUP RMV GERANNFREQGROUP	None	None	Meaning: Indicates the cell ID of the local cell. It uniquely identifies a cell within an eNodeB. GUI Value Range: 0~255 Unit: None Actual Value Range: 0~255 Default Value: None
GeranNfreqGroup	BcchGroupId	ADD GERANNFREQGROUP LST GERANNFREQGROUP MOD GERANNFREQGROUP RMV GERANNFREQGROUP	LBFD-00201803 / TDLBFD-00201803 LOFD-001020 / TDLOFD-001020	Cell Selection and Re-selection PS Inter-RAT Mobility between E-UTRAN and GERAN	Meaning: Indicates a GERAN carrier frequency group. In the cell reselection procedure, the GERAN carrier frequencies are organized in groups and the cell reselection parameters are provided per group of GERAN carrier frequencies. GUI Value Range: 0~31 Unit: None Actual Value Range: 0~31 Default Value: None
GeranNfreqGroup	GeranVersion	ADD GERANNFREQGROUP MOD GERANNFREQGROUP LST GERANNFREQGROUP	LOFD-001020 / TDLOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN	Meaning: Indicates the standard that is supported by the current carrier frequency group. GUI Value Range: GSM, GPRS, EDGE Unit: None Actual Value Range: GSM, GPRS, EDGE Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
GeranNfreqGroupArfcn	LocalCellId	ADD GERANNFREQGROUPARFCN LST GERANNFREQGROUPARFCN RMV GERANNFREQGROUPARFCN	None	None	Meaning: Indicates the cell ID of the local cell. It uniquely identifies a cell within an eNodeB. GUI Value Range: 0~255 Unit: None Actual Value Range: 0~255 Default Value: None
GeranExternalCell	Mcc	ADD GERANEXTERNALCELL LST GERANEXTERNALCELL MOD GERANEXTERNALCELL RMV GERANEXTERNALCELL	LOFD-001020 / TDLOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN	Meaning: Indicates the mobile country code of the external cell. The PLMN consists of the MCC and the MNC. The MCC consists of three digits. The MNC consists of two to three digits. For example, MCC = 123, MNC = 45, PLMN = 12345. GUI Value Range: 3 characters Unit: None Actual Value Range: 000~999 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
GeranExternalCell	Mnc	ADD GERANEXTERNALCELL LST GERANEXTERNALCELL MOD GERANEXTERNALCELL RMV GERANEXTERNALCELL	LOFD-001020 / TDLOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN	Meaning: Indicates the mobile network code of the external cell. The PLMN consists of the MCC and the MNC. The MCC consists of three digits. The MNC consists of two to three digits. For example, MCC = 123, MNC = 45, PLMN = 12345. GUI Value Range: 2~3 characters Unit: None Actual Value Range: 00~99,000~999 Default Value: None
GeranExternalCell	GeranCellId	ADD GERANEXTERNALCELL LST GERANEXTERNALCELL MOD GERANEXTERNALCELL RMV GERANEXTERNALCELL	LOFD-001020 / TDLOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN	Meaning: Indicates the cell ID of the external GERAN cell. It uniquely identifies a GERAN cell within the area specified by the location area identity (LAI). An LAI consists of the MCC, MNC, and LAC. GUI Value Range: 0~65535 Unit: None Actual Value Range: 0~65535 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
GeranExternalCell	Lac	ADD GERANEXTERNALCELL LST GERANEXTERNALCELL MOD GERANEXTERNALCELL RMV GERANEXTERNALCELL	LOFD-001023 / TDLOFD-001023 LOFD-001069	SRVCC to GERAN CS Fallback with LAI to GERAN	Meaning: Indicates the LAC of the external GERAN cell. It uniquely identifies a location within a PLMN. GUI Value Range: 1~65533,65535 Unit: None Actual Value Range: 1~65533,65535 Default Value: None
GeranExternalCell	RacCfgInd	ADD GERANEXTERNALCELL MOD GERANEXTERNALCELL LST GERANEXTERNALCELL	LOFD-001020 / TDLOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN	Meaning: Indicates whether to set the routing area code (RAC) of the external GERAN cell. This parameter is required if the external GERAN cell supports packet switched (PS) services. GUI Value Range: NOT_CFG(Not configure), CFG(Configure) Unit: None Actual Value Range: NOT_CFG, CFG Default Value: NOT_CFG(Not configure)

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
GeranExternalCell	Rac	ADD GERANEXTERNALCELL MOD GERANEXTERNALCELL LST GERANEXTERNALCELL	LOFD-001034 / LOFD-001034 LOFD-001053 LOFD-001020 / TDLOFD-001020	CS Fallback to GERAN Flash CS Fallback to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN	Meaning: Indicates the routing area code. GUI Value Range: 0~255 Unit: None Actual Value Range: 0~255 Default Value: 0
GeranExternalCell	BandIndicator	ADD GERANEXTERNALCELL MOD GERANEXTERNALCELL LST GERANEXTERNALCELL	LOFD-001020 / TDLOFD-001020 LOFD-001023 / TDLOFD-001023 LOFD-001034 / TDLOFD-001034 LOFD-001046 / TDLOFD-001046 LOFD-001073 / TDLOFD-001073	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN Service based inter-RAT handover to GERAN Distance based inter-RAT handover to GERAN	Meaning: Indicates the GERAN band indicator. If the GERAN ARFCN is a value in the range of 512 to 810, the value of this parameter is used to indicate whether the GERAN ARFCN is in the 1800 MHz band or the 1900 MHz band. If the GERAN ARFCN is not in either band, this parameter does not take effect and can be set to either value. For details, see 3GPP TS 36.331. GUI Value Range: GSM_dcs1800, GSM_pcs1900 Unit: None Actual Value Range: GSM_dcs1800, GSM_pcs1900 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
GeranExternalCell	GeranArfcn	ADD GERANEXTERNALCELL MOD GERANEXTERNALCELL LST GERANEXTERNALCELL	LOFD-001020 / TDLOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN	Meaning: Indicates the GERAN BCCH ARFCN. For details, see 3GPP TS 45.005. GUI Value Range: 0~1023 Unit: None Actual Value Range: 0~1023 Default Value: None
GeranExternalCell	NetworkColourCode	ADD GERANEXTERNALCELL MOD GERANEXTERNALCELL LST GERANEXTERNALCELL	LOFD-001020 / TDLOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN	Meaning: Indicates the network color code (NCC) of GERAN. It is operator-specific and identifies a network within the whole country.The base transceiver station identity code (BSIC) consists of the NCC and the base station color code (BCC). For details, see 3GPP TS 23.003. GUI Value Range: 0~7 Unit: None Actual Value Range: 0~7 Default Value: None
GeranExternalCell	BaseStationColourCode	ADD GERANEXTERNALCELL MOD GERANEXTERNALCELL LST GERANEXTERNALCELL	LOFD-001020 / TDLOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN	Meaning: Indicates the base station color code (BCC) of an external GERAN cell. This parameter is used to discriminate external GERAN cells working at the same frequency.The base transceiver station identity code (BSIC) consists of the BCC and the network color code (NCC). For details, see 3GPP TS 23.003. GUI Value Range: 0~7 Unit: None Actual Value Range: 0~7 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
GeranExternalCell	DtmInd	ADD GERANEXTERNALCELL MOD GERANEXTERNALCELL LST GERANEXTERNALCELL	LOFD-001020 / TDLOFD-001020 LOFD-001046 / TDLOFD-001046	PS Inter-RAT Mobility between E-UTRAN and GERAN Service based inter-RAT handover to GERAN	Meaning: Indicates whether the cell supports the dual transmission mode. If the cell supports the dual transmission mode, both CS and PS services can be handed over to the cell at the same time. Otherwise, only the CS service or the PS service can be handed over to the cell. GUI Value Range: DTM_AVAILABLE, DTM_NOT_AVAILABLE Unit: None Actual Value Range: DTM_AVAILABLE, DTM_NOT_AVAILABLE Default Value: DTM_NOT_AVAILABLE
GeranExternalCellPlmn	GeranCellId	ADD GERANEXTERNALCELLPLMN LST GERANEXTERNALCELLPLMN RMV GERANEXTERNALCELLPLMN	LOFD-001036 / TDLOFD-001036 LOFD-001037 / TDLOFD-001037	RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier	Meaning: Indicates the identity of the external GERAN cell. It uniquely identifies a GERAN cell within a PLMN. GUI Value Range: 0~65535 Unit: None Actual Value Range: 0~65535 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
GeranExternalCellPlmn	Lac	ADD GERANEXTERNALCELLPLMN LST GERANEXTERNALCELLPLMN RMV GERANEXTERNALCELLPLMN	LOFD-001036 / TDLOFD-001036 LOFD-001037 / TDLOFD-001037	RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier	Meaning: Indicates the LAC of the external GERAN cell. GUI Value Range: 1~65533,65535 Unit: None Actual Value Range: 1~65533,65535 Default Value: None
GeranExternalCellPlmn	Mcc	ADD GERANEXTERNALCELLPLMN LST GERANEXTERNALCELLPLMN RMV GERANEXTERNALCELLPLMN	LOFD-001036 / TDLOFD-001036 LOFD-001037 / TDLOFD-001037	RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier	Meaning: Indicates the mobile country code of the external cell. A PLMN ID is comprised of an MCC and an MNC. The MCC consists of three digits. The MNC consists of two to three digits. For example, if the MCC is 123 and the MNC is 45, then the PLMN ID is 12345. GUI Value Range: 3 characters Unit: None Actual Value Range: 000~999 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
GeranExternalCellPlmn	Mnc	ADD GERANEXTERNALCELLPLMN LST GERANEXTERNALCELLPLMN RMV GERANEXTERNALCELLPLMN	LOFD-001036 / TDLOFD-001036 LOFD-001037 / TDLOFD-001037	RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier	Meaning: Indicates the mobile network code of the external cell. A PLMN ID is comprised of an MCC and an MNC. The MCC consists of three digits. The MNC consists of two to three digits. For example, if the MCC is 123 and the MNC is 45, then the PLMN ID is 12345. GUI Value Range: 2~3 characters Unit: None Actual Value Range: 00~99,000~999 Default Value: None
GeranExternalCellPlmn	ShareMcc	ADD GERANEXTERNALCELLPLMN LST GERANEXTERNALCELLPLMN RMV GERANEXTERNALCELLPLMN	LOFD-001036 / TDLOFD-001036 LOFD-001037 / TDLOFD-001037	RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier	Meaning: Indicates the mobile country code of the external GERAN cell that is shared among multiple operators. A PLMN ID is comprised of an MCC and an MNC. The MCC consists of three digits. The MNC consists of two to three digits. For example, if the MCC is 123 and the MNC is 45, then the PLMN ID is 12345. GUI Value Range: 3 characters Unit: None Actual Value Range: 000~999 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
GeranExternalCellPlmn	ShareMnc	ADD GERANEXTERNALCELLPLMN LST GERANEXTERNALCELLPLMN RMV GERANEXTERNALCELLPLMN	LOFD-001036 / TDLOFD-001036 LOFD-001037 / TDLOFD-001037	RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier	Meaning: Indicates the mobile network code of the external GERAN cell that is shared among multiple operators. A PLMN ID is comprised of an MCC and an MNC. The MCC consists of three digits. The MNC consists of two to three digits. For example, if the MCC is 123 and the MNC is 45, then the PLMN ID is 12345. GUI Value Range: 2~3 characters Unit: None Actual Value Range: 00~99,000~999 Default Value: None
GeranNcell	LocalCellId	ADD GERANNCELL DSP GERANRIMINFO LST GERANNCELL MOD GERANNCELL RMV GERANNCELL	None	None	Meaning: Indicates the cell ID of the Local cell. It uniquely identifies a cell within an eNodeB. GUI Value Range: 0~255 Unit: None Actual Value Range: 0~255 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
GeranNcell	Mcc	ADD GERANNCELL LST GERANNCELL MOD GERANNCELL RMV GERANNCELL DSP GERANRIMINFO	LOFD-001020 / TDLOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN	Meaning: Indicates the mobile country code of the neighboring GERAN cell. The PLMN consists of the MCC and the MNC. The MCC consists of three digits. The MNC consists of two to three digits. For example, MCC = 123, MNC = 45, PLMN = 12345. GUI Value Range: 3 characters Unit: None Actual Value Range: 000~999 Default Value: None
GeranNcell	Mnc	ADD GERANNCELL LST GERANNCELL MOD GERANNCELL RMV GERANNCELL DSP GERANRIMINFO	LOFD-001020 / TDLOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN	Meaning: Indicates the mobile network code of the neighboring GERAN cell. The PLMN consists of the MCC and the MNC. The MCC consists of three digits. The MNC consists of two to three digits. For example, MCC = 123, MNC = 45, PLMN = 12345. GUI Value Range: 2~3 characters Unit: None Actual Value Range: 00~99,000~999 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
GeranNcell	Lac	ADD GERANNCELL LST GERANNCELL MOD GERANNCELL RMV GERANNCELL DSP GERANRIMINFO	LOFD-001023 / TDLOFD-001023 LOFD-001053 LOFD-001069 LOFD-001020/ TDLOFD-001020	SRVCC to GERAN Flash CS Fallback to GERAN CS Fallback with LAI to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN	Meaning: Indicates the LAC of the GERAN neighboring cell. It uniquely identifies a location within a PLMN. GUI Value Range: 1~65533,65535 Unit: None Actual Value Range: 1~65533,65535 Default Value: None
GeranNcell	GeranCellId	ADD GERANNCELL LST GERANNCELL MOD GERANNCELL RMV GERANNCELL DSP GERANRIMINFO	LOFD-001020 / TDLOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN	Meaning: Indicates the cell ID of the GERAN neighboring cell. It uniquely identifies a GERAN cell within the area specified by the location area identity (LAI). An LAI consists of the MCC, MNC, and LAC. GUI Value Range: 0~65535 Unit: None Actual Value Range: 0~65535 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
CellStandardQci	InterRatHoGeranGroupId	MOD CELLSTANDARDQCI LST CELLSTANDARDQCI	LOFD-001020 / TDLOFD-001020 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN	Meaning: Indicates the ID of a parameter group related to inter-RAT handover to GERAN. GUI Value Range: 0~9 Unit: None Actual Value Range: 0~9 Default Value: 0
CellExtendedQci	InterRatHoGeranGroupId	ADD CELLEXTENDEDQCI MOD CELLEXTENDEDQCI LST CELLEXTENDEDQCI	LOFD-001020 / TDLOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN	Meaning: Indicates the ID of a parameter group related to inter-RAT handover to GERAN. GUI Value Range: 0~9 Unit: None Actual Value Range: 0~9 Default Value: 0
EutranInterNFreq	ConnFreqPriority	ADD EUTRANINTERNFREQ MOD EUTRANINTERNFREQ LST EUTRANINTERNFREQ	LBFD-0201802 / TDLBFD-00201802 LBFD-0201804 / TDLBFD-00201804	Coverage Based Inter-frequency Handover Distance Based Inter-frequency Handover	Meaning: Indicates the frequency priority based on which the eNodeB selects a target frequency for blind redirection or contains a frequency in a measurement configuration. If a blind redirection is triggered and the target neighboring cell is not specified, the eNodeB selects a target frequency based on this priority. If a measurement configuration is to be delivered, the eNodeB preferentially delivers a frequency with the highest priority. If this priority is set to 0 for a frequency, this frequency is not selected as the target frequency for a blind redirection. A larger value indicates a higher priority. GUI Value Range: 0~8 Unit: None Actual Value Range: 0~8 Default Value: 0

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
GeranRanShare	LocalCellId	ADD GERANRANSHARE LST GERANRANSHARE MOD GERANRANSHARE RMV GERANRANSHARE	None	None	Meaning: Indicates the cell identity of the local cell. It uniquely identifies a cell within an eNodeB. GUI Value Range: 0~255 Unit: None Actual Value Range: 0~255 Default Value: None
GeranRanShare	BcchGroupId	ADD GERANRANSHARE LST GERANRANSHARE MOD GERANRANSHARE RMV GERANRANSHARE	LOFD-001036 / TDLOFD-001036 TDLOFD-001112 LOFD-070206	RAN Sharing with Common Carrier MOCN Flexible Priority Based Camping Hybrid RAN Sharing	Meaning: Indicates the index of the GERAN carrier frequency group. It uniquely identifies a GERAN carrier frequency group. GUI Value Range: 0~31 Unit: None Actual Value Range: 0~31 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
GeranRanShare	Mcc	ADD GERANRANSHARE LST GERANRANSHARE MOD GERANRANSHARE RMV GERANRANSHARE	LOFD-001036 / TDLOFD-001036 TDLOFD-001112 LOFD-070206	RAN Sharing with Common Carrier MOCN Flexible Priority Based Camping Hybrid RAN Sharing	Meaning: Indicates the mobile country code (MCC) of the operator that shares the neighboring GERAN carrier frequency group. A public land mobile network (PLMN) ID is comprised of an MCC and a mobile network code (MNC). The MCC consists of three digits. The MNC consists of two to three digits. For example, if the MCC is 123 and the MNC is 45, then the PLMN ID is 12345. GUI Value Range: 3 characters Unit: None Actual Value Range: 000~999 Default Value: None
GeranRanShare	Mnc	ADD GERANRANSHARE LST GERANRANSHARE MOD GERANRANSHARE RMV GERANRANSHARE	LOFD-001036 / TDLOFD-001036 TDLOFD-001112 LOFD-070206	RAN Sharing with Common Carrier MOCN Flexible Priority Based Camping Hybrid RAN Sharing	Meaning: Indicates the mobile network code (MNC) of the operator that shares the neighboring GERAN carrier frequency group. A public land mobile network (PLMN) ID is comprised of a mobile country code (MCC) and an MNC. The MCC consists of three digits. The MNC consists of two to three digits. For example, if the MCC is 123 and the MNC is 45, then the PLMN ID is 12345. GUI Value Range: 2~3 characters Unit: None Actual Value Range: 00~99,000~999 Default Value: None

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MO	Parameter ID	MML Command	Feature ID	Feature Name	Description
InterRatHoGeranGroup	LocalCellId	ADD INTERRATHOGERANGROUP LST INTERRATHOGERANGROUP MOD INTERRATHOGERANGROUP RMV INTERRATHOGERANGROUP	None	None	Meaning: Indicates the local ID of the cell. It uniquely identifies a cell within a BS. GUI Value Range: 0~255 Unit: None Actual Value Range: 0~255 Default Value: None
InterRatHoGeranGroup	InterRatHoGeranGroupId	ADD INTERRATHOGERANGROUP LST INTERRATHOGERANGROUP MOD INTERRATHOGERANGROUP RMV INTERRATHOGERANGROUP	LOFD-001020 / TDLOFD-001020 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN	Meaning: Indicates the ID of the parameter group related to inter-RAT handover to GERAN. GUI Value Range: 0~9 Unit: None Actual Value Range: 0~9 Default Value: None

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10 Counters

Table 10-1 Counters

Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526729475	L.RIM.Load.W2E.Req	Number of times the eNodeB receives a load information request from a WCDMA network	Multi-mode: MRFD-090211 GSM: None UMTS: None LTE: LOFD-070203	eCoordinator based RIM process optimization RIM Based LTE Target Cell Selection
1526729476	L.RIM.Load.W2E.Resp	Number of times the eNodeB sends a load information response to a WCDMA network	Multi-mode: MRFD-090211 GSM: None UMTS: None LTE: LOFD-070203	eCoordinator based RIM process optimization RIM Based LTE Target Cell Selection
1526729477	L.RIM.Load.W2E.Update	Number of times the eNodeB sends a load information update to a WCDMA network	Multi-mode: MRFD-090211 GSM: None UMTS: None LTE: LOFD-070203	eCoordinator based RIM process optimization RIM Based LTE Target Cell Selection
1526729478	L.RIM.Load.W2E.Update.ACK	Number of times the eNodeB receives a load information update acknowledgment from a WCDMA network	Multi-mode: MRFD-090211 GSM: None UMTS: None LTE: LOFD-070203	eCoordinator based RIM process optimization RIM Based LTE Target Cell Selection

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526727317	L.E-RAB.NormRel.HOOut.QCI.1	Number of normal E-RAB releases for outgoing handovers of services with the QCI of 1 in a cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LBFD-00201801 LBFD-00201802 TDLOFD-001019 TDLBFD-00201801 TDLBFD-00201802	PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover
1526727318	L.E-RAB.NormRel.HOOut.QCI.2	Number of normal E-RAB releases for outgoing handovers of services with the QCI of 2 in a cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LBFD-00201801 LBFD-00201802 TDLOFD-001019 TDLBFD-00201801 TDLBFD-00201802	PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526727319	L.E-RAB.NormRel.HOOut.QCI.3	Number of normal E-RAB releases for outgoing handovers of services with the QCI of 3 in a cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LBFD-00201801 LBFD-00201802 TDLOFD-001019 TDLBFD-00201801 TDLBFD-00201802	PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover
1526727320	L.E-RAB.NormRel.HOOut.QCI.4	Number of normal E-RAB releases for outgoing handovers of services with the QCI of 4 in a cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LBFD-00201801 LBFD-00201802 TDLOFD-001019 TDLBFD-00201801 TDLBFD-00201802	PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526727321	L.E-RAB.NormRel.HOOut.QCI.5	Number of normal E-RAB releases for outgoing handovers of services with the QCI of 5 in a cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LBFD-00201801 LBFD-00201802 TDLOFD-001019 TDLBFD-00201801 TDLBFD-00201802	PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover
1526727322	L.E-RAB.NormRel.HOOut.QCI.6	Number of normal E-RAB releases for outgoing handovers of services with the QCI of 6 in a cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LBFD-00201801 LBFD-00201802 TDLOFD-001019 TDLBFD-00201801 TDLBFD-00201802	PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526727323	L.E-RAB.NormRel.HOOut.QCI.7	Number of normal E-RAB releases for outgoing handovers of services with the QCI of 7 in a cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LBFD-00201801 LBFD-00201802 TDLOFD-001019 TDLBFD-00201801 TDLBFD-00201802	PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover
1526727324	L.E-RAB.NormRel.HOOut.QCI.8	Number of normal E-RAB releases for outgoing handovers of services with the QCI of 8 in a cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LBFD-00201801 LBFD-00201802 TDLOFD-001019 TDLBFD-00201801 TDLBFD-00201802	PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526727325	L.E-RAB.NormRel.HOOut.QCI.9	Number of normal E-RAB releases for outgoing handovers of services with the QCI of 9 in a cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LBFD-00201801 LBFD-00201802 TDLOFD-001019 TDLBFD-00201801 TDLBFD-00201802	PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover
1526727326	L.E-RAB.AbnormRel.HOOut.QCI.1	Number of abnormal E-RAB releases for outgoing handovers of services with the QCI of 1 in a cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LBFD-00201801 LBFD-00201802 TDLOFD-001019 TDLBFD-00201801 TDLBFD-00201802	PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526727327	L.E-RAB.AbnormRel.HOOut.QCI.2	Number of abnormal E-RAB releases for outgoing handovers of services with the QCI of 2 in a cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LBFD-00201801 LBFD-00201802 TDLOFD-001019 TDLBFD-00201801 TDLBFD-00201802	PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover
1526727328	L.E-RAB.AbnormRel.HOOut.QCI.3	Number of abnormal E-RAB releases for outgoing handovers of services with the QCI of 3 in a cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LBFD-00201801 LBFD-00201802 TDLOFD-001019 TDLBFD-00201801 TDLBFD-00201802	PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526727329	L.E-RAB.AbnormRel.HOOut.QCI.4	Number of abnormal E-RAB releases for outgoing handovers of services with the QCI of 4 in a cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LBFD-00201801 LBFD-00201802 TDLOFD-001019 TDLBFD-00201801 TDLBFD-00201802	PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover
1526727330	L.E-RAB.AbnormRel.HOOut.QCI.5	Number of abnormal E-RAB releases for outgoing handovers of services with the QCI of 5 in a cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LBFD-00201801 LBFD-00201802 TDLOFD-001019 TDLBFD-00201801 TDLBFD-00201802	PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526727331	L.E-RAB.AbnormRel.HOOut.QCI.6	Number of abnormal E-RAB releases for outgoing handovers of services with the QCI of 6 in a cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LBFD-00201801 LBFD-00201802 TDLOFD-001019 TDLBFD-00201801 TDLBFD-00201802	PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover
1526727332	L.E-RAB.AbnormRel.HOOut.QCI.7	Number of abnormal E-RAB releases for outgoing handovers of services with the QCI of 7 in a cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LBFD-00201801 LBFD-00201802 TDLOFD-001019 TDLBFD-00201801 TDLBFD-00201802	PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526727333	L.E-RAB.AbnormRel.HOOut.QCI.8	Number of abnormal E-RAB releases for outgoing handovers of services with the QCI of 8 in a cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LBFD-00201801 LBFD-00201802 TDLOFD-001019 TDLBFD-00201801 TDLBFD-00201802	PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover
1526727334	L.E-RAB.AbnormRel.HOOut.QCI.9	Number of abnormal E-RAB releases for outgoing handovers of services with the QCI of 9 in a cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LBFD-00201801 LBFD-00201802 TDLOFD-001019 TDLBFD-00201801 TDLBFD-00201802	PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728246	L.E-RAB.NormRel.HOOut	Total number of normal E-RAB releases for outgoing handovers in a cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LBFD-00201801 LBFD-00201802 TDLOFD-001019 TDLBFD-00201801 TDLBFD-00201802	PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover
1526728247	L.E-RAB.AbnormRel.HOOut	Total number of abnormal E-RAB releases for outgoing handovers in a cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LBFD-00201801 LBFD-00201802 TDLOFD-001019 TDLBFD-00201801 TDLBFD-00201802	PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover PS Inter-RAT Mobility between E-UTRAN and UTRAN Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526729562	L.IRATHO.NCell.E2G.PrepAttOut	Number of handover attempts from a specific E-UTRAN cell to a specific GERAN cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 LOFD-001023 LOFD-001034 TDLOFD-001020 TDLOFD-001023 TDLOFD-001034	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN
1526729563	L.IRATHO.NCell.E2G.ExecAttOut	Number of handover executions from a specific E-UTRAN cell to a specific GERAN cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 LOFD-001023 LOFD-001034 TDLOFD-001020 TDLOFD-001023 TDLOFD-001034	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN
1526729564	L.IRATHO.NCell.E2G.ExecSuccOut	Number of successful handovers from a specific E-UTRAN cell to a specific GERAN cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 LOFD-001023 LOFD-001034 TDLOFD-001020 TDLOFD-001023 TDLOFD-001034	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526729565	L.IRATHO.NCell.E2W.PrepAttOut	Number of handover attempts from a specific E-UTRAN cell to a specific WCDMA cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN
1526729566	L.IRATHO.NCell.E2W.ExecAttOut	Number of handover executions from a specific E-UTRAN cell to a specific WCDMA cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN
1526729567	L.IRATHO.NCell.E2W.ExecSuccOut	Number of successful handovers from a specific E-UTRAN cell to a specific WCDMA cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526727034	Handover from E-UTRAN to WCDMA Success Rate	Success rate of inter-RAT handovers from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN
1526727035	Handover from E-UTRAN to GERAN Success Rate	Success rate of inter-RAT handovers from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 LOFD-001023 TDLOFD-001020 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN
1526728866	L.IRATHO.G2E.PrepAttIn	Number of inter-RAT handover attempts from GERAN to E-UTRAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 LOFD-001023 TDLOFD-001020 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728867	L.IRATHO.G2E.PrepInSucc	Number of successful inter-RAT handover preparations from GERAN to E-UTRAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 LOFD-001023 TDLOFD-001020 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN
1526728868	L.IRATHO.G2E.ExecSuccIn	Number of successful inter-RAT handovers from GERAN to E-UTRAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 LOFD-001023 TDLOFD-001020 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN
1526728869	L.IRATHO.G2E.PrepInFail.TgtNotAllow	Number of inter-RAT handover preparation failures from GERAN in the target cell due to Handover Target not allowed	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 LOFD-001023 TDLOFD-001020 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728870	L.IRATHO.T2E.PrepAttIn	Number of inter-RAT handover attempts from TD-SCDMA network to E-UTRAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN
1526728871	L.IRATHO.T2E.PrepInSucc	Number of successful inter-RAT handover preparations from TD-SCDMA network to E-UTRAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN
1526728872	L.IRATHO.T2E.ExecSuccIn	Number of successful inter-RAT handovers from TD-SCDMA network to E-UTRAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728873	L.IRATHO.T2E.PrepInFail.TgtNotAllow	Number of inter-RAT handover preparation failures from TD-SCDMA network in the target cell due to Handover Target not allowed	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN
1526728874	L.IRATHO.W2E.PrepAttIn	Number of inter-RAT handover attempts from WCDMA network to E-UTRAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN
1526728875	L.IRATHO.W2E.PrepInSucc	Number of successful inter-RAT handover preparations from WCDMA network to E-UTRAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728876	L.IRATHO.W2E.ExecSuccIn	Number of successful inter-RAT handovers from WCDMA network to E-UTRAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN
1526728877	L.IRATHO.W2E.PrepInFail.TgtNotAllow	Number of inter-RAT handover preparation failures from WCDMA network in the target cell due to Handover Target not allowed	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728173	L.HHO.Ncell.PingPongHo	Number of ping-pong handovers between two specific cells	Multi-mode: None GSM: None UMTS: None LTE: LBFD-00201801 LBFD-00201802 LOFD-001019 LOFD-001020 LOFD-001021 TDLBFD-00201801 TDLBFD-00201802 TDLOFD-001019 TDLOFD-001020	Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and CDMA2000 Coverage Based Intra-frequency Handover Coverage Based Inter-frequency Handover PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and GERAN
1526728742	L.MeasCtrl.WCDMA.Add	Number of measurement control messages delivered to WCDMA network triggered by poor coverage in a cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 TDLOFD-001019	PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728743	L.MeasCtrl.GERAN.Add	Number of measurement control messages delivered to GERAN triggered by poor coverage in a cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 TDLOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN
1526726989	L.IRATHO.E2W.PrepAttOut	Number of inter-RAT handover attempts from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN
1526726990	L.IRATHO.E2W.ExecAttOut	Number of inter-RAT handover executions from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526726991	L.IRATHO.E2W.ExecSuccOut	Number of successful inter-RAT handovers from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN
1526726992	L.IRATHO.E2G.PrepAttOut	Number of inter-RAT handover attempts from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 LOFD-001023 LOFD-001034 TDLOFD-001020 TDLOFD-001023 TDLOFD-001034	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN
1526726993	L.IRATHO.E2G.ExecAttOut	Number of inter-RAT handover executions from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 LOFD-001023 LOFD-001034 TDLOFD-001020 TDLOFD-001023 TDLOFD-001034	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526726994	L.IRATHO.E2G.ExecSuccOut	Number of successful inter-RAT handovers from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 LOFD-001023 LOFD-001034 TDLOFD-001020 TDLOFD-001023 TDLOFD-001034	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN
1526727226	L.IRATHO.E2T.PrepAttOut	Number of inter-RAT Handover Attempts from E-UTRAN to TD-SCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 TDLOFD-001019 TDLOFD-001022	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN
1526727227	L.IRATHO.E2T.ExecAttOut	Number of inter-RAT handover executions from E-UTRAN to TD-SCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 TDLOFD-001019 TDLOFD-001022	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526727228	L.IRATHO.E2T.ExecSuccOut	Number of successful inter-RAT handovers from E-UTRAN to TD-SCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 TDLOFD-001019 TDLOFD-001022	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN
1526728306	L.IRATHO.E2G.Prep.FailOut.MME	Number of inter-RAT handover preparation failures from E-UTRAN to GERAN because of faults on the MME side	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001034 TDLOFD-001034 LOFD-001020 TDLOFD-001020 LOFD-001046 TDLOFD-001046 LOFD-001073 LOFD-001023 TDLOFD-001023	CS Fallback to GERAN CS Fallback to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN Service based inter-RAT handover to GERAN Service based inter-RAT handover to GERAN Distance based inter-RAT handover to GERAN SRVCC to GERAN SRVCC to GERAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728307	L.IRATHO.E2G.Prep.FailOut.NoReply	Number of inter-RAT handover preparation failures from E-UTRAN to GERAN because of no responses from GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001034 TDLOFD-001034 LOFD-001020 TDLOFD-001020 LOFD-001046 TDLOFD-001046 LOFD-001073 LOFD-001023 TDLOFD-001023	CS Fallback to GERAN CS Fallback to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN Service based inter-RAT handover to GERAN Service based inter-RAT handover to GERAN Distance based inter-RAT handover to GERAN SRVCC to GERAN SRVCC to GERAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728308	L.IRATHO.E2G.Prep.FailOut.PrepFailure	Number of inter-RAT handover preparation failures from E-UTRAN to GERAN due tobecause GERAN cells send handover preparation failure messages	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001034 TDLOFD-001034 LOFD-001020 TDLOFD-001020 LOFD-001046 TDLOFD-001046 LOFD-001073 LOFD-001023 TDLOFD-001023	CS Fallback to GERAN CS Fallback to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN Service based inter-RAT handover to GERAN Service based inter-RAT handover to GERAN Distance based inter-RAT handover to GERAN SRVCC to GERAN SRVCC to GERAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728309	L.IRATHO.E2T.Prep.FailOut.MME	Number of inter-RAT handover preparation failures from E-UTRAN to TD-SCDMA network because of faults on the MME side	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001033 TDLOFD-001033 LOFD-001019 TDLOFD-001019 LOFD-001043 TDLOFD-001043 LOFD-001072 LOFD-001022 TDLOFD-001022	CS Fallback to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN Service based inter-RAT handover to UTRAN Service based inter-RAT handover to UTRAN Distance based inter-RAT handover to UTRAN SRVCC to UTRAN SRVCC to UTRAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728310	L.IRATHO.E2T.Prep.FailOut.NoReply	Number of inter-RAT handover preparation failures from E-UTRAN to TD-SCDMA network due tobecause of no responses from TD-SCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001033 TDLOFD-001033 LOFD-001019 TDLOFD-001019 LOFD-001043 TDLOFD-001043 LOFD-001072 LOFD-001022 TDLOFD-001022	CS Fallback to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN Service based inter-RAT handover to UTRAN Service based inter-RAT handover to UTRAN Distance based inter-RAT handover to UTRAN SRVCC to UTRAN SRVCC to UTRAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728311	L.IRATHO.E2T.Prep.FailOut.PrepFailure	Number of inter-RAT handover preparation failures because TD-SCDMA network cells send handover preparation failure messages	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001033 TDLOFD-001033 LOFD-001019 TDLOFD-001019 LOFD-001043 TDLOFD-001043 LOFD-001072 LOFD-001022 TDLOFD-001022	CS Fallback to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN Service based inter-RAT handover to UTRAN Service based inter-RAT handover to UTRAN Distance based inter-RAT handover to UTRAN SRVCC to UTRAN SRVCC to UTRAN
1526728312	L.IRATHO.BlindHO.E2W.ExecAttOut	Number of inter-RAT blind handovers executions from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728313	L.IRATHO.BlindHO.E2W.ExecSuccOut	Number of successful inter-RAT blind handovers from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN
1526728314	L.IRATHO.BlindHO.E2G.ExecAttOut	Number of inter-RAT blind handovers executions from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 LOFD-001023 LOFD-001034 TDLOFD-001020 TDLOFD-001023 TDLOFD-001034	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN
1526728315	L.IRATHO.BlindHO.E2G.ExecSuccOut	Number of successful inter-RAT blind handovers from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 LOFD-001023 LOFD-001034 TDLOFD-001020 TDLOFD-001023 TDLOFD-001034	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728316	L.IRATHO.E2W.Prep.FailOut.MME	Number of inter-RAT handover preparation failures from E-UTRAN to WCDMA network because of faults on the MME side	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001033 TDLOFD-001033 LOFD-001019 TDLOFD-001019 LOFD-001043 TDLOFD-001043 LOFD-001072 LOFD-001022 TDLOFD-001022	CS Fallback to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN Service based inter-RAT handover to UTRAN Service based inter-RAT handover to UTRAN Distance based inter-RAT handover to UTRAN SRVCC to UTRAN SRVCC to UTRAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728317	L.IRATHO.E2W.Prep.FailOut.PrepFailure	Number of inter-RAT handover preparation failures from E-UTRAN to WCDMA network because WCDMA network cells send handover preparation failure messages	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001033 TDLOFD-001033 LOFD-001019 TDLOFD-001019 LOFD-001043 TDLOFD-001043 LOFD-001072 LOFD-001022 TDLOFD-001022	CS Fallback to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN Service based inter-RAT handover to UTRAN Service based inter-RAT handover to UTRAN Distance based inter-RAT handover to UTRAN SRVCC to UTRAN SRVCC to UTRAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728318	L.IRATHO.E2W.Prep.FailOut.NoReply	Number of inter-RAT handover preparation failures from E-UTRAN to WCDMA network because of no responses from WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001033 TDLOFD-001033 LOFD-001019 TDLOFD-001019 LOFD-001043 TDLOFD-001043 LOFD-001072 LOFD-001022 TDLOFD-001022	CS Fallback to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN Service based inter-RAT handover to UTRAN Service based inter-RAT handover to UTRAN Distance based inter-RAT handover to UTRAN SRVCC to UTRAN SRVCC to UTRAN
1526728326	L.RRCRedirection.E2W	Number of redirections from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001033 TDLOFD-001019 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN CS Fallback to UTRAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728327	L.RRCRedirection.E2G	Number of redirections from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 LOFD-001034 TDLOFD-001020 TDLOFD-001034	PS Inter-RAT Mobility between E-UTRAN and GERAN CS Fallback to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN CS Fallback to GERAN
1526728328	L.IRATHO.BlindHO.E2W.PrepAttOut	Number of inter-RAT blind handover attempts from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001033 TDLOFD-001033 LOFD-001019 TDLOFD-001019	CS Fallback to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN
1526728329	L.IRATHO.BlindHO.E2G.PrepAttOut	Number of inter-RAT blind handover attempts from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001034 TDLOFD-001034 LOFD-001020 TDLOFD-001020	CS Fallback to GERAN CS Fallback to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728330	L.RRCRedirection.E2W.PrepAtt	Number of redirection preparations from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001033 TDLOFD-001019 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN CS Fallback to UTRAN
1526728331	L.RRCRedirection.E2G.PrepAtt	Number of redirection preparations from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 LOFD-001034 TDLOFD-001020 TDLOFD-001034	PS Inter-RAT Mobility between E-UTRAN and GERAN CS Fallback to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN CS Fallback to GERAN
1526728441	L.CCO.E2G.ExecAttOut	Number of CCO executions from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 TDLOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN
1526728442	L.CCO.E2G.ExecSuccOut	Number of successful CCOs from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 TDLOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728468	L.IRATHO.E2W.NoNRT	Number of inter-RAT initiation failures from E-UTRAN to WCDMA network because the adjacent cell is not on the NRT of the source cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-002002 TDLOFD-001019 TDLBFD-002002 LOFD-001022 TDLOFD-001022	PS Inter-RAT Mobility between E-UTRAN and UTRAN Inter-RAT ANR PS Inter-RAT Mobility between E-UTRAN and UTRAN Transport Channel Management SRVCC to UTRAN SRVCC to UTRAN
1526728469	L.IRATHO.E2T.NoNRT	Number of inter-RAT initiation failures from E-UTRAN to TD-SCDMA network because the adjacent cell is not on the NRT of the source cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-002002 TDLOFD-001019 TDLBFD-002002 LOFD-001022 TDLOFD-001022	PS Inter-RAT Mobility between E-UTRAN and UTRAN Inter-RAT ANR PS Inter-RAT Mobility between E-UTRAN and UTRAN Transport Channel Management SRVCC to UTRAN SRVCC to UTRAN
1526728470	L.IRATHO.E2G.NoNRT	Number of inter-RAT initiation failures from E-UTRAN to GERAN because the adjacent cell is not on the NRT of the source cell	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 LOFD-002002 TDLOFD-001020 TDLBFD-002002 LOFD-001023 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and GERAN Inter-RAT ANR PS Inter-RAT Mobility between E-UTRAN and GERAN Transport Channel Management SRVCC to GERAN SRVCC to GERAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728560	L.IRATHO.E2W.NoData.ExecAttOut	Number of inter-RAT handover executions from E-UTRAN to WCDMA network triggered for UEs that do not transmit or receive data	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033 LOFD-001105 TDLOFD-001105	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN Dynamic DRX Dynamic DRX
1526728561	L.IRATHO.E2W.NoData.ExecSuccOut	Number of successful inter-RAT handovers from E-UTRAN to WCDMA network triggered for UEs that do not transmit or receive data	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033 LOFD-001105 TDLOFD-001105	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN Dynamic DRX Dynamic DRX

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728746	L.IRATHO.BlindHO.E2T.PrepAttOut	Number of inter-RAT blind handover attempts from E-UTRAN to TD-SCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001033 TDLOFD-001033 LOFD-001019 TDLOFD-001019	CS Fallback to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN
1526728747	L.IRATHO.BlindHO.E2T.ExecAttOut	Number of inter-RAT blind handover executions from E-UTRAN to TD-SCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN
1526728748	L.IRATHO.BlindHO.E2T.ExecSuccOut	Number of successful inter-RAT blind handovers from E-UTRAN to TD-SCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728751	L.RRCRedirection.E2T.PrepAtt	Number of redirection preparations to TD-SCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001033 TDLOFD-001019 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN CS Fallback to UTRAN
1526728752	L.RRCRedirection.E2T	Number of redirections to TD-SCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001033 TDLOFD-001019 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN CS Fallback to UTRAN
1526728891	L.IRATHO.E2W.MMEAbnormRsp	Number of responses for abnormal causes received by the eNodeB from the MME during inter-RAT handover executions from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 TDLOFD-001019 LOFD-001022 TDLOFD-001022	PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN SRVCC to UTRAN
1526728892	L.IRATHO.E2G.MMEAbnormRsp	Number of responses for abnormal causes received by the eNodeB from the MME during inter-RAT handover executions from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 TDLOFD-001020 LOFD-001023 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN SRVCC to GERAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728893	L.IRATHO.E2T.MMEAbnormRsp	Number of responses for abnormal causes received by the eNodeB from the MME during inter-RAT handover executions from E-UTRAN to TD-SCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 TDLOFD-001019 LOFD-001022 TDLOFD-001022	PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN SRVCC to UTRAN
1526728897	L.CCO.E2G.MMEAbnormRsp	Number of responses for abnormal causes received by the eNodeB from the MME during CCO executions from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 TDLOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN
1526729432	L.IRATHO.E2W.TimeAvg	Average handover duration from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526729433	L.IRATHO.E2G.TimeAvg	Average handover duration from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 LOFD-001023 LOFD-001034 TDLOFD-001020 TDLOFD-001023 TDLOFD-001034	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN
1526729434	L.IRATHO.E2T.TimeAvg	Average handover duration from E-UTRAN to TD-SCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN
1526729485	L.IRATHO.E2W.Coverage.PrepAttOut	Number of coverage-based handover preparation attempts from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 TDLOFD-001019 LOFD-001022 TDLOFD-001022	PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN SRVCC to UTRAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526729486	L.IRATHO.E2W.Coverage.ExecAttOut	Number of coverage-based handover execution attempts from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 TDLOFD-001019 LOFD-001022 TDLOFD-001022	PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN SRVCC to UTRAN
1526729487	L.IRATHO.E2W.Coverage.ExecSuccOut	Number of successful coverage-based handover executions from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 TDLOFD-001019 LOFD-001022 TDLOFD-001022	PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN SRVCC to UTRAN
1526729488	L.IRATHO.E2G.Coverage.PrepAttOut	Number of coverage-based handover preparation attempts from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 TDLOFD-001020 LOFD-001023 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN SRVCC to GERAN
1526729489	L.IRATHO.E2G.Coverage.ExecAttOut	Number of coverage-based handover execution attempts from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 TDLOFD-001020 LOFD-001023 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN SRVCC to GERAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526729490	L.IRATHO.E2G.Coverage.ExecSuccOut	Number of successful coverage-based handover executions from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 TDLOFD-001020 LOFD-001023 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN SRVCC to GERAN
1526729491	L.RRCRedirection.E2W.Coverage	Number of coverage-based redirections from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 TDLOFD-001019	PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN
1526729492	L.RRCRedirection.E2G.Coverage	Number of coverage-based redirections from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 TDLOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN
1526730018	L.IRATHO.E2W.Service.PrepAttOut	Number of service-based handover preparation attempts from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001043 TDLOFD-001043 LOFD-001022 TDLOFD-001022	Service based inter-RAT handover to UTRAN Service based inter-RAT handover to UTRAN SRVCC to UTRAN SRVCC to UTRAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526730019	L.IRATHO.E2W.Service.ExecAttOut	Number of service-based handover execution attempts from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001043 TDLOFD-001043 LOFD-001022 TDLOFD-001022	Service based inter-RAT handover to UTRAN Service based inter-RAT handover to UTRAN SRVCC to UTRAN SRVCC to UTRAN
1526730020	L.IRATHO.E2W.Service.ExecSuccOut	Number of successful service-based handover executions from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001043 TDLOFD-001043 LOFD-001022 TDLOFD-001022	Service based inter-RAT handover to UTRAN Service based inter-RAT handover to UTRAN SRVCC to UTRAN SRVCC to UTRAN
1526730021	L.IRATHO.E2G.Service.PrepAttOut	Number of service-based handover preparation attempts from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001046 TDLOFD-001046 LOFD-001023 TDLOFD-001023	Service based inter-RAT handover to GERAN Service based inter-RAT handover to GERAN SRVCC to GERAN SRVCC to GERAN
1526730022	L.IRATHO.E2G.Service.ExecAttOut	Number of service-based handover execution attempts from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001046 TDLOFD-001046 LOFD-001023 TDLOFD-001023	Service based inter-RAT handover to GERAN Service based inter-RAT handover to GERAN SRVCC to GERAN SRVCC to GERAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526730023	L.IRATHO.E2G.Service.ExecSuccOut	Number of successful service-based handover executions from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001046 TDLOFD-001046 LOFD-001023 TDLOFD-001023	Service based inter-RAT handover to GERAN Service based inter-RAT handover to GERAN SRVCC to GERAN SRVCC to GERAN
1526730024	L.RRCRedirection.E2W.Service	Number of service-based redirections from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001043 TDLOFD-001043	Service based inter-RAT handover to UTRAN Service based inter-RAT handover to UTRAN
1526730025	L.RRCRedirection.E2G.Service	Number of service-based redirections from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001046 TDLOFD-001046	Service based inter-RAT handover to GERAN Service based inter-RAT handover to GERAN
1526730026	L.IRATHO.E2W.ULquality.PrepAttOut	Number of UL-quality-based handover preparation attempts from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 TDLOFD-001019 LOFD-001022 TDLOFD-001022	PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN SRVCC to UTRAN
1526730027	L.IRATHO.E2W.ULquality.ExecAttOut	Number of UL-quality-based handover execution attempts from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 TDLOFD-001019 LOFD-001022 TDLOFD-001022	PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN SRVCC to UTRAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526730028	L.IRATHO.E2W.ULquality.ExecSuccOut	Number of successful UL-quality-based handover executions from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 TDLOFD-001019 LOFD-001022 TDLOFD-001022	PS Inter-RAT Mobility between E-UTRAN and UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN SRVCC to UTRAN
1526730029	L.IRATHO.E2G.ULquality.PrepAttOut	Number of UL-quality-based handover preparation attempts from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 TDLOFD-001020 LOFD-001023 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN SRVCC to GERAN
1526730030	L.IRATHO.E2G.ULquality.ExecAttOut	Number of UL-quality-based handover execution attempts from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 TDLOFD-001020 LOFD-001023 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN SRVCC to GERAN
1526730031	L.IRATHO.E2G.ULquality.ExecSuccOut	Number of successful UL-quality-based handover executions from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 TDLOFD-001020 LOFD-001023 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN SRVCC to GERAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526730034	L.IRATHO.E2W.Distance.PrepAttOut	Number of distance-based handover preparation attempts from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001072 TDLOFD-001072 LOFD-001022 TDLOFD-001022	Distance based inter-RAT handover to UTRAN Distance based inter-RAT handover to UTRAN SRVCC to UTRAN SRVCC to UTRAN
1526730035	L.IRATHO.E2W.Distance.ExecAttOut	Number of distance-based handover execution attempts from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001072 TDLOFD-001072 LOFD-001022 TDLOFD-001022	Distance based inter-RAT handover to UTRAN Distance based inter-RAT handover to UTRAN SRVCC to UTRAN SRVCC to UTRAN
1526730036	L.IRATHO.E2W.Distance.ExecSuccOut	Number of successful distance-based handover executions from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001072 TDLOFD-001072 LOFD-001022 TDLOFD-001022	Distance based inter-RAT handover to UTRAN Distance based inter-RAT handover to UTRAN SRVCC to UTRAN SRVCC to UTRAN
1526730037	L.IRATHO.E2G.Distance.PrepAttOut	Number of distance-based handover preparation attempts from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001073 TDLOFD-001073 LOFD-001023 TDLOFD-001023	Distance based inter-RAT handover to GERAN Distance based inter-RAT handover to GERAN SRVCC to GERAN SRVCC to GERAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526730038	L.IRATHO.E2G.Distance.ExecAttOut	Number of distance-based handover execution attempts from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001073 TDLOFD-001073 LOFD-001023 TDLOFD-001023	Distance based inter-RAT handover to GERAN Distance based inter-RAT handover to GERAN SRVCC to GERAN SRVCC to GERAN
1526730039	L.IRATHO.E2G.Distance.ExecSuccOut	Number of successful distance-based handover executions from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001073 TDLOFD-001073 LOFD-001023 TDLOFD-001023	Distance based inter-RAT handover to GERAN Distance based inter-RAT handover to GERAN SRVCC to GERAN SRVCC to GERAN
1526730848	L.IRATHO.BlindHO.E2W.MMEAbnormRsp	Number of responses with abnormal causes received by the eNodeB from the MME during inter-RAT blind handover executions from E-UTRAN to WCDMA network	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019	PS Inter-RAT Mobility between E-UTRAN and UTRAN
1526730849	L.IRATHO.BlindHO.E2G.MMEAbnormRsp	Number of responses with abnormal causes received by the eNodeB from the MME during inter-RAT blind handover executions from E-UTRAN to GERAN	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020	PS Inter-RAT Mobility between E-UTRAN and GERAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728377	L.IRATHO.E2W.PrepAttOut.PLMN	Number of inter-RAT handover attempts from E-UTRAN to WCDMA network for a specific operator	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033 LOFD-001036 LOFD-001037 TDLOFD-001036 TDLOFD-001037 LOFD-070206	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier Hybrid RAN Sharing

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728378	L.IRATHO.E2W.ExecAttOut.PLMN	Number of inter-RAT handover executions from E-UTRAN to WCDMA network for a specific operator	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033 LOFD-001036 LOFD-001037 TDLOFD-001036 TDLOFD-001037 LOFD-070206	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier Hybrid RAN Sharing

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728379	L.IRATHO.E2W.ExecSuccOut.PLMN	Number of successful inter-RAT handovers from E-UTRAN to WCDMA network for a specific operator	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033 LOFD-001036 LOFD-001037 TDLOFD-001036 TDLOFD-001037 LOFD-070206	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier Hybrid RAN Sharing

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728380	L.IRATHO.E2G.PrepAttOut.PLMN	Number of inter-RAT handover attempts from E-UTRAN to GERAN for a specific operator	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 LOFD-001023 LOFD-001034 TDLOFD-001020 TDLOFD-001023 TDLOFD-001034 LOFD-001036 LOFD-001037 TDLOFD-001036 TDLOFD-001037 LOFD-070206	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier Hybrid RAN Sharing

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728381	L.IRATHO.E2G.ExecAttOut.PLMN	Number of inter-RAT handover executions from E-UTRAN to GERAN for a specific operator	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 LOFD-001023 LOFD-001034 TDLOFD-001020 TDLOFD-001023 TDLOFD-001034 LOFD-001036 LOFD-001037 TDLOFD-001036 TDLOFD-001037 LOFD-070206	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier Hybrid RAN Sharing

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728382	L.IRATHO.E2G.ExecSuccOut.PLMN	Number of successful inter-RAT handovers from E-UTRAN to GERAN for a specific operator	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 LOFD-001023 LOFD-001034 TDLOFD-001020 TDLOFD-001023 TDLOFD-001034 LOFD-001036 LOFD-001037 TDLOFD-001036 TDLOFD-001037 LOFD-070206	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier Hybrid RAN Sharing

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728389	L.IRATHO.BlindHO.E2W.PrepAttOut.PLMN	Number of inter-RAT blind handover attempts from E-UTRAN to WCDMA network for a specific operator	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033 LOFD-001036 LOFD-001037 TDLOFD-001036 TDLOFD-001037 LOFD-070206	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier Hybrid RAN Sharing

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728390	L.IRATHO.BlindHO.E2G.PrepAttOut.PLMN	Number of inter-RAT blind handover attempts from E-UTRAN to GERAN for a specific operator	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 LOFD-001023 LOFD-001034 TDLOFD-001020 TDLOFD-001023 TDLOFD-001034 LOFD-001036 LOFD-001037 TDLOFD-001036 TDLOFD-001037 LOFD-070206	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier Hybrid RAN Sharing

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728391	L.IRATHO.BlindHO.E2W.ExecAttOut.PLMN	Number of inter-RAT blind handover executions from E-UTRAN to WCDMA network for a specific operator	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033 LOFD-001036 LOFD-001037 TDLOFD-001036 TDLOFD-001037 LOFD-070206	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier Hybrid RAN Sharing

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728392	L.IRATHO.BlindHO.E2W.ExecSuccOut.PLMN	Number of successful inter-RAT blind handover executions from E-UTRAN to WCDMA network for a specific operator	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001022 LOFD-001033 TDLOFD-001019 TDLOFD-001022 TDLOFD-001033 LOFD-001036 LOFD-001037 TDLOFD-001036 TDLOFD-001037 LOFD-070206	PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN SRVCC to UTRAN CS Fallback to UTRAN RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier Hybrid RAN Sharing

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728393	L.IRATHO.BlindHO.E2G.ExecAttOut.PLMN	Number of inter-RAT blind handover executions from E-UTRAN to GERAN for a specific operator	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 LOFD-001023 LOFD-001034 TDLOFD-001020 TDLOFD-001023 TDLOFD-001034 LOFD-001036 LOFD-001037 TDLOFD-001036 TDLOFD-001037 LOFD-070206	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier Hybrid RAN Sharing

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728394	L.IRATHO.BlindHO.E2G.ExecSuccOut.PLMN	Number of successful inter-RAT blind handover executions from E-UTRAN to GERAN for a specific operator	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 LOFD-001023 LOFD-001034 TDLOFD-001020 TDLOFD-001023 TDLOFD-001034 LOFD-001036 LOFD-001037 TDLOFD-001036 TDLOFD-001037 LOFD-070206	PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN SRVCC to GERAN CS Fallback to GERAN RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier Hybrid RAN Sharing

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526729503	L.RRCRedirection.E2W.PLMN	Number of redirections from E-UTRAN to WCDMA network for a specific operator	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-001033 TDLOFD-001019 TDLOFD-001033 LOFD-001036 LOFD-001037 TDLOFD-001036 TDLOFD-001037 LOFD-070206	PS Inter-RAT Mobility between E-UTRAN and UTRAN CS Fallback to UTRAN PS Inter-RAT Mobility between E-UTRAN and UTRAN CS Fallback to UTRAN RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier Hybrid RAN Sharing
1526729504	L.RRCRedirection.E2G.PLMN	Number of redirections from E-UTRAN to GERAN for a specific operator	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 LOFD-001034 TDLOFD-001020 TDLOFD-001034 LOFD-001036 LOFD-001037 TDLOFD-001036 TDLOFD-001037 LOFD-070206	PS Inter-RAT Mobility between E-UTRAN and GERAN CS Fallback to GERAN PS Inter-RAT Mobility between E-UTRAN and GERAN CS Fallback to GERAN RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier Hybrid RAN Sharing

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526730042	L.IRATHO.E2W.NoNRT.PLMN	Number of inter-RAT handover initiation failures from E-UTRAN to WCDMA network because the adjacent cell is not on the NRT of the source cell for a specific operator	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001019 LOFD-002002 TDLOFD-001019 TDLBFD-002002 LOFD-001036 LOFD-001037 TDLOFD-001036 TDLOFD-001037 LOFD-070206 LOFD-001022 TDLOFD-001022	PS Inter-RAT Mobility between E-UTRAN and UTRAN Inter-RAT ANR PS Inter-RAT Mobility between E-UTRAN and UTRAN Transport Channel Management RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier Hybrid RAN Sharing SRVCC to UTRAN SRVCC to UTRAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526730043	L.IRATHO.E2G.NoNRT.PLMN	Number of inter-RAT handover initiation failures from E-UTRAN to GERAN because the adjacent cell is not on the NRT of the source cell for a specific operator	Multi-mode: None GSM: None UMTS: None LTE: LOFD-001020 LOFD-002002 TDLOFD-001020 TDLBFD-002002 LOFD-001036 LOFD-001037 TDLOFD-001036 TDLOFD-001037 LOFD-070206 LOFD-001023 TDLOFD-001023	PS Inter-RAT Mobility between E-UTRAN and GERAN Inter-RAT ANR PS Inter-RAT Mobility between E-UTRAN and GERAN Transport Channel Management RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier RAN Sharing with Common Carrier RAN Sharing with Dedicated Carrier Hybrid RAN Sharing SRVCC to GERAN SRVCC to GERAN
1526728744	L.UECNTX.Release.DurMeasU	Number of UE context releases during WCDMA network measurement	Multi-mode: None GSM: None UMTS: None LTE: LBFD-002008 LOFD-001019 TDLBFD-002008 TDLOFD-001019	Radio Bearer Management PS Inter-RAT Mobility between E-UTRAN and UTRAN Radio Bearer Management PS Inter-RAT Mobility between E-UTRAN and UTRAN

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Counter ID	Counter Name	Counter Description	Feature ID	Feature Name
1526728745	L.UECNTX.Release.DurMeasG	Number of UE context releases during a GERAN measurement	Multi-mode: None GSM: None UMTS: None LTE: LBFD-002008 LOFD-001020 TDLBFD-002008 TDLOFD-001020	Radio Bearer Management PS Inter-RAT Mobility between E-UTRAN and GERAN Radio Bearer Management PS Inter-RAT Mobility between E-UTRAN and GERAN

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11 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

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12 Reference Documents

1. 3GPP TS 36.331, "RRC Protocol Specification"
2. 3GPP TS 36.300, "E-UTRAN Overall description"
3. 3GPP TS 36.423, "X2 Application Protocol"
4. 3GPP TS 36.413, "S1 Application Protocol"
5. 3GPP TS 23.203, "Policy and charging control architecture"
6. 3GPP TS 36.104, "Base Station (BS) radio transmission and reception"
7. 3GPP TS 23.401, "General Packet Radio Service (GPRS) enhancements for Evolved Universal Terrestrial Radio Access Network (E-UTRAN) access"
8. *Idle Mode Management*
9. *ANR Management*
10. *MLB*
11. *ICIC*

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